Annual Report
2017-18

भाकृअनुप
ICAR

ICAR- Indian Institute of Vegetable Research

(An ISO 9001:2015 Certified Institute)

(Indian Council of Agricultural Research)

Varanasi - 221 305

INDIAN INSTITUTE OF VEGETABLE RESEARCH

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PREFACE

The vegetable scenario of the country is changing rapidly, both in terms of production and productivity covering 41.3 per cent of total horticulture cropped area and contributing 59.3 per cent of production. The focused attention and high priority accorded to the research and development of vegetables in the country has led to increase in production and productivity and opened new vistas for export of fresh and processed vegetable products. The Government of India too has identified vegetable crops as a means of diversification for making agriculture more profitable by efficient soil management through 'Soil Health Card' Scheme, efficient land and water use through 'More Crop Per Drop' Scheme and creating skilled employment for rural masses.

The research and development activities of ICAR-Indian Institute of Vegetable Research are being carried out under 06 mega-programmes, namely Integrated Gene Management, Seed Enhancement, Productivity Enhancement through Better Resource Management, Plant Health Management, Post-Harvest Management and Value Addition and Prioritization of R&D Needs & Impact Analysis of Technologies Developed by IIVR and 20 externally funded projects. Presently, the institute has research programme on 42 vegetable crops with a major focus on solaneceous, leguminous, cucurbitaceous, cruciferous, malvaceous and root crops along with underutilized and aquatic vegetables.

Out of 64 improved varieties/ hybrids developed so far in 18 different vegetable crops, two varieties in tomato viz., Kashi Amul carrying Ty-3 gene with high resistance against ToLCV for cultivation in Karnataka, Tamil Nadu and Kerala and Kashi Aman for cultivation in Punjab, UP, Bihar and Jharkhand was notified by CVRC during 2017-18. Under biotechnological interventions, CRISPPR-Cas9 genome editing in tomato and okra has been initiated, Agro-infectious clone has been developed for rapid screening of chilli against chilli leaf curl disease. Bio- formulation of PGPR having multicide action against pest, disease and nematode besides growth promotion activity have been identified which can be stored at ambient temperature upto 120 days. Self-life extension of capsicum using biopolymer under MAP has been standardized. The Institute has also produced 400.70 quintals of TL seeds and 32.89 quintals of breeder seed of developed vegetable varieties and supplied to different stakeholders. Non-exclusive license of 04 hybrids and 06 open-pollinated varieties in chilli, brinjal, okra, tomato and cowpea of worth Rs. 19.6 lakh have been executed with 14 private seed companies for further multiplication and marketing.

The institute has organized 278 training programme for 7060 stakeholders which includes 01 International training for 21 executives from 11 Afro-Asian countries. Efforts have also been made for transfer of technologies in 62 adopted villages under TSP Component, Farmer FIRST Programme, Mera Gaon Mera Gaurav and Sansad Adarsh Gaon in Varanasi, Mirzapur, Chandauli, Ghazipur, Jaunpur and Mau districts in Uttar Pradesh and East Champaran district in Bihar through need based training, farmers' meetings and demonstrations of vegetables along with kitchen garden which showed a positive impact in the region in terms of quality, productivity and market value.

At this juncture, I express my deep sense of gratitude and reverence to the Dr. T. Mohapatra, Secretary DARE & Director General, ICAR, Dr. Anand Kumar Singh, DDG (Horticultural Science) and Dr. Ashok Kumar Singh, DDG (Agricultural Extension), ICAR for their continuous support, encouragement and guidance to meet the research, extension and physical targets of the institute during 2017-18. All the scientists, technical, administrative and other staffs of our vegetable family are duly acknowledged for their untiring cooperation, coordination, compilation of information and finally bringing out this document

Varanasi
June 30, 2018

(Bijendra Singh)
Director

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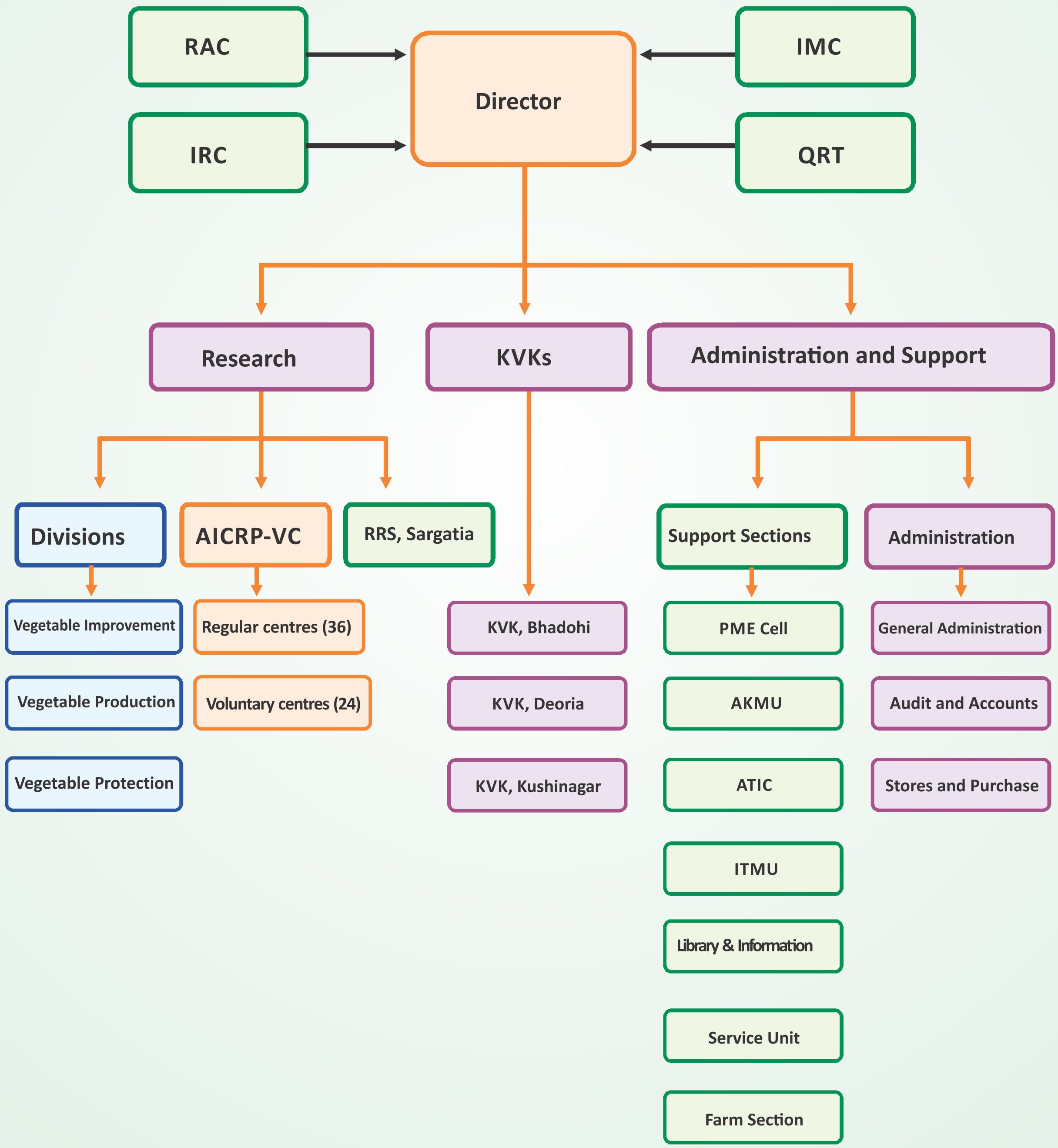
ABBREVIATIONS

a.i.	Active Ingredient
AICRP(VC)	All India Coordinated Research Project (Vegetable Crops)
AIR	All India Radio
ASCI	Agriculture Skill Council of India
ATIC	Agricultural Technology Information Centre
ATMA	Agricultural Technology Management Agency
AU	Astronomical Unit
BOLD	Barcode of Life Database
B-S PE	Black-Silver Polyethylene Mulch
CAPS	Cleaved Amplified Polymorphic Sequences
CD	Critical Difference
CDD	Conserved Domain Database
CMS	Cytoplasmic Male Sterility
CT	Conservation Tillage
CTC	Co-toxicity Coefficient
CV	Coefficient of Variation
DAI	Days After Inoculation
DAS	Days After Sowing
DAT	Days After Transplanting
DDG	Deputy Director General
DFF	Days Required to First Flowering
DNA	Deoxyribonucleic Acid
DS	Drought Stress
DSI	Drought Sensitivity Index
DTPA	Diethylene Triamine Pentaacetic Acid
DW	Dry Weight
DWR	Directorate of Weed Research
EC	Emulsifiable Concentrate
EDTA	Ethylene Diamine Tetraacetic Acid
EPN	Entomopathogenic Nematodes
FD	Fruit Diameter
FL	Fruit Length
FLD	Front Line Demonstration
FSB	Fruit & Shoot Borer
FW	Fresh Weight
GDD	Growing Degree Days
GDP	Gross Domestic Product
GMS	Genetic Male Sterility
GMV	Golden Mosaic Virus
HAT	Hours After the Treatment
IAA	Indole Acetic Acid
IC Numbers	Indigenous Collection Numbers
ICAR	Indian Council of Agricultural Research
IIVR	Indian Institute of Vegetable Research
INLFH	Inter Node Length at First Harvest
IRM	Insecticide Resistance Management
KVK	Krishi Vigyan Kendra
LC50	Lethal Concentration 50
MI	Mycelial Growth Inhibition

MTA	Material Transfer Agreement
MtCOI	Mitochondrial Cytochrome Oxidase I
NAIP	National Agricultural Innovation Project
NBAIR	National Bureau of Agricultural Insect Resources
NFP	Number of Fruits per Plant
NNFH	Number of Node at First Harvest
NPTC	Network Project on Transgenic Crops
NT	Not Tested
NUE	Nutrient Use Efficiency
OC	Organic Carbon
OD	Optical Density
OFT	On Farm Trials
PBNV	Peanut bud necrosis virus
PCR	Polymerase Chain Reaction
PDI	Per cent Disease Index
PHI	Pre Harvest Interval
PLW	Physiological Loss in Weight
PPM	Parts Per Million
PPOC	Per cent Protection Over Control
PPP	Public Private Partenership
PR	Percent Reduction
PRP	Proline Rich Protein
PTC	Pre-Treatment Count
QTL	Quantitative Trait Loci
R&D	Research and Development
RAPD	Randomly Amplified Polymorphic DNA
RBD	Randomized Block Design
RH	Relative Humidity
RILs	Recombinant Inbred Lines
RNA	Ribonucleic Acid
RT	Reduced Tillage
Sc	Sclerotia
SD	Standard Deviation
SDI	Sub-surface Drip Irrigation
SEM	Standard Error of Mean
SNPs	Single Nucleotide Polymorphism
SPS	Single Plant Selection
SR	Survival Rate
SSDI	Sub Surface Drip Irrigation
SSR	Simple Sequence Repeat
TI	Tolerance Index
ToLCV	Tomato Leaf Curl Virus
TSS	Total Soluble Sugars
WBNV	Watermelon bud necrosis virus
WEY	Wheat Equivalent Yield
WFPP	Weight of Fruit Per Plant
WG	Water Dispersible Granules
WUE	Water Use Efficiency
YVMV	Yellow vein mosaic virus
ZT	Zero Tillage

ORGANOGRAM



RAC
IRC
Director
IMC
QRT
Research
KVKs
Administration and Support
Divisions
AICRP-VC
RRS, Sargatia
Support Sections
Administration
Vegetable Improvement
Vegetable Production
Vegetable Protection
Regular centres (36)
Voluntary centres (24)
KVK, Bhadohi
KVK, Deoria
KVK, Kushinagar
PME Cell
AKMU
ATIC
ITMU
Library & Information
Service Unit
Farm Section
General Administration
Audit and Accounts
Stores and Purchase

Executive Summary

The research, extension and development activities of ICAR- Indian Institute of Vegetable Research, Varanasi are being carried out under six Mega-programmes, viz. (1) Integrated Gene Management (2) Seed Enhancement in Vegetables (3) Productivity Enhancement through Better Resource Management (4) Post Harvest Management and Value Addition (5) Prioritization of R&D Needs and Impact Analysis of Technologies Development by IIVR (6) Integrated Plant Health Management and 20 externally funded projects. In each Mega- programme a number of projects have been formulated with specific objectives.

Under the project on Management of vegetable genetic resources including under-utilized crops, a total of 6507 accessions of 44 major and minor vegetable crops (including 188 accessions of 46 wild/related species in 11 vegetables) were maintained and 336 new germplasm in 25 vegetable crops were augmented. A total of 18 breeding materials (13 in radish and 5 in carrot) developed were assigned IC numbers from NBPGR, New Delhi. Besides, 624 accessions of 22 sp. vegetable crops and fungus cultures were shared with 28 organizations for use in research after signing the material transfer agreement (MTA).

In genetic improvement of solanaceous vegetables, the segregating lines, VRNRT-6, VRNRT-5, VRNRT-3 and VRNRT-4 in tomato were superior for maximum number of fruits and high yield, whereas, the lines VRNRT-7 and VRNRT-8 were superior for quality and nutritional values. In cherry tomato, VRCTH-17-64, VRCTH-17-46, VRCTH-17-51 hybrids were superior for high yield and TSS, while advance lines in red cherry tomato VRCRT-8 and VRCRT-9 were better for nutritional attributes. Yellow fruited advance lines VRCYT-11 in cherry type and VRTKB-8 in tomato were superior for high β-carotene, ascorbic acid and acidity. In brinjal, two hybrids IVBHL-22 (long fruit) and IVBHR-18 (round fruit) and two advance lines IVBL-27 (long fruit) and IVBR-19 (round fruit) were selected for multi-location testing. Kashi Himani (IVBL-26), a medium long, white fruited advance line was identified through Institute Technology Identification Committee (ITIC) and proposal was submitted for release by State Varietal Release Committee (SVRC) of Uttar Pradesh. In chilli, an inbred line (PT12-3xBJ-PT-13-3-2, referred as VRC-14) derived from the cross of PT-12-3 x Bhut Jolokia and a CMS based hybrid, CCH-10 were identified for multi-locational testing. The hybrid combinations involving VR-339 as the pollen parent were found to be tolerant for leaf curl disease under field condition along with high pungency in fruits. The F_5 families derived from an inter-specific cross of chilli such as GT-125, GT-127 and GT-246 showed good potential with respect to ChiLCV resistance.

Among leguminous vegetables, advance lines in cowpea, namely 112-4, 167-2 and 167-3 were superior for yield attributes and tolerance to cowpea golden mosaic virus under field conditions. The varieties Kashi Kanchan and Kashi Shyamal were superior for antioxidants activities and secondary metabolites, respectively. In vegetable pea, a multi-podded genotype VRPM-901-5 produced four or five flowers in each flowering node in its 60% population. In mid-maturity group, three advance lines VRPM-915, VRPM-901-3 and VRPM-903 were promising for earliness and yield attributes. In snap bean, EC792393 and FMGCV1187 were promising for yield and quality of pods. In dolichos bean, two advance lines in bush-type VRBSEM-206 and VRBSEM-207 were promising for yield and DYMV resistance.

In gourds, advance lines of bitter gourd VRBTG-3 and OBG-11-1 in small segment, VRBTG-15 and VRBTG-12 in medium segment, VRBTG-47-1 and VRBTG-5 in long segment, and VRBTG-10 in extra-long segment were promising for yield attributes. Out of 23 bitter gourd hybrids evaluated for different attributes, 9 combinations were promising for yield attributes. Three genotypes IC212504, VRBTG-10 and VRBTG-11-1 showed mild-resistance towards root knot nematode. In bottle gourd, maximum yield was recorded in line VRBG-61. Among 41 bottle gourd hybrids evaluated for different attributes, the hybrids were 2 in long fruited group, 2 in round fruited group, and 2 in segmented leaf group were promising for yield attributes. Further, a genotype VRBOG-63-Sel-02 of bottle gourd having small and good quality fruits was suitable for winter season. In pointed gourd, 139 female clones were evaluated for traits of economic importance and a long fruited genotype VRPG-215 was promising with fruit yield potential of 12.50 kg/vine. Diversity analysis of 72 hybrid female clones derived from two cross-combinations (Kashi Alankar × Male-1 and Kashi Suphal × Male-1) revealed 61.42% of the variation through PC1 and PC2. A putative seedless fruit specific marker *i.e.* 850 bp amplicon from UBC-840 was identified linked to seeded clone. Among 32 advanced lines of sponge gourd, 9 were promising for various horticultural traits, and showed tolerance to downy mildew and virus diseases under field condition. In teasle gourd, maximum fruit yield per plant was harvested in VRSTG-6. The genotypes VRSEG-9, VRSEG-7 and VRISEG-4 of spine gourd, and VRIG-4, VRIG-3 and VRIG-6 of ivy gourd excelled for yield and quality.

Under genetic improvement of melons, pumpkin and cucumber, on the basis of overall performance (green fruit yield, earliness, fruit shape and size), advance lines in pumpkin, VRPK-18-01, VRPK-230, VRPK-222-2-1 and VRPK-09-01, and hybrids VRPKH-16-06, VRPKH-16-05 and VRPKH-16-04 were promising. In cucumber, advance lines VRCU-Sel.-12-02 and VRCU-Sel.-12-03 in summer squash VRSS-65 and VRSS-66 in muskmelon VRMM-170 and VRMM-186 in watermelon VRW-514 and VRW-516 in round melon VRM-1 and VRM-5 and in long melon VRLM-01 and VRLM-40 were found superior for fruit colour, appearance and yield. Six cross-combinations of muskmelon involving two monoecious × monoecious, one andromonoecious × hermaphrodite and three andromonoecius × monoecious were attempted to widen the genetic base. Further, sixty RILs (F_7) from cross Kashi Madhu × B-159 were evaluated for various traits of economic importance. Among them the RILs 711, 272, 320 and 523 were distinctly placed from the others.

Total 64 accessions of wild relatives of okra were evaluated for YVMV and OLCV disease reaction. The advance lines VRO-110, VRO-112-1 and VRO-113, and hybrids 307-10-1 × VRO-115, VROB-178 × 416-10-1 and VRO-115 × VRO-110 were promising for yield and viral diseases (YVMV and OLCV) reaction during *kharif* season. VRO-109 was screened against begomovirus using specific primers, and 9 plants out of 24 showed symptoms and 18 amplified band with Rojas Primer at 88 day, when crop was harvested. A total of 260 F_2 seeds of cross VROR-156 × VRO-5 (red fruited × green fruited) were sown which followed 3:1 ratio for red and green colour fruits. Two promising lines Kashi Chaman (VRO-109), Kashi Lalima (VROR-157) and a hybrid Kashi Shristi (VROH-12) were identified through the ITIC and proposal was submitted for release by SVRC of Uttar Pradesh.

Under cole crops and root crops, among 52 genotypes of cauliflower, two genotypes VRCF-86 and VRCF-201 were potential yielder in mid-October maturity; VRCF-50, VRCF-102 and VRCF-113 in mid-November maturity; and VRCF-104, VRCF-22 and VRCF-202 in late-November to mid-December maturity group. A genotype VRKALE-1 of kale induced bolting and flowering, and sets seeds in the North Indian plain. In root crops, 61 genotypes of carrot having different root colour (red, orange, black, yellow and rainbow) were evaluated, and VRCAR-186 and VRCAR-201for (red root); VRCAR-126 and VRCAR-89-1 (black root); VRCAR-91-1 and VRCAR-91-2 (orange root); and VRCAR-153 and VRCAR-178 (yellow root) were promising for root yield and quality. A black colour carrot genotype, VRCAR-126 was a good source of antioxidants, rich in anthocyanins (250-280 mg/100 g) and able to produce 65-70 kg/ha anthocyanins. In radish, among 56 genotypes, VRRAD-150 and VRRAD-203 (white root); VRRAD-131-2 and VRRAD-170 (red root); and VRRAD-131 and VRRAD-134 (purple root) were promising for yield and root quality. In summer trial of radish, higher root yield (230-270 q/ha) was obtained in genotype VRRAD-203 and VRRAD-200 during mid-April to May. To facilitate heterosis breeding in radish, two Ogura-CMS lines i.e. VRRAD-198 and VRRAD-201 were developed at ICAR-IIVR through back-crossing. For transfer of CMS trait in cole crops and root crops, >45 BC populations have been advanced as BC_1-BC_5 stages in various backgrounds through backcrossing in cauliflower, broccoli, radish and carrot.

Under transgenic and regeneration protocols, *in-planta* transformation of okra in the cultivar Kashi Kranti and transformation protocol were optimized. The presence of *npt* II gene confirmed in only 3 T_1 events by PCR using *npt* II specific primers. Selfing was performed on fully grown plants for multiplication and T_1 seeds of mature selfed fruits from three plants were harvested and stored. In regeneration studies of cauliflower genotype VRCF-50, maximum frequencies of callus induction i.e. 41.11 and 64.45% were observed in MSB5 medium supplemented with BAP (8.9 µM) and TDZ (9.1 µM) alone, respectively. Moreover, the frequency of callus induction increased to 55.56% on MSB5 medium containing 8.9 µM BAP along with 0.5 µM NAA, and increased to 85.56% on MSB5 medium containing 9.1 µM TDZ augmented with 0.5 µM NAA. Highest rooting response (71.11%), the maximum number of roots per shoot (7.67), and the maximum root length (11.33 mm) were obtained on half strength MS medium supplemented with 4.9 µM IBA.

Under biotechnological interventions for improvement of selected vegetable crops, CRISPR-Cas9 mediated genome editing in tomato was undertaken. To progress the work towards the development of vector for plant transformation, the guide RNA construct targeting replicase (*rep*) gene of Tomato leaf curl virus (ToLCV) was artificially synthesized. The target region of replicase gene was identified manually by multiple sequence alignment of five major strains of ToLCV to achieve broad spectrum resistance. For the purpose of transgenic male sterility in tomato, pollen specific gene *SlCRK*1 was selected as a target gene and gRNA construct was designed with the help of CRISPR direct tool. Further, a total of 30 hybrids developed based on combinations of *Ty-2* and *Ty-3* lines were evaluated, superior combinations were VRT16-11 X VRT16-12, VRT16-12 X VRT18-1 and VRT16-7 X VRT16-13. Kashi Amul, a semi-determinate tomato variety has been notified for release in zone VIII having yield potential of 50-60 tonnes/ha. In brinjal, a total of 258 markers (SSR, STMS, SCoT and ISSR) were found polymorphic between parental lines, which were used for genotyping of RILs. Further, 2182 and 2348 genes were identified and classified into 60 transcription factor families in the transcriptome sequences of *S. melongena* and *S. incanum*, respectively. To understand the architecture of leucine-rich repeat family protein in Ramnagar Giant (*S. melongena*) and W-4 (*S. incanum*), application of SWISS-MODEL homology -a modelling software is in progress.

In leafy vegetables, basella genotypes VRB-17 with trailing type growth habit and VRB-31 with bushy growth habit were promising. The photometric results obtained for total betalains exhibited maximum pigment content in ripened fruits of VRB-30 (200.93 mg/100 g FW) and VRB 3 (150.87 mg/100 g FW), which could be exploited as a source of betalain for industrial use. Charcoal rot disease caused by *Macrophomina phaseolina* in basella has been reported for the first time from ICAR-IIVR and the same was confirmed using elongation factor (EF-1a) gene specific primers TEF1-983F and TEF1-2218R. The quinoa, genotypes were assessed for horticultural traits and yield, which showed wide variability for various traits. The biomass yield of leafy chenopod or bathua was realized 408, 494 and 369 q/ha, respectively for VRCHE-2, VRCHE-4 and Pusa Bathua-1. Among moringa germplasms, the genotypes VRMO-1 to VRMO-15 started flowering in 140-150 days after sowing which is 15-40 days early to PKM-1 and PKM-2.

The genotype VRWC-1 of water chestnut and VRWS-1, VRWS-4, VRWS-8 and VRWS-9 of water spinach were promising for various horticultural traits. A total of 47 germplasm of baby corn were screened for banded sheath leaf blight (BSLB) caused by *Rhizoctonia solani* under field as well artificial conditions, which revealed that lines BC-13 and BC-62 were resistant and BC-16 was moderately resistant.

Under seed enhancement in vegetables, the overall seed production programme (breeder & truthful labelled) was undertaken in 39 varieties of 21vegetable crops. A total of 30358.40 kg seed of different vegetables were produced, including 3289.15 kg breeder seed. A quantity of 2876.00 kg breeder seed produced against the target of 2058.00 kg as per the National indents. In addition, 414.50 kg breeder seeds and 14.85 kg hybrid seed of different varieties/hybrids of ICAR-IIVR were also produced. Additionally, TL seed and planting materials were also produced at RRS Sargatia, where 3894.00 kg seeds of 13 varieties in 11 vegetable crops, 198 q of paddy, 525 q of lentil, 125 q of turmeric and 75 q of elephant foot yam were produced. A total of 1,28,979 seed packets were prepared for sale/distribution. In priming, coating, pollination and ovule conversion studies, seed treatment of basella with 200 ppm nano-Zinc particles exhibited the potential ability to alleviate the lead toxicity. The application of 200 ppm nano-Zinc + 200 ppm nano-Iron + 200 ppm NAA enhanced the conversion of ovules to seed as well as seed quality of okra cv Kashi Kranti. In pollination studies, application of 5% sugar + 5% jaggery + multivitamin + 200 ppm nano-Zinc + 0.1% boron boosted seed yield and quality of bottle gourd and sponge gourd. Seed priming with 200 ppm iron-nano particles enhanced the germination, root growth, chl a, chl b, total chlorophyll and carotenoids, and reduced proline and H_2O_2 in the stressed seedlings of tomato and amaranth. Moringa seed without wing kept in between rolled towel paper produced healthy and vigorous seedling.

Studies were carried out for standardizing technologies for protected vegetable production in which 4 parthenocarpic cucumber hybrids (Pant Cucumber 2 & 3, Multi-star and King-star) and one open-pollinated cultivar (Damini) were evaluated under naturally ventilated polyhouse. The maximum number of fruits (25.33/ plant) and yield (3.35 kg/plant) was obtained in cultivar Multi-star followed by King-star (2.59 kg/ plant). In addition to this, three varieties of tomato *i.e.* two indeterminate hybrids-NS 4266 and GS 600 and one determinate open pollinated cultivar Kashi Aman were evaluated in two protected structures *i.e.* naturally ventilated polyhouse and net house conditions. Maximum plant height (267.26 cm), fruit number (97.0/ plant) and yield (13.66 kg/plant) was recorded with NS 4266 with an increase of 135.5% yield over GS-600. In net house condition, also this cultivar noticed an increase of 132% in yield over GS-600. Three hybrid capsicums *i.e.* Almirante, Swarna and Natasha were grown both in naturally ventilated polyhouse and nethouse condition. The maximum yield of 2.19 kg/plant was reported in cultivar Swarna followed Almirante (1.47 kg/plant) and Natasha (1.24 kg/plant). The yields of Swarna, Almirante and Natasha in nethouse condition were 1.98, 1.76 and 1.56 kg/plant, respectively. Under both protected structures, Swarna produced maximum yield over other cultivars, however it performed better under naturally ventilated polyhouse and registered 49% and 76.6% higher yields, respectively over Almirante and Natasha.

Under residual effect of N applied in tomato on growth and yield of subsequent crop cowpea cv. Kashi Nidhi, the maximum plant height (88.40cm), number of pods per plant (38.50), pod weight (17.70g) and yield (142.67q/ha) were obtained with residual effect of 200 kg N/ha. However, maximum nitrogen use efficiency (0.4368q/kg N) was noticed with 160kgN/ha. Under a study on vegetable based cropping system, ten different cropping systems were evaluated and the system productivity was compared. The highest productivity during kharif season was obtained with brinjal crop having 202.80 q/ha of Rice equivalent yield. During winter season, productivity was compared on the basis of wheat equivalent yield. The highest WEY was obtained with Pea+ Radish (109.62 q/ha) crop followed by Tomato crop (108.30 q/ha).

Effect of organic and inorganic management systems on vegetable productivity, quality and soil health revealed that highest yield of bottle gourd (27.66 t/ha) and okra (7.48 t/ha), was recorded with application of FYM @ 25t/ha, which was *at par* with inorganic fertilizer application. In cowpea the highest yield (6.96 t/ha) was recorded with application of NADEP compost @ 25t/ha. Organic management systems recorded higher organic carbon and available N in soil. Among the different organic management systems, the highest increase in OC was noted under application of NADEP compost @ 25t/ha. There was decline in OC in absolute control treatment, while there was no change in OC of inorganic treatment. The total microbial activity in terms of fluorescein diacetate hydrolysis was also higher under organic systems as compared to inorganic. The soil microbial activity was minimum under inorganic and maximum under application of FYM @ 25t/ha. The study on nematode population dynamics in Okra revealed that % reduction of nematode reproduction/ population was more by FYM (76-87%) followed by vermicompost (76-82 %) and NADEP compost (69-74%) over absolute control. During rabi season, increasing rates of organic sources increased the total yield of brinjal. The highest dose of all the three organic sources produced fruit yield of brinjal comparable to the yield level obtained with application of recommended dose of inorganic fertilizer. However, marketable yield of brinjal in inorganic source was significantly higher than organic sources due to less infestation of fruit borer. In pea, the organic sources produced significantly higher green pod yield over control and inorganic source. It was also observed that increasing dose of all the three organic sources did not increase the green pod yield. In French bean the highest green pod yield was obtained with application of 25t/ha FYM which was significantly more than inorganic source. Among the three organic sources FYM was superior. In cabbage, all the three sources produced significantly higher yield over control. The highest yield was obtained with application of NADEP compost @25t/ha or Vermicompost @10t/ha which was comparable with inorganic source. In cabbage, it was observed that increasing dose of all the three sources increased the yield. The highest broccoli yield was recorded with application of 25t/ha NADEP compost which was *at par* to inorganic. The quality of vegetables in terms of vitamin C content was better under organic system as compared to inorganic system in brinjal, pea and cabbage.

Resource conservation studies in vegetable production revealed that the maximum yield of 9.63 t /ha, in cowpea, 11.55 t/ha in okra and 53.84 t/ha in tomato was obtained with ZT which was significantly superior to conventional tillage. Zero tillage increased the OC content of the soil by 13.33 % and reduced the Bulk Density of the soil. The net return and benefit cost ratio was also maximum in this treatment. The economics was better in the ZT due to increased yield and also lower cost of cultivation. Residue retention/ incorporation, in general improved the yield in all the crops. The yield increase in cowpea, okra and tomato was 25.63, 9.52 and 4.03 % due to residue retention/ incorporation over its removal. The increase in organic carbon (%) of soil due to residue incorporation/ retention was 6.5% higher over residue removal.

Drip fertigation scheduling studies carried out in tomato (cv. Kashi Aman) to optimize method and quantity of fertilizer application revealed that maximum number of fruits (54.33/plant), fruit weight (147.22 g) and fruit yield (6.11 kg/plant and 68.86 t/ha) was obtained with 100% NPK through WSF. In this treatment, 83.1% and 53.7% higher yield was recorded, respectively over control and 100% NPK application through soil. The maximum nutrient use efficiency (2.375 q yield/ kg NPK) was also reported under this treatment.

In spring-summer okra, an experiment was conducted to optimize drip irrigation scheduling with or without mulch which revealed that drip irrigation and mulching significantly affected the okra production. Among the irrigation schedulings, drip irrigation daily or 2-day intervals had significant enhancement in number of fruits (30.89 and 30.78/plant), fruit yield (546.0 and 525.22 g/plant and 103.92 & 101.99 q/ha). Among the mulch used, the maximum plant height (91.03 cm), fruit number (31.92/ plant) and yield (577.58 g/plant, 100.98 q/ha) was observed under organic mulching. Under organic mulch, 34.2% and 7.3% higher yield were registered over unmulched control and black-silver mulch, respectively. Studies conducted to assess the performance of vegetables under varying levels of water application through sub-surface drip irrigation, showed that the yield of tomato cv. Kasha Aman was maximum (50.4 t/ha) with 100 % ET, that was 54.1% higher than control. The Water Use Efficiency (WUE) enhanced 1.7-2.45 times of control /furrow irrigation that was maximum 2.316 t/ha-cm for SDI at 80% ET.

Micronutrient study in cauliflower and broccoli revealed that combined spray of B 50 ppm + Mo 25 ppm or B 100 ppm + Mo 50 ppm in broccoli, and sole spray of Mo 50 ppm thrice at 10 days intervals in cauliflower significantly enhanced the yield. Significant effect of micronutrients application on the growth and yield of bitter gourd was also recorded. The highest fruit yield of 127.3 q/ha was recorded under combined application of micronutrients which was *at par* to sole application of B and Zn @ 100 ppm alone. Besides this, some crop-group specific micronutrient formulations for Solanaceous (5 nos.) and Cole crops (4 nos.) were prepared in the laboratory and were applied three times @ 1.5g/l at 10 days intervals after 30 days after planting of the crops. Among different formulations, Micromix B proved more effective as compared to the others. There was a significant effect of micronutrients foliar application on the growth and yield of tomato. Among different micronutrient formulations, Micromix A proved slightly better recording maximum plant height (116.5 cm), number of fruits per plant (65.1) and total fruit yield (450.9 q/ha).

In the trial on Effect of post and pre-emergence herbicides on pod yield of French bean, highest pod yield was attained with weed free (13.5 t/ha),which was *at par* with application of combination of pendimethalin (pre-emergence) *fb* sodium acifluorfen 16.5 % + clodinafop- propargyl 8 % EC (post emergence) at 25 DAS (12.4 t/ha) and pendimethalin @ 750g/ha (pre-emergence) *fb* imazethapyr @ 100 g/ha (post emergence) at 25 DAS (11.9 t/ha). In cowpea, significantly lower population of grass and broad leaf weed were recorded with the application pendimethalin *fb* imazethapyr, imazethapyr + imazemox, whereas, highest pod yield was recorded with weed free (13.8 t/ha) *fb* application of pendimethalin *fb* imazethapyr (12.3 t/ha) and imazethapyr + imazemox (12.2 t/ha).

In post-harvest technology trials, shelf life of capsicum was studied in expanded polyethylene biopolymer films under modified atmospheric packaging storage (MAP) in small size polymer (325 g ±10%) and big size polymer(940 ±10%) during storage at 3 and 10°C. Oxygen concentration (20%) remained static after 5 days of MAP storage while carbon di oxide concentration ranged to 0.4-0.5% during MAP storage at 3°C. Moisture content, firmness, total chlorophyll content, total sugar content, total phenol content and antioxidant activity decreased during storage at 3°C and 10°C in both small and big size polymer.

In the project Research prioritization of vegetable crops, priority areas were identified as per the perception analysis of vegetable experts from ICAR institutes, SAUs and KVKs from all over India in the major areas of genetic resource management of vegetable crops, seed, production, protection, post-harvest and entrepreneurship development in vegetable enterprises. Impact of IIVR technologies were analysed in terms of production and productivity enhancement of improved vegetable varieties and income generation through its adoption. Several outreach programmes were also carried out in 50 adopted villages under Mera Gaon Mera Gaurav, Sansad Adarsh Gaon scheme in 8 districts of Uttar Pradesh and Bihar where nearly 5200 farmers benefitted though promotion of kitchen garden, getting improved vegetable varieties, training, exposure visit, kisan mela, gosthi and need based advisory services. Special efforts were given for low cost protected cultivation at village level and farm waste management by producing vermicompost.

Under integrated plant health management, biointensive module comprising spraying of Nimbeicidine, Bt, *Lecanicillium lecanii, Beauveria bassiana*, Nimbeicidine at 10 days intervals was found superior in terms of reducing adult red pumpkin beetle (62.93%), plume moth reduction (71.50%) and mirid bug reduction (51.88%) which was at par with chemical module.

Under toxicological investigation, the laboratory bioassay of novel molecules against *Spodoptera litura*, revealed that Indoxacarb, Chlorantraniprole were found effective at very low concentration in comparison to Emamectin benzoate, Quinalphos and Deltamethrin. In case of *Spilosoma obliqua* lab bioassay using direct spray method, it was found that Indoxacarb was found the most effective followed by Chlorantraniprole, Emamectin benzoate, Novaluron, Fenvelerate and Quinalphos at 24 HAT. Various novel insecticides were tested by three bioassay methods viz., Leaf dip, direct spray and larval dip against third instar larvae of *S. litura*. *S. litura* larvae were highly susceptible to Indoxacarb in direct spray method, Chlorantraniprole in leaf dip method and Cypermethrin in leaf dip method. For most of the insecticides, leaf dip method was found the most suitable method to determine the susceptibility of S. litura.

Under biological control, a polyphagous aphid *Rhopalosiphum nymphaeae* was identified feeding on water chestnut and lotus in Varanasi, Uttar Pradesh. In large colonies that develop on water lilies aggregate along the leaf veins and infest leaves as well. To control the polyphagous aphid *Myzus persicae* feeding cole crops commonly used neonictinoids (Imidacloprid, Thiamethoxam and Acetamiprid) and biopesticides viz., *Beauveria bassiana*, *Metarhizium anisopliae*, *Lecanicillium lecanii* were tested at half of their recommended doses and found compatible. Combination of Acetamiprid and *L. lecanii* took the lowest median lethal time (22.61 hour) with co-toxicity coefficient (CTC) value (1.08). Molecular characterization of indigenous entomopathogenic nematode strains of Uttar Pradesh was done and the identity of indigenous EPN strains was confirmed by using the ITS-rDNA region. The sequence of EPN strains such as IIVR EPN03 (MG976754), IIVR JNC01 (MH208855) and IIVR JNC02 (MH208856) showed maximum identity with *Steinernema siamkayai* (99%). Pathogenicity of indigenous entomopathogenic nematode strain, *Steinernema siamkayai* IIVR JNC01 (MH208855) compared with ICAR- NBAIR entomopathogenic strain *Heterorhabditis indica* NBAIIH38 (KX950751) against third instar larvae of *Spoladea recurvalis, Spodoptera litura* and *Spilosoma obliqua*. *Steinernema siamkayai* (MH208855) IIVR strain caused mortality between 30-85%, 60-100% and 62.5-100% in *S. recurvalis, S. litura* and *S. obliqua*, respectively, while *H. indica* caused mortality between 40-100%, 50-100% and 65- 100% in *S. recurvalis, S. litura* and *S. obliqua*, respectively at 25-200 IJ's concentration. Among the EPN species, irrespective of IJ concentration, *H. indica* caused significantly ($P < 0.05$) greater mortality only in case of *Spoladea recurvalis* compared with IIVR strain.

For the management of various fungal diseases in tomato integrated module (T4) was best for maximum seedling stand 80% and almost same 79% in T5. Weed population mostly *Echinocloa* spp. was reduced about 60 times in solarized nursery beds. Maximum marketable yield was 807.4 q/ha in T4-IDM followed by 760.7/ha in T1-chemical in comparison to control (655.3q/ha). The unmarketable yield varied from 195.2q/ha to 275.8 in different treatment modules which was about 24% of total yield. The unmarketable yield comprises of 60% bird damage, 30% insect damage and 10% diseased. Minimum diseased fruits 3.7% were recorded in T3-GAP while maximum 9.9% in control. Among diseased fruits 85.9% to 100% was infected only by early blight with minimum in chemical module (T1). The fruit rot by *Rhizoctonia solani* was recorded from first harvest to ninth harvest as second major pathogen after *A. solani*. Early blight infected fruits were recorded during 13th to 16th harvest in the month of March-April.

Under bioprospecting of microorganisms, the minimum Percent Diseaase Index of cercospora leaf spot (47.77) and highest yield (55.43 t/ha) were recorded in bottle gourd (cv. Kashi Ganga) with seed treatment @ 10 g/kg and foliar spray @1% *T. asperellum* (2.0 x 107 cfu/g) in compared to control. A total 56 out of them 33 different morphotypes of actinomycetes were recovered from vermicompost, and 23 from NADEP sample. Colony Farming Unit count was comparably higher in Vermicompost (23 X 104cfu/g) compared with NADEP (13 X 104cfu/g). Among different isolates of actinomycetes, IIVR strain N1.2 was found to be promising against *Macrophomina phaseolina*, *Sclerotium rolfsii* and *Scleritonia sclerotiorum.*

Under diagnostic and management of viral diseases, bipartite begomovirus causing mosaic disease on sponge gourd is identified as Tomato leaf curl New Delhi virus (ToLCNDV). Two distinct strains of ToLCNDV is found associated with sponge gourd causing mosaic disease based on DNA A (clone SP 1A and clone SP 22A) having 88.5% nucleotide identity among themselves. They are having maximum identity of 99% with the already reported strains from India. DNA B (clone SP 7B) had 96% identity with ToLCNDV isolate reported on snake gourd from Varanasi in BLAST analysis. In addition, it is also associated with Tomato leaf curl New Delhi alphasatellite (ToLCNDA) and Tobacco leaf curl Patna betasatellite (TbLCPB).

In Roving survey conducted in tomato fields across 11 villages in Varanasi and Mirzapur districts of Uttar Pradesh 25 out of 34 samples showing leaf curl showed positive for begomovirus. Among them 12 were infected with ToLCNDV, 3 with ToLCKV and 1 with ToLCPalV whereas none of them were infected with ToLCBV. Interestingly, mixed of more than one begomoviruses combinations such as ToLCNDV+ToLCGV, ToLCNDV+ToLCKV, ToLCNDV+ToLCGV+ToLCKV and ToLCNDV+ToLCPalV+ToLCKV were also observed. Some samples were found associated with β- and α-satellites. Symptomatic squash samples collected in IIVR farm were found infected with PRSV, CGMMV and Polerovirus in RT-PCR assay. Amplified fragment of ~1300bp from the nested PCR of brinjal little leaf sample collected from IIVR farm had 99% identity with Karnataka isolate. In phylogenetic analysis, sequence revealed phytoplasma infecting brinjal in IIVR farm belongs to 16Sr VI group. In silico virtual restriction analysis showed 16Sr VI-D is found infecting the brinjal.

Under nematode management, root knot nematode species infecting vegetable crops in ICAR-IIVR, Varanasi research farm was identified through molecular characterization by using ITS 5S-18S ribosome region specific marker 194 using molecular technique. The sequence of root knot nematode species showed maximum identity with *Meloidogyne incognita* (94%). Further, the same root knot nematode species was again confirmed by using specific SCAR marker Inc-K14-F and Inc-K14-R. Three antinemic PGPR isolates *Bacillus marisflavi* (CRB2), *Bacillus subtilis* (CRB7) and *Bacillus subtilis* (CRB9) showed for egg hatching inhibition of root knot nematode (> 90%). In okra, (cv. Kashi Pragati) under pot condition, among three PGPR isolates CRB7 and CRB2 exhibited maximum antinemic activity against root knot nematode. Further, under field condition, integrated module involving seed treatment 20g/kg of seed : soil application 5 kg/2 tonnes of vermicompost : soil drenching at 1% CRB2 and CRB7 exhibited maximum antinemic activity by reducing final nematode soil population with lesser gall index.

The seasonal incidence of cucurbit fruit fly, *Bactrocera cucurbitae* infesting bottle gourd was observed almost throughout its growth period of during April, 2017 to March, 2018 in and around Varanasi. The fruit fly population was higher during Oct-Nov and April –June months. Highest number of fruit fly (119.33 per trap) was recorded during 16thSMW (3rd week of April, 2017) followed by 43rd SMW (4th Week of October, 2017) i.e., 118/trap.

Considering on-going institute activities during current year, two varieties in Tomato viz., Kashi Amul for cultivation in Karnataka, Tamil Nadu & Kerala and Kashi Aman for cultivation in Punjab, UP, Bihar and Jharkhand was notified by CVRC along with 04 production and 02 protection technologies as well as 04 processes for selflife extension & value addition were developed and standardized. Besides, 40.0 tons truthful label seeds and 3.3 tons breeder seeds of 39 varieties in 21 different vegetable crops were also produced and provided to stakeholders. Non-exclusive licenses of 10 varieties/hybrids in tomato, brinjal, chilli, okra and cowpea were made with 14 private seed companies for further multiplication and marketing in the country. The institute organized 278 training programme including 01 international training for executives of 11 Afro-Asian countries. Under various government schemes viz., Scheduled Tribes Component, Farmer FIRST Programme, Mera Gaon Mera Gaurav and Sansad Adarsh Gaon, technological interventions through various extension means in 62 adopted villages of Eastern Uttar Pradesh and East Champaran in Bihar, showed positive impact in the region in terms of quality, productivity and market value. 101 research papers in national/international journals, 10 technical bulletin/ mannual, 09 books, 23 book chapters and 50 Popular articles 38 extension folders, were also published. National Conference on “Food and Nutritional Security through Vegetable Crops in Relation to Climate Change” and North Zone Regional Agriculture Fair, Tribal Farmers Fair were organized for the benefit of farmers and other stakeholders.

Division of Vegetable Improvement

Research Achievements

MEGA PROGRAMME 1: INTEGRATED GENE MANAGEMENT

Project 1.1: Management of Vegetable Genetic Resources

Status and augmentation of vegetable germplasm: The institute is maintaining 6,507 germplasm accessions of 44 different major and minor vegetable crops. These include, tomato (1219), brinjal (380), chilli (346), capsicum (24), French bean (214), cowpea (384), pea (506), Indian bean (129), cluster bean (230), winged bean (95), vegetable soybean (126), okra (658), cauliflower (78), kale (1), cabbage (5), broccoli (7), radish (105), carrot (90), ivy gourd (8), bitter gourd (115), bottle gourd (82), sponge gourd (98), ridge gourd (56), satputia (45), snake gourd (21), pumpkin (92), ash gourd (60), cucumber (112), watermelon (65), long melon (30), round melon (5), broad bean (130), water chestnut (4), water spinach (23), pointed gourd (139), fenugreek (21), muskmelon (174), sweet gourd (50), spine gourd (16), basella (72), amaranth (212), chenopod (161), quinoa (14), lotus (1), sweet corn (42) and baby corn (62). The institute is also maintaining 188 accessions of 46 related wild species in 11 vegetable crops.

During the year, 336 new germplasm in 25 vegetable crops were augmented to enrich the germplasm holding of the institute. These include 10 in brinjal from West Bengal and Odisha; 5 in chilli from NEH region; 7 in pea from BHU, Varanasi and a private company; 15 in cluster bean from Eastern Uttar Pradesh; 40 in vegetable soybean from ICAR-NBPGR, Bhowali; 41 in okra from Kalahandi, Nuapada, Nawarangpur district of Odisha, Junagadh district of Gujarat including 13 accessions of *Abelmoschus ficulneus,* 9 accessions of *A. tuberculatus* , 7 *A. tetraphyllus,* 1 *A. crinitus,* 11 accession of *A. caillei* type; 4 cultivated okra, 2 in cauliflower, 1 in cabbage and 2 in broccoli from local market, 2 in carrot, 6 in sponge gourd from Newada, Saidpur, Gazipur; Rajatalab, Varanasi and Belaisha, Azamgarh; 6 in ridge gourd from Sikhad Block, Mirzapur and Vidyapeeth Block, Varanasi; 1 in satputia from Sevarahi, Kushinagar; 5 in snake gourd including wild types, 3 in ivy gourd, 5 in watermelon from Rajasthan, Delhi and Varanasi; 12 in long melon from CRB-Leg, France; 1 in round melon from local collection; 3 in pointed gourd wild related species *Trichosanthes bracteata* from Nawrangpur, Koraput and Gajapati district of Odisha; 22 in muskmelon from Odisha; 83 in amaranth from Malda, Dakshin Dinajpur and Uttar Dinajpur; 1 in chenopod, 1 in lotus, 11 in water spinach from Eastern UP and Bihar; 5 in sweet gourd, 22 in *Cucumis melo* var *agrestris*, 19 in *C. callosus* and one in snapmelon. An exploration trip was conducted in collaboration with ICAR-NBPGR RC, Cuttack to collect wild relatives of okra and *Cucumis* from parts of Eastern Ghat and Kalahandi, Nuapada, Nawarangpur districts of Odisha (Fig. 1 & 2).

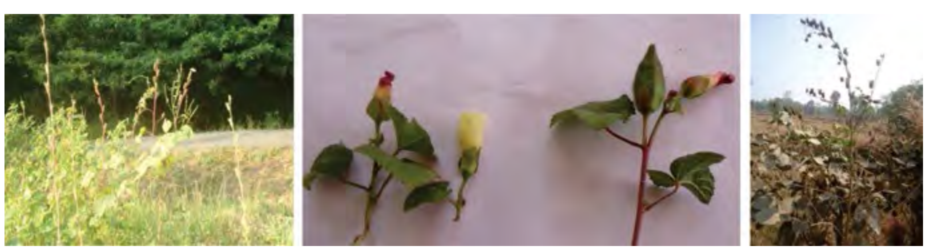


Fig. 1: Different species of wild okra collected from Odisha

C. sativus var. *hardwickii* — *C. melo* var *agrestris* — *T. bracteata*

Fig. 2: Different forms of *Cucumis* and *Trichosanthes* collected from Odisha

Characterization, screening and evaluation of germplasm

Chilli: The germplasm collections in the year included, two genetic and nine cytoplasmic male sterile lines and others. The maintenance of all the germplasm was done using the nylon net bags or cages for true to type seeds. The 34 stuff-type chillies evaluated and maintained comprised of collections from stuff type hotspot areas (Saidpur, Paterwa and other local areas) and some selected lines from amongst the breeding populations (Table 1). Red ripe fruit yield of about 1 kg/plant was obtained in several lines such as Saidpur Coll.-13, Saidpur Coll.-4, Paterwa pickle type and Pickle Type-12 (Fig. 3), depicting thereby an estimated average yield potential of ≥250 q/ha.

Pickle Type-12-8 Saidpur Collection-13

Fig. 3: Promising stuff pickle type chilli accessions

Brinjal: 96 germplasm accessions obtained from ICAR-NBPGR, New Delhi were evaluated for yield and related traits. Promising accessions targeting various market segments were identified and are listed in Table 2.

Table 1: Evaluation of stuff (pickle) type chilli germplasm

Genotype	Fruit length (cm)	Fruit width (cm)	Fruits/pl (No.)	Fruit wt./pl (g)	Pl. height (cm)
Saidpur Coll.-13	8.8	2.2	57.0	1026.0	58.8
Saidpur Coll.-2	6.9	2.5	61.0	854.0	59.6
Saidpur Coll.-4	8.5	3.2	50.0	1050.0	72.2
Pickle Type-6	9.1	2.1	47.0	846.0	69.0
Saidpur Coll.-5	8.8	2.2	57.0	855.0	58.8
Pickle Type-12	7.4	2.9	35.0	960.0	66.8
Paterwa Pickle Type	8.9	3.0	35.0	1050.0	59.2

Table 2: Promising accessions targeting various market segments of brinjal

Accession	50% Flowering (DAT)	Accession	No. of fruit /plant	Accession	Yield (q/ha)
Light purple long					
IC-0111328	43.12	IC-90061	216.00	EC-169769-A	573.91
EC-305013	43.25	EC-316277	148.00	EC-316277	476.86
IC-112293	44.12	IC-0111328	112.00	IC-112293	467.30
IC-127239	44.25	EC-305013	112.00	IC-126711	348.92
IC-0112339	44.37	IC-0112339	98.00	IC-0112339	344.37
Kashi Taru ©	48.32	Kashi Taru ©	93.25	Kashi Taru ©	331.21
Green long					
EC-0169763	42.53	IC-0126930	156.00	IC-0126930	572.69
IC-0510417	44.11	IC-112636	110.00	IC-144073	553.09
IC-137695	44.32	IC-144073	102.00	IC-137695	377.89
IC-144073	44.12	IC-0112779	90.00	IC-144093	354.58
IC-0126930	45.25	IC-0510459	84.00	IC-90978	289.11
Kashi Taru ©	48.32	Kashi Taru ©	93.25	Kashi Taru ©	331.21
Light purple round					
IC-0127150	43.35	EC-316264-1	92.00	IC-0127150	533.85
IC-136490	44.12	IC-136511	86.00	EC-316264-1	497.84
IC-136511	44.23	IC-90934	76.00	IC-0136213	468.38
EC-169761-1	44.25	IC-0089911	72.00	IC-136511	400.68
IC-0111065	44.55	IC-0127150	64.00	IC-135912	390.69
KS-224 ©	50.32	KS-224 ©	37.77	KS-224 ©	297.64
Dark purple round					
IC-0127150	43.25	IC-136511	86.00	IC-600615	573.23

IC-136511	44.35	IC-90934	76.00	IC-0127150	533.85
IC-136490	44.45	IC-600615	74.00	IC-0136213	468.38
EC-169761-1	44.47	IC-0089911	72.00	IC-136511	400.68
IC-0111065	44.65	IC-0127150	64.00	IC-135912	390.69
KS-224 ©	50.32	KS-224 ©	37.77	KS-224 ©	297.64
Purple oblong					
IC-0126898	43.25	IC-89955	62.00	IC-0126898	446.60
IC-89955	46.11	IC-0126898	60.00	IC-89955	393.27
IC-0090126	45.33	IC-0510419	58.00	IC-0510445	369.81
IC-99665-X	47.07	IC-510461	56.00	IC-0126906	355.66
IC-510461	50.11	IC-0510445	54.00	IC-0510419	354.87
Kashi Taru ©	48.32	Kashi Taru ©	93.25	Kashi Taru ©	331.21
Dark purple oblong					
IC-136100	43.11	EC-169765	134.00	IC-510476	574.90
IC-111010	50.22	IC-111010	82.00	EC-169765	478.39
IC-510476	50.47	IC-510476	74.00	IC-0126898	446.60
EC-169765	53.71	IC-89955	62.00	IC-89955	393.27
IC-0090126	45.22	IC-0126898	60.00	IC-0126906	355.66
Kashi Taru ©	48.32	Kashi Taru ©	93.25	Kashi Taru ©	331.21

Pea: A set of 25 lines were characterized for various horticultural traits and some lines were identified as promising (Table 3).

Table 3: Promising germplasm lines of Pea for different horticultural traits

Trait	Promising lines
Earliness (DTF ≤ 40 Days)	DARL-407,VRPE-62, VRPE-11, VRPE-72, KS-257, VRPE-31, VRP-719, VRP-81 and VRP-107
Number of branches (2.6-2.8)	CHP-2, VRPE-58, VRPE-31, VRP-719 and VRP-107
Pod length (9.2 cm-10.3cm)	DARL-407, CHP-2, VRPE-100, VRPE-101 and KS-257
Number of pod per plant (15.2-26.0)	DARL-407, CHP-2, VRPE-101, VRPE-100,VRPE-31, VRP81, VRP-113, VRP-56 and VRP-107
10-pod weight (80-90 g)	DARL-407, CHP-2, VRPE-101, VRPE-100 and KS-257
Number of seed per pod (7.4-8.2)	VRPE-100, VRPE-101, CHP-2, DPP-94106 and KS-257
Pod yield per plant (104-125g/plant)	DARL-407, CHP-2, VRPE-100, VRPE-101 and VRP-56

Screening against powdery mildew and pea rust: Six new germplasm lines augmented from Banaras Hindu University, Varanasi were screened for powdery mildew under normal field conditions. Score range between 0–4 was classified as resistant whereas, scale range from 5–9 was classified as susceptible. In addition, the above genotypes were also scored for rust by following 0–4 scale of Stakman *et al.* (1962); values 0–2 were considered as indicative of resistance while 3–4 of susceptible. Genotype HUDP-15 was found resistant to powdery mildew while genotype EC866019 was found to be resistant to pea rust (Fig. 4).

Fig. 4: Field resistance of HUDP-15 against powdery mildew

Winged bean: With the aim to identify the winged bean germplasm lines suitable for vegetable purpose, 87 new accessions of winged bean were characterized for various horticultural traits *viz.*, pod colour (light green, green, dark green and purple-green colour), number of pod per cluster (2-4), days to 50 percent flowering (53. 24-107.5 days), days to edible maturity (17.34-30.75), days to first picking (63.55- 118.35), pod length (15.25 -28.0 cm),

Fig. 5: Winged bean

pod width (2.2- 3.9 cm), average pod weight (7.6-13.75 g), number of seed per pod (5.24-10.23) and pod yield per plant (1.35-3.75 kg), tuber length (7.5 -22.30 cm), tuber width (1.45 – 2.75 cm), number of tuber per plant (2-12) and tuber yield per plant (37.0-350 g). The promising genotypes identified were MWBS-16-11, MWBS-16-17, MWBS-16-19, MWBS-16-20, EC-27884, AmbikaWB-13-16 and RWB-13 (Fig. 5).

Cluster bean: A total of 155 germplasm lines of cluster bean (Fig. 6) were characterized for different horticultural traits *viz*., pod colour (light green, green and dark green), number of pod per cluster (6.5- 11.4), days to 50 percent flowering (37.35 - 45.25 days), days to first picking (45.36- 60.35), pod length (6.25 -15.80 cm), pod width (2.2- 3.9 cm) and pod yield per plant (276–500 g) and 100 seed weight (3.3-4.9 g). The promising genotypes identified were VRCB-5, VRCB- 7 and PNB (Earliness), VRCB-2, VRCB-5, VRCB-7, VRCB-14 and PNB (Days to Ist pod set), VRCB-3, VRCB-5, VRCB-7, VRCB-14 and PNB (Number of pod per cluster), VRCB-3, VRCB-4, VRCB-12, VRCB-13 and PNB (More pod length), VRCB-3, VRCB-5, VRCB-10, VRCB-12 and PNB for more pod yield/plant and VRCB-14 and Pusa Navbahar for 100 seed weight.

Fig. 6: Cluster bean

Water chestnut: Four genotypes of water chestnut were grown in a pond of size of 7.5 m (L) x 5.0 m (W) x 1.2 m (D) and were characterized for different horticultural traits. Number of leaves per plant varies between (24.8-35.0), number of fruit per plant (3.2-5.8), leaf length (4.28-4.57 cm), leaf width (5.41-6.78 cm), fruit pedicel length (4.96-5.46 cm), number of spine per fruit (2.0-2.0), fresh fruit weight (9.78-16.76 g), shelled fruit weight (4.34 -15.48 g), dry fruit weight (0.46-1.89 g), dry matter content (10.6-26.6 %), TSS (3.0-3.9 °Brix), and fruit yield per pond (21.88-30.20 kg). Among the all genotypes, VRWC-1 was promising for dry matter content and fruit yield. Among minerals, zinc content varies from 37.4-44.8 ppm and 28.6-127.6 ppm, iron from 155.6-199.0 ppm and 400.6–1133.0 ppm, and manganese from 33.1-43.6 ppm and 192.9-320.50 ppm in shelled fruit and fruit rind, respectively (Fig. 7).

Fig. 7: Water chestnut

Indian bean: A total of 33 pole type Indian bean germplasm were evaluated for early, high yield and DYMV tolerance. The germplasm VRSEM-950 and VRSEM-101 were early in terms of flowering and recorded maximum yield per plant. The genotypes VRSEM-797, VRSEM-734, VRSEM-703, VRSEM-890 and

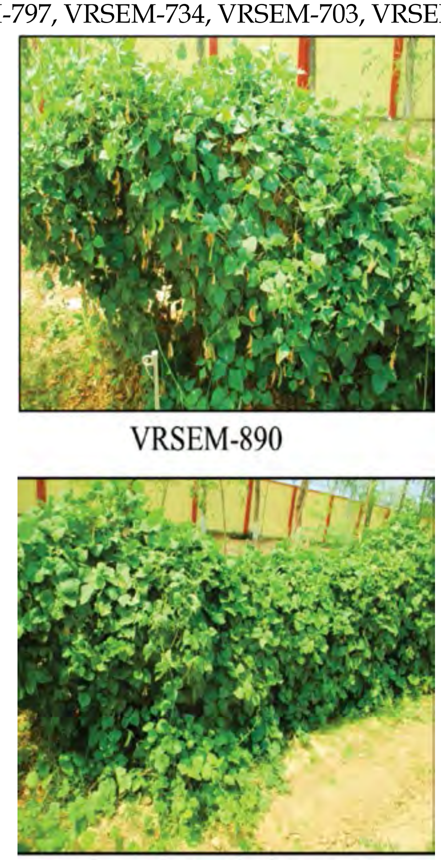


Fig. 8: DYMV tolerant lines of Indian bean

VRSEM-891 were marked as tolerant against DYMV compared to other genotypes (Table 4 and Fig. 8)

Table 4: Performance of pole type genotypes of Indian bean

Characters	Entries
Early and high yield	VRSEM-101 (3.37 kg/plant), VRSEM-950 (3.12 kg/plant),
Pod availability up to 35 ± 2ºC	VRSEM-936, VRSEM-1000, VRSEM-601 and VRSEM-307
DYMV tolerant	VRSEM-797, VRSEM-734, VRSEM-891, VRSEM-890 (Data recorded in second week of April)
Green with purple pod	VRSEM-11, VRSEM-207, VRDB-01, VRSEM-1001, VRSEM-941 A
White pod	VRSEM-601, VRSEM-307 and RP-08-50
Purple pod	VRSEM-936, VRSEM-109, VRSEM-703 and VRSEM-734
Green pod	VRSEM-1000, VRSEM-45, VRSEM-101 and VRSEM-201

Vegetable soybean: With the aim to identify the soybean germplasm lines suitable for vegetable purpose, 40 new accessions of soybean were characterized for various horticultural traits and seed were also multiplied. 18 lines *viz.*, EC-771-148, EC-771-221, AGS-339, AGS-406, AGS-447, AGS-457, AGS-459, AGS-460, AGS-461, AGS-328, AGS-423, AGS-429, AGS-430, AGS-456, AGS-466, AGS-469, AGS-472 and Swarna Vasundhara were found promising. The line AGS-339 was earliest in flowering (29 days) whereas, Swarna Vasundhara took 46 days for first flowering under Varanasi conditions. The genotypes AGS-339 and AGS-460 took minimum 62.5 days, whereas Swarna Vasundhara took 82.0 days to reach R6 stage. Pod length varied from 4.0 cm (AGS-339, AGS-406 and EC-771-221) to 5.2 cm in AGS-472. The desirable horticultural traits like pod length, width, number of seeds per pod, pod yield per plant was found highest in the genotypes EC-771-148 (120.0 g), AGS-429 (95.0 g) and Swarna Vasundhara (103.8 g) (Fig. 9).

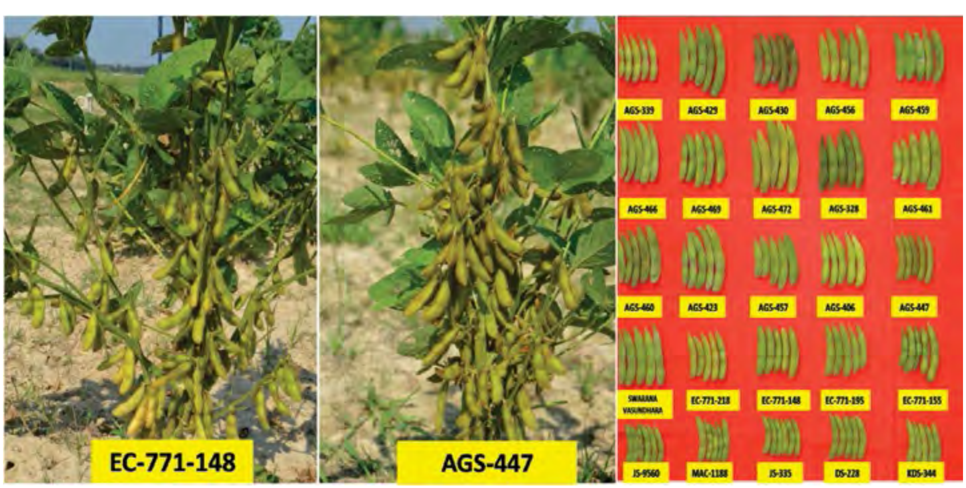


Fig. 9: Field performance of EC-771-148, AGS-447 and variability for pod length and pod width in vegetable soybean accessions.

***Molecular characterization using SCoT markers*:** Fifty start codon targeted (SCoT) markers were used to assess genetic diversity among selected soybean germplasm lines including the vegetable type, pulse type along with its wild relative *Glysine soja*. Among these, 26 markers were found to be polymorphic (Fig. 10).

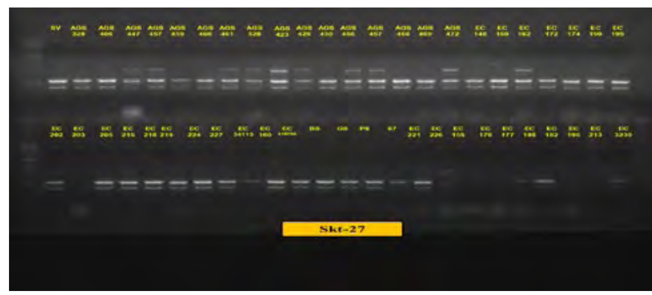


Fig. 10: Amplification products generated by SCoT (SKT-27) primer with 48 DNA templates of vegetable soybean

***Screening for tolerance to Spodoptera spp.*:** Total 120 lines were screened for tolerance to *Spodoptera* (Table 5).The incidence was recorded during the vegetative and pod formation stage of the crop. Screening was done on the basis of leaf damage rating scale for *Spodoptera* spp., which varies from 0-9 (0-No visible leaf damage, 9-Leaves destroyed on 70% of leaves).The line EC-39054 was to be tolerant; 73 were found moderately susceptible and 46 were highly susceptible.

***Estimation of protein and total soluble sugars*:** Twenty soybean accessions (harvested at R6 stage) were analysed for protein content by using Biuret Protein Assay and it was found to vary from 28.8-41.95% (on dry basis). Thirty accessions were analysed for total soluble sugar which found to vary from 9.9-17.5 mg/g on dry basis.

Watermelon: Sixty lines were evaluated for yield and quality attributes and 65 lines including identified/released varieties were maintained as active collections and multiplied.

Table 5: Screening of soybean germplasm for Spodoptera spp. damage

Insect reaction	Genotypes
Tolerant (1-3)	EC-39054
Moderately Susceptible (4-6)	Swarna Vasundhra, EC-771-155, EC-771-160, EC-162, EC-771-170, AGS-459, AGS-460, AGS-461, EC-771-174, EC-771-177, EC-771-205, EC-771-213, EC-771-226 , EC-771-215, EC-771-224, EC-771-227, AGS-328, AGS-456, AGS-457, AGS-472, EC-771-182, EC-771-195, EC-771-199, EC-771-202, EC-771-224, EC-177-227, AGS-328, EC-771-219, EC-771-146, EC-771-147, EC-771-166, EC-771-159, EC-771-176, EC-771-186, EC-771-187, EC-771-192, EC-771-197, EC-771-192, EC-771-208, EC-771-220, EC-771-222, EC-771-230, EC-771-240, JS-9305, DS-228, JS-335, JS-9560, MAC-1172, P . Agrani, EC-34113, EC-34121, EC-34354, EC-36998, EC-37111, EC-37115, EC-37184, EC-39043, EC-39049, EC-39824, IC-279776, IC-316142, IC-316194, IC-316196, IC-317428, IC-338716, IC-341322,IC-393199, IC-393218, IC-393230, EC-7711213, EC-771221, EC-785680, VLS-54
Highly Susceptible (7-9)	EC-771-148, EC-771-172, AGS-406, EC-771-188, EC-771-218, AGS-423, AGS-429, AGS-430, AGS-466, AGS-469, EC-771-190, EC-771-203, EC-771-221, EC-771-151, EC-771-156, EC-771-161, EC-771-167, EC-771-173, EC-771-179, EC-771-182, EC-771-184, EC-771-194, EC-771-209, EC-771-217, EC-771-223, EC-771-229, EC-771-232, EC-771-234, EC-771-233, EC-771-236, KDS-344, IC-281641, EC-39061, IC-273990, IC-279342, IC-282885,IC-355999, IC-356021, IC-391361, IC-393183, IC-393195, IC-393197, IC-418345, IC-419766, EC-771118, VLS-47.

Long melon: Thirty lines/germplasm including identified/released varieties of institute were maintained as active collections. Early flowering was observed in VRLM-28 (25 days after sowing) followed by VRLM-01 (28 days after sowing). Maximum fruit length was recorded in VRLM-40 (50 cm) and minimum in VRLM-11-1 (25 cm). The genotypes VRLM-01, VRLM-40, VRLM-28 and VRLM-3 were found to be superior for yield and quality attributes.

Cole crops: Fifty-two lines of Indian cauliflower were evaluated for various traits of horticultural importance. The genotypes were characterized for maturity group (October, November and December), plant growth habit (spreading, semi-spreading, semi-erect and erect leaf), frame size (small, medium and large), leaf pubescence (present or absent), stalk length (short, medium and long), maturity period (short, medium and long), curd shape (flat, round and pointed), curd colour (yellow-white, cream-white, white and snow-white), curd compactness (loose, medium compact and compact), riceyness (present, weak or absent), leafiness (present or absent), flower colour (yellow and cream), and self-incompatibility (absent, weak and strong). Among them, genotypes VRCF-86 and VRCF-201 were found to be yield potential in October maturity; VRCF-50, VRCF-102, VRCF-27, VRCF-75-1 and VRCF-113 in mid-November maturity; and VRCF-104, VRCF-202, VRCF-22 and VRCF-77 in late-November to mid-December maturity group. Further, a genotype of tropical kale, five of tropical cabbage and seven of tropical broccoli were also evaluated for various traits.

Cucumber: Total 43 germplasm/genotypes including 9 released varieties of different organizations were evaluated for flowering, yield and related traits. The results indicated that the number of days required for anthesis of first female flower ranged from 33.0 (VRCU. Sel. 12-03) to 53.0 (VRCU-26) and number of days required for anthesis of 50% female flower ranged from 35.0 (VRCU Sel. 13.05) to 53.0 (Swarna Sheetal). The average fruit weight of 5 fruits for 43 genotypes ranged from 100 to 350 g. Yield per plant ranged from 710.20 (IIHR-03) to 1425.50 (VRCU-27).

Pumpkin: Fifty five germplasm were evaluated for yield and quality attributes. A total of 110 lines including identified/released varieties were maintained as active collections. Variability for different characteristics was observed in evaluated lines. The evaluated lines were grouped based on the rind colour and shape of the fruits. Fruit yield per plant ranged between 1.63 kg (GS-81) to 6.16 kg (VRPK-18-01). Number of fruits per plant varied between 1.2 (GS-81) to 4.30 (VRPK-18-01). Individual fruit weight ranged from 0.8 kg (VRPK-230) to 2.35 kg (VRPK-Sel-10-15) at edible green mature stage. All the lines have been maintained through selfing / sibbing for their further utilization.

Muskmelon: Seventy five diverse genotypes of muskmelon were evaluated for various important horticultural traits. Significant genetic variability observed for the above mentioned traits in the genotypes evaluated (Table 6). The genotype VRMM-170 and VRMM-186 found promising with a yield of 3.85 kg and 3.50 kg per vine, respectively.

Bitter gourd: During the year, 7 collected genotypes (BTG-47, BTG-47-1, BTG-47-2, BT-1-A, BT-1-B and BT-1-C) were evaluated for different horticultural traits. Among them line BT-1-A collected from Bhubaneswar (Odisha) gave maximum fruit yield of 2100 g/plant and its individual fruit weight was also found maximum (100.0 g). Another line, BTG-47 was found next best yielder (1494 g/plant). The minimum yield was recorded in BTG-47-1 (235 g/plant) having the individual fruit weight of 35.0 g (Table 7).

Table 6: Variation in horticultural traits in muskmelon

Traits	Range	Mean	Deviation
Vine length (cm)	148.00 – 300.00	155.23	2.02
Number of primary branch	3.00 – 6.00	3.75	2.00
Days to first productive flower anthesis	40 .00 – 65.00	52.35	1.62
Node at which first productive flower appear	4.00 – 12.00	6.50	3.00
Fruit length (cm)	15.20 – 29.12	20.30	1.91
Fruit diameter (cm)	9.00 – 20.50	16.55	2.27
Average fruit weight (g)	350.00 – 1100.00	690.36	3.14
Pericarp thickness (cm)	0.75 – 4.00	2.17	5.33
TSS(°Brix)	6.00 – 12.60	10.25	2.10
Yield/plant (kg)	1.55 – 3.85	1.75	2.48

Table 7: Performance of collected germplasm of bitter gourd

Genotypes	Place of collection	F. length (cm)	Fruit circumference (cm)	Fruit wt. (g)	No. fruits/ plant	Fruit yield/ plant (g)
BTG-47	Bhubaneswar	17.1	13.4	80.00	18	1494
BTG-47-1	Bhubaneswar	9.2	7.0	35.00	8	235
BTG-47-2	Bhubaneswar	8.2	10.2	36.00	12	432
BT-1-A	New Delhi	14.8	14.0	100.00	21	2100
BT-1-B	New Delhi	11.6	12.7	75.00	10	750
BT-1-C	New Delhi	14.5	15.2	95.00	12	1140

Bottle Gourd: Ten promising lines were identified on the basis horticultural traits (Table 8). Maximum yield was recorded in VRBG-61 (9.12 kg/plant). This genotype attains the fruit length of 37.9 cm, and fruit circumference of 35 cm. DR-2017 Bagpat was noted the next high yielder (8.36 kg/plant).

Table 8: Performance of promising genotypes of bottle gourd

Germplasm	Fruit length (cm)	Fruit circumference (cm)	Weight (g)	No. of fruits/ plant	Yield (kg/ plant)	Plant length (cm)	Fruit colour
DVBG-15-1	29.2	24.5	750.00	6.4	4.80	602	Green
VRBG-61	37.9	35	800.00	11.4	9.12	705	Green (cut leaf)
DR-2017 BAGPAT	45.2	12.5	950.00	8.8	8.36	535	DG star- spotted
DVBG-15-2	19.7	44.9	650.00	6.7	4.35	591	Green
VRBG-61-3	16.5	38.8	690.00	10.6	7.314	950	cut leaf
VRBG-4-1	19.8	41.7	500.00	5.3	2.65	640	Green
VRBG-9-1	18.5	27.1	650.00	11.5	7.475	575	Green
VRBG-71	14.7	36.0	520.00	5.8	3.01	1145	DG (spotted)
IC-594544	20.1	45.2	825.00	5.5	4.53	685	Green
IC-594545	18.5	44.3	750.00	5.7	4.27	690	Green

Carrot: Sixty-one lines of tropical carrot were evaluated for traits like root colour (red, orange, black, yellow and rainbow), gross plant weight, root weight, root length and self-colour core. The genotypes varied considerably for horticultural traits such as 112.9-231.3 g for gross plant weight, 62.6-191.8 g for root weight, 15.6-27.3 cm for root length, 2.68-4.42 cm for shoulder diameter, 49.8-69.8% for harvest index and 165-428 q/ha for root yield. The genotypes VRCAR-186, VRCAR-201, VRCAR-112 and VRCAR-185 (red root); VRCAR-126 and VRCAR-89-1 (black root); VRCAR-91-1 and VRCAR-91-2 (orange root); and VRCAR-153, VRCAR-178 and VRCAR-127 (yellow root) were found to be promising for yield potential.

Radish: Fifty-six genotypes were characterized for various horticultural traits such as root colour (white/ red/purple), plant biomass, root length, root diameter, root shape (triangular/cylindric/elliptic/ spheric), number of leaves, leaf division incision (lyrate/ sinuate/entire/lacerate) and petiole colour (green/ pink/purple). The characterized genotypes varied considerably for anthocyanin content, responsible for red/purple colour (negligible to 210 µg/g), gross plant weight (98.7-281.6 g), shoot weight (40.1-125.5 g), number of leaves (8.1-13.4), leaf length (28.5-43.4 cm), shoulder diameter (2.6-4.8 cm), root length (9.3-28.1 cm) and root weight (58.6-231.5 g). The promising genotypes were VRRAD-150, VRRAD-203, VRRAD-200, VRRAD-202 and VRRAD-216 (white root); VRRAD-131-2, VRRAD-170 and VRRAD-173 (red root);

and VRRAD-131, VRRAD-134 and VRRAD-151 (purple exterior).

Sponge gourd: Out of 76 germplasms ten i.e. VRSG-50, VRSG-68, VRSG-18, VRSG-9, VRSG-136, VRSG-195, VRSG-49-1, VRSG-171, VRSG-13, VRSG-142-1, VRSG-3-13 and VRSG-40 were found promising for horticultural traits and were free from Sponge Gourd Mosaic Virus (SGMV) disease symptoms under field conditions (Table 9 & Fig. 11).

VRRG-35 VRRG-6A VRRG-46-2016 VRRG-7-2016

Fig. 12: Promising lines of ridge gourd

Satputia: Out of 37 germplasms of satputia, five lines VRS-3-17, VRS-24-1, VRS-1-17, VRS-36 and VRS-11 were found promising for horticultural traits (Fig. 13).

Table 9: Characteristics of promising germplasm of sponge gourd

Genotypes	Days to 1st male flower appeared	Days to 1st female flower appeared	Days to first harvesting	Fruit colour	Fruit length (cm)	Fruit diameter (cm)	No. of fruits/ plant	Average fruit wt.(g)	Yield/ plant (kg)
VRSG-50	44	42	50	Green	23.46	3.34	8.5	130.2	1.106
VRSG-68	40	43	51	Green	21.0	3.3	11	103.0	1.133
VRSG-18	63	59	70	Green	23.6	3.20	9	135	1.215
VRSG-9	45	44	52	Dark green	29.0	2.9	7.43	165.2	1.227
VRSG-136	32	31	39	Dark green	24.36	3.4	8.3	147.6	1.23
VRSG-195	36	37	45	Dark green	25.45	3.60	9.50	153.6	1.46
VRSG-49-1	37	35	43	Dark green	25.52	3.42	7.5	143.5	1.090
VRSG-171	38	38	45	Green	27.76	3.2	8.5	162.5	1.381
VRSG-13	67	66	75	Green	21.5	3.7	7	146	1.022
VRSG-142-1	41	40	48	Green	21.1	3.7	9.6	123	1.180
VRSG-3-13	51	52	60	Green	21.9	3.05	10	120	1.200
VRSG-40	45	47	55	Light green	20.9	3.12	11	114	1.254

VRSG-171 VRSG-195 VRSG-136

Fig. 11: Promising germplasm of sponge gourd

Ridge gourd: Out of 46 germplasms of ridge gourd, ten lines VRRG-88, VRRG-6A, VRRG-26, VRRG-35, VRRG-5A, VRRG-110, VRRG-7-2016, VRRG-42-2016, VRRG-46-2016 and VRRG-1-17 were found promising for horticultural traits and were free from Sponge Gourd Mosaic Virus (SGMV) and downy mildew disease symptoms under field conditions (Fig. 12).

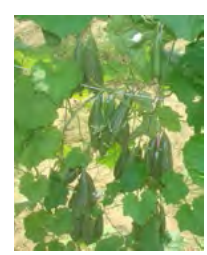


Fig. 13: VRS-24-1

Minor gourds: Germplasm of teasel gourd (55), spine gourd (16) and ivy gourd (08) were collected, planted and being maintained at RRS, Sargatia, Kushinagar.

Lotus: With the aim to identify the suitability of lotus cultivation in this region, a germplasm line was augmented from Kashmir, (J&K). Standardization of lotus seed germination was done. Nursery was raised in pots and lotus plants were planted in pond (Fig. 14).

Water spinach: Eleven germplasm lines (VRWS-12, VRWS-13, VRWS-14, VRWS-15, VRWS-16, VRWS-17, VRWS-18, VRWS-19, VRWS-20, VRWS-21, VRWS-22), most suitable for vegetable purpose, were augmented from different parts of country (Fig. 15). These lines were characterized for various horticultural traits and

Seedling preparation of lotus

Transplanting of lotus in Pond

Fig. 14: Lotus seedling and transplanting in main pond

Transplanting of water spinach

VRWS-1

VRWS-8

VRWS-9

Fig. 15: Transplanting and promising lines of water spinach

plants were multiplied for next season. A total of 23 germplasm lines of water spinach were characterized for different horticultural traits *viz*., leaf length (3.5-9.5 cm), petiole length (3.15-6.0 cm), leaf width (1.25-6.25 cm), number of vine/plant (3.25-5.0), vine length (35.0-85.0 cm), inter-nodal length (3.25-7.15 cm), number of nodes/vine (6-14), number of cuttings/month (2-3) and fresh weight of 50 leaves (35.50- 60.25 g). The genotypes VRWS-1, VRWS-4, VRWS-8 and VRWS-9 were found promising.

Documentation of germplasm

Five genotypes of carrot (IC 0625057 (VRCAR-124), IC 0625058 (VRCAR-107-1), IC 0625059 (VRCAR-107-2), IC 0625060 (VRCAR-171-1) and IC 0625061 (VRCAR-203) and 13 genotypes of radish (IC 0625062 (VRRAD-199), IC 0625063 (VRRAD-200), IC 0625064 (VRRAD-201), IC 0625065 (VRRAD-202), IC 0625066 (VRRAD-203), IC 0625067 (VRRAD-214), IC 0625068 (VRRAD-216), IC 0625069 (VRRAD-130), IC 0625070 (VRRAD-136), IC 0625071 (VRRAD-170), IC 0625072 (VRRAD-130-2), IC 0625073 (VRRAD-132) and IC 0625074 (VRRAD-135) were documented with ICAR-NBPGR, New Delhi.

Germplasm distribution

Promising germplasm and released varieties/ hybrids being maintained at the institute were supplied to various organizations for research, evaluation and demonstration purpose after signing the material transfer agreement (MTA). During 2017-18, 624 accessions in 22 species of vegetable crops and cultures of pathogenic fungi were supplied to 28 organizations. Details of the germplasm supplied along with the recipient organization are given in the (Table 10).

Table 10: Crop-wise details of the germplasm supplied to other organizations

Crop/species	Organization
Tomato (131)	IGKVV, Raipur (3); BHU (30), SHIATS (10); JNKVV (1); SKUAST (15); BBAU, Lucknow (20); ICAR-CAZRI (30); DRPCAU, Pusa (22)
Brinjal (163)	BCKVV (8); JNKVV (2); YSPUA&T (25); VNMKV, Parbhani (40); VCSGUHF, Pauri-Garhwal (25); MGCGV, Satna (35); AKS U, Satna (25); IARI, New Delhi (5)
Chilli (41)	JNKVV (1); Eagle Seeds and Biotech Ltd. (2); MGCGV, Satna (18); BBAU, Lucknow (20)
Okra (81)	Annamalai Univ. (26); SKNAU, Jobner (15); BHU (21); BBAU, Lucknow (18); Dinkar Seeds Pvt. Ltd. (1)
Pea (35)	SKUAST, Srinagar (35)
Cowpea (40)	SHIATS(20); VNMKV, Parbhani (20)
Indian Bean (49)	RBS College, Agra (19); VNMKV, Parbhani (30)
Bitter gourd (12)	RMVCC, Rahara (5); UHS, Bagalkot (7),
Bottle gourd (6)	ICAR-RCER, Patna (6);
Pumpkin (5)	Hort. Coll Res. Inst., Tirichurapalli (5)
Muskmelon (51)	SKUAST, Srinagar (10); Mandor Agri. Univ. (16), TNAU (25)
Radish (2)	Kulbhaskar Ashram PG College (2)
Fusarium oxysporium and *F. solani* (2)	UP College, Varanasi (1 each)
22 species (624)	**28 organizations**

Project: 1.2: Genetic Improvement of Solanaceous Vegetables

Tomato

Evaluation of segregating populations for quality traits: Ten segregating populations comprising of F_4-F_5 generations were advanced and also evaluated for yield, quality and nutritional attributes**.** The segregating lines *viz.*, VRNRT-6 (YPP-3.98 kg), VRNRT-5 (YPP-3.67 kg), VRNRT-3 (YPP-3.55 kg), VRNRT-4 (YPP-2.55 kg) were superior for maximum number of fruits and high yield, whereas, the lines like, VRNRT-7 (Lycopene-7.53 mg, β-carotene-6.40 mg) and VRNRT-8 lycopene-6.21 mg, β-carotene-2.75 mg) were superior from nutritional rich point of view (Table 11).

Evaluation of F_1 cherry tomato: Fifteen cross combinations (developed in 2017) were evaluated for yield and quality traits in open field conditions. F_1s namely VRCTH-17-64 (YPP- 4.54 kg), VRCTH-17-46 (YPP 4.32 kg), VRCTH-17-51 (YPP-3.97 kg), VRCTH-17-57 (YPP-3.51 kg) and VRCTH-17-45 (YPP-3.47 kg) were superior in respect to high yield with high TSS (Table 12 and Fig. 16).

Table 11: Performance of segregating tomato populations for quality

Genotypes	Growth habit	FPC	FS	YPP (kg/plant)	TSS (°Brix)	Lycopene (mg/100 g FW)	β-carotene (mg/100 g FW)	Ascorbic acid (mg/100 g FW)	Acidity (%)
VRNRT -3	Determinate	4-5	Round	3.55	5.8	4.50	1.27	8.95	2.21
VRNRT -4	Determinate	3-4	Round	2.55	5.8	5.80	2.66	12.3	2.43
VRNRT -5	Determinate	2-3	Round	3.67	6.7	5.98	1.95	11.1	2.74
VRNRT -6	Determinate	3-4	Round	3.98	7.1	3.15	1.40	9.56	1.82
VRNRT -7	Determinate	3-4	Round	2.95	5.0	7.53	6.40	15.3	2.76
VRNRT -8	Determinate	4-5	Round	2.52	6.4	6.21	2.75	13.4	2.59
CD at 5%	-	0.62	-	1.53	0.25	0.53	0.27	1.05	0.26

FPC: Fruit per cluster. FS: Fruit shape, YPP: Yield per plant

Table 12: Performance of promising F_1s of cherry tomato in open field conditions

Hybrids	Growth habit	FPC	FS	FW (g)	YPP (kg/plant)	PT (cm)	TSS (°Brix)
VRCTH-17-45	Indeterminate	6-7	Round	10-12	3.47	0.24	6.8
VRCTH-17-51	Indeterminate	5-6	Round	8-10	3.97	0.41	5.4
VRCTH-17-46	Determinate	6-7	Round	6-8	4.32	0.23	7.7
VRCTH-17-57	Indeterminate	6-7	Round	12-14	3.51	0.24	6.8
VRCTH-17-64	Indeterminate	6-8	Round	14-16	4.54	0.31	6.4
CD at 5%	-	1.29	-	0.83	1.39	0.05	0.12

FPC: Fruit per cluster. FS: Fruit shape, YPP: Yield/plant .

Figure 16: Promising F_1s of cherry tomato

Evaluation of red fruited advanced cherry tomato lines: Fourteen red fruited cherry tomato (F_6 generations) were evaluated and advanced. The advanced lines *viz*., VRCRT-2 (YPP-3.50 kg), VRCRT-9 (YPP- 3.20 kg) and VRCRT-8 (YPP-3.09 kg) were superior in terms of total yield and contributing characters. However, the advanced lines *viz*., VRCRT-8 (Lycopene-3.54 mg) and VRCRT-9 (Lycopene-4.22 mg) were superior for containing high nutritional attributes (Table 13 and Fig.17).

VRCYT-6 VRCYT-10 VRCYT-11

Figure 18: Promising cherry lines (yellow fruited).

Table 13: Performance of red fruited advanced cherry tomato lines in open field.

Genotypes	Growth habit	FPC	FS	TSS (°Brix)	YPP (kg/ plant)	Lycopene (mg/100 g FW)	β-carotene (mg/100 g FW)	Ascorbic acid (mg/100 g FW)	Acidity (%)
VRCRT-2	Indeterminate	7-8	Round	4.6	3.50	5.10	1.21	13.0	1.92
VRCRT-8	Determinate	5-6	Oval	7.2	3.09	3.54	1.17	13.7	2.51
VRCRT-9	Determinate	5-6	Oblong	7.2	3.20	4.22	1.35	15.8	1.34
CD at 5%	-	0.66	-	0.15	1.10	0.16	0.09	0.59	0.19

FPC: Fruit per cluster. FS: Fruit shape, YPP: Yield per plant . Note: In earlier reports the code used for the lines was VRTCH

VRCRT-2 VRCRT-8 VRCRT-9

Fig. 17: Promising cherry lines (red fruited)

Evaluation of yellow fruited advanced cherry tomato: Eleven yellow fruited cherry tomato (F_6 generations) were evaluated and advanced. Out of 11, the advanced lines *viz*., VRCYT-10 (YPP-3.61 kg), VRCYT-6 (2.98 kg) and VRCYT-7 (YPP-2.30 kg) were superior in high yield and contributing characters. However, an advance line namely VRCYT-11 (β-carotene-2.51 mg) was superior for containing high β-carotene and also maintained substantial amount of ascorbic acid and acidity (Table 14 and Fig. 18).

Evaluation of advanced β-carotene lines: Thirteen β-carotene tomato lines (F_7-F_8 generation) were evaluated and advanced. Out of 13 lines, the advanced lines *viz*., VRTKB-8 (YPP-4.34 kg), VRTKB-14 (YPP-3.75 kg) and VRTKB-4 (YPP-3.30 kg), were superior in term of high yield and advanced lines like VRTKB-12 (β-carotene-4.42 mg) and VRTKB-14 β-carotene-4.07 mg) were superior in containing high β-carotene, ascorbic acid and acidity (Table 15 and Fig. 19).

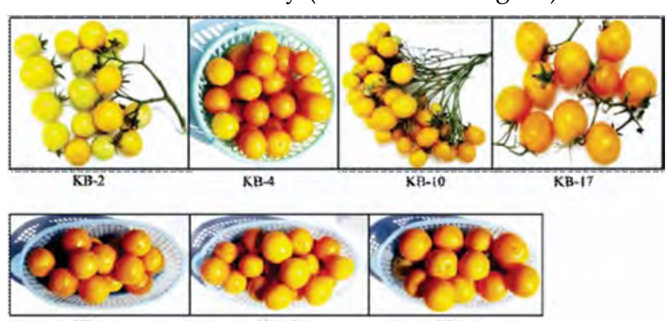


Fig. 19: β-carotene rich lines of tomato.

Table 14: Performance of yellow fruited cherry tomato advance lines in open field

Genotypes	Growth habit	FPC	FS	TSS (°Brix)	YPP (kg/ plant)	Lycopene (mg/100 g FW)	β-carotene (mg/100 g FW)	Ascorbic acid (mg/100 g FW)	Acidity (%)
VRCYT-6	Determinate	8-10	Round	6.8	2.98	0.81	1.91	6.33	0.61
VRCYT-7	Determinate	3-4	Round	4.0	2.30	0.94	1.88	7.92	0.65
VRCYT-10	Determinate	9-10	Round	5.2	3.61	0.80	2.01	6.86	0.94
VRCYT-11	Determinate	4-5	Round	5.4	1.85	0.74	2.51	7.30	0.90
CD at 5%	-	0.87	-	0.22	1.02	0.10	0.11	0.25	0.18

Table 15: Performance of high β-carotene tomato advance lines

Genotypes	Growth habit	FPC	FS	TSS (°Brix)	YPP (kg/ plant)	Lycopene (mg/100 g FW)	β-carotene (mg/100 g FW)	Ascorbic acid (mg/100 g FW)	Acidity (%)
VRTKB-4	Determinate	3-4	Round	4.4	3.30	0.88	1.35	10.5	0.72
VRTKB-8	Determinate	4-5	Round	5.1	4.34	1.37	4.57	8.81	0.74
VRTKB-12	Determinate	3-4	Round	5.2	2.07	1.07	4.42	10.7	0.86
VRTKB-14	Determinate	2-3	Oval	4.5	3.75	0.98	4.07	12.7	0.93
CD at 5%	-	1.12	-	0.23	1.25	0.20	0.27	0.64	0.09

FPC: Fruit per cluster. FS: Fruit shape, YPP: Yield per plant

Evaluation of advanced lines for cultivation in rainy season: Twenty two advance lines comprising of different generations from F_8 to F_{12} were evaluated for cultivation in rainy season (transplanting in second week of July). Out of 22, the advanced lines VRT-50 (YPP-7.12kg), VRT-51 (YPP-6.90 kg), VRT-34 (YPP-6.29 kg) VRT-30 (YPP-5.80 kg), VRT-32 (YPP-5.03 kg) and VRT-20 (YPP-5.40 kg) showed ability to give high yield with low pressure of tomato leaf curl virus incidence in rainy season (Table 16 and Fig. 20).

Evaluation of VRT-02 tomato advance line in field and pot culture: The VRT-02 line of tomato was evaluated in both field and pot culture. The performance of this line is as follows (Table 17 and Fig. 21)

Table 16: Performance of tomato advance lines grown in rainy season (transplanting in second week of August)

Genotypes	Growth habit	FPC	FS	FW (g)	FL (cm)	Fw (cm)	PT (cm)	TSS (°Brix)	YPP (kg/plant)
VRT-20	Indeterminate	2-3	Round	55-60	4.5	5.2	0.42	4.4	5.40
VRT-30	Determinate	2-3	Round	35-40	4.1	4.9	0.52	4.2	5.80
VRT-50	Semi-determinate	2-3	Round	80-90	4.4	4.1	0.59	5.2	7.12
VRT-51	Semi-determinate	2-3	Round	90-100	5.2	4.7	0.55	4.8	6.90
VRT-32	Determinate	3-4	Oblong	35-40	4.8	2.7	0.42	4.3	5.03
VRT-34	Determinate	2-3	Oval	65-70	4.6	4.0	0.48	4.7	6.29
CD@5%	-	0.66	-	5.38	0.32	0.41	0.06	0.31	0.41
CV	-	10.4	-	4.65	3.37	4.38	6.24	3.36	4.51

FPC: Fruit per cluster. FS: Fruit shape, FW: Single fruit weight, FL: Fruit length, Fw: Fruit width, PT: Pericarp thickness, YPP: Yield per plant (kg/plant).

VRT-34 VRT-50 VRT-51 VRT-20 VRT-30

Fig. 20: Promising advance lines suitable for cultivation in rainy season

Table 17: Performance of VRT-02- an advance line of tomato field and pot cultivation

Characters	Field condition	Pot condition
Growth habit	Determinate	Determinate
Number of fruits/ plant	355-380	225-275
Fruits/cluster	6-8	6-8
Single fruit weight (g)	15-20	15-18
Fruit length (cm)	3.4-3.9	3.0-3.2
Fruit width (cm)	2.9-3.5	2.8-3.0
Fruit shape	Oblong	Oblong
Pericarp thickness (cm)	0.32-0.36	0.32-0.36
TSS (°Brix)	4.2-4.5	4.2-4.5

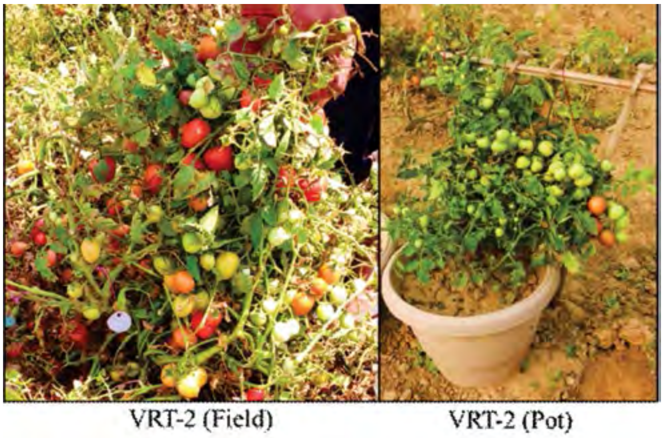


Fig. 21: Tomato line VRT-02 in field and pot condition.

Brinjal

Evaluation of hybrids: Fifty four F_1 hybrids were evaluated for earliness, yield and quality traits along with leading hybrids of private seed companies available in the market. Yield parameters of some promising hybrids in different segments are given in (Table 18).

Hybrids selected to multi-location testing: Based on the two year station trial, the hybrid IVBHL-22 and IVBHR-18 were selected for multi-location testing through AICRP (VC) trials (Table 19 and Fig. 22).

Fig. 22: Promising hybrids of brinjal identified for multi-location testing

Hybridization and generation advancement: In cultivated brinjal, 18 cross-combinations in round shape, 20 cross-combinations in long shape and 1 cross-combination in oblong shape have been attempted utilizing promising parental line. A total of 386 segregating populations (58: F_1 to F_2; 52: F_2 to F_3; 102: F_3 to F_4; 58: F_4 to F_5; 25: F_5 to F_6; 31: F_6 to F_7; 10: F_7 to F_8; 50: F_8 to F_9) were advanced to next higher generation.

Table 18: Performance of brinjal hybrids

CODE NAME	Days to 50% Flowering	No. of Fruit/ plant	Fr. length (cm)	Fr. dia (cm)	Av. Fr. Wt (g)	Fruit Shape	Fruit Colour	Yield	
								q/ha	kg/plant
Round fruited									
HRB/B3-31	48	48	8.86	10.06	305.67	Round	Light purple	494.15	14.67
IVBHR-16	48	44	9.20	9.56	270.00	Round	Light purple	400.12	11.88
HRB/B3-21	48	42	11.10	8.72	252.67	Round	Dark purple	357.41	10.61
HRB/B3-13	50	39	9.48	9.40	286.33	Round	Dark purple	376.10	11.17
HRB/B3-25	47	38	9.18	9.66	259.00	Round	Purple	331.48	9.84
Pusa Hybrid-6©	52	45	10.30	9.10	322.00	Round	Dark purple	488.02	14.49
Kashi Sandesh©	55	54.23	11.8	9.16	264.25	Round	Light purple	482.64	14.33
Long fruited									
HLB/B4-17	44	110	18.44	3.54	95.00	Long	Dark purple	351.96	10.45
IVBHL-21	44	86	20.30	3.02	110.33	Long	Dark purple	319.58	9.49
HLB/B4-15	47	65	21.86	4.00	140.00	Long	Purple	306.49	9.10
HLB/B4-5	45	78	13.54	3.30	121.67	Long	Light purple	319.62	9.49
HLB/B4-7	48	81	17.30	3.86	110.21	Long	Light purple	300.66	8.93
Navina ©	54	55	12.68	5.52	185.00	Long	Light purple	342.69	14.57

Table 19: Performance of promising brinjal hybrids

	Accession	Days to 50% flowering	Fruits/ plant (No.)	Fruit length (cm)	Fruit width (cm)	Fruit Shape and colour	Calyx colour	Fruit weight (g)	Yield/ plant (kg)
Long fruited									
(2016-17)	IVBHL-22	43	138.25	18.32	3.42	Long, Dark purple	Green	102.23	14.13
	Navina (c)	50	75.92	12.67	5.22	Long, Light purple	Green	130.25	9.89
(2017-18)	IVBHL-22	44	129.21	18.44	3.54	Long, Dark purple	Green	95.00	12.27
	Navina (c)	51	72.51	13.12	5.25	Long, Light purple	Green	132.67	9.62
Round fruited									
(2016-17)	IVBHR-18	45	65.21	8.65	10.12	Round, Light purple	Green	298.25	19.45
	Pusa Hybrid-6 ©	52	48.51	11.58	9.45	Round, Dark purple	Purple	345.37	16.75
(2017-18)	IVBHR-18	48	59.65	8.86	10.06	Round, Light purple	Green	305.67	18.23
	Pusa Hybrid-6 ©	54	45.75	11.25	9.62	Round, Dark purple	Purple	355.26	16.25

Entries selected for AICRP (VC) trials: Two promising advance lines IVBL-27 in long-fruited type and IVBR-19 in round-fruited type in F_8 generation were selected for high yield and better fruit quality (Table 20 & Fig. 23). The selected hybrids and lines shall be used for yield trial at station before submitting it for multi-location testing in AICRP (VC).

IVBL-27 IVBR-19

Fig. 23: Promising advance lines selected for multi-location testing through AICRP (VC) trials

Kashi Himani (IVBL-26): A medium long, soft white fruited variety developed by selection from local collection has been identified by the Institute Technology Identification Committee of ICAR-IIVR, Varanasi and submitted for release through State Varietal Release Committee of Uttar Pradesh (Fig. 24).

Fig. 24: Kashi Himani

Maintenance breeding: Seeds of Kashi Sandesh (400 g), Kashi Taru (500 g), Kashi Komal (200 g), Kashi Prakash (400 g) and Kashi Uttam (500 g) were multiplied for distribution to farmer and multi-location demonstration /evaluation by public and private organizations. Parental line of hybrids *viz-* PR-5 (250 g), CHBR-2 (150 g), Pant Rituraj (200 g), Uttara (150 g), Punjab Barsati (150 g), ADM-190 (150 g) and IVBL-22 (300 g) were also multiplied.

Chilli and Capsicum

Utilization and maintenance of germplasm: Cytoplasmic genetic male sterile lines and nuclear genetic male sterile lines were maintained and used in

Table 20: Promising advance lines of brinjal

Year	Accession	Days to 50% flowering (DAT)	Fruits/ plant (No.)	Fruit length (cm)	Fruit width (cm)	Fruit Shape and colour	Calyx colour	Fruit weight (g)	Yield/ plant (kg)
Long fruited									
(2016-17)	IVBL-27	55	115.63	16.85	3.27	Long, Purple	Green	110.25	12.75
	Punjab Sadabahar (c)	53	87.96	20.25	3.22	Long, Black	Green	120.32	10.58
(2017-18)	IVBL-27	56	107.33	17.14	3.5	Long, Purple	Green	101.24	10.87
	Punjab Sadabahar (c)	55	88.54	20.36	3.28	Long, Black	Green	110.25	9.76
Round fruited									
(2016-17)	IVBR-19	49	65.28	11.36	9.68	Round, Dark purple	Green	305.28	19.93
	Swarnamani (c)	54	42.14	10.25	8.1	Round, Dark purple	Purple	370.68	15.62
(2017-18)	IVBR-19	53	64.88	11.14	9.4	Round, Dark purple	Green	300.67	19.51
	Swarnamani (c)	52	48.14	10.26	8.25	Round, Dark purple	Purple	362.25	17.44

crossing programme for the development of superior hybrid combinations. Male sterile lines were maintained by crossing with maintainer lines and in heterozygous state in case of genetic male sterility. Nucleus seeds of Kashi Anmol (700 g), Kashi Gaurav (300 g), Kashi Sinduri (400 g), and Pusa Jwala (400 g) and sufficient amount of parental lines were produced.

Germplasm creation: In order to create variability in chillies with respect to plant and fruit morphology, the cross of Kashi Sinduri and AKC-89-38 was advanced through selfing of phenotypes up till F_8 generation and a stable population has been established. A total of 142 populations were characterized for their morphological traits. The population exhibited wide variability for various traits like fruit length (2.5 - 12 cm) fruits per plant (5 - 65) and ten fruit weight (25 - 85 g). The fruit colour varied from light green to dark green with erect or pendant orientation of fruits on the plant. Similarly, another set of 78 advanced lines of F_8 generation derived from the cross between PT-12-3 (a sweet pepper genotype) and Bhut Jolokia (highly pungent, resistant to leaf curl disease) was also morphologically characterized. The fruit length of these lines varied from 5.6 - 12 cm, and 20 – 80 fruits per plant. The lines PT12-3xBJ-13-3-2, PT12-3xBJ-16-7-2, PT12-3xBJ-12-3 and PT12-3xBJ-16-7 were found promising (Fig. 25) and the line PT12-3xBJ-13-3-2 was identified for multilocational testing under AICRP Vegetable crops.

Evaluation of hybrids: Total 47 F_1 hybrids including eight commercially grown hybrids from the private seed sector were evaluated for various characters (Table 21 & Fig. 26). The performance evaluation of the hybrids revealed that A1 x VR-339, A5 x KDCS-810, A7 x EC-519636, A4 x Jayanti manifested superior yield potential of 200, 178, 175 and 170 q/ha. Fruit length was found better in the CMS based cross combinations of A5 x KDCS-810 (11.80), A7 x EC-519625 (11.42), A1 x Dabbi (11.32), A1 x EC-519625 (11.14), A1 x Japani Longi (10.64). Regarding fruit number per plant the hybrids A4 X Kalyanpur Chanchal (194), A9 x VR-339 (152), Kashi Anmol x J. Longi (144), A4 x Jayanti (142), A1 x VR-339 (130) were found promising. The hybrid combinations involving VR-339 as the pollen parent were found to be tolerant for leaf curl disease under field condition along with high pungency in the fruits.

PT12-3xBJ-PT-13-3-2

PT12-3xBJ-16-7

Fig. 25: Promising advanced lines of chilli

A1 x VR-339

A7 x EC-519636

Fig. 26: Promising CMS-based F_1 hybrids of chilli

Table 21: Performance of selected F_1 chilli hybrids for different traits

Traits	F_1 hybrids
Fruit length (cm)	A5 x KDCS-810 (11.80), A7 x EC-519625 (11.42), A1 X B. Dabbi (11.32), A1 x EC-519625 (11.14), A1 x J. Longi (10.64)
Fruit number/ plant	A4 X K. Chanchal (194), A9 x VR-339 (152), K. Anmol x J. Longi (144), A4 x Jayanti (142), A1 x VR-339 (130)
Fruit yield (q/ha)	A1 x VR-339 (200), CW x EC-519636 (199), A5 x KDCS-810 (178), A4 x Jayanti (170),
Plant height (cm)	A4 x BS-79 (89), A9 x Jayanti (88.8), A9 X IIVRC-462 (88.3), A1 x VR-339 (87.8), A1X IIVRC-462 (87.2)

During 2017-18, a total of 55 new cross combinations including 39 wild relatives based crosses have been developed involving the cytoplasmic male sterile lines and elite pollen parents in order to develop chilli hybrids having better yield potential along with other desirable traits such as resistance and qualities.

Early generation screening of an interspecific chilli population for leaf curl disease resistance: In chillies, there is no resistant cultivar available against leaf curl disease. Earlier, BS-35, a natural interspecific derivative of *C. frutescens* and *C. chinense* was selected for developing base population. In order to explore the resistance source, a population of 109 families in F_4 generation of the inter-specific cross of Kashi Sinduri x BS-35 was screened for the reaction to chilli leaf curl disease caused by whitefly transmitted begomoviruses. The scoring was done as per the standard procedure and 0 – 4 grades were given based on the symptom appearance on the plants under field condition (0 for no symptom; 1 for Curling of leaves up to 25%, no stunted growth, with normal flowering and fruiting; 2 for curling of leaves 26 to 50%, mild stunted growth, reduced flowering with normal sized fruits; 3 for curling of leaves 51 to 75%, stunted growth, distortion of leaves, reduction and malformation of fruit size, plant continue to grow and 4 for curling of leaves more than 75%, severe stunting of plants, no flowering and fruit formation, malformation of entire plants, plant growth stops). Samples collected from field were subjected to molecular screening using universal begomovirus specific primer pair (PALIc1960 /PALIr772). Six lines were completely free from virus in both field and molecular screening and four lines showing symptoms in field were not detected with any virus in molecular screening. PDI was calculated as sum of all score x 100/ Total number of plants observed x maximum grade. The population is further being monitored for the expression of ChiLCD and efforts are on to isolate resistant chilli pepper lines.

Categorization of F_4 families based on PDI: One hundred nine F_4 families were screened twice under field condition and based on average scoring; numbers of families were grouped in different categories. Five plants were scored individually and almost all grades, from 1 to 5, were observed in each family. No families wer found to be immune (PDI =0). One family was resistant (PDI =0.1 to 10), 10 were moderately resistant (PDI =10.1 to 25); 54 were susceptible (PDI =25.1 to 50) and 44 were highly susceptible (PDI =>50). Plants falling under 0 and 1 grade in different families were selected for molecular screening and these a total of 32

Table 22: Scoring of chilli families under field and artificial screening

F_4 Family	Field screening	Molecular Screening
GT123-1	0	+
GT167-3	1	+
GT194-2	1	+
GT212-4	1	+
GT136-2	1	+
GT109-5	1	+
GT109-4	1	+
GT127-3	0	-
GT115-4	0	-
GT128-1	1	+
GT144-3	0	+
GT193-2	1	+
GT108-1	1	-
GT106-4	1	+
GT140-2	0	+
GT241-2	0	-
GT156-4	0	+
GT193-5	1	-
GT260-2	1	+
GT113-5	1	+
GT132-4	0	+
GT204-2	1	-
GT193-4	0	-
GT130-1	1	+
GT177-3	1	+
GT123-2	0	+
GT182-4	1	+
GT127-1	0	-
GT135-3	0	-
GT125-2	1	+
GT109-3	0	+
GT126-4	1	-

individual plants were found under 0 and/or 1 grade (Table 22).

A total of 10 plants were completely free from virus as well as least infected by thrips and mites. Some plants showing minimum curling under field condition and are negative for molecular based probing, might be due to insects' damage. The families such as GT-125, GT-127 and GT-246 (Fig. 27) showed good potential with respect to ChiLCV resistance.

GT-125

GT-127

Fig. 27: Plants showing highly resistant reaction against ChiLCV disease

Pre-breeding and line development: A total of 220 lines including 142 RILs derived from the cross Kashi Sinduri x AKC-89-38 and 78 lines derived from PT-12-3 x Bhut Jolokia have been characterized in F_8 generations. These populations have some important genotypes which may be utilized as inbreds or cultivars after validation/multilocational testing. Using the wild species and natural interspecific derivatives, advance population have been maintained in different generations for various traits such as 116 lines in generation, 17 lines in F_6 and 29 lines in F_7 generation.

In the line development programme, different numbers of families are being advanced to subsequent generation. Overall, a total of 56 populations were maintained in F_2 generation and advanced to F_3. Similarly 24 F_3 families were advanced to F_4 and 145 F_4 families to F_5. In generation, 250 families of a cross Pusa Jwala x IIVRC-452 and 109 population of Kashi Sinduri x BS-35, besides 70 families of various combinations have been advanced to F6 generation. Over 150 families of F6 generation and 110 of F7 generation were furthered to subsequent generations.

Regarding line development in sweet pepper, inbred lines or varieties were crossed in order to create variability in capsicum (Fig 28). In that sequence, the F3 generations of 14 crosses/hybrids were advanced to next higher generation.

Entry for multi-location testing under AICRP (VC): An inbred line derived from the cross of PT-12-3 x Bhut Jolokia has been identified after two years (2016-

Fig. 29: VRC-14: An inbred line of chilli with high yield potential

Hyb. Navnita-derived line (F_3)

California Wonder x Bell Orange (F_3)

California Wonder x Nishat-1 (F_3)

Fig. 28: Variability in capsicum

Table 23: Performance of VRC-14, an inbred line of chilli for various morphological traits

Genotype	Year	Frt. Len (cm)	Frt. Wid (cm)	Frt. No./pl	10 Frt. Wt (g)	Pl Ht (cm)	Frt. Yd. (q/ha)
VRC-14	2016-17	9.58	1.18	85.33	71.50	66.20	167.78
	2017-18	10.04	1.18	87.00	79.00	67.25	187.28
Kashi Anmol	2016-17	7.72	1.06	82.60	40.00	44.60	90.86
	2017-18	7.60	1.10	96.00	38.00	42.30	100.32
Pusa Jwala	2016-17	10.85	1.02	49.60	41.00	58.40	55.92
	2017-18	10.90	1.03	58.00	46.00	56.40	73.37

17 & 2017-18) of station evaluation (Table 23) for multi-locational testing under AICRP (Vegetable Crops) during 2018-19. The line, PT12-3xBJ-PT-13-3-2, hereafter referred as VRC-14 (Fig. 29) bears more than 80 fruits of about 9.6 cm long having average yield potential of 175 q/ha of green fruits. The fruits are green, pungent and pendent in orientation.

Project 1.3: Genetic improvement of legume vegetables

Cowpea

Hybridization: Parents from available germplasm were selected on the basis of their earliness, growth habit, yield and yield attributing traits, pod quality and resistance to cowpea golden mosaic virus and 11 F_1s were made during *kharif*, 2017.

Advancement of generation: 11 BC_1F_1 and 11 BC_1F_2 were advanced to next generation. 32 SPS were done in 10 F_3 combinations on the basis of bushy growth, earliness, higher yield, better pod quality and CGMV resistance. 31 SPS were done in 10 F_4 combinations (32 SPS) and in 9 combinations on the basis of desirable horticultural traits and CGMV resistance. 70 SPS were done in 20 combinations of 21 BC_1 combinations (61 SPS) for further improvement and selection. 53 SPS were done in 19 combinations of 20 BC_1F_6 combinations (70 SPS) for desirable horticultural traits and CGMV resistance. 75 SPS were done in 31 F_7 combinations (76 SPS) for further improvement and selection. 4 lines (VRCP-182-4, VRCP-186-2, VRCP-187-5 and VRCP-191-1) were selected on the basis of desirable horticultural traits and CGMV resistance from 75 SPS in 31 F_8 combinations.

Evaluation of advance breeding lines: Fourteen dwarf and bush type advance lines along with one national check (Kashi Kanchan) were evaluated for various growth characters, earliness, yield attributing traits, yield and CGMV resistance during *kharif*, 2017 (Table 24). Line 96-4 flowered earliest and took minimum days to 50% flower (38.0 DAS) followed by line 68-2 (39.0 DAS). The maximum number of branches/plant was recorded in line 159-4 (5.5) followed by 68-2 (5.4). The longest peduncle was found in line 65-8 (38.8 cm) followed by Kashi Kanchan (37.8 cm). The maximum number of peduncles/plant was obtained from line 167-3 (28.7) followed by Kashi Kanchan (27.5). However, the maximum number of pods/plant was recorded in line 167-3 (37.0) followed by line 167-2 (32.5). The longest pod was obtained from line 112-4 (37.6 cm) followed by line 79-4 (36.9 cm). Similarly, the heaviest pod was recorded in line 144-5 (15.5 g) followed by line 79-4 (15.1 g). The maximum number of seeds/pod was observed in line 79-4 (14.7) followed by line 71.1 (14.4). The highest pod yield/plant was recorded in line 112-4 (445.5 g) followed by line 167-2 (417.4 g) and 167-3 (391.5 g). Overall line 112-4, 167-2 and 167-3 were found superior with regard to number of pods/plant, pod length and pod yield plant^{-1}. All the advance material showed resistance to cowpea golden mosaic virus under field condition except VRCP-112-4 (moderately susceptible) and VRCP-144-5 (highly susceptible).

Nutritional quality of immature green pod of IIVR developed cowpea varieties: The immature green pods of five released varieties of vegetable cowpea, developed by this institute, were analyzed for nutritional, secondary metabolites and antioxidant activities (Table 25). Total chlorophyll and carotenoid content was recorded maximum in variety Kashi Nidhi followed by Kashi Kanchan and Kashi Gauri. Kashi Unnati and Kashi Shyamal showed high level of protein content. Overall, Kashi Kachan was found to be best among all the varieties from antioxidants point of view. Kashi Shyamal was found to be best for secondary metabolites.

Genetic divergence for yield and yield related traits: Out of nine characters studied in seventy nine genotypes, the characters that contributed maximum towards divergence through PCA analysis were pod number, peduncle number, number of primary branches per plant and yield per plant.

Entries in AICRP (VC) trial: Two promising cowpea lines (VRCP-49-5 and VRCP-112-4) were in

Table 24: Performance of advance breeding lines of cowpea during *Kharif*, 2017

Advance line	Days to 50% flower (No)	Plant height (cm)	Branches plant^{-1} (No)	Peduncle length (cm)	Peduncles plant^{-1} (No)	Pods plant^{-1} (No)	Pod length (cm)	Pod weight (g)	Seeds pod^{-1} (No)	Pod yield (g plant^{-1})
65-8	45.0	45.0	3.0	38.8	14.8	20.4	38.2	14.4	12.3	290.5
66-4	46.3	50.3	3.6	35.2	16.3	22.5	30.8	13.8	14.0	313.2
68-2	39.0	62.6	5.4	26.6	16.0	24.8	32.8	12.7	13.4	317.8
71-1	42.7	54.4	3.2	36.8	23.5	32.2	28.1	10.5	14.4	341.5
79-4	46.3	57.0	4.2	34.3	18.4	24.5	36.9	15.1	14.7	365.6
96-4	38.0	44.8	3.0	22.0	18.8	28.4	32.5	11.3	12.0	324.2
98-4	41.7	49.2	4.0	35.4	15.5	23.2	28.9	12.1	13.0	284.6
112-4	47.3	48.6	4.4	33.8	24.4	32.0	37.6	13.8	12.5	445.5
144-5	44.0	45.7	3.5	31.7	15.3	21.5	34.7	15.5	12.8	338.3
147-2	43.3	44.3	3.8	34.5	21.2	31.3	27.6	9.9	11.6	311.8
158.3	40.0	45.5	4.6	33.2	19.3	29.2	30.5	12.1	11.2	348.7
159-4	42.7	42.1	5.5	35.6	23.0	32.5	30.4	10.3	10.9	338.8
167-2	41.7	51.7	4.3	31.8	22.3	32.7	32.6	12.6	13.8	417.4
167-3	43.0	38.4	4.7	30.6	28.7	37.0	30.2	10.5	12.8	391.5
K. Kanchan	47.0	58.6	4.8	37.8	27.5	28.2	32.4	13.1	13.6	365.2
CD (P=0.05)	4.15	6.73	0.48	3.14	2.46	2.72	2.74	1.23	1.37	46.47
CV (%)	6.51	7.26	6.84	5.73	6.30	6.59	5.58	6.32	5.72	10.36

Table 25: Biochemical analyses of fresh pods of vegetable cowpea varieties developed by ICAR-IIVR (based on fresh weight basis)

Biochemical parameters	ICAR-IIVR released cowpea varieties				
	Kashi Shyamal	Kashi Gauri	Kashi Unnati	Kashi Kanchan	Kashi Nidhi
Total Chlorophyll (mg/g)	0.51	0.66	0.45	0.73	0.78
Carotenoid (mg/g)	0.15	0.24	0.10	0.24	0.32
Total soluble protein (mg/g)	57.90	46.34	60.72	51.93	37.25
Total sugar (mg/100 g)	2.47	2.40	2.17	1.54	2.14
Phenol (mg/g)	7.84	6.87	5.99	4.37	6.51
Phenol : Total sugar ratio	3.17	2.86	2.76	2.84	3.04
Flavonoids (mg/g)	0.41	0.26	0.37	0.24	0.40
Antioxidant activities					
FRAP (µmol TE/g)	8.93	4.96	4.05	6.92	3.87
CUPRAC (µmol TE/g)	11.64	10.82	11.53	14.45	8.38
DPPH (µmol TE/g)	3.53	7.65	4.29	7.82	2.95
TEAC (µmol TE/g)	11.34	17.56	14.32	17.32	8.48

FRAP: Ferric reducing ability of plasma, CUPRAC: Cupric reducing antioxidant capacity, DPPH: Diphenyl picryl hydrazyl, TEAC: Trolox equivalent antioxidant capacity.

AVT-I trial for their multi-location evaluation through AICRP (VC).

Maintenance breeding: The maintenance breeding of ICAR-IIVR developed cowpea varieties *viz*. Kashi Shyamal, Kashi Gauri, Kashi Unnati, Kashi Kanchan and Kashi Nidhi were done through pure line selection.

Pea

Hybridization: A total of 43 F_1 crosses were attempted for targeted traits *viz*., earliness, high yield and resistant to powdery mildew and rust. To get higher number of pod per plant in early group, the genotypes *viz.*, Kashi Nandini, Kashi Uday, AP-3, Arkel, Kashi Ageti were crossed with multi-flowering genotypes *viz.*, VRP-500 (triple flowers) and VRPM-901(triple flowers). Similarly, F_1s were also made by utilizing the parents HUDP-15 and Kashi Samridhi having resistance to Powdery mildew.

Performance of triple podded lines: During the year 2016-17, five new stable triple flowered/podded genetic stocks *viz.*, VRPM-501, VRPM-502', VRPM-503 and VRPM-901-3 were reported. Single plant harvested seeds of these genotypes were grown in rows under normal field conditions in *Rabi* 2017-18. Furthermore, the genotypes that uniformly showed high frequency for triple podded trait were evaluated for their yield potential along with the VRP-500 (INGRI5009) for triple pod trait (Table 26). Among these, the genotype 'VRPM-901-3' was found to be most promising having higher number of multi-pods. The frequency of appearing three flowers and pods on each node is very high in this genotype compared to other triple genetic stocks.

plant families (SPS families) with fixation of this trait in 60 % population (Fig. 30). The genotypes showed consistent behaviour for appearance of penta- flower raceme and on an average it appeared on 22.3th node. In previous studies, it was observed that beside the genetic background, appearance of multi-flower in this line is also regulated by low temperature. In present crop season, the penta- flowered nodes were also observed during 9th -11th week after sowing (second week of January, 2018), surprisingly this year also, this period coincides with low temperature of the season i.e. 13.1°C. For further investigation regarding stability of this trait, seeds from multi - flowered plants of this genotype were maintained through SPS for evaluation in next generation.

Inheritance of multi-flower and multi-pods phenotypes: A total of nine crosses were attempted during 2016-17 to study the inheritance of multi-flowering by involving the parental lines *viz.*, VRP-500, VRPM-901, No-17 and VRP-386. F_1s seeds of these crosses were space planted and advanced to F_2 generation.

Table 26: Characterization of stable three flowered/podded genetic stocks in pea.

Trait	VRP-500	VRPM-501	VRPM-502	VRPM-901-3
Days to 50 % flowering	55.0	50.3	54.0	48.7
First multi-flower node	16.2	16.5	16.6	19.1
Pod length (cm)	7.72	9.20	7.12	8.0
Pods per plant (No.)	21.60	18.27	23.73	32.4
10-pod weight (g)	75.33	70.00	53.00	55.0
Seeds per pod	6.5	6.3	6.2	6.7
Pod yield per plant (g)	145.6	125.5	141.2	173.0

Stability for four & five flowers in VRPM-901-5: To check the stability of five flowered/podded trait in genotype 'VRPM-901-5', single plant harvest seeds (SPS) from 15 plants families (capable of producing four to five flowers at some of flowering nodes) were space planted under normal field conditions. Four to five flowered/podded plants were observed in 9

Fig. 30: Five- & four-podded racemes at seed maturity in VRPM-901-5.

Identification of advance promising lines: Among the mid- maturity group, three advance lines *viz.* PC-531 × PMR-32 and VL-8 × PC-531 were found to be promising with respect to days to 50% flowering, Number of pod per plant, average pod weight (g), pod length (cm) and pod yield per plant (Table 27).

Table 27: Promising advance breeding lines of pea for mid maturity

Trait	VRPM-915
Days to 50 % flowering	57.0
Number of pod per plant	16.3
Average pod weight (g)	8.5
Pod length (cm)	10.5
Number of seed per pod	7.0
Pod yield per plant (g)	135.5

Generation advancement of breeding material: Among total 86 advanced lines, 39 lines were advanced to F_2 population, 13 lines to F_5, 20 to F_6, 6 to F_7 and 10 to F_8.

Entries in AICRP trials: A new entry 'VRPE-105' was submitted for IET (early) of AICRP (VC) trial. Seed multiplication was done for the entries namely 'VRPE-103' in early group and 'VRPM-905' in mid group which were submitted earlier for AICRP (VC) trials.

Maintenance breeding: The maintenance breeding was done through true to type SPS of pea varieties *viz.* Kashi Uday, Kashi Nandini, Kashi Ageti, Kashi Mukti, Kashi Samrath, Kashi Shakti, and Kashi Samridhi.

French bean

Advancement of generations: 15 F_1s were advancement to F_2 and 2 F_2 populations advance to F_3 generations. For snap beans, the pods should be free of parchment. So, parchment free genotypes *viz.*, EC792393, FMGCV1187 and FMGCV1006 were identified for developing snap bean cultivars.

Entries in AICRP (VC) trial: Two promising bush type French bean lines (VRFBB-2 and VRFBB-91) are in AVT-I trial and two promising pole type French bean lines (VRFBP-44 and VRFBP-14) are in AVT-II trial for their multi-location evaluation through AICRP (VC).

Indian bean (*Dolichus* bean)

Generation advancement: A total of 45 cross combinations in different generation F_2 (15), F_4 (3), F_5 (7), F_6 (10) and F_{10} (10) were advanced for bushy growth habit, earliness, higher yield and DYMV tolerance. The advanced line *viz.*, VRBSEM-08, VRBSEM-206 and VRBSEM-207 were superior, liked by farmers and will be entered in AICRP (VC) 2018-2019 trial (Table 28).

Table 28: Performance of selected bush type genotypes

Advance line	Days to 50% flowering	Pod width (cm)	Pod length (cm)	Single pod weight (g)	Pod Yield (kg/ plant)*
VRBSEM -206	51	1.6	11.5	11.5	2.49
VRBSEM -207	47	1.4	12.5	10.2	2.82
VRBSEM -08	55	2.3	10.2	9.7	2.75

*Three pickings start from second week of January to first week of March

Maintenance breeding: Sixteen parental lines of bush type Indian bean were maintained.

Project: 1.4: Genetic improvement of gourds

Bitter Gourd

Evaluation of advance lines: During the year advanced breeding lines in three segments (small, medium and long) were evaluated for important horticultural traits. In the previous year (2016-17) three small fruited lines VRBTG-17, VRBTG-18 and VRBTG-33 were found to be yielder and recorded better for different traits. In small segment, two lines i.e. VRBTG-3 and OBG-11-1 gave maximum yield of 1030 and 1197 g/plant with the 19 and 45 number of fruits/plant, respectively. In medium type, advance line VRBTG-23 and VRBTG-21 performed better with respect to yield and fruit quality. In medium segment, this year two lines i.e. VRBTG-15 and VRBTG-12 gave maximum yield of 1200 g and 1644 g per plant, respectively. In long type, VRBTG-47-1 and VRBTG-5 gave maximum yield 1644 g and 1350 g per plant, respectively (Table 29 and Fig. 31). Due to consistence yield and acceptable fruit quality, VRBTG-5 identified from the Institute Technology Release Committee as "Kashi Mayuri". In extra-long fruiting group promising line VRBTG-10 gave 1186 g yield per plant (Fig. 32). This line is very much suitable for processing due to desirable attributes like uniform shape and size, less seeds, uniform length etc. In this evaluation programme, VRBTG- 12 and VRBTG-6-1 were also noted very promising for horticultural traits.

Development of superior hybrids: Total 23 hybrids developed in green fruited segment and evaluated with two leading check hybrids i.e. Sagar and

Table 29: Performance of advance lines of bitter gourd

Types	Genotypes	Fruit length (cm)	Fruit circum. (cm)	Fruit wt. (g)	Fruit/ plant (No.)	Yield/ plant (g)
Small	VRBTG-3	10.1	6.0	52.3	19	1030
	OBG-11-1	7.6	3.7	26.6	45	1197
Medium	VRBTG-15	12.0	5.5	75.0	16	1200
	VRBTG-12	13.5	6.2	78.3	21	1644
Long	VRBTG-5	17.9	5.3	75.0	18	1350
	VRBTG-6-1	16.5	5.0	74.0	15	1110
Extra Long	VRBTG-10	19.7	4.4	51.6	23	1186
	VRBTG-EL-1	20.6	4.8	82.0	21	1722
	VRBTG-8	19.8	4.9	85.0	20	1700

VRBTG-5 (Kashi Mayuri)

VRBTG-12

VRBTG-6-1

Fig. 31: Promising advance lines of bitter gourd

Fig. 32: VRBTG-10- Promising line of bitter gourd suitable for processing

Mohini for horticultural attributes and among them 9 promising hybrids (VRBTG-10 x VRBTG-47, VRBTG-10 x VRBTG-47-1, VRBTG-12 x BT-1, VRBTG-8 x IC212504, VRBTG-3 x VRBTG-5, VRBTG-5 x VRBTG-2-1, VRBTG-23 x VRBTG-4-1, VRBTG-21 x VRBTG-4-1 and VRBTG-37 x VRBTG-36) were selected. Cross combination VRBTG-5 x VRBTG-2-1 gave maximum yield of 4267.7 g/plant followed by VRBTG-10 x VRBTG-47-1 (4070.2 g/plant). Both the hybrids will be evaluated again and will be multiplied. Further, 4 cross combinations i.e. VRBTG-15 X IC212504, VRBTG-12 X BT-3-C, VRBTG-12 X VRBT-15 and VRBT-12 X BT-1B were developed and will be evaluated in the next season (Table 30).

Table 30: Performance of hybrids in bitter gourd

Cross combination	Fruit Length (cm)	Fruit Circu. (cm)	Fruit wt. (g)	Fruit/ plant (No.)	Yield/ plant (g)
VRBTG-10 x VRBTG-47-1	12.5	4.33	86.6	47	4070.2
VRBTG-5 x VRBTG-2-1	15.4	5.9	101.6	42	4267.7
VRBTG-23 x VRBTG-4-1	13.1	6.2	91.6	32	2931.2
VRBTG-37 x VRBTG-36	16.0	6.5	110.0	19	2090.0
Sagar©	12.5	5.9	85.0	25	2125.0
Mohini ©	12.5	5.3	63.3	22	1392.6

Advancement of generation: During the year, 18 from F_1 to F_2, 15 from F_2 to F_3, 13 from F_3 to F_4 and 11 from F_4 to F_5 number of population were advanced.

Screening of bitter gourd germplasms for root knot nematode: The experiment was conducted in CRD with 3 replications and 13 genotypes. For screening of bitter gourd genotypes soil sterilization was done at 21 °C and sowing of 2-3 viable seeds in sterilized soil filled 2 kg pots were done on 28.10.2017. Nematode eggs mass collected from cultured tomato plants and inoculated on 13.11.2017 in seedlings received 2000 juveniles per plant at root zone. Suitable conditions were ensured for good plant growth and data were collected on 22.01.2018 by uprooting the plants. The observations were presented (Table 31 & Fig. 33).

Table 31: Reaction of bitter gourd genotypes against root knot nematode

Genotypes	Number of galls (Mean ± SE)	Gall index (0-5 Scale)	Resistance reaction (S/MR/R)
VRBTG5-1	31.33 (5.6)^{cd} ± 1.18	4	S
VRBTG11-1	23.0 (4.8)^{c} ± 1.69	3	MR
VRBTG47	55.0 (7.4)^{ef} ± 3.09	4	S
VRBTG29-1	45.67 (6.8)^{e} ± 3.06	4	S
IC44428	4.67 (2.29)^{a} ± 1.78	2	R
BBGS-09-1	37.0 (6.16)^{de} ± 1.41	4	S
IC44438	4.34 (2.27)^{a} ± 0.98	2	R
VRBTG43	43.0 (6.60)^{de} ± 4.32	4	S
IC212504	23.67 (4.9)^{c} ± 1.78	3	MR
VRBTG-10	12.0 (3.5)^{b} ± 1.41	3	MR
VRBTG-15	32.67 (5.7)^{d} ± 3.13	4	S
VRBTG-1-1	33.34 (5.8)^{d} ± 1.96	4	S
VRBTG47-1 (susceptible check)	110.0 (10.52)^{g} ± 4.71	5	HS
SE (m)	0.27		
CD	0.8		
CV	8.54		

Gall index scale 0-5 (Gaur *et al.*, 2001). Ratings were categorized as 1= 0 galls/ no galls (HR: Highly resistance), 2.0= 1-10 galls (R: Resistance), 3.0= 11-30 galls (MR: Moderately resistance), 4.0 = 31-100 galls (S: Susceptible) 5.0= More than 100 galls (HS: Highly Susceptible)

IC44438 (R) VRBTG-47-1 (HS) IC44428 (R)

Fig. 33: Reaction of genotypes of bitter gourd against root knot nematode

Bottle gourd

Hybridization and evaluation of hybrids: During the year, 41 hybrids developed and evaluated for different agro-horticultural traits. In long group, VRBG- 8 x VRBG- 5 cross combination gave the yield of 11.400 kg/plant and next best combination was VRBG-8 x VRBG- 6 (11.200 kg/plant). In round fruited group, VRBG-4 x VRBG- 59 and VRBG-27 x VRBG-34 gave maximum yield of 20.35 and 19.00 kg/plant, respectively. In cut leaf segment, cross combinations VRBG-67 x VRBG-61, Cut-L and VRBG9-1-1 x VRBG-61.3 CL gave maximum yield of 8.250 and 18.900 kg/plant, respectively (Table 32 & Fig. 34).

VRBG-4 x VRBG- 59 VRBG-8 x VRBG- 5 VRBG-8 x VRBG- 6

Fig. 34: Promising cross-combinations of bottle gourd

Advancement of generation: During the year, 7 population from F_1 to F_2, 10 from F_2 to F_3, 6 from F_3 to F_4, 2 from F_4 to F_5 and 7 from F_5 to F_6 were advanced next generation.

Evaluation of winter fruited bottle gourd during rainy season: A total of 4 advance lines were evaluated along with Kashi Ganga and one hybrid for yield and horticultural traits during rainy season. Maximum number of fruits was recorded in advance line VRBOG-63-Sel-02 (10.3) followed by Kashi Ganga (8.25). Similarly, the yield/plant was maximum in advance line VRBOG-63-Sel-02 (9.79 kg) followed by Kashi Ganga (9.5 kg). The advance line VRBOG-63-Sel-02 has small fruit and good edible quality. The yield potential of same variety is better in winter season due to long duration of fruiting as compare to rainy season crop.

Pointed gourd

Evaluation of clones for various yield contributing traits: One hundred thirty nine female clones of pointed gourd were evaluated during 2017 for different economic important traits. Number of days required for the anthesis of the 1st flower varied from 55-70 days with mean of 65 days and number of node at 1st harvest ranged from 6-11 with a mean of 9.22. Substantial variation was also observed among the clones for internode length at first harvest (10.10-13.50 cm with mean value of 11.75 cm), fruit length (4.50 -12.25 cm, mean: 9.82 cm), fruit diameter (2.75 - 4.00 cm, mean: 3.65 cm), fruit weight (15.00-45.00 g, mean: 35 g), number of fruits per plant (75.00 -300, mean: 225) and weight of fruit per vine (4.00 -12.50 kg, mean: 6.50). Among the 139 female clones VRPG-215 found promising with 12.50 kg fruit yield per vine and long fruits (10-12.25 cm).

Kashi Amulya (VRPG-89): VRPG-89 is identified by the ITIC as Kashi Amulya (Fig. 35). It is a less seeded unique yield potential female clone.

Table 32: Best performing hybrids of bottle gourd

Hybrids	Fruit length (cm)	Fruit circum. (cm)	Fruits/plant (No.)	Fruit weight (g)	Yield per plant (kg)
Long group					
VRBG-8 x VRBG- 6	35.0	22.0	14	800.0	11.200
VRBG-8 x VRBG- 5	33.0	21.0	15	760.0	11.400
Pusa Samridhi ©	27.0	24.0	9	625.0	5.625
Varad-20 ©	25.5	18.20	6	400.00	4.800
Round group					
VRBG-4 x VRBG- 59	20.0	37.0	22	925.00	20.35
VRBG-27 x VRBG-34	27.2	56.3	20	950.00	19.00
Cut leaf base hybrids					
VRBG-67 x VRBG-61, Cut-L	22.3	19.5	15	550	8.250
VRBG9-1-1 x VRBG-61.3 CL	20.9	48.5	18	1050	18.900

Fig. 35: Fruits of Kashi Amulya (VRPG-89)

Evaluation of hybrid female clones: Seventy two hybrid female clones derived from two intra specific cross-combinations Kashi Alankar × Male-1 and Kashi Suphal × Male-1 was evaluated for 7 agro-morphological traits including yield per vine. The principal component analysis using neighbour joining clustering was performed. First two major axis of differentiation (PC1 and PC2) explained 61.42% of the total variation. Further 72 hybrid clones were classified into 3 major groups. The hybrid clones Hyfc-27, Hyfc-30, Hyfc-35 and Hyfc-42 were noticeably located from rest of the hybrid female clones.

Identification of molecular marker linked to seedless fruit development: With an objective to identify trait linked molecular marker in pointed gourd, DNA was isolated from male clone, seeded female clones and seedless female clone (IIVR-105) and 90 ISSR primers were used to screen trait specific polymorphism. A putative seedless fruit specific marker *i.e.* 850 bp amplicon from UBC-840 was identified as linked to seeded clone. Same maker amplified 625 bp band which is present in seeded female clone and is absent in seedless female clone. Further, this amplicom will be sequenced to develop either STS or SCAR marker for identification of trait in female clones for seedless fruit development (Fig. 36).

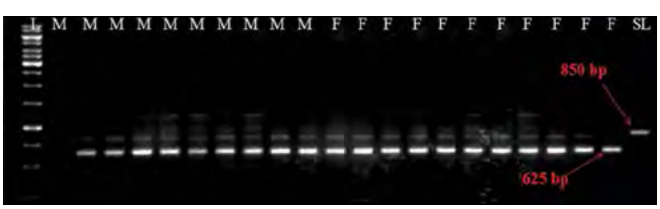


Fig. 36: ISSR pattern generated by primer UBC-840

Planting material and clonal multiplication of selected clones: About 5000 planting materials of Kashi Alankar, Kashi Suphal and Kashi Amulya were produced for distribution to the farmers. All the selected clones of pointed gourd were clonally multiplied to enhance the plant population. Beside this, approximately 100 numbers of planting materials of VRPG-141, VRPG-103, VRPG-05, VRPG-17, VRPG-215 and VRPG-85 were also produced.

Ash gourd

Evaluation of segregating lines of ash gourd: Fifteen lines of wax gourd including waxless lines along with 3 released varieties were planted. The performance of waxless segregating lines was better as compared to previous year. The variation was observed in fruit shape and size. The weight of waxless lines ranged from 2.3-6.9 kg. The storability of waxless lines was at par with the waxy lines.

Multiplication and maintenance of seeds of released varieties: One kg seeds of each Kashi Dhawal, Kashi Surbhi and Kashi Ujwal were produced and SPS were selected for maintenance of the variety.

Sponge gourd

One hybrid and one variety were identified by Institute Technology Identification Committee-

a. Kashi Saumya (VRSGH-3): An early high yielding hybrid with dark green foliage, good fruit quality and resistant to sponge gourd mosaic virus and tolerant to downy & powdery mildew under field condition (Fig. 37). The proposal of this hybrid has been submitted to SVRC, Lucknow, Uttar Pradesh. Three kg seed has been produced of the hybrids.

Fig. 37: Kashi Saumya

b. Kashi Jyoti (VRSG-17-1): An early high yielding variety with medium vine length (2.55- 4.5 m.) and good fruit quality (Fig. 38). It is identified at the Institute level on 17th May, 2017 for cultivation in North Indian plains particularly in Uttar Pradesh. The proposal of this variety has been submitted to SVRC, Lucknow, Uttar Pradesh. A total of 2.250 kg nucleus seed has been produced.

Fig. 38: Kashi Jyoti

Promising F_1 hybrids: Among the 48 F_1 hybrids, VRSG-17-3 × VRSG-171, VRSG-57 × VRSG-195, Phule Prajkta × VRSG-195, VRSG-2-12 × VRSG-214, VRSG-2-12 × VRSG-195, VRSG-195 × VRSG-2-12 and VRSG-57×VRSG-194 were found promising for various horticultural traits over the checks and showed tolerance against downy mildew and viral disease under field conditions (Table 33 & Fig. 39).

Promising advanced breeding lines: Among the 32 advanced breeding lines, VRSG-17-1, VRSG-17-2, VRSG-17-3, VRSG-17-4, VRSG-17-5, VRSG-17-10, VRSG-17-11, VRSG-17-12, VRSG-17-14 were found promising for various horticultural traits and showed tolerance against downy mildew and virus disease under field conditions (Fig. 40).

Table 33: Performance of promising F_1 hybrids of sponge gourd

F_1 hybrids	Days to Ist male flower appeared	Days to Ist female flower appeared	Days to first harvesting	Fruit colour	Fruit Length (cm)	Fruit diameter (cm)	No. of fruits/ plant	Average fruit wt.(g)	Yield/ plant (kg)
VRSG-17-3 × VRSG-171,	40	45	56	Dark green	29.54	3.8	17.53	198.8	3.485
VRSG-57×VRSG-195	36	35	43	Green	23.18	3.42	11.28	150.0	1.692
VRSG-2-12 × VRSG-214	39	41	48	Dark green	36.0	3.6	10.5	186	1.95
VRSG-2-12 × VRSG-195	44	41	51	Green	28.0	2.5	9.5	171	1.62
Kashi Saumya - ©	37	33	41	Dark green	19.74	3.3	14.67	112.2	1.65
Priya (Golden Seeds) - ©	39	40	51	Green	23.30	3.16	10.27	140.2	1.439
KSP-1125 (Kalash Seeds) - ©	45	43	51	Dark green	25.60	3.06	12.83	165.2	2.119
Utsav (Clause Seeds) - ©	41	42	51	Green	26.2	3.32	6.30	166	1.046
VNR Alok (VNR Seeds) - ©	40	40	47	Dark green	25.60	3.14	8.6	129.0	1.109


VRSG-17-3xVRSG-171


VRSG-57x VRSG-195


VRSG-57 × VRSG-171


VRSG-171× VRSG-1-12


VRSG-57 x VRSG-194


VRSG-195 × VRSG-2-12

Fig. 39: Promising F_1 hybrids of sponge gourd


VRSG-17-2


VRSG-17-4


VRSG-17-5

Fig. 40: Advance lines of sponge gourd

Generation advancement: Eleven F_{10}, 14 F_{9}, 13 F_{8}, 4 F_7 and 7 F_6 population of sponge gourd were advanced to F_{11}, F_{10}, F_9, F_8 and F_7, respectively. Whereas the individual population of VRSG-136 xVRS-1 and V RSG-136 xVRRG-27 were advance (*Luffa cylindrica syn. Luffa aegyptiaca x Luffa acutangula* var. Satputia syn. *Luffa hermaphrodita)* advanced from F_5 to F_6 (160 plants).

Teasle gourd

Forty lines of Teasle gourd were characterized for various horticultural traits at ICAR-IIVR-RRS, Sargatia, Kushinagar, Uttar Pradesh during summer and rainy season, 2017. A high genetic variability has been observed in germplasm for the traits like node number to first pistillate flower appearance, days to first pistillate flower anthesis, days to first fruit harvest, peduncle length, polar circumference of fruit, equatorial circumference of fruit, length of blossom end, number of fruits per plant, average fruit weight and fruit yield per plant (Fig. 41). Based on the initial screening, some of the potential genotypes identified for various horticultural traits are given (Table 34). Maximum fruit yield per plant found in VRSTG-6 (1.81 kg) followed by VRSTG-20 (1.55 kg) and VRSTG- 38 (1.53 kg).

Spine gourd

Ten lines of spine gourd were evaluated for various horticultural traits at ICAR-IIVR-RRS, Sargatia, Kushinagar, Uttar Pradesh during summer and rainy season, 2017. A high genetic variability has been observed in germplasm for the traits like node number to first pistillate flower appearance, days to first pistillate flower anthesis, peduncle length, polar circumference of fruit, number of fruits per plant, average fruit weight and fruit yield per plant (Fig. 42). Based on the primary screening, some of the potential genotypes have been identified for various horticultural traits that are given (Table 35). The genotype VRSEG-9 (1.86kg) gave maximum fruit yield per plant followed by VRSEG-7 (1.81 kg) and VRISEG-4 (1.81 kg).

Table 34: Potential genotypes of teasle gourd for various horticultural traits

Characters	General Mean	Range		Promising lines
		Min.	Max.	
Node number to first pistillate flower appearance	19.40	10.70	32.70	VRSTG-12 (10.67), VRSTG-7 (11.33), VRSTG-10 (11.33)
Days to first pistillate flower anthesis	117.36	57.33	153.33	VRSTG-20 (57.33), VRSTG-6 (88.67), VRSTG-10 (93.00)
Days to first fruit harvest	139.92	81.67	178.67	VRSTG-20 (81.67), VRSTG-10 (108.67), VRSTG-6 (109.00)
Peduncle length (cm)	10.96	6.67	17.67	VRSTG-53 (14.60), VRSTG-49 (17.00), VRSTG-28 (17.67)
Polar circumference of fruit (cm)	17.77	12.67	21.53	VRSTG-49 (20.67), VRSTG-53 (20.93), VRSTG-44 (21.53)
Equatorial circumference of fruit (cm)	13.32	11.00	16.33	VRSTG-49 (16.33), VRSTG-11 (16.17), VRSTG-25 (15.47)
Length of blossom end (cm)	1.37	0.50	2.30	VRSTG-13 (2.30), VRSTG-1 (2.17), VRSTG-9 (2.07)
Number of fruits per plant	13.20	5.33	35.00	VRSTG-15 (35.00), VRSTG-20 (35.00), VRSTG-6 (34.33)
Average fruit weight(g)	48.28	27.94	70.89	VRSTG-49 (70.89), VRSTG-43 (68.33), VRSTG-11 (65.44)
Fruit yield per plant (kg)	0.640	0.204	1.81	VRSTG-6 (1.81), VRSTG-20 (1.55), VRSTG-38 (1.53)

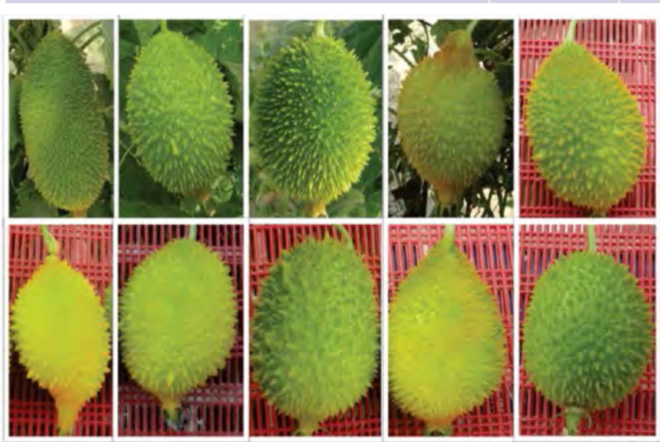


Fig. 41: Variability in fruit shape of teasle gourd

Table 35: Potential genotypes of spine gourd identified for various horticultural traits

Characters	General Mean	Range		Promising lines
		Min.	Max.	
Node number to first pistillate flower appearance	20.73	14.00	27.00	VRSEG-10 (14.00), VRSEG-2 (15.33), VRSEG-8 (20.33)
Days to first pistillate flower anthesis	62.00	55.67	74.67	VRSEG-7 (55.67), VRSEG-1 (57.67), VRISEG-4 (59.33)
Days to first fruit harvest	81.40	76.00	93.33	VRSEG-7 (76.00), VRISEG-4 (76.33), VRSEG-1 (77.67)
Peduncle length (cm)	2.09	1.33	3.00	VRSEG-8 (1.33), VRSEG-1 (1.40), VRSEG-10 (1.77)
Polar circumference of fruit (cm)	11.75	10.67	16.27	VRSEG-5 (16.27), VRSEG-10(12.20), VRSEG-11 (11.75)
Equatorial circumference of fruit (cm)	8.87	7.00	10.63	VRSEG-5 (10.63), VRISEG-4 (9.17), VRSEG-10 (9.13)
Length of blossom end (cm)	1.20	1.00	2.10	VRSEG-1 (1.00), VRSEG-3 (1.00), VRISEG-2 (1.07)
Number of fruits per plant	155.17	94.33	195.33	VRSEG-9 (195.33), VRISEG-4 (190.33), VRSEG-7 (184.67)
Average fruit weight (g)	9.32	7.58	11.20	VRSEG-10 (11.20), VRSEG-2 (10.25), VRSEG-5 (10.02)
Fruit yield per plant (kg)	1.45	0.922	1.86	VRSEG-9 (1.86), VRSEG-7 (1.81) , VRISEG-4 (1.81)

Fig. 42: Variability for fruit shape in spine gourd

Ivy gourd

Eight lines of Ivy gourd were collected and evaluated for various horticultural traits at ICAR-IIVR-RRS, Sargatia, Kushinagar, Uttar Pradesh during summer and rainy season, 2017. A high genetic variability has been observed in germplasm for the traits like node number to first pistillate flower appearance, days to first pistillate flower anthesis, days to first fruit harvest, peduncle length, polar circumference of fruit, equatorial circumference of fruit, number of fruits per plant, average fruit weight and fruit yield per plant. Based on the primary screening, some of the potential genotypes have been identified for various horticultural traits (Table 36 & Fig. 43). Highest fruit yield per plant was observed in line VRIG-4 (8.35 kg) followed by VRIG-3 (6.77 kg), VRIG-6 (5.49 kg).

Table 36: Potential genotypes of ivy gourd for various horticultural traits

Characters	General Mean	Range		Promising lines
		Min.	Max.	
Node number to first pistillate flower appearance	16.00	10.67	20.67	VRIG-6 (10.67), VRIG-5 (14.00), VRIG-12 (15.33)
Days to first pistillate flower anthesis	71.96	56.67	87.33	VRIG-11 (56.67), VRIG-4 (57.33), VRIG-14 (68.00)
Days to first fruit harvest	85.13	69.67	101.33	VRIG-11 (69.67), VRIG-4 (71.67), VRIG-14 (78.67)
Peduncle length (cm)	3.14	1.75	3.87	VRIG-11 (1.75), VRIG-8 (2.67), VRIG-12 (2.70)
Polar circumference of fruit (cm)	13.89	12.53	16.47	VRIG-8 (16.47), VRIG-3 (14.92), VRIG-4 (14.54)
Equatorial circumference of fruit (cm)	8.57	7.80	9.76	VRIG-4 (9.76), VRIG-3 (9.39), VRIG-12 (8.83)
Number of fruits per plant	173.42	81.33	280.67	VRIG-4 (280.67), VRIG-3 (275.00), VRIG-6 (270.33)
Average fruit weight (g)	22.64	16.37	29.74	VRIG-4 (29.74), VRIG-11 (25.26), VRIG-14 (25.16)
Fruit yield per plant (kg)	4.03	1.61	8.35	VRIG-4 (8.35), VRIG-3 (6.77), VRIG-6 (5.49)

Fig. 43: Variability in fruit shape of ivy gourd

Project 1.5: Genetic improvement of melons, pumpkin and cucumber

Cucumber

Development and evaluation of hybrids: Selected five F_1 cross combinations have been evaluated. Yield per plant was maximum in hybrid VRCUH-16-01 followed by VRCUH-16-02. Seed of these hybrids as well as parental lines were produced. Other yield contributing traits of promising hybrids are given below (Table 37).

Table 37: Yield and contributing traits of selected hybrids in cucumber

Hybrids	Fruit length (cm)	Fruit diameter (cm)	Fruit/ plant (No.)	Average fruit weight (g)	Yield/ plant (g)
VRCUH-16-01	20.72	4.52	6.56	264.83	1730
VRCUH-16-02	23.24	4.76	6.24	225.25	1410

Evaluation of advance lines: Five advance lines along with check PCUC-09 have been evaluated for yield and its contributing traits in mottle green segment. Fruits of these lines were non-bitter in taste. The best performing lines based on the fruit colour, appearance and yield were VRCU-Sel.-12-02 followed VRCU-Sel.-12-03 based on yield data (Table 38).

Table 38: Yield and contributing traits of selected advance lines in cucumber

Hybrids	Fruit length (cm)	Fruit diameter (cm)	Fruit/ plant (No.)	Average fruit weight (g)	Yield/ plant (g)
VRCU-Sel - 12-03	18.5	4.8	4.27	250	1067.5
VRCU-Sel - 12-02	21.1	4.4	6.0	200	1200.0
PCUC-9	23.0	5.0	3.6	230	828.0

Advancement of breeding material: A total of 37 segregating lines which includes F_5 (10), F_6 (7), and F_7 (20) were evaluated; selfed and further selection were made to advance as next generation.

Pumpkin

Isolation and maintenance of inbred: A total of 9 inbred have been selected for hybridization. The selected inbred have been maintained for purity through selfing. The target of hybrid development is in mottle green and flat round/round segment.

Advancement of breeding material: A total of 60 segregating lines which includes F_2 (09), F_3 (11), F_4 (11), F_5 (7), F_6 (8), F_7 (2) and F_8 (12) were evaluated; selfed and further selection were made to advance as next generation.

Evaluation of advance lines: Five advance breeding lines have been evaluated for important horticultural traits. Maximum yield per plant was reported in VRPK 18-01 (6.16 kg/plant) followed by VRPK-230 (4.67 kg/plant). Maximum individual fruit weight was observed in VRPK-Sel-10-15 (2.35 kg) followed by VRPK-63 (2.0 kg) at green edible stage. On the basis of overall performance VRPK-18-01, VRPK-230, VRPK-222-2-1 and VRPK-09-01 were found promising (Fig. 44).

Fig. 44: VRPK-18-01

Development and evaluation of hybrids: Nine selected parents namely VRPK-63, VRPK-230, VRPK-310, HAPK-1, Arka Chandan, Narendra Agrim, Anand Pumpkin-1, Azad Pumpkin-1 and Kashi Harit were crossed in half diallel technique excluding reciprocals. Developed F_1 hybrids along with parents evaluated for their yield and yield attributing traits.

Maximum green fruit yield per plant was recorded in hybrid VRPKH-17-03 (9.23kg) followed by VRPKH-17-04 (9.11 kg) and VRPKH-17-05 (9.05 kg). Based on the overall performance like green fruit yield, earliness, fruit shape and size, hybrid VRPKH-16-06 is recorded as most promising. The detail of the hybrids are given (Table 39 & Fig. 45).

Fig. 45: VRPKH-16-06

Table 39: Yield and contributing traits of pumpkin hybrids

Hybrids	Characters				
	Fruit equatorial circumference (cm)	Fruit polar circumference (cm)	Number of fruit/plant	Av. fruit weight (kg)	Fruit yield/ plant (kg)
VRPKH-17-03	66.40	62.40	3.72	2.50	9.23
VRPKH-17-04	57.67	52.13	2.68	3.52	9.11
VRPKH-17-05	67.20	59.20	4.30	2.13	9.05
VRPKH-16-06	64.62	55.80	3.38	2.18	7.37

Multiplication and maintenance of released variety and advance lines: Two kg seeds of Kashi Harit variety of pumpkin were produced and 32 SPS were selected for maintenance of the variety.

Summer squash

Maintenance and evaluation of advance lines: Four promising advance lines and one check of *Cucurbita pepo* (summer squash) and 62 segregating lines (5 F_3, 7 F_4 and 50 F_5) were evaluated. Among these, the advance lines VRSS-65 and VRSS-66 were found promising. The segregating lines of summer squash have major variation in colour and shape. All these segregants are resistant to viruses. These lines were advanced through selfing and further selected for next generation. Among the segregating population, one line VRSS-17-05 showed the stability towards high frequency femaleness. The high frequency female line has been maintained by sibing/selfing. Seeds of Kashi Subhangi were also multiplied and 1 kg seed has been produced.

Muskmelon

Hybridization and generation advancement: A total of six cross combinations were attempted which conclude two monoecious × monoecious, one andromonoecious × hermaphrodite and three andromonoecius × monoecious and crossed seeds were harvested. Four F_1, two F_2 and three F_3 advanced to F_2, F_3 and F_4 generations, respectively. Beside, one monoecious line VRMM-170 identified in previous season also self-pollinated to develop monoecious inbreed.

Evaluation of recombinant inbred lines: Sixty RILs (F_7) of Kashi Madhu × B-159 evaluated for various traits of economic important. Sex form of the RILs is either andromonoecious or monoecious. Fruit shape varies from oblong to flattish round (Fig. 46). Days and node to first productive flowers ranged from 40-55 days and 4-9 nodes. Fruit weight varied from 200-750g with wide variation in TSS ranged from 4.5-10 °Brix. Late occurrence of downy mildew also observed in the RILs family. On the basis of eight traits PCA was done. Results of the analysed data shows that the RILs 711, 272, 320 and 523 were distinctly placed from rest of the RILs.

Fig. 46: Variability for fruit traits in RILs of muskmelon

Watermelon

Isolation and maintenance of inbreds: The selected inbreds have been maintained for purity through selfing. The aim of identification of inbred is to hybrid development in red fleshed with less seeded, thin pericarp and high TSS (>13 ˚Brix) in Ice-box/Mini watermelon segment.

Evaluation of advance lines: Sixty lines were evaluated for yield and quality attributes and a total of 65 lines including identified/released varieties maintained as active collections and multiplied. There was severe incidence of thrips, tospovirus and watermelon bud necrosis virus in both seasons under open field conditions. As a result, the fruiting was very poor and did not reach maturity. Hence, fourteen advanced lines

Table 40: Mean performance of various genotype under net house condition

Traits	General mean	Range		Genotype
		Mini	Max.	
Node of 1st female flower	22.76	14	33	VRW-508(14),VRW-9-1(14), VRW-507(15)
Days to 50% flowering	37.12	33	41.3	VRW-507(33),VRW-509(34), VRW-511(35)
Number of fruits	3.01	2.2	3.8	VRW-14-1(3.8),VRW-514(3.6), VRW-516(3.4)
Fruit length (cm)	13.50	9.50	17.33	VRW-509(16.50),VRW-513(17.33),
Fruit diameter (cm)	11.84	9.57	15.35	VRW-511(15.35),VRW-515(14.36)
Fruit weight (g)	1204	700	1816	VRW-509(1816),VRW-514(1713),VRW-57(1646)
Yield per plant (kg)	3.58	2.17	6.15	VRW-514(6.15),VRW-516(4.45),VRW-57(4.39), VRW-14 1(4.37)

were evaluated for important horticultural traits under Net/Poly house conditions. Maximum yield per plant was observed in VRW-514 (6.15 kg/plant) followed by VRW-516 (4.45 kg/ plant). Maximum individual fruit weight under small segment was observed in VRW-509 (1.8 kg) followed by VRW-514 (1.7 kg) at mature stage. On the basis of overall performance VRW-514, VRW-516, VRW-57 and VRW-14-1 were found promising (Table 40 and Fig. 47).

VRW-516

VRW-514

Fig. 47: Promising genotypes of watermelon VRW-516, VRW-514 and VRW-514-2

Round melon

Eight lines of *Praecitrullus fistulosus* (round melon) were evaluated under poly house due to severe incidence of viruses in open field. Maximum yield was found in VRM-1 followed by VRM-5 which has medium green colour and flatty-round to round fruit shape (Fig. 48). All these lines are susceptible to viruses under field condition and were advanced through selfing.

Fig. 48: VRM-1

Table 41: Characterization and evaluation of long melon germplasm for various horticultural traits

Characters	Potential genotypes
Earliness (<30 Days)	VRLM-01, VRLM-28, VRLM-38, VRLM-40, VRLM-02, VRLM-21, VRLM-29 and VRLM-29-1
Node to 1st female flower (<7)	VRLM-01, VRLM-02, VRLM-3, VRLM-34, VRLM-11, VRLM-13-1, VRLM-18, VRLM-21, VRLM-23, VRLM-29 and VRLM-29-1
Fruit length (30-50 cm)	VRLM-40, VRLM-01, VRLM-3, VRLM-7-1 , VRLM-6, VRLM-14, VRLM-24-1, VRLM-28, VRLM-2, VRLM-38 and VRLM-16
Fruit diameter (>3.5 cm)	VRLM-37, VRLM-4-1, VRLM-7, VRLM-18 , VRLM-11-1 etc.
Average weight (>100g)	VRLM-37, VRLM-4-1, VRLM-7, VRLM-16, VRLM-11-1, and VRLM-38
No. of fruits/ plant (>5)	VRLM-16, VRLM-40, VRLM-7-1, VRLM-01, VRLM-11-1, VRLM-28, VRLM-29-1, VRLM-02, and VRLM-24-1
High yield lines	VRLM-28, VRLM-24-1, VRLM-16, VRLM-11-1, VRLM-16, VRLM-01, VRLM-40, VRLM-3, VRLM-2 and VRLM-7-1
Dark green fruit color	VRLM-7-1, VRLM-4-1, VRLM-24-1 and VRLM-9
No. of branches (>5)	VRLM-13-1, VRLM-02, VRLM-16, VRLM-38, VRLM-8 VRLM-29 and VRLM-11-1

Long melon

Thirty diverse genotypes were maintained and evaluated for different horticultural traits. Early flowering was observed in VRLM-28 (25 DAS) followed by VRLM-01 (28 DAS). Maximum fruit length was recorded in VRLM-40 (50 cm) and minimum in VRLM-11-1 (25 cm). From the present study, the genotypes VRLM-01, VRLM-40, VRLM-28 and VRLM-3 were found to be superior for yield and quality attributes. The genotypes were categorised according to their morphological parameters (Table 41). All the lines are being maintained through selfing.

Project 1.6: Genetic Improvement of Okra

Evaluation of hybrids: Thirty six F_1 hybrids involving seventeen diverse parents were evaluated during *kharif* season of 2017 for yield and YVMV and OLCV disease reaction. Pusa Sawani was used as susceptible check. The hybrid VRO109 x VRO115 was earliest, took 42 day for 50% flowering. Highest yield/plant was harvested from 307-10-1 × VRO-115 (580g) and the lowest PDI for YVMV was also recorded in 307-10-1 × VRO-115 (0.00) (Table 42). The YVMV PDI and OLCV PDI in susceptible check ranged from 85-95% for YVMV and 50-60 % for OELCV.

Table 42: Agronomic performance and disease incidence of 10 promising hybrids

Promising hybrids	Day to 50% flowe-ring	Plant height (cm)	Number of fruits/ plant	Fruit yield/ plant (g)	YVMV PDI	OLCV PDI
307-10-1 × VRO-115	45	178	43.70	580	0.00	6.66
VROB-178 × 416-10-1	44	103	42.70	553	5.57	10.85
VRO-115 × VRO-110	46	156	32.1	503	0.00	11.28
VRO-109 × VRO-115	42	130	35.8	437	0.00	8.25
VRO-115 × VRO-109	43	127	32.7	403	0.00	5.28
VRO-110 × VRO-112	44	121	31.9	355	0.00	8.12
VRO-115 × VRO-114	46	161	29.9	343	2.25	6.00

Evaluation of advance lines: A total of 35 advance lines were evaluated during *kharif* season for yield and viral disease (YVMV and OLCV) reaction. VRO-110, VRO-112-1 and VRO-113 were found promising, among these advance lines, for various horticultural traits and disease reaction (Table 43 & Fig. 49).

VRO-113 VRO-110 VRO-112-1

Fig. 49: Promising advance lines of okra

Screening of hybrids against major sucking pest: Thirty six F_1s of okra were screened against its major sucking pest *viz.* jassids (*Amrasca biguttula biguttula*) and whitefly (*Bemisia tabaci*) during *Kharif* season of 2017-18. Amongst these test hybrids, VRO-114×VRO-102 harboured lowest whitefly population (0.82 whitefly/leaf) followed by VRO-109×VRO-115 (0.84whitefly/leaf). In case of jassids, lowest population were recorded in VRO-114×VRO-112 (2.75 jassids/ leaf) followed by VRO-102×Kashi Kranti (2.98).

Molecular screening of VRO-109 for begomoviruses: Okra genotype VRO-109 screened against begomovirus through PCR amplification using begomovirus specific primers. For PCR amplification leaf samples were taken from randomly selected 24 plants at 7 days interval from 30 days after sowing (DAS) to 90 DAS. Up to 37 days all the 24 selected plants did not show any disease symptoms and zero amplification of begomovirus specific primers observed. Though the selected plants remained symptom less up to 74 days,

Table 43: Characteristics of promising lines VRO-110, VRO-112-1 and VRO-113

Advance line	Days to first flowering	Node to first flower	Fruit color	Fruit length (cm)	Fruit diameter (cm)	Fruits per plant	Resistance	Yield Potential (q/ha)
VRO-110	39-42	4-5	Dark green	11-12	1.3-1.4	20-25	YVMV & OELCV	160-170
VRO-112-1	41-43	4-5	Dark green	11-12	1.4-1.5	25-30	YVMV & OELCV	175-180
VRO-113	40-42	5-6	Dark green	12-15	1.4-1.5	19-24	YVMV & OELCV	150-160

faint bands with Rojas Primer observed in 6, 7 and 7 samples at 60, 67 and 74 days after sowing. At 81 days, Rojas Primer amplified in 11 sample but no symptoms was observed under field condition. Out of 24 plant 9 plant showed symptoms and 18 produce band with Rojas Primer at 88 day when crops is almost harvested.

Evaluation of wild species: A total 64 accessions of wild relatives of okra which include *A.caillei*, *A. ficulneus*, *A. tuberculatus, A. angulosus* var *grandiflorus, A. tetraphyllus, A. manihot, A. crinitus, A. moschatus, A. moschatus* subsp. *tuberosus* were evaluated for various descriptor traits and YVMV and OLCV disease reaction. Among these, *A. crinitus, A. angulosus* var *grandiflorus* (IC- 470751, IC203834), *A. moschatus* subsp *tuberosus* (IC-470750) and *A. moschatus* (IC-140985, IC-141013) remained free from both YVMV and OLCV disease under field condition. All the wild species had 5 sepal and petal, though number of epicalyx varies from 4 (*A. angulosus*) to 11 (*A. crinitus*). Petal colour is white in *A. ficulneus*, creamish white in *A. tuberculatus,* yellow in *A. crinitus* and *A. moschatus* and red in *A. moschatus* subsp *tuberosus.* Fruit length, fruit diameter and average fruit weight varied from 3.50-11.20 cm, 1.20--2.90 cm and 2.11-14.00 g, respectively. Number of seeds per fruits ranged from 20-70.

Evaluation of GCD and GED series lines from NBPGR-RS, Thrissur: A total 33 okra accessions which includes 19 GCD and 14 GED series lines evaluated for horticultural traits and YVMV and OLCV disease reaction. All the lines showed susceptible reaction to both the viral diseases (YVMV and OLCV). Among these 33 lines GCD-574, GCD-548, GCD-493, GCD-588, GCD-483 and GCD-460 found as round fruited lines, while GED-159, GCD-150 and GED-12, produced red and white coloured fruit, respectively.

Evaluation of wild *Abelmoschus* derived genetic materials from Thrissur: Eighty wild *Abelmoschus* derived genotypes from Thrissur evaluated for viral disease resistance. These lines include 15 cultivated okra × *A. angulosus* var *grandiflorus*, 40 cultivated okra × *A. mizoramensis sp. nova,* 14 cultivated okra × *A. tetraphyllus* var. *tetraphyllus and* 10 *A. caillei* × *A. angulosus* var *grandiflorus.* All the wild derived materials having *A. mizoramensis sp. Nova* as male parent showed high degree of resistance to both the diseases (YVMV and OLCV). Few lines of cultivated okra × *A. mizoramensis* sp. *nova* also set fruits during December –January.

Development of RILs of VROR-156 × VRO-5: In the summer season of 2017, a total of 260 F_2 seeds sown to develop recombinant inbred lines of VROR-156 (red fruited) × VRO-5 (green fruited). In F_2 generation, segregation of 260 plants followed 3:1 ratio for red and green colour fruits. Each individual of 260 F_2 plants will form a RIL after fixing of trait under consideration. All the plants were selfed and F_3 seeds harvested from single fruit. In the rainy season, the freshly harvested seeds of 260 RILs F_3 sown in single row and selfed to produce F_4. Seeds were harvested from single fruit of F_3 plant.

Generation advancement: A number of progeny families in various stages (generations) of inbred development were grown, single plants selection was done for dark green fruit colour, small fruit size, YVMV/ OELCV resistance/ tolerance and seeds were collected for further advancement of generation [26 lines of F_2, 10 lines (62 SPS) of F_3, 13 lines (54 SPS) of F_4, 10 lines (49 SPS) of F_5, 13 lines (52 SPS) of F_6, 12 lines (51 SPS) of F_7, 8 lines (37 SPS) of F_8, 6 lines (27 SPS) of F_9 and 4 lines (21 SPS) of F_{10}].

Maintenance breeding: The maintenance breeding of different okra varieties released by ICAR-IIVR *viz.* Kashi Kranti, Kashi Pragati, Kashi Sathdhari, Kashi Lila, Kashi Vibhuti and Kashi Vardaan were done through true to type single plant selection. The fruits of the selected plants were covered with butter paper bag and seeds were harvested. Further, hybrid seeds of Kashi Bhairov were produced and its parental lines were also maintained by selfing.

Kashi Chaman, Kashi Lalima and Kashi Shrishti: Two promising varieties (VRO-109 and VROR-157) and one hybrid okra (VROH-12) identified by the ITIC of the institute. VRO-109 and VROR-157 are identified as Kashi Chaman and Kashi Lalima while VROH-12 as Kashi Shristi (Fig. 50). Kashi Chaman has dark green fruits, highly resistant to both YVMV and OLCV disease and yield potential of 150-160q/ha. Kashi Lalima is a red fruited genotype, field tolerant to both YVMV and OLCV and having a yield potential of 140-150q/ha,

Kashi Chaman Kashi Lalima Kashi Shristi

Fig. 50: Promising lines of okra identified at Institute level

while Kashi Shristi is dark green fruited yield potential (180-190q/ha) hybrid having resistance to YVMV disease. The proposals were submitted to SVRC, Uttar Pradesh.

Project 1.7: Genetic improvement of cole crops and root crops

Cauliflower

A total of fifty-two, including eleven promising lines and forty-one germplasm/accessions, of cauliflower with different maturity groups such as October (22-32 °C), November (16-28 °C) and late-November to mid-December (11-22 °C) were evaluated and characterized for various traits of economic importance. Lines having high yield potential are VRCF-86 and VRCF-201 in mid-October maturity; VRCF-50 (Fig. 51), VRCF-102, VRCF-27, VRCF-75-1 and VRCF-113 in mid-November maturity; and VRCF-104, VRCF-202, VRCF-22 and VRCF-77 in late-November to mid-December maturity group.

Fig. 51: VRCF-50 curd

Genotype VRCF-86 realized better marketable yield potential of 175-200 q/ha having small frame size of 45-50 cm, short in duration 60-70 days, 425-475 g net curd weight, 500-575 g marketable curd weight of medium-compact cream-white curds. Further, VRCF-50 realized marketable yield potential of 225-250 q/ha during first fortnight of November having net curd weight of 525-560 g, marketable curd weight of 625-675 g, maturity period of 70-75 days of medium compact to compact and white curds. A VRCF-22 showed better yield potential of 250-275 q/ha during mid-December possessing net curd weight of 600-675 g, marketable curd weight of 725-800 g, maturity period of 70-80 days, and self-blanched compact and snow-white curds. Nucleus seed of VRCF-50 (2.5 kg) has been produced in nylon-net cage for further multiplication (Fig. 52). Six plants having orange curd have been identified and their seeds have been multiplied for further evaluation.

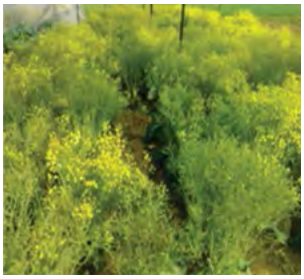


Fig. 52: VRCF-50- Nucleus seed production in nylon-net cage

Kale

With respect to transfer of cytoplasmic male sterility (CMS) system in cauliflower, 15 BC populations have been advanced (BC_1F_1-BC_4F_1) in different curd maturity groups (Early, Mid and Mid-late maturity) through back-crossing. Three best performing genotypes namely VRCF-86, VRCF-50 and VRCF-102 are in multi-location testing under AICRP-VC.

Fig. 53: VRKALE-1

A genotype of tropical kale 'VRKALE-1' induces bolting and flowering, sets seeds in the North Indian plain and don't require any vernalization (Fig. 53). This genotype initiates bolting and flowering during third week of February in North Indian plain i.e. mean temperature for 60 days before flowering is 12-23 °C. The leaves are ready for first picking in 23-28 days after transplanting and thereafter at 7-10 days interval. The leaves measured 20-26 cm in length, weighed 12.5-15.0 g and contain 13-14% dry matter. A plant produces 100-125 leaves in 10-12 pickings. It has leaf yield potential of about 50 t/ha.

Cabbage and Broccoli

Five genotypes of tropical cabbage (Fig. 54) and seven germplasm of tropical broccoli which induce robust bolting and flowering at 10-22 °C have been evaluated for various traits of horticultural importance and advanced to next generation. As like Indian cauliflower, back-crosses have been made in four backgrounds to transfer CMS system in broccoli.

Fig. 54: Tropical cabbage

Carrot

Sixty-one genotypes of tropical carrot, including thirteen promising lines and forty-eight germplasm, having various root colour (red, black, orange, yellow and rainbow) were evaluated for various economic traits (Fig. 55). Among these, the following lines were found to be potential root yielder having better quality traits (self-coloured core, fewer secondary roots, lesser root scars) such as VRCAR-186, VRCAR-201, VRCAR-112, VRCAR-109 and VRCAR-185 (red root); VRCAR-126, VRCAR-89-1 and VRCAR-124 (black root); VRCAR-91-1 and VRCAR-91-2 (orange root); VRCAR-153, VRCAR-178 and VRCAR-127 (yellow root); and VRCAR-107-1, VRCAR-107-2 and VRCAR-171-1 (rainbow-type root).

Fig. 55: Red, black, rainbow, orange and yellow carrots

Two genotypes of red carrot namely VRCAR-186 and VRCAR-185 performed well (>98% uniformity for root shape and colour) having respective root yield potential of 400-425 q/ha and 370-380 q/ha, root weight of 175-180 g and 160-165 g, root length of 22-23 cm and 21-22 cm, marketable roots of 91-93% and 90-91% and self-coloured roots of 93-94% and 91-92%; and are also in multi-location testing under AICRP-VC. Furthermore, a black colour genotype 'VRCAR-126' was found to be promising with potential yield of 260-280 q/ha, root weight of 140-150 g, root length of 21-23 cm, marketable roots of 85-88% and self-coloured roots of 90-92%. VRCAR-126 is good source of antioxidants, and rich in anthocyanins (250-280 mg/100 g) i.e. able to produce 65-70 kg/ha anthocyanins. Nucleus seed (5 kg) of VRCAR-186 has been produced in nylon-net cage for further seed multiplication (Fig. 56). Population has been developed by crossing temperate and tropical genotypes to combine the traits like smooth and scar free roots in tropical carrot.

The generations of 11 back-cross population have been advanced to different stages (BC_1F_1-BC_3F_1) in red, black, orange, rainbow and yellow coloured carrots to transfer petaloid-CMS system for facilitating heterosis breeding (Fig. 57).

Fig. 56: VRCAR-186- Nucleus seed production in nylon-net cage

Fig. 57: Petaloid CMS in red carrot (BC_2F_1)

Radish

Fifty-six promising lines/varieties/germplasm have been evaluated for various traits of economic traits such as root colour (white, red, purple), leaf morphology (lyrate, sinuate, entire), root shape (tapering, blunt) and flower colour (white, purple, dark purple). Among them; twelve genotypes were found to be promising for yield and root quality (uniform root shape, smooth root surface and fewer secondary roots) such as VRRAD-150, VRRAD-203, VRRAD-200, VRRAD-202 and VRRAD-216 (white root); VRRAD-131-2, VRRAD-170, VRRAD-171 and VRRAD-173 (red root); and VRRAD-131, VRRAD-134 and VRRAD-151 (purple exterior) (Fig. 58).

Fig. 58: VRRAD-131-2, Red radish

In summer trial (2017) of seven genotypes, highest root yield (230-270 q/ha) was harvested for genotype VRRAD-203 and VRRAD-200 during mid-April to May. Additionally, VRRAD-203 showed delayed bolting habit during winter season and there was no bolting during summer trials. Two promising genotypes i.e. VRRAD-150 and VRRAD-131-2 are in multi-location testing under AICRP-VC trial. Nucleus seed of two varieties Kashi Shweta (2.00 kg) and Kashi Hans (0.500 kg) has been produced in nylon-net cage for further seed multiplication.

Coloured radishes are good source of anthocyanins, ascorbic acid and phenolics content that varied considerably such as total phenolics ranged from 13–70 mg/100 g FW, anthocyanins content from 5–175 µg/g FW, FRAP value from 1.5–6.0 µmol/g FW, CUPRAC value from 3–12 µmol/g FW and ascorbic acid from 12–27 mg/100 g FW.

For developing robust Ogura-CMS lines in radish, 16 back-cross population have been advanced to various stages (BC_1F_1-BC_5F_1) in different backgrounds (root colour, leaf morphology and root shape) to harness heterotic potential. Two Ogura-CMS lines i.e. VRRAD-198 and VRRAD-201 have been developed at IIVR, Varanasi (First time from Public Sector) through back-crossing, which are very similar to their respective maintainer for leaf morphology; plant growth habit;

colour, shape and weight of root; flowering duration and flower colour; and seed maturity, colour and test weight. Both CMS lines namely VRRAD-198 and VRRAD-201 and their maintainers possessing leaf morphology (leaf division incision) of sinuate type, white and triangular root, bear whitish-purple flower, ready to seed harvest in about 4 months after transplanting of stecklings, and details of quantitative data are given in Table 44.

Three CMS-based promising F_1 hybrids (VRRAD-201×VRRAD-90, VRRAD-201×VRRAD-150 and VRRAD-201×VRRAD-4) with yield potential of 700-750 q/ha have been identified among 28 hybrids evaluated (Fig. 59).

Fig. 59: VRRAD-201×VRRAD-90 (F_1 hybrid)

were raised from 32 T_0 events in pots under containment proof insect house and 20 days old seedlings were sprayed with 100 mg/l of kanamycin. After five to six successive sprays the *Bt*-positive plants survived but the non-transgenic plants died. Further, from survived plants total DNA was extracted, the presence of *npt* II gene confirmed in only 3 T_1 events by PCR using *npt* II specific primers. Selfing was performed on fully grows plants for multiplication and *Ti* seeds of mature selfed fruits from three plants were harvested and stored.

Regeneration studies in cauliflower: Cauliflower (*Brassica oleracea* L. var. *botrytis*) is an important cole crop of the Brassicaceae family and is cultivated for its curd. VRCF-50, a November maturity genotype with a whitish, semi-dome and compact curd was selected for regeneration studies. Seeds were surface sterilized by rinsing with 70% (v/v) ethanol for 2 min followed by treatment with 0.1% (w/v) mercuric chloride for 1 min. and inoculated onto half-strength and incubated in darkness at 25 ± 2°C for germination. MSB5 medium of agar supplemented with 3% (w/v) sucrose and solidified with 0.8% (w/v) was used as the culture medium. Leaf explant excised from 2-wk old seedlings

Table 44: Root and seed traits of radish CMS lines and their maintainers

Trait	VRRAD-198		VRRAD-199		VRRAD-201		VRRAD-202	
	2016-17	2017-18	2016-17	2017-18	2016-17	2017-18	2016-17	2017-18
Plant wt. (g)	265.4	248.3	242.5	238.1	275.0	268.3	270	264.1
Root wt. (g)	166.8	156.1	160.0	157.1	190.0	185.4	182.5	178.5
Root length (cm)	23.7	22.2	22.8	22.4	26.0	25.4	24.3	23.8
Shoot length (cm)	32.4	30.6	34.0	33.6	38.4	37.5	37.1	36.6
Root dia (cm)	3.2	3.0	3.7	3.6	3.6	3.5	3.7	3.7
Leaf number	9.8	9.2	10.6	10.4	10.7	10.4	11.1	10.9
DTFH	52.3	48.9	53.6	50.8	52.4	47.8	52.9	49.7
Marketable yield	581	543	530	521	602	587	591	578
DTFF	34.2	32.5	28.9	27.1	38.1	35.8	31.5	30.1
No. of pods/plant	359.1	371.2	333.2	360.7	381.6	405.2	362.1	381.6
No. of seeds/pod	3.9	4.0	3.8	3.8	4.2	4.3	4.0	4.0
1000 seed weight (g)	13.4	13.5	12.2	12.4	14.3	13.8	13.2	12.9

DTFH: days to first harvest (root), DTFF: days to 50% flowering

Project 1.8: Transgenic and Regeneration Protocols

***In-planta* transformation in okra:** Tissue culture independent, *in-planta* transformation of okra in the cultivar Kashi Kranti was initiated. For *Agrobacterium* mediated *in-planta* transformation a vertical cut was made at the junction of cotyledonary leaves, superficially along the length of the shoot apex, partially bisecting the shoot tip and exposing meristem cells, without damaging the apical meristem. A total of 264 seedlings

and cultured initially exhibited tissue expansion after 2–3 d on culture medium callus induction was observed at the base of the leaf petiole and from the cut edges of the leaf explants (Fig. 60a.). Calluses induced from leaf explants were green or pale yellow and friable in nature. The maximum frequencies of callus induction on MSB5 medium supplemented with BAP (8.9 µM) or TDZ (9.1 µM) alone was 41.11 and 64.45%, respectively. However, further increase in the concentration of cytokinins (22.2 µM BAP and 22.7 µM TDZ) resulted in reduced callus induction. Moreover, the frequency of

a) Leaf explants on SIM b) Multiple shoot regeneration c) Shoot elongation on SIM d) Rooting on root induction medium e) Acclimatization

Fig. 60: Stages of regeneration in cauliflower

callus induction increased to 55.56% on MSB5 medium of agar containing 8.9 µM BAP along with 0.5 µM NAA. The frequency of callus induction increased to 85.56% on MSB5 medium of agar containing 9.1 µM TDZ augmented with 0.5 µM NAA (Fig. 60b.). No significance response was noted with GA3. The result revealed that TDZ induced a better response for callus induction than BA. Shoots were induced after 6 wk on culture medium from leaf-derived callus. The maximum frequency of shoot induction on MSB5 medium containing 8.9 µM BAP alone was 22.22%. However, on MSB5 medium supplemented with 8.9 µM BAP along with 0.5 µM NAA, the frequency increased to 24.45%. In case of TDZ-supplemented medium, the maximum frequency of shoot induction (55.44%) was obtained with 4.5 µM TDZ with 0.5 µM NAA (Fig. 60c.), and the minimum frequency (40%) was obtained with 4.5 µMTDZ alone. Low concentrations of TDZ (0.9–2.3 µM) stimulated shoot formation, but also minimized the number of shoots per explant. In contrast, higher concentrations of TDZ (4.5–9.1 µM) increased the frequency of callus formation but resulted in hyperhydrated shoots, which became normal in 3 wk when cultured on MSB5 medium with a low level of TDZ (0.9 µM) or even on PGR-free MSB5 medium. Further, increase in TDZ level up to 22.7 µM did not result in shoot induction. No significance change was observed with the addition of GA3 to the medium. Roots were induced from shoots after 2 wk of culture and well-developed roots were observed after 3 wk of growth on root induction medium, frequency of rooting, number of roots per shoot, and the root length for each explant were recorded (Fig. 60d.). The highest rooting response (71.11%), the maximum number of roots per shoot (7.67), and the maximum root length (11.33 mm) were obtained on half strength MS medium supplemented with 4.9 µM IBA. Rooting was inhibited at a higher concentration of IBA (9.85 µM). Well-rooted plants were removed from the medium, washed with sterile, distilled water, and used for hardening. For acclimatization, a mixture of sterile coco pit and Boro kit in 1:1 ratio used and drenched with a fungicide, Bavistin (0.5% w/v (Fig. 60e.). After about 2 wk, the acclimatized plants were transferred to pots containing soil for further growth and transfer to Glass house.

Project 1.9: Biotechnological interventions for improvement of selected vegetable crops

Genome editing in tomato: CRISPR/Cas9 mediated genome editing work in tomato was undertaken to progress the work towards the development of vector for plant transformation. The pRGEB31 vector procured from Addgene (USA) was earlier used to modify its gRNA sequence to target our gene of interest. However, due to technical reasons, the Cas9 gene from pRGEB31 vector was removed and cloned into pORE-O4 vector along with CaMV35S promoter. The guide RNA construct targeting replicase (re) gene of Tomato leaf curl virus (ToLCV) was artificially synthesized. The target region of replicase gene was identified manually from a conserved sequence by multiple sequence alignment of five major strains of ToLCV to achieve broad spectrum resistance.

Male sterility is an important trait in crops for hybrid development programme. Therefore, work on development of male sterile line in tomato has been initiated. For this purpose, pollen specific genes were identified in tomato, mutation of which could potentially lead to male sterility due to developmental defects in pollens. SlCRK1 gene was finally selected as a target gene and gRNA construct was designed with the help of CRISPR direct tool. Both these gRNA constructs targeting SlCRK1 and *Rep* gene of ToLCV will be cloned in pORE-O4 vector which is already having Cas9 with CaMV35S promoter (Fig. 61).

Tomato variety Kashi Amul notified: A semi-determinate tomato variety Kashi Amul has been notified for release in zone VIII comprising Karnataka, Tamil Nadu and Kerala states. Average yield of 50-60 tonnes/ha could be realised with Kashi Amul. The fruits of this variety are round and firm with pericarp thickness of 0.5-0.6 cm. This variety has also shown high level of resistance in artificial screens and field tests conducted over years in disease hot spot at ICAR-Indian Institute of Vegetable Research, Varanasi (Fig. 62).

Promising tomato line with prominent green shoulder: A total of 5 lines were selected following

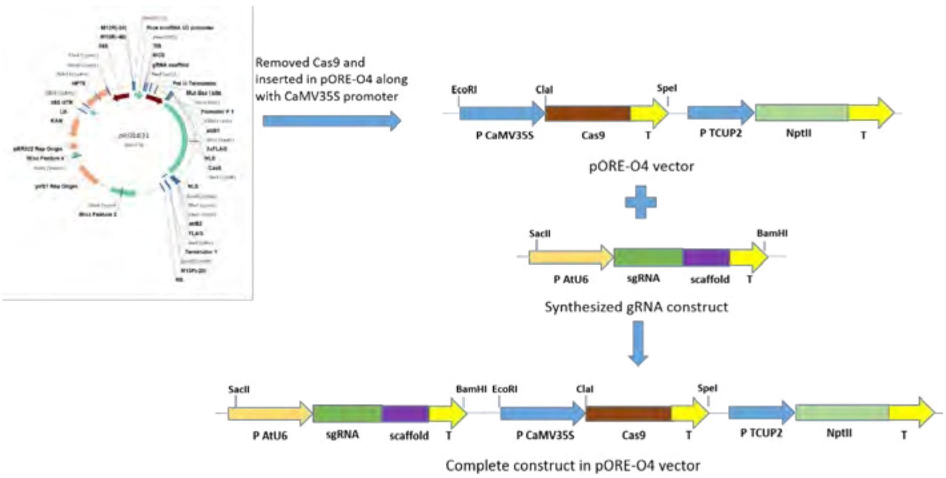


Fig. 61: Complete construct in pORE-O4 vector

marker assisted selection and pedigree selection were evaluated. Marker assisted selection was performed using *Ty*-3 linked markers. Based on performance of the lines, VRT18-1 was selected for inclusion in the AICRP determinate variety trial of tomato (Fig. 63). This line is early maturing and fruits ripening starts at 70 days post planting. Fruit yield of 52 t/ha was realised in the evaluation trial with an average fruit weight of 75 g. The fruits have prominent green shoulder and thick pericarp (0.6-0.7 cm).

Promising tomato hybrid: A total of 30 hybrids developed based on combination of *Ty*-2 and *Ty*-3 lines were evaluated again during early (August-December) and main tomato growing seasons (planted in October). The parental lines used in the generation of these hybrids included previously developed *Ty*-2 and *Ty*-3 carrying lines. A total of 10 hybrids of commercial companies were also used as commercial checks. Based on the cumulative trials conducted over two years and three seasons, test hybrid VRT16-11 X VRT16-12 was selected for multilocation trial under AICRP (VC). The commercial checks used comprised hybrids Abhilash, NS585, Devika and TO3150. The test hybrid VRT16-11 X VRT16-12 yielded 75 tonnes/ha followed by VRT16-12 X VRT18-1 and VRT16-7 X VRT16-13. The fruits of VRT16-11 X VRT16-12 hybrid show medium firmness with a pericarp thickness of 0.5-0.6 cm and record average fruit weight of 80-110 g (Fig. 64).

Brinjal

Screening of parental lines for identification of polymorphic markers and genotyping of mapping population: For identification of polymorphic markers to be used in QTL mapping, 186 new SSRs were developed during the year 2017-18 using the transcriptome sequence of Ramnagar Giant and W_4. 57 SSRs were identified to be polymorphic among the parental lines. Out of these, 12 markers were further used for genotyping of the mapping population of 114 RILs through PAGE (polyacrylamide gel electrophoresis). Till date, a total of 258 markers (which includes SSRs, STMS, SCoT, and ISSR) have been found as polymorphic among the parental lines.

Analysis of transcription factor classes: Overall, 60 different transcription factors (TF) were identified in responses to biotic and abiotic stresses which play a pivotal role in expression and developmental pathways. Total of 2182 and 2348 genes were identified classified into these 60 TF families in transcriptome

Fig. 62: Kashi Amul

Fig. 63: VRT18-1

Fig. 64: Promising tomato hybrid VRT16-11 X VRT16-12

sequences of *S. melongena* and *S. incanum* respectively. Sequence similarity search was done using Plant TFDB (transcription factor database), NCBI-CDD (conserved domain database) and tBLAST*n* algorithm taking *S. lycopersicon* sequence data as the reference.

Prediction of protein structure of Leucine-rich repeat family protein of brinjal: An attempt to understand the architecture of leucine-rich repeat family protein of brinjal lines including Ramnagar Giant and W-4 is under progress using SWISS-MODEL homology modelling software. This family is known play significantly important role in elicitor-induced plant defense mechanism in response to attack by insects and pathogens. For this, *S. melongena* transcripts encoding for Leucine-rich repeat family protein was searched using TSA-BLAST. Total 12 transcripts were identified from which transcript sequence based on e-value was selected. The transcript sequence was subjected to Expasy-translate tool for identification of its amino acid sequence which was used as query to build a probable protein structure using SWISS MODEL-Homology modelling software. Four models were built for the structure of brinjal Leucine-rich repeats protein using SWISS MODELLING software data. Primary amino acid sequence for which models were built was:

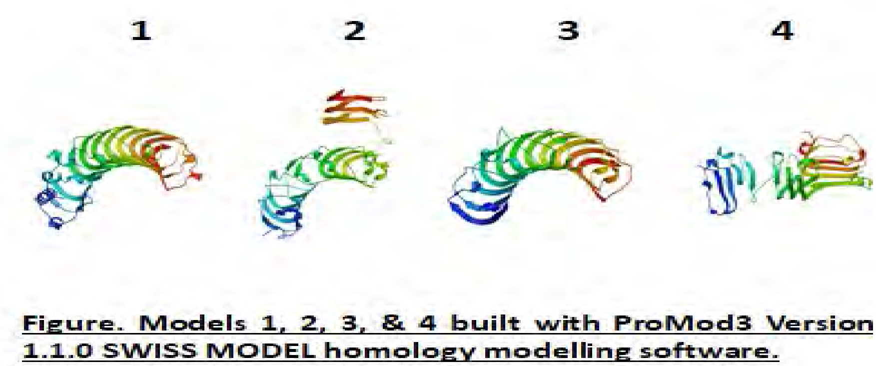


Fig. 65: Models 1, 2, 3 and 4 built with ProMod3 Version 1.1.0 SWISS MODEL homology modelling software

Project 1.10: Genetic improvement of leafy vegetables

Basella

Wide range of variation was recorded for morphological traits like leaf and stem characteristics among 24 genotypes. The edible shoot length ranged from 19-182.2 cm, leaf length 5.5-10.84 cm, leaf width 3.48 -8.64 cm, days to 50% flowering from 81-152 and yield per plant ranged from 207-542 g. There were two groups of plants namely, trailing and bushy types. Among the trailing types of genotypes, VRB-17 recorded highest yield of 542 g/plant, while among the bushy genotypes VRB-31 yielded 350 g/plant.

VRB-17: This genotype produces long succulent green shoot of ~60cm with the leaf length and width of 10.3 cm and 7.10 cm, respectively. Yield was recorded to be 542g per plant. This genotype is one of the most promising genotype to grow in the kitchen garden of the household. Flowering in this genotype starts after four months of planting. This genotype is suitable for 6-7 harvesting at 15 to 20 days interval (Fig. 66).

Fig. 66: VRB-17

VRB-31: VRB-31 is the bushy genotype which produces small bunched shoot of 22.5 cm length, with leaf length and width of 5.8 cm and 4.0 cm respectively. It produces flower after 113 days after planting. On an average per plant yield of 350 g was recorded. This genotype is suitable for high density planting due to bushy nature of the plant.

Betalain estimation in different basella genotypes: In *Basella alba* var. *alba* the stem colour is green, while in *Basella alba* var. *rubra* stem colour is red/ purple. However, flower colour of both red and green types are with pink/purple tinge and mature fruits are enriched with dark purple fruit juice having ample amount of betalains. During characterization and evaluation, colours comprising purple, pink, white with pink or purple tip were observed (Fig. 67). The fruits were fleshy, stalk less, ovoid or nearly

Fig. 67: Variability in flower color and morphology among basella genotypes.

spherical and the colour varies from green, red violet to black during ontogeny. However, the colour of matured fruit is black with bright purple fruit juice in both green and red types (Fig. 68).

Fig. 68: Variability for immature fruit colour among Basella genotypes

In order to quantify the observations, betalain content was estimated in the unripe and ripe fruits of both red and green genotypes. The photometric results obtained for total betalains (betanins and vulgaxanthin) exhibits maximum pigment content in ripened fruits of VRB-30 (200.93 mg/100 g FW), followed by VRB 3 (150.87 mg/100 g FW). The mature fruits of VRB 48-1 showed the lowest betalain (10.48 mg/100 g FW) (Table 45). Hence, genotypes with high betalain content may be exploited as a source of betalain for indusrial/ pharmaceutical use. However, the genotype with colourless fruit juice or with low expression of betalain can never be ignored as it may serve as the key to unravel biosynthetic pathway of betalain synthesis. Besides, it may serve as the basic material for understanding the genetics of betalain in basella, may be used as parent for generation of bi-parental mapping population and further used for mapping of the betalain gene on basella genome. These identified genomic regions can be delineated to further identify the candidate gene(s) associated with the betalain production.

First report of charcoal rot disease in basella: During the period of August to October 2016-17 and 2017-18, the basella plants were observed with charcoal rot symptoms and 30-40% plants were found to be infected. Symptom consisted of brownish to black discoloration at the collar region of the stem and branches that progressed into wilting and drying of entire plant. Infected plant stems appeared shredded and contained black microsclerotia. Under a compound microscope, black round to oblong or irregular shaped black colored microsclerotia with mycelial attachment were observed. The average diameter of microsclerotia was 76.26 ± 8.06 µm (n=50). The isolated pathogen was inoculated to plant under artificial conditions with three cuttings were planted in each pot filled with inoculated and mock inoculated soil (Fig. 69). Symptoms typical of charcoal rot on collar region were first observed 12 days after planting. Whereas, plants in mock inoculated soil remained healthy. On the basis of morphological characteristics and pathogenicity test, the isolated charcoal rot causing pathogen was identified as *Macrophomina phaseolina.*

Fig. 69: Collection of symptomatic plant, isolation of pathogen and artificial screening of charcoal rot

Table 45: Betalain content in different genotypes of Basella

Genotype	Fruit Stage	Betanin (mg/100 g FW)	Vulgaxanthin (mg/100 g FW)	Total Betalains (mg/100 g FW)
VRB-48-1	Immature	3.987±0.181	7.184±0.141	11.171±0.322
	Mature	4.688±0.088	5.801±0.067	10.489±0.021
VRB-3	Immature	3.770±0.233	5.338±0.260	9.108±0.027
	Mature	115.173±1.058	35.700±0.354	150.873±0.705
VRB-48	Immature	2.349±0.105	4.930±0.035	7.279±0.141
	Mature	120.200±0.884	24.335±0.237	144.535±0.647
VRB-30	Immature	2.274±0.052	2.560±0.223	4.834±0.170
	Mature	154.823±0.247	46.113±0.090	200.936±0.337

Values are mean ± S.D

Further, confirmation was done using elongation factor (EF-1a) gene specific primers TEF1-983F (5′GCYCCY GGHCAYCGTGAYTTYAT3′) and TEF1-2218R (5′ATGACACCRACR GCRACRGTYTG3′) of *Macrophomina phaseolina.* PCR products were sequenced and submitted to GenBank with accession number MG733372. In BLAST analysis, EF1a gene showed 100% sequence homology with *M. phaseolina* (DQ677929).

Amaranth

An exploration trip was organised in collaboration with ICAR-NBPGR for collection of leafy amaranth from northern part of West Bengal. High variability is evident in the leafy vegetable in these areas. Collections include mainly of *Amaranthus tricolor* germplasm, besides wild germplasms. Good variability in *A. tricolor* in terms of part used as vegetable (stem/leaf), branching pattern, leaf size, shape and pigmentation has been observed and collected. Apart from augmenting interesting variability in *A. tricolor* (like long-petioled type, small-leaved green form) and *A. tristis*, a sample showing intermediary characters between *A. viridis* and *A. tristis* has been collected. A total of 71 accessions were collected, majority being *A. tricolor* (58 acc.), *A. tristis* (6) and *A. blitum* (7).

Quinoa

Twelve quinoa genotypes were grown during 2017-18 rabi season. These genotypes were assessed for horticultural traits and yield. Plant height was ranging from 31.2 to 102.1 cm. Leaf length and leaf width varied from 3.76 to 6.38 cm and 2.56 to 3.66 cm, respectively. All the genotype started flowering 30 days after sowing. Yield per plant was recorded to be in range of 3.2 to 53g.

Chenopodium (Bathua)

Biomass yield of three genotypes of bathua namely VRCHE-2 (green leaves), VRCHE-4 (purplish-green leaves) (Fig. 70) and Pusa Bathua-1, realized 408, 494 and 369 q/ha, respectively in six cuttings. Moreover, the respective plant growth at different stages was measured 21.7, 22.0 and 20.0 cm at 40 days after sowing (DAS); 43.2, 48.9 and 41.2 cm at 80 DAS; 145.7, 155.0 and 127.9 cm at 120 DAS; and 209.8, 236.4 and 188.9 cm at 160 DAS. Fresh edible biomass of VRCHE-2, VRCHE-4 and Pusa Bathua-1 possessed 15.3, 16.4 and 15.5% of dry matter; 150.7, 129.9 and 120.3 mg/100 g of ascorbic acid; and 43.6, 34.2 and 29.3 µmol TE/g of CUPRAC antioxidant potential, respectively. Two genotypes, namely VRCHE-2 and VRCHE-4 are in the multi-location varietal trial of AICRP-VC and their seeds (8 kg of each) have been produced.

Fig. 70: VRCHE-4

Moringa

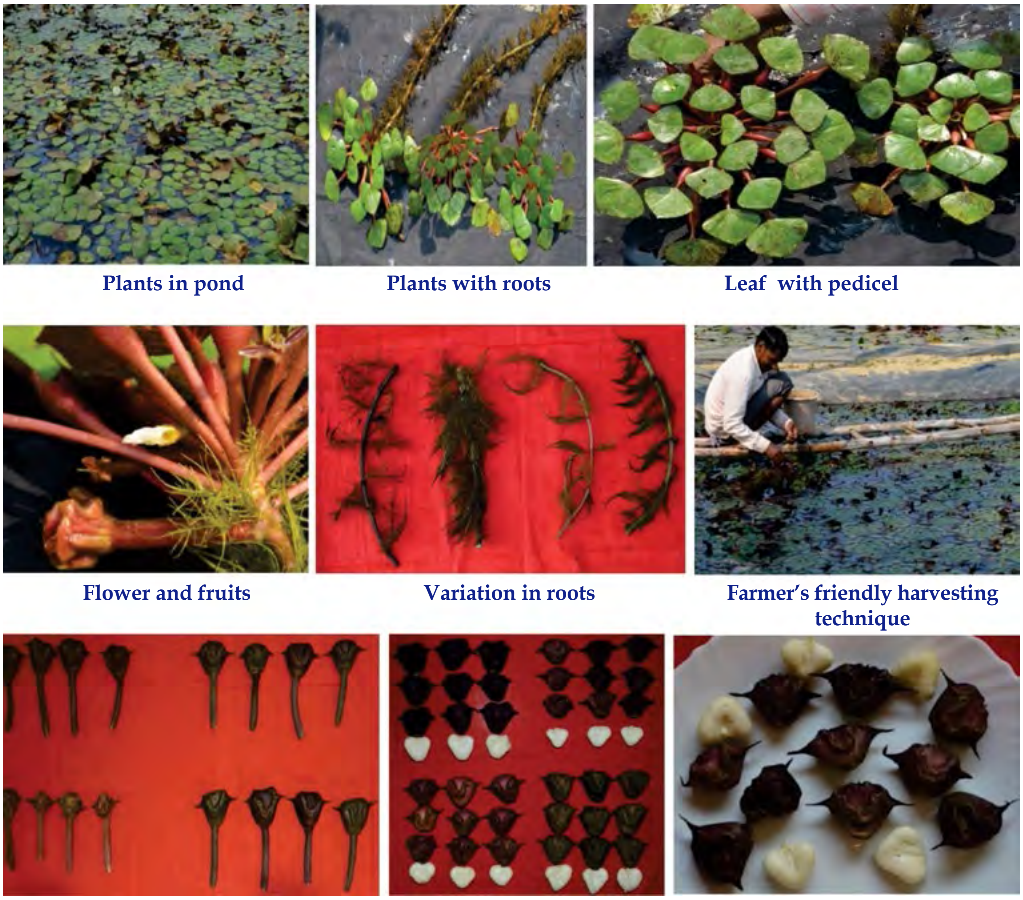
A total of 1200 plants were raised from the seeds and transplanted in the field. Of the total 1200 plants, 15 plants started flowering 140-150 days after sowing, which is 15-20 and 40 days early, respectively to PKM-1 and PKM-2 (Fig. 71). The early plants started setting fruits during the month of October. These plants were evaluated for horticultural and fruit traits. The fruit length varies from 50 to 79 cm, fruit diameter of edible fruits are around 1 cm. The average fruit weight ranged from 30 to 46 g per fruit. The number of fruits/ plant varied much ranging from 10-168. These selected genotypes will be further evaluated and multiplied through seed and cuttings.

Fig. 71: Moringa selection with heavy fruiting

Project 1.11: Genetic Improvement of Aquatic Vegetables

Water chestnut

Four genotypes of water chestnut were grown in water pond with size of 7.5 m (L) x 5.0 m (W) x 1.2 m (D) and were characterized for different horticultural traits (Fig. 72) *viz.*, average number of leaves per plant,



Plants in pond — Plants with roots — Leaf with pedicel

Flower and fruits — Variation in roots — Farmer's friendly harvesting technique

Fruits with pedicel — Colour variation in fruits — Fruit and shelled fruit

Fig.72: Morphological variations in water chestnut

average number of fruit per plant, average leaf length (cm), average leaf width (cm) , average fruit pedicel length (cm), number of spine per fruit, average fresh fruit weight (g), average shelled fruit weight (g), dry weight (g), dry matter content (%), TSS (°Brix) and fruit yield/pond (kg) varied significantly . Average number of leaves per plant varies between (24.8-35.0), number of fruit per plant (3.2-5.8), average leaf length (4.28-4.57 cm), average leaf width (5.41-6.78 cm), average fruit pedicel length (4.96-5.46 cm), number of spine per fruit (2.0-2.0), average fresh fruit weight (9.78-16.76 g), average shelled fruit weight (4.34 -15.48 g), dry fruit weight (0.46-1.89 g), dry matter content (10.6-26.6 %), TSS (3.0-3.9 °Brix), and fruit yield per pond (21.88-30.20 kg). Among the all genotypes VRWC-1 adjudged as promising genotype for dry matter content and fruit yield too. Proximate micronutrient composition of shelled fruit and fruit rind was also estimated. Zn content varies between 37.4-44.8 ppm in shelled fruit while 28.6-127.6 ppm in fruit rind. Fe varies between 155.6-199.0 ppm in shelled fruit while 400.6-1133.0 ppm in fruit rind. However, Mn content varies between 33.1-43.6 ppm and 192.9-320.50 ppm in shelled fruit and fruit rind, respectively.

Project 1.12: Genetic Improvement of baby corn and sweet corn

During the *kharif* season of 2017-18, total 47 germplasm lines of baby corn were screened against the banded sheath and leaf blight (BSLB) disease caused by *Rhizoctonia solani* under field conditions as well under artificial condition following the artificial inoculation. The inoculum was prepared using the *Typha latifolia* leaves cutting of 2.5cm in length. Cutting were autoclaved twice and inoculated with 5 days old culture of *Rhizoctonia solani.* 15 days old inoculum was inoculated on the based leaf sheath (3 leaves below the cob). Data were recorded 15 days post inoculation (Table 46 & Fig. 73).

Table 46: Scaling of baby corn germplasm under different resistance grades

Resistant	Moderately resistant	Moderately susceptible	Susceptible
BC-13, 62	BC-16	BC-6, 7, 8, 25, 35, 41,	BC-1, 2, 3, 4, 5, 9, 10. 12, 19, 23, 28, 29, 30, 31, 32, 33, 34, 36, 37, 38, 39, 40, 43, 45, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61

Promising lines from the field study were further evaluated for confirmation under control conditions in growth chamber where most favourable temperature and humidity were maintained and artificial inoculated plants were kept for 15 days. Screening of germplasm against the BLSB caused by *Rhizoctonia solani* the germplasm under field and artificial conditions revealed that line BC-13, 62 were resistant, BC-16 was moderately resistant and rest of the germplasm lines were susceptible (Fig. 74).

Fig. 73: Evaluation of baby corn germplasm against BSLB under filed conditions

Fig. 74: Evaluation of Baby corn germplasm against BSLB under artificial conditions

MEGA PROGRAMME-2: SEED ENHANCEMENT IN VEGETABLES

Project 2.1: Priming, Coating, Pollination and Ovule Conversion Studies

Alleviation of lead (Pb) toxicity through ZnO nano-particles seed priming: Heavy metal contamination due to different anthropogenic activities (sewage water, industrialization) is a serious problem especially near the urban area, and these are mainly vegetables producing areas. This study was conducted to observe lead (Pb) stress effect on seeds and seedlings of Basella and to alleviate its effect through nano particles seed priming (16 hrs at 25°C). Results revealed that increasing concentration of Pb showed significant toxicity of Pb to the seed and seedling growth mainly on seed germination and root growth. Seeds treated with 200 ppm nano Zn particles exhibited the potential ability to alleviate the Pb toxicity by increasing the secondary roots and root volume under the Pb stress conditions. Root scanning analysis also showed the reduced root mass and volume as the concentration of Pb increased from 0 mM to 20 mM. However, seeds primed with 200 ppm nano Zn showed the increased root mass and volume in control as well as in Pb stressed seedlings (Fig. 75-76).

Conversion of ovules to seeds: An experiment was conducted to enhance conversion of ovules to seed in okra cv Kashi Kranti. In okra, during the initial stages, the conversion seemed to be high but as the pods progressed towards maturity, many of the ovules remained rudimentary (Fig. 77). It was observed that application of 200 ppm nano Zn+200 ppm nano Fe+200 ppm NAA significantly enhanced the conversion of ovules to seed as well as quality of seed over the control (Fig. 78).

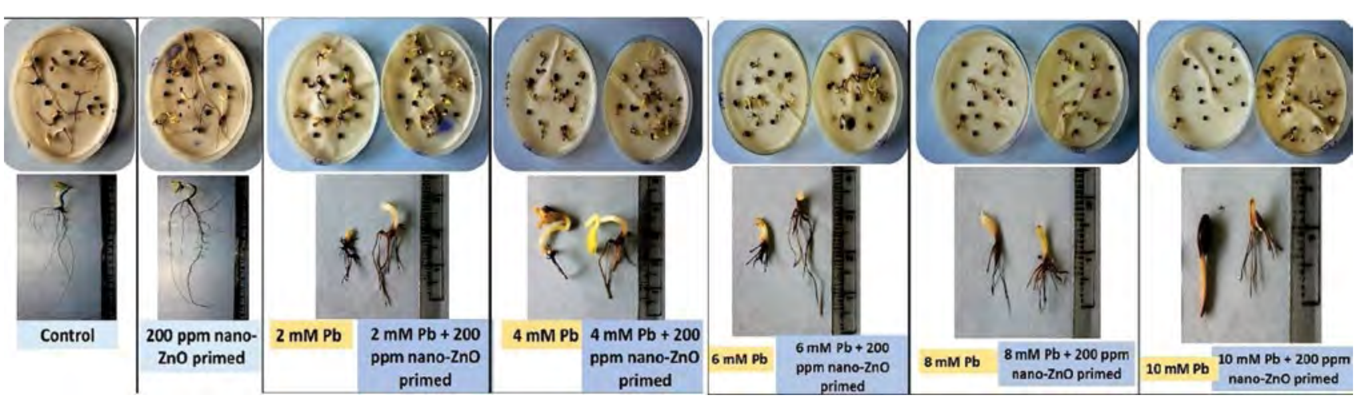


Fig. 75: Effect of Pb stress on basella seedling and its alleviation through nano-particle seed priming

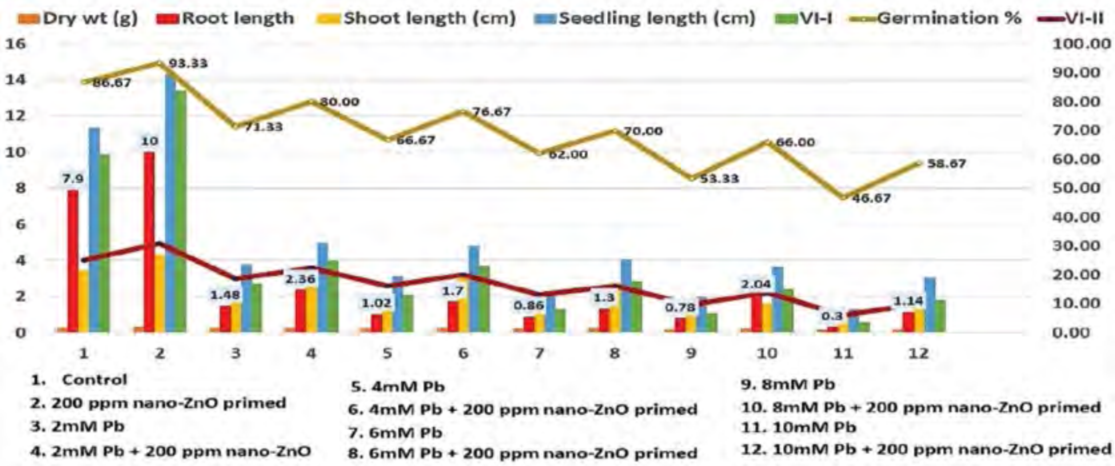
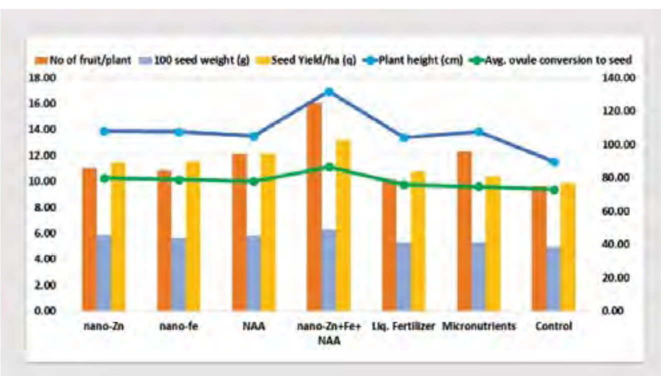


Fig. 76: Effect of Pb stress and nano particle seed priming on seed quality parameters of basella



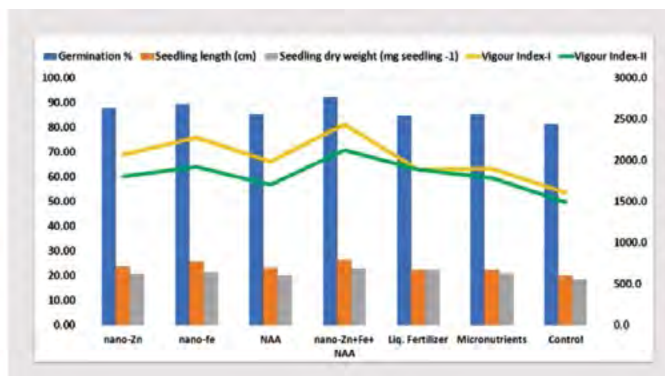


Fig. 77 and 78: Effect of application of different treatments on seed yield and seed quality parameters in okra cv. Kashi Kranti

Pollination efficiency and seed filling enhancement through nano-particles based pollinator attractant: In bottle gourd and sponge gourd, the development of higher number of under developed seeds is a major problem due to improper pollination and seed filling. To overcome this problem, an experiment was conducted with the spray of Zn nano-particle based pollinator attractant. Results revealed that application of 5% sugar+ 5% jaggery + multivitamin + 200 ppm nano Zn+0.1% boron significantly enhanced the seed yield and quality by enhancing pollinator activity and seed filling in both the cucurbits (Table 47-48 & Fig. 79).

Heavy metal (Pb) stress study: Among the increasing pollutants, the heavy metals are one of the important ones affecting the vegetables. To have preparedness about this increasing menace, a study was conducted to observe its effect on vegetable seeds. Since leafy vegetables are considered as indicator plants

Fig. 79: Seed morphology of bottle gourd and sponge gourd (1- 10% sugar + MV+ nano-Zn + 0.1% B, 2- 10% Jaggery + MV + nano-Zn + 0.1% B, 3- 5% Suger + 5% Jaggery + MV+ nano-Zn + 0.1% B, 4- Hand pollinated, 5- Control

Table 47: Effect of nano-particle based pollinator attractant on seed yield and quality of bottle gourd

Treatment	No. of fruits / vine	100 seed wt. (g)	Seed Yield (kg/ha)	Germin-ation %	Seedling length (cm)	Seedling dry weight (mg / seedling)	Vigour Index-I	Vigour Index-II	under dev. seed %
10% sugar + MV	3.0	14.51	452.3	86.7	20.72	35.80	1794.16	3102.1	24.62
10% Jaggery. + MV	2.9	14.68	467.0	87.3	22.37	37.37	1952.80	3265.3	23.84
5% sugar+5% Jag + MV	3.3	14.54	495.7	88.3	22.58	41.17	1994.92	3636.6	21.32
10% sugar + MV+ 200 ppm nano-Zn+0.1% B	3.1	15.31	505.7	92.7	22.52	42.80	2086.79	3965.7	20.62
10% Jag + MV+ 200 ppm nano-Zn+0.1% B	3.5	16.07	517.0	92.0	24.93	45.37	2294.52	4174.5	18.55
5% sugar +5% Jag + MV+ 200 ppm nano-Zn+0.1% B	4.2	17.37	564.3	96.7	29.71	49.00	2869.90	4738.0	9.09
HP	3.0	14.26	395.7	85.3	21.39	32.97	1825.54	2811.5	10.16
Control	2.5	13.65	345.0	83.3	19.58	29.57	1629.54	2464.9	32.55
CD (P=0.05)	0.502	0.857	16.511	3.417	2.446	2.872	214.17	308.646	1.302
CV	10.18	3.29	2.015	2.775	6.151	4.227	6.018	5.066	3.699

Table 48: Effect of nano-particle based pollinator attractant on seed yield and quality of sponge gourd.

Treatment	No. of fruits/ plant	100 seed wt. (g)	Seed Yield kg/ha	No. of seeds/ fruit	Germi-nation %	Seedling length (cm)	Seedling dry wt (mg seedling -1)	Vigour Index-I	Vigour Index-II
10% sugar + MV	6.87	9.30	226.67	206.9	74.33	28.15	35.24	2093.7	2795.7
10% Jaggery + MV	6.80	9.76	229.69	206.2	77.67	29.51	35.13	2293.2	2874.0
5% sugar+5% Jag + MV	6.97	9.96	236.32	230.0	79.33	31.36	36.86	2487.3	3069.1
10% sugar + MV+ 200 ppm nano-Zn+0.1% B	7.87	10.14	239.01	242.7	76.33	30.57	37.74	2334.3	3064.5
10% Jag + MV+ 200 ppm nano-Zn+0.1% B	7.92	10.42	275.64	257.6	81.33	32.33	39.38	2630.0	3394.4
5% sugar +5% Jag + MV+ 200 ppm nano-Zn+0.1% B	8.80	11.00	301.25	280.3	89.33	35.72	43.15	3190.4	4075.4
HP	5.63	9.60	223.33	262.2	73.67	29.25	31.84	2154.2	2409.1
Control	5.20	8.95	218.67	182.3	70.00	26.47	29.08	1853.1	2095.4
CD (P=0.05)	0.589	0.403	14.91	21.872	2.848	1.629	2.605	195.2	255.1
CV	4.801	2.324	3.79	5.346	2.65	3.093	3.967	4.74	4.958

for heavy metals, the lead stress through lead acetate was given to seeds of one leafy (amaranth) and one solanaceous (tomato) vegetable. The results showed that increasing concentration of Pb from 0 mM to 20 mM in tomato (Kashi Aman and H-86) and amaranth reduced the seed germination and root growth in both petri-plate and hydroponics (Fig. 80-81). Interestingly, Kashi Aman showed better tolerance under increasing Pb stress than H-86. However, seed primed with 200 ppm Fe nano particles enhanced the germination and root growth in amaranth and in both tomato varieties. Similar results were shown by root scanning analysis where increasing concentration of Pb exhibited toxicity to root mass and volume while nano primed seed showed significantly increased root mass and volume in both control and Pb stressed seedlings. Biochemical analysis revealed that increasing Pb stress reduced chl a, chl b, total chlorophyll, carotenoids and increased proline & H_2O_2 in seedling. While nano particle priming enhanced chl a, chl b, total chlorophyll, carotenoids and reduced proline & H_2O_2 in stressed seedlings (Fig. 82)

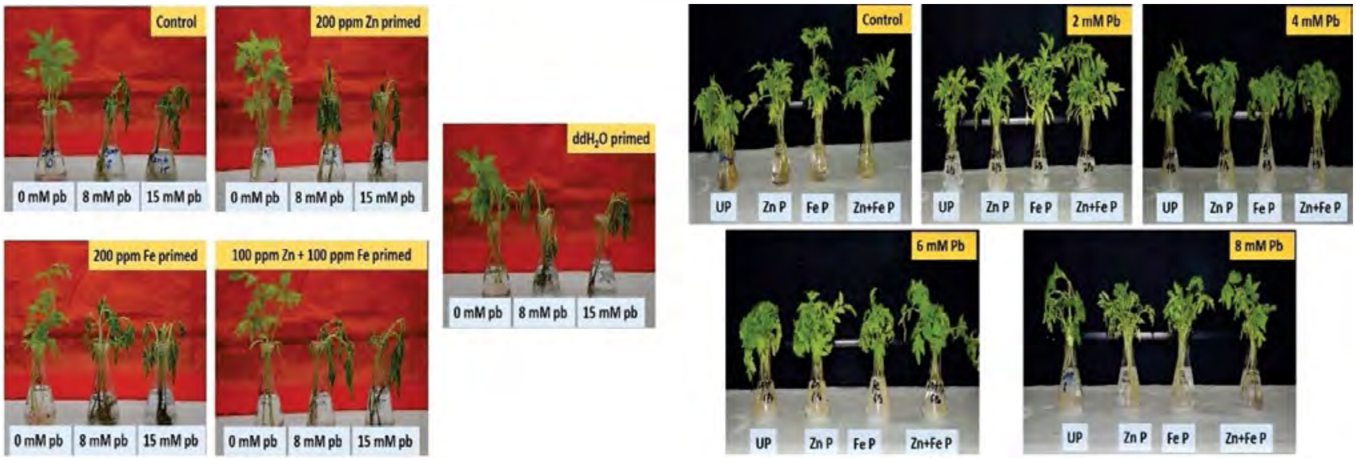


Fig. 80: Effect of pb stress and its alleviation through nano-particle seed priming in hydroponics in tomato seedlings

Seed quality enhancement through priming: In preliminary study, enhancement in vigour of okra cv. Kashi Kranti seeds was observed when seeds were primed in solutions of poly ethylene glycol 6000 (-0.1 MPa) and sorbitol (5% and 6%) for 24hrs. Further, field trial was conducted during summer 2017 with primed seeds. Seeds primed with sorbitol 6% solution recorded significantly higher field emergence (71.4%) and seed yield per plant (11.2 g). All other details are depicted in table 49 & 50.

Standardization of germination method in moringa: There are no standards available for seed germination testing and methods for moringa therefore, an experiment was conducted to standardize the germination method in moringa. It was observed that seed without wing kept in between paper (rolled towel method) produce healthy and vigorous seedling (Table 51).

Quality of moringa seed at different position in fruits: Seed of moringa at different position in fruit were collected and observed for seed quality parameters. Observations recorded have been presented below (Table 52 & Fig. 83) indicate that seed quality is higher in the seeds position at middle of pod.

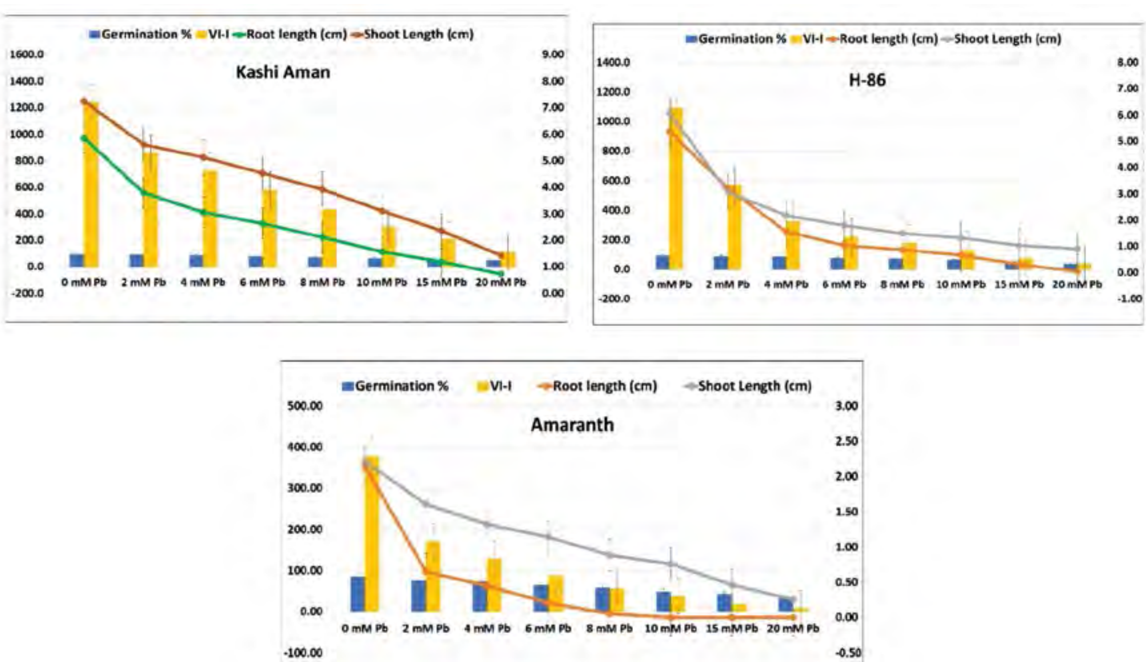


Fig. 81: Seed quality parameters of amaranth and tomato (Kashi Aman, H-86) under Pb stress condition

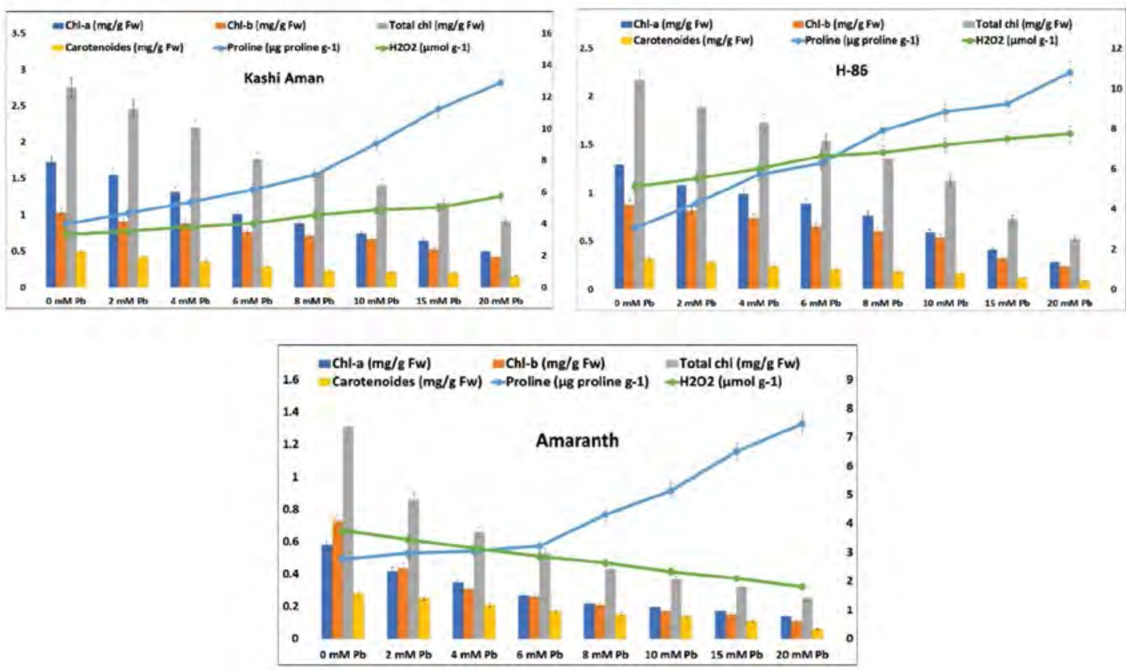


Fig. 82: Biochemical parameters of amaranth and tomato (Kashi Aman, H-86) under Pb stress condition

Table 49: Effect of priming treatments on okra cv. Kashi Kranti in field condition

Treatments	Germination (%) of primed seed	Field emergence (%) of primed seed	Plant height (cm)	Pods / plant (No.)	Seed yield (g / plant)	Seed yield (q / ha)
PEG-1.0 Mpa	81 (64.3)	66.8 (54.8)	45.8	6.04	9.6	7.94
Sorbitol 5%	83 (65.6)	67.4 (55.2)	48.6	6.30	10.2	8.4
Sorbitol 6%	84 (66.6)	71.4 (57.6)	49.8	6.36	11.2	8.78
Distilled water	73 (58.7)	62.2 (52.0)	41.3	4.86	8.4	7.28
Dry control	70 (57.0)	51.6 (45.9)	40.2	4.92	8.1	6.98
CD at 5%	2.8**	1.97**	1.63**	0.45**	0.58**	0.48**
CV (%)	3.4	2.8	2.7	6.1	4.7	4.6

(Arcsine transformed vales of germination per cent are given in parenthesis)

Table 50: Quality of harvested seeds of okra from seed priming field trial

Treatments	100 seed weight	Germination (%)	Vigour index I	Vigour index II
PEG-1.0 Mpa	5.02	83 (65.9)	2311	2024
Sorbitol 5%	5.12	85 (67.2)	2446	2138
Sorbitol 6%	5.26	87 (68.7)	2661	2236
Distilled water	4.82	81 (64.1)	2180	1884
Dry contorl	4.80	80 (63.4)	2120	1779
CD at 5%	0.31*	3.4*	172.1**	180.8**
CV (%)	4.76	3.89	5.6	6.8

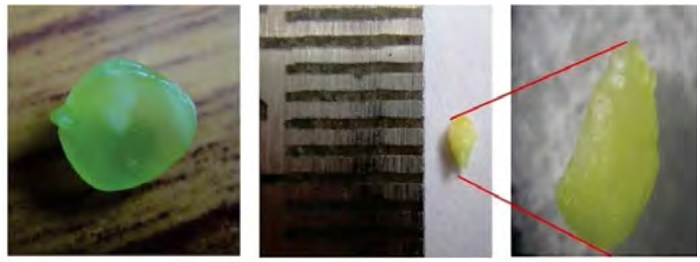
(Arcsine transformed vales of germination per cent are given in parenthesis)

Table 51: Standardization of moringa seed germination method

Treatments	Germination %	Remarks
Seed with wing-BP	50	Severe fungus growth
Seed without wing-BP	65	Healthy and vigorous growth
Seed with wing-TP	38	Fungus growth + abnormal seedling
Seed without wing-TP	20	Produce abnormal Seedling
Without seed coat-TP	15	Severe fungus growth
Without seed coat-BP	20	Severe fungus growth
With seed coat- sand	60	Took 10 days to come out
without seed coat-sand	0	Not germinated
Brushed- BP	51	Healthy growth
Brushed seed-TP	40	Fungal growth
24 hrs soaked with seed coat- TP	10	Fungal growth
24 hrs soaked with seed coat- BP	35	Produce Normal seedling
24 hrs soaked without seed coat- TP	0	Not germinated
24 hrs soaked without seed coat- BP	0	Not germinated

TP-Top of the paper method; BP- Between Paper (Rolled towel method); temperature was 25°C.

Seed development in moringa



Cotyledon + Embryonic axis

Embryonic axis

Fig. 83: Cotyledon and embryonic axis of moringa seed

Table 52: Quality parameters of moringa seeds taken from different position in fruit

Parameters	Position of seed in Fruit		
	Proximal Part	Middle part	Distal part
Girth of fruit (cm)	7.56	8.42	7.12
No. of seeds/ fruit	7.2	9.2	6.4
Germination %	37	51	22
Seed Moisture %	8.69	7.48	6.81
Root length (cm)	10.75	12.5	11.8
Shoot length (cm)	15.8	15.35	15.33
Seedling length (cm)	26.55	27.85	27.13
Seedling Dry weight (g)	2.20	2.71	1.64
VI-I	982.35	1420.35	596.86
VI-II	81.4	138.21	36.08

Lotus seed germination: In lotus, seed germination is problematic due to hard seed coat. To overcome this problem, an experiment was conducted to enhance the seed germination. Among different treatments scarification of seed coat could give germination up to 60% (Fig. 84).

Fig. 84: Seed germination in lotus after seed scarification

Project 2.2: Breeder and TL Seed Production of important Vegetable Crops

Vegetable seed production: At ICAR-IIVR farm, the overall seed production programme (Breeder+TL) was undertaken in 39 varieties of 21 vegetable crops. A total

Table 53: TL seed production at ICAR-IIVR, Varanasi

Crop	Variety	Quantity produced (kg)
Okra	Kashi Pragati	225.00
Cowpea	Kashi Kanchan	1340.00
	Kashi Nidhi	920.00
	Kashi Gauri	2.00
Pea	Kashi Nandini	9200.00
	Kashi Udai	9060.00
	Kashi Ageti	1640.00
	Kashi Samridhi	150.00
	Kashi Mukti	1830.00
Brinjal	Kashi Uttam	150.00
	Kashi Taru	5.50
Tomato	Kashi Vishesh	44.50
	Kashi Aman	163.00
	Kashi Adarsh	35.50
	Kashi Anupam	3.50
	Kashi Amrit	0.15
Chilli	Kashi Anmol	63.50
Sponge gourd	Kashi Divya	42.70
Ridge gourd	Kashi Shivani	16.00
Satputiya	Kashi Khushi	3.00
Bottle gourd	Kashi Ganga	300.50
Bitter gourd	Kalyanpur Baramasi	4.60
Pumpkin	Kashi Harit	50.00
Cucumber	Swarna Ageti	15.00
Muskmelon	Kashi Madhu	0.75
Palak	All Green	300.00
Ash gourd	Kashi Surabhi	11.00
	Kashi Dhawal	16.00
Indian bean	Kashi Haritma	490.00
French bean	Kashi Sampann	190.00
	Kashi Rajhans	480.00
Carrot	Kashi Arun	178.00
Radish	Kashi Hans	41.00
	Kashi Shweta	79.00
Cauliflower	Kashi Gobhi-25	13.80
Chenopod/ Bathua	Kashi Bathua-2	2.50
	Kashi Bathua-4	2.75
	Total	**27069.25**


Fig. 85: Monitoring of seed production crops at farmers field


Fig. 86: Monitoring of breeder seed during 2017-18 at ICAR-IIVR

of about 30358.4 kg seeds of different vegetables were produced which includes 3289.15 kg breeder seeds also. The TL seeds are given in table-53. Quantity of 2875.00 kg breeder seeds produced against the target of 2058.00 kg as per National indents from Deputy Commissioner (Seeds) (Table 54). In addition to it, 414.15 kg breeder seeds of different varieties of IIVR were also produced (Table 55).

Table 54: Quantity of breeder seed produced as per National indents from Deputy Commissioner (Seeds)

Crop	Variety	Indent (kg)	Quantity produced (kg)
Cowpea	Kashi Kanchan	194	260.00
	Kashi Gauri	20	*
Okra	Kashi Pragati	515	750.00
	Kashi Kranti	20	200.00
	Kashi Vibuthi	0.5	1.00
Pumpkin	Kashi Harit	2	32.00
Bottle gourd	Kashi Ganga	5	33.00
Muskmelon	Kashi Madhu	5	*
Pea	Kashi Nandini	290	500.00
	Kashi Uday	740	840.00
	Kashi Mukti	240	250.00
Tomato	Kashi Vishesh	2.5	3.00
Chilli	Kashi Anmol	4	4.00
Radish	Kashi Hans	8	1.0 *
	Kashi Shweta	12	1.0 *
		2058	**2875.00**

* In progress

Table 55: Quantity of breeder seed of ICAR-IIVR varieties (other than National indent)

Crop	Variety	Quantity produced (kg)
Cowpea	Kashi Nidhi	250.00
Ashgourd	Kashi Dhawal	5.00
	Kashi Surbhi	2.00
Pea	Kashi Ageti	50.00
	Kashi Samridhi	50.00
Brinjal	Kashi Taru	1.00
Tomato	Kashi Aman	5.00
	Kashi Adarsh	0.50
	Kashi Amrit	0.15
	Kashi Anupam	0.50
Dolichus bean	Kashi Haritima	50.00
		414.15

To augment the seed availability, the participatory seed production programme at farmers' field was also undertaken. Through this programme, 6012.00 kg seeds of vegetable pea, 665.00 kg seeds of cowpea and 2430.00 kg seeds of okra were produced during the year (Fig. 85-86).

Hybrid seed production: Under hybrid seed production programme at ICAR-IIVR, a total of 14.85

kg hybrid seed of different vegetable was also produced under protected conditions during 2017-18 (Table 56).

Table 56: Seed production of F_1 hybrids

Crop	Hybrid	Quantity produced (kg)
Brinjal	Kashi Sandesh	8.25
Tomato	Kashi Abhiman	0.60
Chilli	Kashi Tej	6.00
	Total	**14.85**

Seed packaging and distribution: Since vegetables are the main source of nutritional security, hence total of 1,28,979 seed packets were prepared for distribution/ sale which includes 12,024 kitchen garden packets (10 small packets in each kitchen garden packet) of 17 crops.

Seed production at RRS Sargatia: The seed production programme was also undertaken at Regional Research Station, Sargatia where 3894.00 kg TL seeds of 13 varieties of different 11 vegetables were produced. In addition to it, RRS also produced 125 q planting material of turmeric, 75 q of elephant foot yam, 198 q seeds of paddy and 525 q of lentil (Fig. 87).

Chilli var. Kashi Anmol

Brinjal var. Kashi Uttam

Sponge gourd var. Kashi Divya

Okra var. Kashi Kranti

Fig. 87: Seed production at RRS, Sargatia

Project 2.3: Drying and Storage Studies Including Modified Atmosphere Storage

Seed storage study with zeolite beads: Zeolite beads are micro porous material and having specific pore size of 3Å and it can absorb the moisture from the storage environment. Seed storage study with zeolite beads initiated with the aim of maintaining the vigour of the seed for longer period. Seeds of cowpea cv. Kashi Nidhi, okra cv. Kashi Kranti, pumpkin cv. Kashi Harit and radish cv. Kashi Hans were stored with beads and without beads both in room temperature and cold storage (Fig. 88). For the purpose of comparison, seeds were also stored in polythene bag, cloth bags and in plastic container with silica gel. Initial observation was recoded before storage (Table 57).

Fig. 88: Seed storage without and with Zeolite Beads

Table 57: Initial observations with respect to seed quality before seed storage

Variety	Moisture content (%)	Speed of germination	Germination (%)	Seedling length (cm)	Seedling Dry weight (mg/ seedling)	Vigour index I	Vigour index II
Kashi Nidhi	8.9	14.34	82	21.56	75.5	1768	6191
Kashi Kranti	7.9	16.92	78	15.93	112.0	1243	8736
Kashi Harit	5.5	10.40	79	13.51	95.5	1067	7545
Kashi Hans	5.0	43.92	98	28.4	91.0	2783	8918

Division of Vegetable Production

MEGA PROGRAMME-3: PRODUCTIVITY ENHANCEMENT THROUGH BETTER RESOURCES MANAGEMENT

Project 3.1: Technologies for protected and off-season vegetable production

Protected Cultivation of Cucumber: In cucumber, 4 parthenocarpic lines (Pant Cucumber 2 & 3, Multi-star and King-star) and one open-pollinated cultivar (Damini) were evaluated under naturally ventilated polyhouse during June to October, 2017 (Table 1). Findings revealed that the maximum number of fruits (25.33/ plant) and yield (3.35 kg/plant or 3.50 kg/m^2) was reported in cultivar Multi-star followed by King-star (2.59 kg/plant or 3.03 kg/m^2) (Fig. 1). Multi-Star has yield increase of 29.3% and 54.3%, respectively over King-star and Pant Cucumber-2.

Table 1: Performance of cucumber under protected condition

Variety	No. of fruits/ plant	Yield/ plant (kg)	Yield (kg/ m^2)
Pant Cucumber-2	11.33	2.17	1.64
Pant Cucumber- 3	11.67	2.36	2.08
Multi-star	25.33	3.35	3.50
King-star	19.67	2.59	3.03
Damini	10.00	2.87	2.93
CD $_{0.05}$	2.44	0.37	0.29

Fig.1(a): Parthenocarpic cucumber grown under naturally ventilated polyhouse

(b): Parthenocarpic Cucumber harvested fruits

Tomato: In tomato, two indeterminate hybrids-NS 4266 and GS 600 and one determinate open pollinated cultivar Kashi Aman were evaluated in two protected structures *i.e.* naturally ventilated polyhouse and net house condition (Table 2 & 3; Fig.2). Maximum plant height (267.26 cm), fruit number (97.0/ plant) and yield (13.66 kg/plant) was reported with NS 4266. This cultivar registered an increase of 135.5% yield over GS-600. In net house condition, also this cultivar noticed an increase of 132% in yield over GS-600.

Table 2: Performance of tomato under protected condition

Variety/ Hybrid	Fruits/ plant	Fruit wt. (g)	Fruit length (cm)	Fruit dia. (cm)	Fruits/ cluster	Plant height (cm)	Yield/ plant (kg)
Kashi Aman	41.00	68.94	5.50	5.76	3.20	102.26	2.55
NS-4266	97.00	153.60	5.86	6.92	10.40	267.26	13.66
GS-600	39.60	143.36	5.98	6.98	4.80	151.96	5.80

Table 3: Performance of tomato under net house condition

Tomato varieties	Plant height (cm)	Number of fruits plants	Fruit weight (g)	Yield/ plant (kg)
GS-600	245.60	48.50	95.30	4.17
NS-4266	265.00	73.30	82.50	9.67
Cherry Tomato	238.48	68.72	8.20	1.68

Fig.2 (a): Kashi Aman (b): NS-4266 (c): GS-600

Capsicum: Three hybrid capsicums *i.e.* Almirante, Swarna and Natasha were grown both in naturally ventilated polyhouse and nethouse condition (Table 4 & 5; Fig.3). The maximum yield of 2.19 kg/plant was reported in cultivar Swarna followed by Almirante (1.47 kg/plant) and Natasha (1.24 kg/plant). The yields of Swarna, Almirante and Natasha in nethouse condition were 1.98, 1.76 and 1.56 kg/plant, respectively. Under both protected structures, Swarna produced maximum yield over other cultivars, however it performed best under naturally ventilated polyhouse and registered 49% and 76.6% higher yields, respectively over Almirante and Natasha.

Fig.3(a): Capsicum Hybrid Almirante

(b): Capsicum hybrid Swarna

Table 4: Performance of Capsicum hybrids under naturally ventilated polyhouse

Variety/ Hybrid	Plant height (cm)	No. of fruits/ plant	Fruit weight (g)	Fruit length (cm)	Fruit diameter (cm)	Yield/ plant (kg)
Almirante (Green)	83.06	7.80	193.60	9.32	7.30	1.47
Swarna (Yellow)	73.82	5.60	335.00	12.90	9.28	2.19
Natasha (Red)	71.98	6.80	178.00	8.98	7.16	1.24

Table 5: Performance of Capsicum under net house condition

Capsicum hybrids	Plant height (cm)	Number of fruits	Av. Fruit weight (g)	Yield/ plant (kg)
Almirante	59.30	10.57	166.24	1.757
Swarna	62.20	11.00	170.57	1.976
Natasha	58.13	9.58	163.43	1.565

Project 3.2: Precision farming in vegetable crops

Studies on the residual effect of Nitrogen applied in tomato on growth and yield of cowpea cv. Kashi Nidhi: Field experiment was conducted during summer season to study the performance of cowpea with different levels of nitrogen applied in tomato on growth and yield of cowpea cv. Kashi Nidhi. Data presented in Table 6 revealed that there was an increasing trend in growth, yield and quality parameters of cowpea up to 200kg N/ha. The maximum values for yield and protein content in greenpod were found to be 142.67 q/ha and 2.80 percent. However, N-use efficiency was maximum (0.4368 q/kg) with 160 kg N/ha.

Table 6: Growth, yield and quality of cow pea as affected by residual N levels

Nitrogen Level	Average Plant Height (cm)	No. of pods / plant	Pod length (cm)	Pod weight (g)	Yield (q/ha)	Protein % in green pod	Nitrogen use efficiency (q/kg)
N0	57.80	20.10	20.10	9.70	70.20	1.96	0.00
N40	69.23	24.67	22.83	10.30	82.42	2.17	0.3055
N80	74.75	28.30	23.50	12.40	90.38	2.22	0.2522
N120	81.36	33.20	24.00	14.30	106.67	2.40	0.3039
N160	89.34	37.45	26.32	17.4	140.10	2.62	0.4368
N200	88.40	38.50	25.34	17.70	142.67	2.80	0.3628
N240	83.88	36.00	24.42	17.10	140.23	2.75	0.2916
CD at 5 %	14.87	2.90	1.70	1.36	6.50	N.S.	--

Project 3.3: Development of vegetable based cropping system for sustainability and profitability

Ten different vegetable based cropping systems were evaluated. During kharif season, the yield of kharif crops was compared in terms of rice equivalent yield. The highest productivity during kharif season was obtained with brinjal crop having 202.80 q/ha of Rice equivalent yield. During winter season, the yield of different crops in the cropping system was compared on the basis of wheat equivalent yield. The highest WEY was obtained with Pea+ Radish (109.62 q/ha) crops followed by tomato crop (108.30 q/ha). The grain yield of wheat was in the range of 39.4 to 40.28/q/ha (Table 7).

Table 7: Crop yields under different cropping systems

Cropping sequence	Rice Eq. Yield (q/ha)	Yield(q/ha) of kharif crops	Wheat Eq. yield (q/ha)	Yield (q/ha) of Rabi crops
Paddy–wheat	-	Paddy (HUV-917) -48.62	-	Wheat (RD-2967) Grain -39.48 Straw -53.85
Paddy -wheat-coriander	-	Paddy (HUV-917) -47.50	-	Wheat (RD-2967) Grain-40.28 Straw-52.74
Paddy–tomato-mungbean	-	Paddy (HUV-917) -49.20	72.66	Tomato (Kashi Aman)-327.6
Paddy -broccoli-cowpea	-	Paddy (HUV-917) -49.00	48.54	Broccoli-145.63
Bottle gourd-wheat-radish-amaranth	115.7	Bottle gourd-(Kashi Ganga)-270.46	-	Wheat (RD-2967) Grain -41.31 Straw-50.78
Maize- -pea – radish -pumpkin	40.18	Maize (Cob yield) -140.65	109.62	Pea (Kashi Uday) -95.50, Radish-270.48
Brinjal-cowpea-amaranth	202.8	Brinjal (Kashi Uttam) -315.48	-	-
Okra-tomato-cowpea	56.74	Okra (Kashi Kranti)-132.41	101.92	Tomato (Kashi Aman) 458.65
Paddy -pea-okra	-	Paddy (HUV-917) - 48.52	74.58	Pea (Kashi Mukti)- 89.50
Cowpea-tomato-okra	93.71	Cowpea (Kashi Nidhi)-131.20	108.30	Tomato (Kashi Aman) 487.38

Price of different crops: Rice- Rs.14.5/kg, Wheat- Rs.18.00/kg, Bottle gourd- Rs.6.00/kg, Maize- Rs.4.00/kg, Brinjal - Rs 9.00/kg, Okra-Rs.6.00/kg, Cowpea- Rs.10.00/kg, Tomato - Rs.4.00/kg, Broccoli-Rs.6.00/kg, Pea- Rs.15.00/kg Radish- Rs.2.00/kg,

Project 3.4: Impact of organic and inorganic management systems on vegetable productivity, quality and soil health

ZAID 2017: During zaid season, cowpea (var. Kashi Nidhi), okra (Var. Kashi Kranti) and Bottle gourd (var. Kashi Ganga) were grown under twelve different organic treatments comprising of three organic sources and three rates as given in Table 8. An inorganic treatment where inorganic fertilizer and chemical control of disease pest was applied was taken for comparison along with one absolute control where no fertilizer was applied.

Table 8: Treatment details

T_1 :	FYM @ 15 t/ha
T_2 :	FYM @ 20 t/ha
T_3 :	FYM @ 25 t/ha
T_4 :	Vermicompost @ 5 t/ha
T_5 :	Vermicompost @ 7.5 t/ha
T_6 :	Vermicompost @ 10 t/ha
T_7 :	NADEP compost @ 15 t/ha
T_8 :	NADEP compost @ 20 t/ha
T_9 :	NADEP compost @ 25 t/ha
T_{10} :	FYM @ 10 t/ha + NADEP @ 10 t/ha
T_{11} :	FYM @ 10 t/ha + Vermicompost @3.5 t/a
T_{12} :	NADEP @ 10 t/ha + Vermicompost @ 3.5 t/ha
T_{13} :	Control
T_{14} :	Recommended dose of fertilizer (Inorganic NPK)

The result presented in Fig 4, 5, and 6 revealed that highest yield of bottle gourd (27.66 t/ha) and okra (7.48 t/ha), was recorded with application of FYM @ 25t/ha, which was *at par* with inorganic fertilizer application. In cowpea the highest yield (6.96 t/ha) was recorded

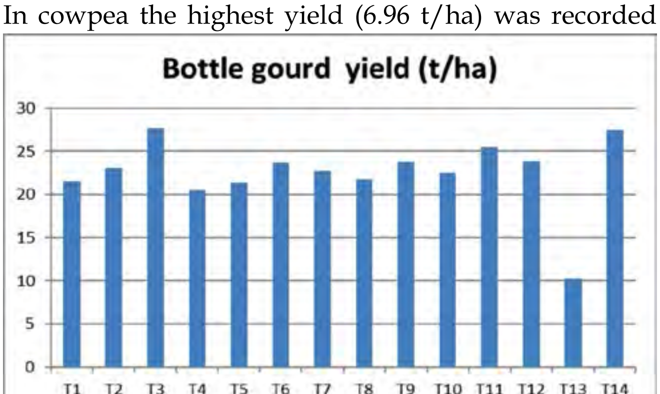


Fig. 4: Yield of bottle gourd as influenced by different treatments under organic system

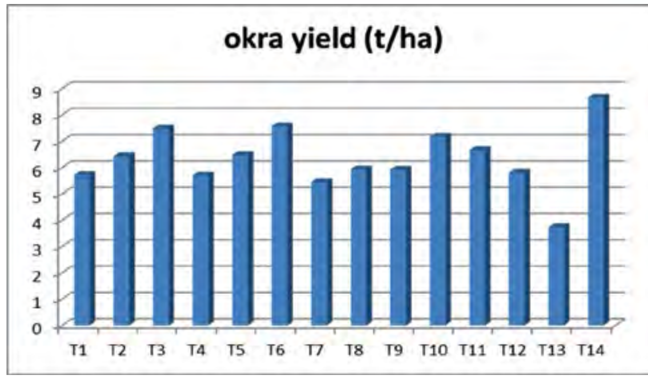


Fig. 5: Yield of okra as influenced by different treatments under organic system

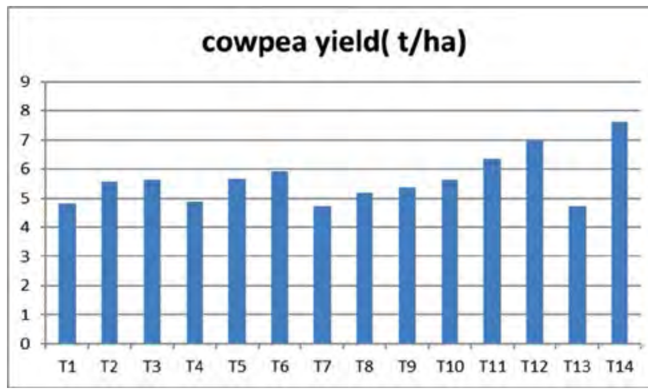


Fig. 6: Yield of cowpea as influenced by different treatments under organic system

with application of NADEP compost @ 25t/ha. The minimum yield was noted under absolute control.

Effect on soil properties: After completion of the cycle, the soil samples were collected and analyzed for organic carbon content and the result is presented in Fig. 7. Organic management systems recorded higher organic carbon and available N in soil. Among the different organic management systems, the highest increase in OC was noted under application of NADEP compost @ 25t/ha. There was decline in OC in absolute control treatment, while there was

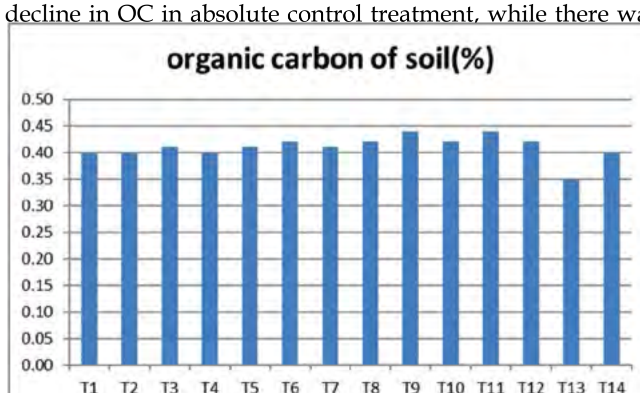


Fig. 7: Organic carbon content of soil as influenced by treatments under organic system

no change in OC of inorganic treatment. The total microbial activity in terms of fluorescein diacetate hydrolysis was also higher under organic systems as compared to inorganic control. The soil microbial activity was minimum under inorganic control and maximum under application of FYM @ 25t/ha. The nematode population dynamics in Okra was also studied. It was found that % reduction of nematode reproduction/population is more by FYM (76-87%) followed by vermicompost (76-82%) and NADEP compost (69-74%) over Absolute control.

Kharif 2016: During kharif season green manure crop, *Dhaincha* was grown and turned down in soil 42 days after sowing. On an average the addition of dry matter through *Dhaincha* was in the range of 3.2 to 3.5 t/ha. The N: P: K: content in *Dhaincha* on dry weight basis is 2.37: 0.64: 1.64 percent respectively. The result is presented in Fig. 8.

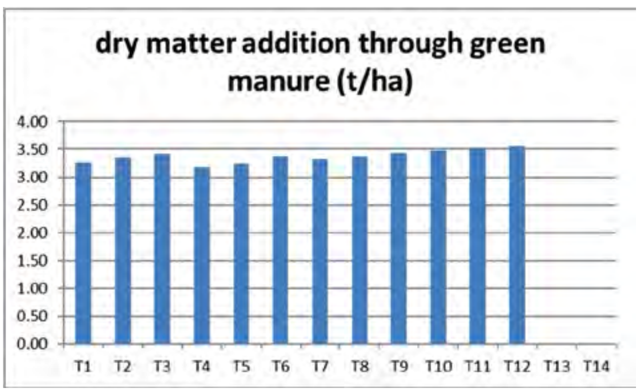


Fig. 8: Dry matter addition through green manure crop of *Dhaincha*

Rabi 2016: During rabi season the crops included in the cropping systems were brinjal, cabbage, broccoli, pea, French bean and leek The organic management treatments remained the same as given in table 8, Four week old seedlings of brinjal variety Kashi Sandesh were transplanted at spacing 75 cm X 70 cm. The organic sources were applied 15 days before transplanting and well mixed in soil during field preparation. Similarly 25 days old seedlings of cabbage cultivar Golden Acre were transplanted at 45 cm X 45 cm spacing, while 21 days old seedling of Broccoli variety LUD-1.was transplanted at 60 X 50 cm. Pea seed of variety Kashi Nandini was sown with seed rate of 150 kg/ha at 30 cm row to row spacing with plant to plant spacing of 10 cm while French bean was sown at seed rate of 80 kg/ha with row spacing of 60 cm and plant to plant spacing of 20 cm. Two hand weeding followed by earthing was done at one month interval after planting in brinjal while one hand weeding was done in cabbage, brocolli and pea and French bean. The crops were raised following recommended protocols and agronomic package of practices for organic farming. Need based plant protection measures comprising of organic formulations was done. The produce of each plot was weighed separately.

Effect of sources and its dose: Crop stand of brinjal, cabbage, broccoli, French bean and pea was unaffected due to organic sources or its dose. The data presented in Fig. 9 indicates that increasing rates of organic sources increased the total yield of brinjal. The highest dose of all the three organic sources produced fruit yield of brinjal comparable to the yield level obtained with application of inorganic fertilizer at recommended dose (T11). However, marketable yield of brinjal in inorganic source was significantly higher than organic sources and absolute control due to less infestation of fruit borer as chemical control measures was more effective than organic control measures. The farm yard manure and NADEP compost and vermicompost were equally effective source of manure for brinjal.

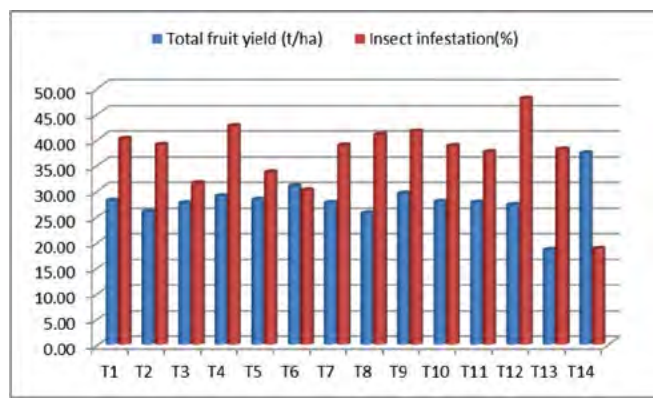


Fig. 9: Yield of brinjal as influenced by different treatments under organic system

In pea, the organic sources produced significantly higher green pod yield over control and inorganic source (Fig. 10). It was also observed that increasing dose of all the three sources did not increase the green pod yield. The increase in yield was associated with increase in number of pods/plant as well as number of grains per pod. However, in French bean the highest green pod yield was obtained with application of 25 t/ha FYM which was significantly more than inorganic source. Among the three sources, FYM was superior source than vermicompost and NADEP. In cabbage, all the three sources produced significantly higher yield over control. The highest yield was obtained with application of NADEP compost @25t/ha or Vermicompost @10t/ha which was comparable with inorganic source (Fig. 11). Among different organic sources, NADEP compost was found superior than other two sources. There was significant difference with regards to yield and head size of cabbage. It was also observed that increasing dose of all the three sources increased the cabbage yield. The combined application of sources improved the cabbage yield. The broccoli yield was influenced by source and dose of organic manure. The highest yield was recorded with application of 25t/ha NADEP compost which was *at par* to inorganic treatment but significantly superior to rest of the treatments.

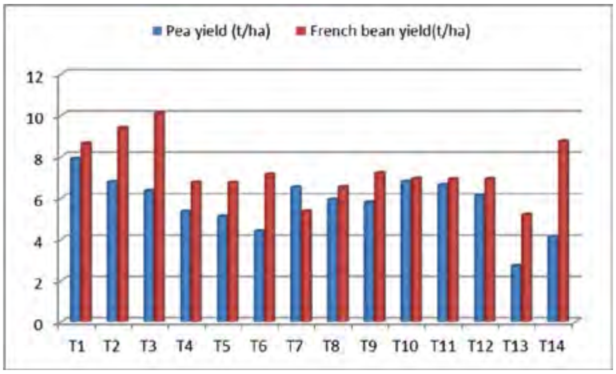


Fig. 10: Yield of Pea and French bean as influenced by different treatments under organic system

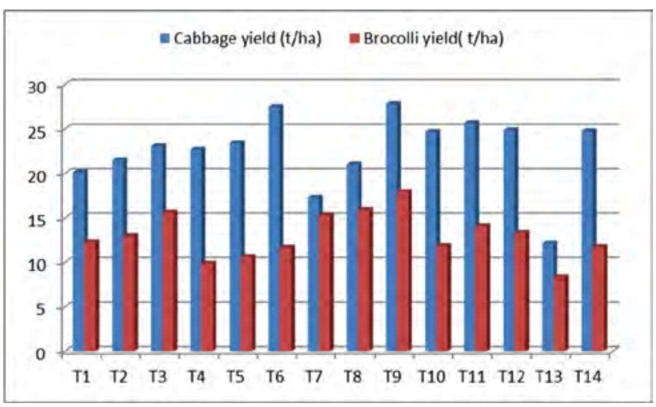


Fig. 11: Yield of cabbage and broccoli as influenced by different treatments under organic system

Quality parameters: The quality of vegetables in terms of vitamin C content was better under organic system as compared to inorganic system in brinjal, pea and cabbage (Table 9). There was no consistent trend in colour and texture in cabbage, pea and brinjal.

Table 9: Quality of Pea as influenced by different treatments under organic system

Treatment	Moisture (%)	TS	Ascorbic Acid (mg/100g dm.)	Total Phenol (mg/100g dm)	Antioxidant (%)
T1	74.79	24.11	262.04	177.86	16.96
T2	71.24	27.66	215.63	141.16	19.21
T3	71.54	27.36	224.63	125.30	19.12
T4	72.40	26.50	228.85	138.15	9.44
T5	72.10	26.80	248.88	167.73	10.92
T6	74.08	24.82	309.39	121.34	16.56
T7	64.98	33.92	207.02	128.23	18.00
T8	71.06	27.84	291.38	127.77	16.96
T9	69.44	29.46	179.68	114.09	21.18
T10	71.45	27.45	228.58	135.04	17.92
T11	70.57	28.33	199.03	117.33	17.24
T12	70.54	28.36	215.13	125.44	14.07
T13	68.97	29.93	235.13	121.72	14.15
T14	69.99	28.91	220.79	135.36	11.65

Project 3.5: Improving soil health and carbon sequestration in vegetable production system through conservation tillage and residue incorporation

The experiment on resource conservation in vegetable production is continuing since last four years at the same site on fixed plots. During 2017-18 also, experiment was conducted during the summer, kharif and rabi season, to study the effect of conservation tillage on the production potential and soil heath in vegetable cowpea-okra-tomato cropping systems. There were three tillage treatments evaluated with or without residue retention/incorporation in split plot design replicated four times. The tillage treatments were T1= Zero tillage, T2=Reduced tillage,T3= conventional tillage in the main plots and two subplots as R1= residue retention/ incorporation and R2= residue removal. The reduced tillage consisted of one cross ploughing with harrow/cultivator while conventional tillage consists of two-three cross ploughing with cultivator/ one harrowing followed by ploughing with cultivator, depending on the crop. Each ploughing was followed by planking to break the clods. Cowpea crop was sown during summer, okra during kharif and tomato during Rabi. The details of package of practice are given in Table 10.

Table 10: Package of practice of crops

	Summer	Kharif	Rabi
Crop	Cowpea	Okra	Tomato
Variety	Kashi Kanchan	Kashi Kranti	Kashi Aman
Seed rate	30 kg/ha	20kg/ha	Seedling transplanted
Spacing	60 cm X 15cm	60cm X 20cm	75 cm X 50 cm
Fertilizer rate	25:60:40	120:60:60	150:60:100

The recommended dose of fertilizer for the crop was applied at the time of field preparation in conventional (CT) and reduced tillage (RT) treatment while in zero tillage (ZT), fertilizer was dibbled in soil at the time of sowing. In ZT, the residues were retained on surface by cutting the residues of crops and spreading it on soil surface. While in reduced tillage and conventional tillage the residues of the crop were incorporated in the soil of respective plot by ploughing immediately after the completion of the crop. Recommended weedicides were applied to the crops after sowing to control weeds. Need based one to two hand weeding was done in the crop. The removed weeds of the plots were spread between rows on the soil surface. The crops were raised following recommended agronomic package of practices. Need based plant protection measures were adopted to save the crop from insect pests and diseases. Bulk density (BD) of the top soil 0-20 cm was estimated after completion of the cycle. Top soil (0-20 cm depth) was analysed for soil pH and organic carbon (OC).

It is evident from the perusal of the data In Table 11, 12 and 13, that the maximum yield of 9.63 t /ha, in cowpea, 11.55 t/ha in okra and 53.84 t/ha in tomato was obtained with ZT which was significantly superior to conventional tillage treatments. The yield increase in cowpea, okra and tomato was 15.6, 44.55 and 19.8 percent respectively over conventional tillage practice. The yield increase in cowpea, okra as well as tomato under zero tillage was associated with increase in yield attributing characters in respective crops. Zero tillage increased the OC content of the soil by 13.33 percent and reduced the BD of the soil (Table 14). The net return and benefit cost ratio was also maximum in this treatment.

Table 11: Productivity of cowpea crop as influenced by tillage and residues retention

Treatment	Yield (t/ha)	Plant height (cm)	Pod length (cm)	Pod wt (g)
ZT	9.63	41.55	27.4	13.25
RT	7.63	37.6	26.05	12.4
CT	8.33	40.5	26.45	13
CD(P=0.05)	1.76	NS	NS	NS
Residue removal	9.49	43.30	28.20	13.35
Residue retention	7.57	34.45	25.07	12.13
CD(P=0.05)	1.34	8.12	NS	NS

The economics was better in the ZT due to increased yield and also lower cost of cultivation.

Table 12: Productivity of Okra crop as influenced by tillage and residues retention.

Treatment	Yield (t/ha)	Plant height (cm)	No. fruits/ plant	Average fruit wt. (g)
ZT	11.55	61.54	12.47	10.42
RT	8.79	47.19	11.93	10.75
CT	7.99	47.34	10.93	11.25
CD(P=0.05)	2.14	12.12	NS	NS
Residue removal	9.87	53.54	12.04	10.91
Residue retention	9.01	50.51	11.51	10.70
CD(P=0.05)	NS	NS	NS	NS

Table 13: Productivity of tomato crop as influenced by tillage and residues retention

Treatment	Yield (t/ha)	Plant height (cm)	Average fruit weight (g)
ZT	53.84	170.50	60
RT	53.28	158.50	54
CT	44.91	159.50	49.5
CD(P=0.05)	**9.87**	**NS**	**9.35**
Residue removal	49.72	158.67	52.33
Residue retention	51.63	167.00	56.67
CD(P=0.05)	**NS**	**NS**	**NS**

Residue retention/incorporation, in general improved the yield in all the crops which may be due to its positive influence on weed suppression as well as moisture conservation and increase in organic carbon in the soil. The yield increase in cowpea, okra and tomato was 25.63, 9.52 and 4.03 percent due to residue retention/ incorporation over its removal. The increase in organic carbon (%) of soil due to residue incorporation/retention was 6.5% over residue removal. The increased organic carbon content decreased the bulk density of soil in plots where residues of crops were incorporated /retained over its removal The organic carbon content of soil was in general more in residue retention/incorporation over

Table 14: Bulk density and organic carbon content of soil as influenced by tillage and residues retention

	Organic carbon			Bulk density		
	Residue retention	Residue removable	Average	Residue retention	Residue removable	Average
ZT	0.53	0.49	**0.51**	1.39	1.41	**1.40**
RT	0.47	0.44	**0.46**	1.39	1.42	**1.41**
CT	0.46	0.44	**0.45**	1.43	1.44	**1.43**
Average	**0.49**	**0.46**		**1.40**	**1.43**	
CD(P=0.05) (tillage)			0.01			0.01
Residue	0.02			0.02		
Tillage X residue	0.028			0.034		

residue removal in all the three tillage treatments. It is interesting to note that the OC of soil under under ZT even with residue removal was significantly higher than residue incorporation in other two tillage treatments.

Interaction Effect: The perusal of the data in table 15, 16 and 17 revealed that there was significant difference between ZT and RT in terms of cowpea yield in the plots where residue retention was done while it was *at par* in the plots where residues were removed. The yield of cowpea in ZT was significantly superior over CT only in residue removal while it was *at par* in residue retention. The residue retention/incorporation of previous crop increased the yield of cowpea in all the three tillage treatments as compared to residue removal; however yield increase was statistically not significant only in RT.

In Okra, there was significant difference due to tillage treatments under both the conditions of either residue retention or residue removal. However, yield increase in okra under ZT due to retention of residues was significant over residue removal, while in other two tillage treatments yield of okra was *at par* in both residue removal and residue retention.

In tomato, zero tillage produced significantly higher yield over conventional tillage both under residue retention and residue removal condition. Residue retention in general increased the tomato yield over its removal, though it was not up to the level of significance under all the three-tillage condition. The yield recorded under zero tillage in residue removal was significantly higher than yield recorded under conventional tillage in residue retention condition.

Table 17: Interaction effect of tillage and residues retention on tomato yield

	Yield (t/ha)		Average fruit weight (g)	
	Residue retention	Residue removable	Residue retention	Residue removable
ZT	55.03	52.64	62	58
RT	54.58	51.98	56	52
CT	45.28	44.55	52	47
CD (P=0.05)	6.45		9.21	

Table 15: Interaction Effect of tillage and residues retention on yield and yield attributes of cowpea

	Yield (t/ha)		Plant height (cm)		No. of fruits/plant		Average fruit weight (g)	
	Residue retention	Residue removal	Residue retention	Residue removal	Residue retention	Residue removal	Residue retention	Residue removal
ZT	10.79	8.47	46.7	36.4	30.1	24.7	14.2	12.3
RT	7.87	7.38	42.7	32.5	26.7	25.4	12.9	11.9
CT	9.81	6.85	40.5	35.7	27.8	25.1	13.8	12.2
CD (P=0.05)	2.67		9.58		6.17		2.12	

Table 16: Interaction Effect of tillage and residues retention on yield and yield attributes of okra

	Yield (t/ha)		Plant height (cm)		No. of fruits/plant	
	Residue retention	Residue removable	Residue retention	Residue removable	Residue retention	Residue removable
ZT	12.45	10.65	60.65	62.43	13.2	11.73
RT	8.81	8.76	45.01	49.37	11.73	12.13
CT	8.35	7.62	45.87	48.81	11.2	10.67
CD (P=0.05)	2.63		11.98		2.71	

Project 3.6: Enhancing water and nutrient use efficiency in vegetable crops

Drip fertigation scheduling in tomato: Drip fertigation scheduling was carried out in tomato (cv. Kashi Aman) to optimize method and quantity of fertilizer application. In all treatments, 25% fertilizers were applied in soil through normal fertilizers (Urea, DAP and MOP), while rest of the fertilizers were applied either as water soluble (WSF) or normal fertilizers (NF) alone or in various combinations. A total of 6 treatments including control (surface irrigation and soil application of fertilizers) were used in this study (Table 18). Experimental findings revealed that maximum number of fruits (54.33/plant), fruit weight (147.22 g) and fruit yield (6.11 kg/plant and 68.86 t/ha) was obtained in plants that were fed with 100% NPK through WSF (Fig. 12). In this treatment, 83.1% and 53.7% higher yield

Table 18: Effect of NPK fertigation scheduling on performance of tomato cv. Kashi Aman

Fertigation scheduling	Plant height (cm)	Fruits no./ plant	Fruit wt. (g)	Fruit length (cm)	Fruit dia. (cm)	Yield/ plant (kg)	Yield (t/ ha)	NUE (q/ kg NPK)
100% NPK fertigation through WSF	103.69	54.33	147.22	6.04	5.91	6.11	68.86	2.375
50 % NPK WSF +50% soil appl.	99.17	47.89	140.22	5.92	5.79	5.21	58.54	2.018
100% NPK fertigation normal fertilizers (NF)	89.83	44.00	136.00	5.81	5.68	4.29	50.74	1.750
50 % NPK fertigation NF + 50% soil appl.	84.16	37.56	132.22	5.76	5.61	4.49	48.32	1.666
100% NPK through soil	81.02	33.00	138.69	5.71	5.61	3.65	44.81	1.723
Control-Surface irrigation	74.36	29.03	129.88	5.22	5.33	3.21	37.60	1.446
SEm ±	2.67	2.75	2.36	0.45	0.31	0.20	2.37	-
CD (0.05)	8.23	8.28	7.09	NS	NS	0.59	7.15	-

was recorded, respectively over control and 100% NPK application through soil. The maximum nutrient use efficiency (2.375 q/kg NPK) was also reported under this treatment.

Fig. 12 (a): 100% NPK fertigation through WSF **(b): 100% NPK fertigation through WSF**

Micronutrient study in Cole crops: Micronutrients-boron and molybdenum were used in broccoli and cauliflower to optimize their concentration in cole crops. Two quantities of boron (50 and 100 ppm) and molybdenum (25 and 50 ppm) were sprayed thrice; alone or in different combinations. It was observed that maximum curd weight (462.33 and 429.67 g) and yield (108.17 and 102.80 q/ha) were obtained with foliar spray of B 50 + Mo 25 (T_1) or B 100 + Mo 50 (T_4) (Table 19 and Fig. 13). These two treatments noticed 12.8% and 7.2% higher yield over control (water spray). Micronutrient spray also showed significant effect in cauliflower (cv. Madhubani) (Table 20 and Fig. 14). The maximum curd size (10.26 × 19.60 cm^2), curd weight (1377.78 g) and yield (331.20 q/ha) was obtained with

Table 19: Effect of micronutrient sprays on yield of Broccoli (cv. Shishir)

Treatment	Curd length (cm)	Curd dia. (cm)	Curd weight (g)	Curd yield (q/ha)
T1= B_{50} M_{25}	13.98	15.24	462.33	108.17
T2 = B_{50} M_{50}	13.03	14.46	378.44	90.67
T3 = B_{100} M_{25}	13.40	13.77	362.22	86.93
T4 = B_{100} M_{50}	13.54	14.90	429.67	102.80
T5 = B_{50}	13.07	14.28	387.22	93.73
T6 = B_{100}	13.28	14.79	352.78	84.67
T7 = M_{25}	13.36	14.03	356.22	85.87
T8 = M_{50}	12.39	14.49	353.89	84.93
T9 = Commercial	13.97	15.44	392.78	94.00
T10 = Control	13.67	15.23	389.78	95.90
SEm±	0.42	0.48	21.74	4.19
LSD (0.05)	NS	NS	64.60	12.46

Fig. 13: Broccoli crop under micronutrient trial

Table 20: Effect of micronutrient sprays on yield of Cauliflower (cv. Madhubani)

Treatment	Curd length (cm)	Curd dia. (cm)	Curd wt. (g)	Curd yield (q/ha)
T1= B_{50} M_{25}	9.69	15.41	848.89	203.73
T2 = B_{50} M_{50}	8.49	16.56	867.78	196.93
T3 = B_{100} M_{25}	9.09	16.97	926.89	221.53
T4 = B_{100} M_{50}	10.50	17.33	835.67	177.67
T5 = B_{50}	9.49	20.29	1233.33	296.00
T6 = B_{100}	9.26	18.74	1061.11	254.67
T7 = M_{25}	8.97	17.74	1032.22	247.73
T8 = M_{50}	10.26	19.60	1377.78	331.20
T9 = Commercial	8.61	18.84	931.11	214.13
T10 = Control	9.98	15.87	1035.56	242.80
SEm±	0.26	0.48	64.29	13.39
LSD (0.05)	0.77	1.41	191.00	39.79

Fig. 14: Harvested cauliflower of micronutrient trial

three sprays of Mo @ 50ppm. Boron spray at 50ppm also registered significantly higher yield (296.00 q/ha) over other treatments. There was an increase of 36.4% and 21.9% in yield, respectively under Mo 50 and B 50 over control. From one year study, it may concluded that combined spray of B50 + Mo 25 or B100 + Mo 50 in broccoli, and sole spray or Mo 50 thrice at 10 days interval significantly enhanced the yield in these cole crops.

Studies on micronutrients in vegetables

Micronutrient studies in bitter gourd: A field experiment was conducted during *Kharif*-2017 to study the effect of foliar application of micronutrients on growth and yield of bitter gourd (Hybrid), The treatments consisted of T1-control, T2- Zn @ 100 ppm, T3- Cu @100 ppm, T4- Fe @100 ppm, T5- Mn @ 100 ppm, T6- B @100 ppm, T7- Mo @ 50 ppm, T8- Mixture of all these micronutrients at their respective concentrations. The micronutrients were applied after 30 days of planting three times at 10 days intervals.

The results presented in Fig. 15 indicate that there was a significant effect of micronutrients application on the growth and yield of bitter gourd. The number of fruits was found to be maximum (25/plant) with the combined use of all the micronutrients, however, it was *at par* with the use of boron @ 100 ppm. Almost similar trend was noticed with respect to fruit weight and fruit yield also. The highest fruit yield of 127.3 q/ha was recorded under combined application of

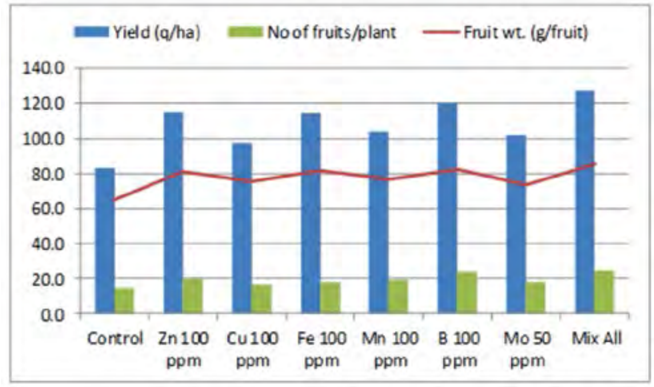


Fig. 15: Response of bitter gourd to foliar spray of micronutrients

micronutrients which was *at par* with those recorded under sole application of B and Zn @ 100 ppm each. The lowest number of fruits (15.0/ plant), fruit weight (65.1 g/ fruit) and fruit yield (83.3 q/ha) were recorded under control.

Preparation and evaluation of micronutrient formulations: Crop-group specific micronutrient formulations for Solanaceous (5 nos.) and Cole crops (4 nos.) were prepared in the laboratory and were evaluated for their efficacy under field conditions during *Rabi*-2017-18 on tomato and broccoli crops, respectively (Fig. 15 and 18). A commercial formulation (Mobomin) from Aries Agro Limited, Mumbai was taken for comparison.

Effect of micronutrient formulations on Broccoli: The four prepared micronutrient formulations (Micromix A, Micromix B, Micromix C and Micromix

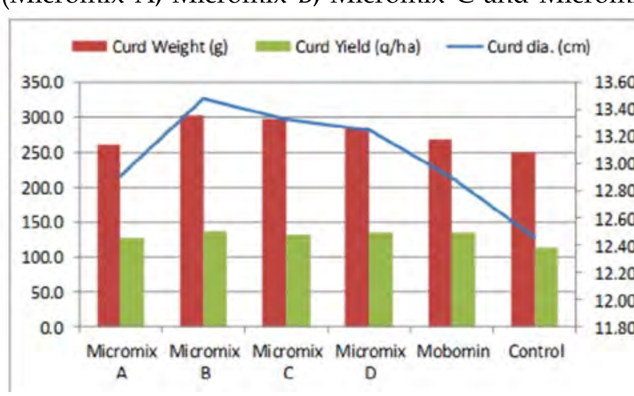


Fig. 16: Response of broccoli to different micronutrient formulations

D) and the commercial formulation (Mobomin) were applied three times @ 1.5g/l at 10 days intervals at 30 days after planting of the crops and growth and yield parameters were recorded at the time of harvest. The results presented in Fig. 16 indicate that all the micronutrient formulations improved the curd size, curd weight and curd yield of broccoli as compared to control. Among different formulations, Micromix B proved more effective as compared to the others.

Effect of micronutrient formulations on Tomato: An experiment was conducted to evaluate the performance of five different micronutrient formulations prepared for solanaceous crops during *Rabi*-2017-18. Tomato var. Kashi Aman was taken as the test crops. For comparision, a commercial micronutrient formulation (Mobomin) was used as check. All the formulations were applied in the form of foliar spray @ 1.5 g/l three times at 15 days intervals after 30 days of planting.

The results presented in Fig. 17 reveal that there was a significant effect of micronutrients foliar application on the growth and yield of tomato. Among different micronutrient formulations, Micromix A proved slightly better recording maximum plant height (116.5 cm), number of fruits per plant (65.1) and total fruit yield (450.9 q/ha) to others, however, these did not differ significantly with each other. The lowest values of all the parameters were recorded under control plot.

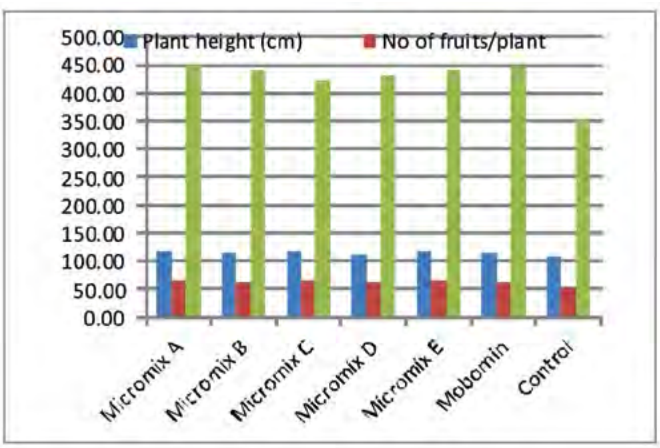


Fig. 17: Response of tomato to different micronutrient formulations

Drip irrigation scheduling and mulching study in okra: In spring-summer okra, an experiment was conducted to optimize drip irrigation scheduling with or without mulch condition (Table 21). In this study, four irrigation schedulings *i.e.* drip irrigation daily (I_1), 2-day interval (I_2), weekly (I_3) and surface irrigation (I_4) were applied in main plot, whereas three mulching

Table 21: Effect of drip irrigation scheduling and mulching on yield attributes of okra

Plant height (cm)				
Treatment	**Organic mulch (M_1)**	**Black-Silver mulch (M_2)**	**No mulch (M_0)**	**Mean**
Drip-daily (I_1)	96.90	90.70	77.53	**88.38**
Drip-2 day (I_2)	90.63	82.07	74.10	**82.27**
Drip- weekly (I_3)	85.83	75.60	63.47	**74.97**
Surface irrigation (I_4)	90.73	79.60	76.57	**82.30**
Mean	**91.03**	**81.99**	**72.92**	
LSD (0.05) Irrigation= NS; Mulch =6.76; I x M = NS				
Fruits no./plant				
Drip-daily (I_1)	34.00	31.67	27.00	**30.89**
Drip-2 day (I_2)	35.67	30.33	26.33	**30.78**
Drip- weekly (I_3)	30.00	25.00	22.67	**25.89**
Surface irrigation (I_4)	28.00	25.67	26.33	**26.67**
Mean	**31.92**	**28.17**	**25.58**	
LSD (0.05) Irrigation= 2.21; Mulch =2.43; I x M = 3.35				
Fruits yield/plant (g)				
Drip-daily (I_1)	647.67	573.00	417.33	**546.00**
Drip-2 day (I_2)	612.67	539.67	423.33	**525.22**
Drip- weekly (I_3)	503.67	451.00	305.00	**419.89**
Surface irrigation (I_4)	546.33	409.67	342.00	**432.67**
Mean	**577.58**	**493.33**	**371.92**	
LSD (0.05) Irrigation= 61.77; Mulch =70.15; I x M = 81.60				
Fruits yield (q/ha)				
Drip-daily (I_1)	120.30	104.57	86.90	**103.92**
Drip-2 day (I_2)	115.60	105.07	85.30	**101.99**
Drip- weekly (I_3)	84.13	88.23	59.73	**77.37**
Surface irrigation (I_4)	83.87	78.43	68.97	**77.09**
Mean	**100.98**	**94.08**	**75.23**	
LSD (0.05) Irrigation= 9.45; Mulch =8.11; I x M = 11.07				

Fig. 18: Spring summer okra under different mulch system

treatments *i.e.* organic mulch @ 7.5 t/ha (M_1), black-silver polythene (M_2) and no-mulch (M_o) were executed in sub-plot of split plot design with 3 replications (Fig. 18). Water under drip irrigation was applied at 100% PE, whereas in surface irrigation it was given at IW/CPE = 1.0. Experimental findings revealed that drip irrigation and mulching significantly affected the okra production. Among the irrigation schedulings, drip irrigation daily or 2-day intervals (I_1 and I_2) had significant enhancement in number of fruits (30.89 and 30.78/plant), fruit yield (546.0 and 525.22 g/plant and 103.92 & 101.99 q/ha). Among the mulch used in present study, the maximum plant height (91.03 cm), fruit number (31.92/ plant) and yield (577.58 g/plant, 100.98 q/ha) was observed under organic mulching. Under organic mulch, 34.2% and 7.3% higher yield were registered over unmulched control and black-silver mulch, respectively

As far as irrigation scheduling and mulching on yield was concerned, significantly higher plant height and yield attributes was recorded with drip irrigation

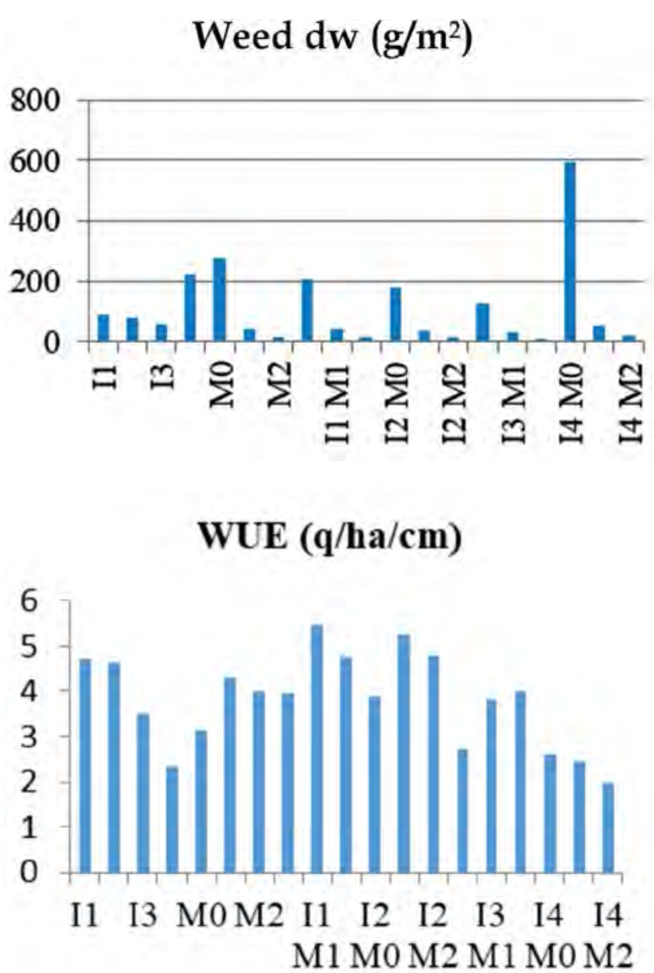


Fig. 19: Effect of irrigation scheduling and mulching weed growth and water use efficiency in okra

daily or two day intervals coupled with organic mulching. These two treatments (I_1M_1 and I_2M_1) had maximum number of fruits (34.0 and 35.7/plant) and yield (617.67 & 612.67 g/plant; 120.30 & 115.60 q/ha). These two treatments registered 74.4% and 67.6% higher yield over surface irrigation without mulch application. Drip irrigation at weekly intervals without mulch had observed the lowest fruit yield (59.73 q/ha). In this experiment, the lowest weed growth *i.e.* 12.3 g dm/m² was reported under I_3M_2 followed by I_2M_2 (Fig. 19). The maximum water use efficiency of 5.47 and 5.25 q/ha/cm water was registered under I_1M_1 and I_2M_1, respectively.

Project 3.8: Performance of vegetable crops under subsurface drip irrigation system

Experiments were conducted to study the performance of vegetable under varying levels of water application through sub-surface drip irrigation (SDI). SDI laterals (16 mm diameter) were placed at 10 cm depths below soil surface. Varying level of water applied to the crop was to the tune of 50% crop evapotraspiration (ET), 60% ET, 80% ET and 100% ET through SDI and 100% ET through surface drip and control furrow irrigation. Kashi Aman variety of tomato was transplanted in the experiment with plant to plant spacing of 50 cm under SDI, surface drip and control furrow irrigation (Fig.20).

Fig. 20: Tomato under sub-surface drip irrigation with 80% and 100% ET

The yield of tomato under experiment was found maximum 50.4 t/ha with 100 % ET, that was 54.1% higher than the yield under control (Fig.21). Tomato yield with SDI at 60% ET, 80% ET and surface drip irrigation were respectively, 13.54, 51.68 and 48.37% higher than control. Tomato yield reduced by 7.7% at 50% ET than control. Irrigation water depth applied was 26.8 cm at 100% ET. The Water Use Efficiency (WUE) enhanced 1.7-2.45 times of control /furrow irrigation that was maximum 2.316 t/ha-cm for SDI at 80% ET.

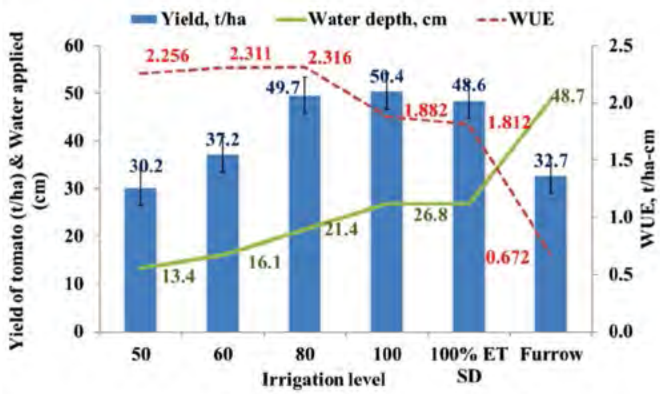


Fig. 21: Yield and WUE of tomato under varying levels of water application

The quadratic best fit regression equation (R^2=0.995) between irrigation level and yield indicated maximized yield of 50.76 t/ha at 95.65% ET (Table 22). Water depth and yield relation (R^2=0.992) resulted in maximized yield of 51.37 t/ha with 25.25 cm depth of applied water. Maximized WUE of 2.581 t/ha-cm can be obtained at 19.75 cm depth of irrigation water using water depth and WUE regression (R^2=0.979) equation.

Table 22: Maximized values of ET, water depth and WUE and yield

Parameters	Best fit regression equation	R^2	Maximized parameters	
1. ET & Yield (Y)	$Y = -0.01\,ET^2 + 1.913ET - 40.72$	0.995	ET_{max} = 95.65%	Y_{max} = 50.76 t
2. Water depth (d) & Yield (Y)	$Y = -0.154\,d^2 + 7.779\,d - 46.86$	0.992	d_{max} = 25.25 cm	Y_{max} = 51.37 t
3. Water depth (d) & WUE	$WUE = -0.006d^2 + 0.237d + 0.241$	0.979	d_{max} = 19.75 cm	WUE_{max} = 2.581 t/ha-cm

Weed management in vegetables

Effect of post and pre-emergence herbicides on pod yield of French bean: In India, French bean is traditionally a crop of temperate regions cultivated mainly for dry pods (rajmash). Though, it is a legume crop, it does not nodulate in roots either with native rhizobia or commercially produced cultures. Thus, it requires higher dose of nitrogen. Due to high moisture and nutrients in rajmash field, weeds become a problem, thus timely weed control is necessary to exploit the yield potential. During its early growth stage, weed competes with it leading to severe competition. Since, initial growth of rajmash is very slow, the initial period of growth (30-45 DAS) is most crucial for crop-weed competition. In addition to slow initial crop growth, wider crop spacing also facilitates crop-weed competition which poses a serious limitation in rajmash production and thus, estimated yield loss may likely to go to the extent of 45-65% under unweeded condition. During winter season, dominance of broad-leaved weeds in the early stages of crop growth period is mainly due to their fast growth and deep root system, which enables them to easily tap soil moisture and nutrients. Manual and mechanical methods of weed control are quite effective, but they are non remunerative and time consuming. Thus, chemical weed control becomes a promising option to control the weeds during crop growth period. Therefore, the present investigation was undertaken with newly introduced low volume and highly effective molecules for development of proper weed control schedule in rajmash.

The perusal of the data revealed that highest pod yield attained with weed free (13.5 t/ha) was *at par* with application of combination of pre and post herbicides of pendimethalin (pre-emergence) *fb* sodium acifluorfen 16.5 % + clodinafop- propargyl 8 % EC (post emergence) at 25 DAS (12.4 t/ha), pendimethalin @ 750g/ha (pre-emergence) *fb* imazethapyr @ 100 g/ha (post emergence) at 25 DAS (11.9 t/ha) (Table 23 and Fig. 22). The post emergence herbicides were selective

Table 23: Effect of different weed management treatments on weed dry weight, WCE and WI and pod yield in French bean

Treatments	Weed dry weight (g/m^2)	WCI	WI	Pods/ plant	Green pod yield (t/ha)
Sodium acifluorfen + clodinafop	4.23	94.3	11.3	24.0	10.5
Pendimethalin fb imazethapyr	3.69	95.0	9.1	27.3	11.9
Pendimethalin fb clodinafop-propargyl	28.4	61.5	47.3	14.3	6.23
Imazethapyr	6.24	91.5	13.1	23.0	10.2
Pendimethalin fb quizalofop-p-ethyl	24.6	66.7	48.9	13.0	6.94
Pendimethalin fb Sodium acifluorfen + clodinafop-propargyl	2.45	96.7	4.0	29.2	12.4
Weed free	0	100.0	0.0	31.4	13.5
Weedy Check	73.8	0.0	76.2	10.2	4.16
LSD at 5 %	3.26			1.7	1.97

Fig. 22 (a) : Lay out of Experimental plot

(b): Lay out of Experimental plot

(c): Crop under weedy check

(d): Crop under weed free traetment

and the harmful effects on crop were not observed. The results showed that efficient weed management of weeds in frenchbean can be achieved with single application of post emergence herbicide

Pre and post emergence herbicides for weed management in cowpea: Cowpea (*Vigna unguiculata* (L.)) cultivated around the world primarily for seed, but also as a vegetable (for leafy greens, green pods, fresh shelled green peas, and shelled dried peas), as cover crop and for fodder. During rainy season the crop suffers severely due to weed infestation resulting into wide range reduction in crop yield. The critical period of crop weed competition in cowpea has been identified as 20-30 days after sowing and presence of weeds beyond this period causes severe reduction in yields. Hence, weed control needs to be undertaken during initial period of crop growth. Though the hand weeding is a well proven effective method of weed control, but non-availability of labour and the cost incurred in it is very high. Keeping in view the fact, the present experiment was conducted to find out suitable and cost effective weed management practice to manage weeds during the critical period of crop weed competition.

Weed management practices significantly reduced the weed population. Significantly lower population of grass and broad leaf weed were recorded with the application pendimethalin *fb* imazethapyr, imazethapyr + imazemox. Highest pod yield was recorded with weed free (13.8 t/ha) *fb* application pendimethalin *fb* imazethapyr (12.3 t/ha) and imazethapyr + imazemox (12.2 t/ha) (Table 24 and Fig. 23).

Fig. 23(a): Cowpea under weedy check

(b): Cowpea treated with Pendimethalin + imazethapyr

Table 24: Effect of different weed management treatments on weed density and yield of cowpea

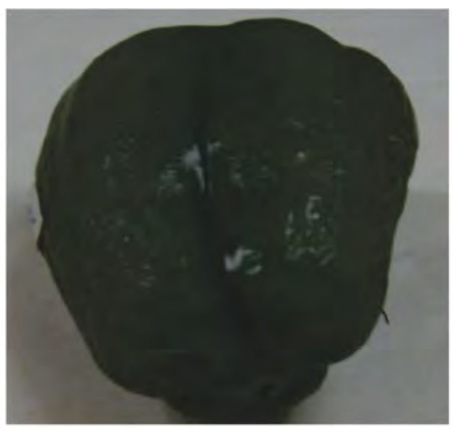
Treatment	Weed population (no/m^2)			Dry wt of weed (g/m^2)	WCI (%)	WI (%)	Yield (t/ha)
	Grasses	Broad leaf	Sedge				
imazethapyr+ imazemox	2.12^C (4.0)	4.80^{CD}(19.9)	4.01 (16.5)	15.6	84.4	7.80	12.2
Pendimethalin *fb* quizalofop	2.20^C(4.3)	5.05^{CD}(22.6)	4.22 (17.9)	16.8	83.2	14.18	11.3
Pendimethalin *fb* clodinafop	2.88^B(7.8)	5.15^C(26.1)	4.35 (18.5)	24.0	74.2	18.02	10.8
Clodinafop	2.93^B(8.1)	6.45^B(41.1)	4.68 (21.5)	52.8	51.5	24.73	9.9
Pendimethalin	2.48^{BC}(5.6)	5.09^{CD}(23.6)	4.28(18.04)	17.6	77.9	14.95	11.2
Quizalofop	2.90^B(7.9)	6.67^{AB}(44.1)	4.63 (21.0)	49.8	54.2	24.07	10.0
Pendi fb imazethapyr	2.08^C(3.8)	4.69^D(21.5)	3.95 (15.8)	14.6	86.5	6.48	12.3
Weedy check	3.94^A (15.0)	7.06^A (49.4)	4.85 (23.2)	108.9	0	39.78	7.9
2 HW	2.02^C(3.6)	4.08^E(16.2)	3.49 (15.2)	13.4	87.7	6.37	12.4
weed free	0.71^D(0)	0.71^F(0)	0.71(0)	0	100	0.00	13.8
LSD at 5%	0.57	0.434	NS	6.16			2.2

MEGA PROGRAMME 4: POST HARVEST MANAGEMENT AND VALUE ADDITION

Project 4.1: Shelf life extension of fresh vegetables

Shelf life extension of capsicum under MAP storage: Capsicum is a rich source of vitamin C, vitamin A and antioxidants such as ß- carotene, lutein, zea xanthin and cryptoxanthin along with rich sources of minerals such as copper, zinc, potassium, magnesium and iron. The shelf life of capsicum is not more than 7-8 days under refrigerated storage condition. However, various other means also extend the shelf life but there has been significant decrease in quality attributes. The shelf life of capsicum can be significantly increased in expanded polyethylene biopolymer under modified atmospheric storage. Modified atmospheric storage is maintained within 3-4 days of packaging at low temperature of 3°C. Freshly harvested capsicum is packaged in small and big size polymer with varying oxygen transmission rate for maintaining desired MAP (Fig. 24). Minimum decrease in flavour score (8.5-6.5) was obtained in small size capsicum after 35 days of storage at 3°C. Similarly colour and appearance score decreased from 8.5-8.0 and from 8.5-5.67 in small and big size capsicum, respectively after 49 days of MAP storage at 3°C. During initial storage of 5-6 days, the oxygen content decreased to 18.8% while carbon dioxide content decreased to 2.0-2,2% in small and big size capsicum at 3°C. The firmness value decreased from 4.71-2.28N and from 4.71-2.06N after storage of small and big size capsicum, respectively at 3°C after 49 days. Total sugar in capsicum under MAP storage decreased from 504.64-42.84 and 504.64-44.12 mg/100 g, dm, in small and big size capsicum respectively after 49 days of storage at 3°C. The decrease in ascorbic acid in capsicum was 74.83% and 70.35% in small and big size capsicum, respectively. Similarly, small and big size capsicum stored at 3°C resulted in significant decrease in total antioxidant activity from 109.85-24.44 and 109.85-26.25 µM TEAC/g,dm, respectively after 49 days of storage.

Fig. 24(a): Freshly harvested capsicum at 0 day of storage

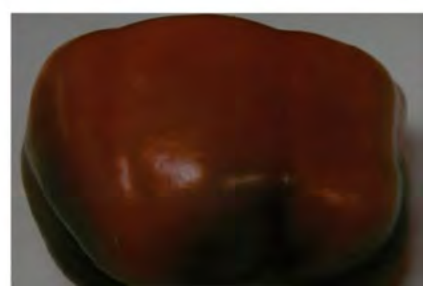


(b): 3°C Fully control capsicum

(c): 10°C Fully control capsicum

(d): 3°C Small package capsicum

(e): 3°C Big package capsicum under MAP storage



(f): 10°C Small and Big package capsicum under MAP storage

MEGA PROGRAMME 5: PRIORITIZATION OF R&D NEEDS AND IMPACT ANALYSIS OF TECHNOLOGIES DEVELOPED BY ICAR-IIVR

Project 5.1 Research Prioritization for Vegetable Crops

In this project future research priorities have been identified through surveying vegetable experts from ICAR institutions, SAUs and KVKs and vegetable farmers of the country. Several research priorities have been rank ordered as per the perception of the experts in the different field of vegetable research like vegetable germplasm collection, production technology, protection technology, seed production, post-harvest management and entrepreneurship development in vegetable enterprise. In the last phase of the project a survey was conducted with the vegetable farmers to analyze the field level constraints they face in day to day operation and formulating research priorities for grass root problem solving.

Table 25: Opinion of farmers on constraints faced by them in vegetable farming

Constraints	Per cent sample respondents		
	Very important	Important	Not important
Timely availability and adequate quantity of quality seeds/variety	66.67	29.41	3.92
Timely availability and adequate quantity of farm machinery / implement	30.39	57.84	11.76
Timely availability and adequate quantity of quality pesticides	51.96	41.18	6.86
Difficult processing/ post-harvest management	22.55	58.82	18.63
Lack of access to credit	67.65	29.41	2.94
Less price for the produce	61.76	33.33	4.90
Conversion of land	59.80	35.29	4.90
Labour shortage	41.18	57.84	0.98
Migration	50.98	35.29	13.73
Irrigation	74.51	19.61	5.88
Drainage	50.00	45.10	4.90
Electricity	57.84	33.33	8.82
Road	50.00	44.12	5.88
Lack of storage facilities	54.90	40.20	4.90
Natural disaster	46.08	43.14	10.78
Soil erosion	42.16	50.98	6.86
Water quantity and quality	50.98	41.18	7.84
Crop loss due to biotic and abiotic factors	57.84	36.27	5.88
Lack of advisory services/ trainings	35.29	59.80	4.90
Marketing- exploitation by middlemen	38.24	51.96	9.80

According to 66.67% vegetable farmers non-availability of quality seed in adequate quantity in proper time is the most important problem. According to 30.39% farmers vegetable cultivation is adversely affected due to non-availability of proper farm implements suitable for small scale farming. Vegetable growers face severe problem in case of pesticide and 51.96% farmers told that they did not get proper recommendation of pesticides for application in their vegetable crops. Likewise, there are several other problems faced by vegetable growers which have been described in the table which need to be addressed on priority basis (Table 25).

Project 5.2 Impact of Improved Vegetable Technologies Developed by ICAR-IIVR

Improving on-farm technology is one of the solutions identified to increase vegetable productivity and the income of vegetable farmers in Eastern Uttar Pradesh. At present, farmers are encountering problems of pests and diseases and declining soil productivity due to continuous cultivation. The most important factors influencing the adoption of improved vegetable varieties developed by ICAR-IIVR were high yields and high market demand. In addition, resistance to pest and diseases, the early maturing characteristics of the variety and high demand were identified as important adoption factors for okra, pea, tomato and cowpea. Considering these facts yield performance of major varieties developed in different vegetables were assessed at farmers' field in different villages of Eastern Uttar Pradesh and results are reported as follows:

Okra (Kashi Kranti) and cowpea (Kashi Nidhi) varieties were assessed for yield performance in 22 villages of Varanasi, Mirzapur and Sonbhadra districts in Uttar Pradesh fetched an average yield of 14.6 t/ha and 12.8 t/ha respectively. Tomato cv Kashi Aman developed by institute was assessed for yield and fruit quality in an area of 51.24 ha in 21 villages of Varanasi, Mirzapur and Sonbhadra districts in Uttar Pradesh fetched an average yield of 41.2 t/ha with distant marketing quality. Pea varieties – Kashi Udai and Kashi Nandini were assessed for yield performance in 24 villages of Varanasi, Mirzapur and Sonbhadra districts in Uttar Pradesh fetched an average yield of 10.8 t/ha. Carrot cv. Kashi Arun developed by the institute was assessed for yield performance in an area of 8.2 ha in 20 villages of Varanasi, Mirzapur and Sonbhadra districts in Uttar Pradesh which fetched an average yield of 11.8 t/ha. Bottle gourd (Kashi Ganga), sponge gourd (Kashi Divya) and pumpkin (Kashi Harit) varieties were assessed in an area of 42.02 ha, 20.54 ha and 26.04 ha respectively in Varanasi, Mirzapur and Sonbhadra districts in Uttar Pradesh fetched an average increase of 31.47% over local cultivar.

As the result of successful extension approaches, most of the varieties developed by institute in different vegetables put a positive impact on the area of adoption. Since, these popular varieties are open-pollinated therefore; growers also started multiplication of some varieties for self-sustainability in vegetable seeds. These trends of seeds production at growers' level was also reflected on the demands of seeds and area covered by the farmers with truthful level seeds produced by the institute (fig 25). It has been observed that in compare to 2016-17, during 2017-18 there was an increasing trend of area covered under institute's TL seeds at the farmers' field in Kashi Aman (49.7%) in tomato; Kashi Anmol (72.02%) in chilli; Kashi Udai (47.62%) in pea; Kashi Nandini (106.4%) in pea and Kashi Nidhi (129.3%). However in some other vegetable varieties decreasing trends of area covered with institute's TL seeds was observed which is mostly due to seed multiplication by the growers.

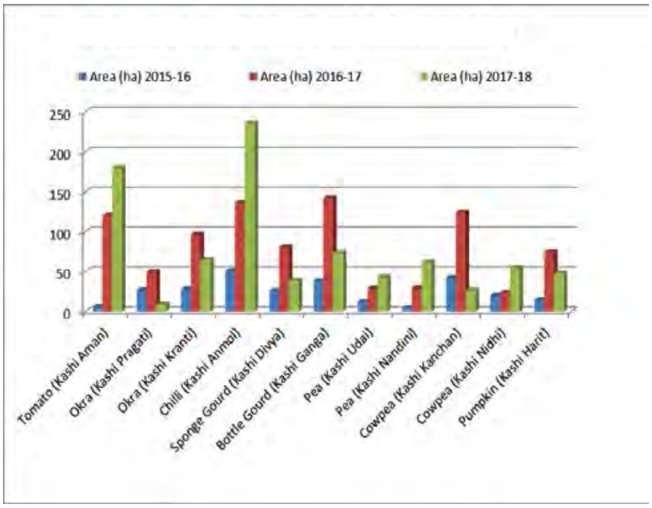


Fig. 25 : Area covered at farmers field by TL seeds produced by ICAR-IIVR




Division of Vegetable Protection

MEGA PROGRAMME 6: INTGERTED PLANT HEALTH MANEGEMENT

Project 6.1: Bio-Intensive Management of Major Insect Pests of Vegetables in the Current Scenario of Climate Change

Evaluation of different pest management modules in bottle gourd: Different pest management modules were evaluated against insect pests of bottle gourd (cv. Kashi Ganga). Among the tested modules, module 1 (M1) i.e. biointensive module comprising spray of Azadirachtin 0.03% @ 10 ml/l during 20 and 30 days after sowing (DAS); spraying of *Bacillus thuringiensis* @ 2 g/l during 40 DAS; spraying *of Lecanicillium lecanii* @ 5 g/l during 50 DAS; spraying of *Beauveria bassiana* @ 5 g/l during 60 DAS and spraying of Azadirachtin 0.03% @ 10 ml/l during 70 DAS was superior in terms of reducing the adult red pumpkin beetle. However, it was also at par with the chemical module (M3) consisting spray of Dichlorovos (DDVP) 76 EC @ 1 ml/l during 20 & 30 DAS; spraying of Chlorantraniliprole 18.5% SC @ 0.25 ml/l during 40 DAS; spraying of Imidacloprid17.8% SL @ 0.4 ml/l during 50 DAS; spraying of Thiacloprid 21.7% SC @ 1 ml/l during 60 DAS and spraying of Cyantraniliprole 10.26% OD @ 1.8 ml/l during 70 DAS as both the modules registered 62.93 per cent reduction over control (PROC). Similarly, maximum plume moth reduction (71.50) was recorded in biointensive module which was at par with chemical module. In case of mirid bug population on leaves as well as in fruits, biointensive module was superior to the other test modules however it was statistically at par with the chemical module (Table 1).

Project 6.2: Toxicological investigations on the novel insecticide molecules and plant origin insecticides against major insect pests of vegetables

Median lethal concentrations of different insecticides against *Spodoptera litura*: A laboratory study was carried out to determine the median lethal concentrations of different insecticides against *Spodoptera litura*. At 24 HAT, Indoxacarb (39 ppm), followed by Chlorantraniliprole (53 ppm) were found highly effective as compared to Emamectin benzoate (269 ppm), Quinalphos (524 ppm) and Deltamethrin (568 ppm). At 48 HAT, Chlorantraniliprole (1.9 ppm) was found most effective followed by Indoxacarb (2.9 ppm), Emamectin benzoate (26.4 ppm), Deltamethrin (336 ppm) and Quinalphos (523 ppm). Amongst all, the novel insecticides viz., Indoxacarb and Chlorantraniliprole proved most effective against *Spodoptera litura* (Table 2).

Table 1: Effect of different pest management modules against insect pests in bottle gourd

Treatments	Red pumpkin beetle (per 5 leaves / plant)			White plume moth (per 5 leaves / plant)			Mirid bugs					
							Per leaves			Per fruit		
	Before spray	After spray	PROC	Before spray	After spray	PROC	Before spray	After spray	PROC	Before spray	After spray	PROC
M1	5.57	1.72	62.93	4.29	1.22	71.50	7.09	1.71	66.70	3.42	1.79	51.88
M2	5.72	2.00	58.90	4.71	1.79	58.18	7.14	2.72	46.35	4.28	2.15	42.20
M3	5.86	1.72	62.93	4.86	1.29	69.86	5.22	2.14	57.79	3.29	1.86	50.00
Control	5.29	4.64	--	4.67	4.28	--	7.29	5.07	--	3.86	3.72	--
SEm (±)			1.07			1.33			1.22			1.39
CD (5%)			2.84			3.89			3.02			4.37

*PR = Percent Reduction

M1 : Spray of Azadirachtin 0.03% @ 10 ml/l during 20 and 30 days after sowing (DAS); spraying of *Bacillus thuringiensis* @ 2 g/l during 40 DAS; spraying *of Lecanicillium lecanii* @ 5 g/l during 50 DAS; spraying of *Beauveria bassiana* @ 5 g/l during 60 DAS and spraying of Azadirachtin 0.03% @ 10 ml/l during 70 DAS

M2 : Spray of Dichlorovos (DDVP) 76 EC @ 1ml/l at 20 & 30 DAS ; spraying of *Bacillus thuringiensis* @ 2 g/l during 40 DAS; spraying of Imidacloprid 17.8% SL@ 0.4 ml/l at 50 DAS; spraying *of Lecanicillium lecanii* @ 5 g/l during 60 DAS; spraying of Azadirachtin 0.03% @ 10 ml/l at 70 DAS.

M3 : spray of Dichlorovos (DDVP) 76 EC @ 1 ml/l during 20 & 30 DAS; spraying of Chlorantraniliprole 18.5% SC @ 0.25 ml/l during 40 DAS; spraying of Imidacloprid17.8% SL @ 0.4 ml/l during 50 DAS; spraying of Thiacloprid 21.7% SC @ 1 ml/l during 60 DAS and spraying of Cyantraniliprole 10.26% OD @ 1.8 ml/l during 70 DAS

Table 2: Bio-efficacy of some newer molecules against *Spodoptera litura*

Treatments	Heterogeneity		Regression equation (Y=)	Median lethal concentration (LC_{50}) (%)	Fiducial limit	LC_{50} (ppm)
	df	χ2				
24 HAT						
Chlorantraniliprole	4	9.456	0.578X + 6.316	0.0053	0.0214 – 0.0013	53
Indoxacarb	5	6.716	2.337X + 10.632	0.0039	0.0029 – 0.0052	39
Emamectin Benzoate	6	2.342	1.082X+6.699	0.0269	0.0415 – 0.0174	269
Deltamethrin	5	1.426	1.027X + 6.279	0.0568	0.1076 – 0.0299	568
Quinalphos	5	1.015	1.784X + 7.284	0.0524	0.0782 – 0.0035	524
48 HAT						
Chlorantraniliprole	4	0.923	0.966X + 8.580	0.00019	0.00035 – 0.00011	1.9
Indoxacarb	5	4.159	3.239X + 13.201	0.00294	0.00745 – 0.0023	2.9
Emamectin Benzoate	6	1.985	3.175X + 13.185	0.00264	0.00346 – 0.0020	26.4
Deltamethrin	5	1.775	1.135X + 6.674	0.03356	0.07277 – 0.0154	336
Quinalphos	5	5.679	2.976x + 8.815	0.05226	0.04113 – 0.0664	523

Determination of Median lethal concentrations of different insecticides against *Spilosoma obliqua*: A laboratory experiment was conducted to determine the Median lethal concentrations of novel insecticides along with few conventional insecticides against third instar larvae of *S. obliqua* by using direct spray method. On the basis of LC_{50} Indoxacarb (0.007) was found to be more effective followed by Chlorantraniliprole (0.009), Emamectin benzoate (0.01), Novaluron (0.012), Fenvelarate (0.019) and Quinalphos (0.025) at 24 HAT. At 48 HAT, Chlorantraniliprole (0.003) was found most effective followed by Emamectin benzoate (0.004), Indoxacarb (0.005) Novaluron (0.0051), Fenvelarate (0.0123) and Quinalphos (0.0143) on the basis of LC_{50} (Table 3).

Table 3: Bio-efficacy of some newer molecules an conventional insecticides against *Spilosoma obliqua*

Treatments	LC_{50} (%)	Fiducial limits	LC_{50} (ppm)
24 HAT			
Fenvelarate	0.01946	0.01382- 0.02743	194.6
Novaluron	0.01205	0.00915-0.01551	120.5
Quinolphos	0.02513	0.01770-0.03680	251.3
Chlorantraniliprole	0.00903	0.00701-0.01135	90.3
Emamectin Benzoate	0.01054	0.00571-0.01740	105.4
Indoxacarb	0.00749	0.00422-0.01156	74.9
48 HAT			
Fenvelarate	0.01234	0.00865-0.01691	123.4
Novaluron	0.00519	0.00371-0.00680	51.9
Quinolphos	0.01436	0.01007-0.01992	143.6
Chlorantraniliprole	0.00381	0.00261-0.00510	38.1
Emamectin Benzoate	0.00496	0.00352-0.00653	49.6
Indoxacarb	0.00501	0.00359-0.00655	50.1

Comparison of efficacy of different insecticides through different bioassay methods against *Spodoptera litura*: Insecticides belonging to different groups were tested by three bioassay methods viz., Leaf dip, direct spray and larval dip against third instar larvae of *S. litura.* A considerable variation was observed in the larval mortality in all the three bioassay methods for the insecticides viz., Indoxacarb, Chlorantraniliprole, Emamectin benzoate and Cypermethrin. *S. litura* larvae was highly susceptible to Indoxacarb in direct spray method, Chlorantraniliprole in leaf dip method and Cypermethrin in leaf dip method. For most of the insecticides, leaf dip method gave the maximum mortality and thus, it could be the most suitable method to determine the susceptibility of *S. litura* (Fig. 1).

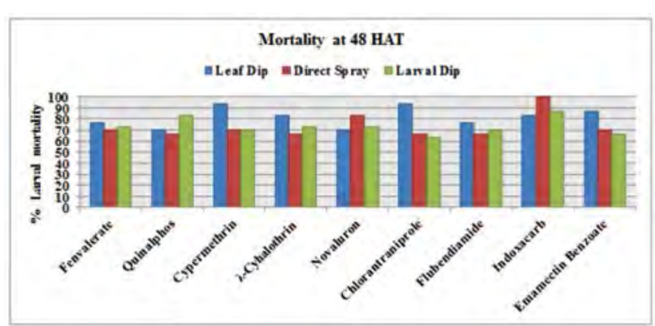


Fig. 1: Efficacy of insecticides through different bioassay methods against *Spodoptera litura*

Project 6.3: Biological Control of major Insect Pests of Vegetable crops

Occurrence of Water chestnut aphid, *Rhopalosiphum nymphaeae* (L.) (Aphididae: Homoptera): A polyphagous, heteroecious, holocyclic aphid *Rhopalosiphum nymphaeae* (Linnaeus, 1761) (Aphididae: Homoptera) was identified on feeding the water chestnut from Varanasi, Uttar Pradesh. In large colonies that develop on water chest nut aggregate along the leaf veins and infest leaves as well. Large colonies of the aphid infest young twigs, leaf petioles and fruit stalks of water chestnut causing curling of host plant leaves. They also secrets the copious amount of honey dew which deposits on the leaf surfaces and produces black sooty moulds which further reduces the photosynthesis of the plants. Apart from water chest nut, its incidence was also recorded on lotus and water lily (Fig. 2, 3 & 4).

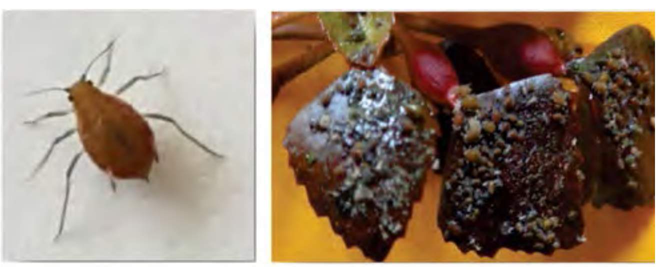


Fig. 2 & 3 : Water lily aphid, *Rhopalosiphum nymphaeae*

Fig. 4: Aggregation of nymphs and adults of aphid on leaves of water chest nut

Mirid bugs as an emerging threat to bottle gourd cultivation: Dynamics and bio- rational management: Apart from the occurrence of regular insect as pests, bottle gourd was infested by the serious incidence of mirid bugs. In the beginning, minute puncture spots with yellow hallow were observed on tender leaves. The damage was more prominent in young fruits. Brown puncture spots on the rind with sap oozing out from the tender fruits formed the characteristic symptoms of these sucking pests (Fig. 5). Several local farmers visiting the institute also reported the same problem. The affected fruits often failed to fetch a good market price. About 70-80 per cent fruits and 30 per cent shoots were damage by these bugs. The mirid bugs duo was identified as *Nesidiocoris cruentatus* (Ballard) (Fig. 6) and *Metacanthus pulchellus* Dallas (Fig. 7). Studies on species composition of duo mirid bugs showed that *N. cruentatus* was dominant species contributing overall 68.63% of the mirid bug population infesting bottle gourd followed by *M. pulchellus* (31.37%) (Fig. 8). *N. cruentatus* exhibits a strong diurnal activity as its incidence increased gradually during day time from 10 AM onwards, with a peak at 1 PM (3.88 bugs/ fruit) (Fig. 9).

Fig. 5: Bottle gourd fruit damaged by *N. cruentatus* and *M. pulchellus*; (inset) *N. cruentatus* adult

Fig. 6: Adult of *Nesidiocoris cruentatus* (Ballard).

Fig. 7: Adult of *Metacanthus pulchellus* Dallas, 1852

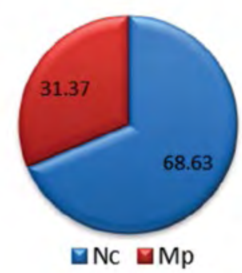


Fig. 8: Overall species composition of mirid bugs on bottle gourd

NC = *N. cruentatus* ; MC = *M. pulchellus*

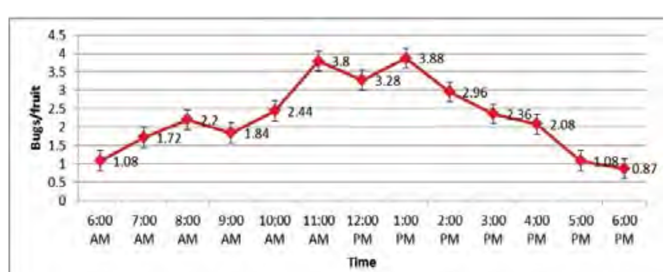


Fig. 9: Diurnal activity of mirid bug, *N. cruentatus* on bottle gourd

Amongst the different plant parts *viz*., apical buds, tender fruits, young leaves, male and female flowers kept in a six-armed olfactometer, the highest number of *N. cruentatus* (26.39%) were moved towards apical buds followed by young leaves (16.67%) and tender fruits (15.28%). The descending order of food preference was apical buds > young leaves > tender fruits > male flowers > female flowers. Similarly, in case of *M. pulchellus* the highest orientation (25.50%) was towards apical buds followed by young leaves (18.25%) and male flower (14.75%).

Comparison of the two popular systems of cultivations *viz*., raised bed and trailing systems indicated that trailing system of cultivation harbored more mirid bugs. Significantly the highest number of bugs per tender shoot (7.2) was recorded in the bottle gourd when grown in trailing system than the raised bed system (1.35). A similar trend was also observed for the young fruits. Higher number of bugs (5.7) per fruit was recorded in bottle gourd when grown in trailing system compared to the conventional raised bed system (3.45).

Amongst the biopesticides tested, neem oil (1%) was most promising with lowest median lethal time (50.31 h) followed by entomopathogenic fungi, *B. bassiana* (52.26 h) and *L. lecanii* (56.59 h) whereas Flonicamid 50 WG and Spiromesifen 22.9 SC were most promising chemicals under field and laboratory conditions.

Compatibility and synergism of major neonicotinoids with different entomopathogenic fungi (EPF) against *Myzus persicae* Sulz.: To control the polyphagous *Myzus persicae* feeding on cole crops, commonly used neonictinoids (Imidacloprid,

Table 4: Median lethal time of neonicotinoid insecticides and EPF alone and their 1:1 combinations against adults of *Myzus persicae*

Biopesticides	Heterogenity		Regression equation (Y=)	LT_{50} (hr)	Fiducial limit	Co-toxicity coefficient
	df	χ^2				
Beauveria bassiana	5	6.876	3.135X – 0.275	48.17	55.83 – 41.57	--
Metarhizium anisopliae	5	5.604	4.782X – 3.199	51.81	57.69 – 46.53	--
Lecanicillium lecanii	5	6.315	3.754X – 1.364	49.57	56.04 – 43.85	--
Imidacloprid 17.8% SL	5	0.641	4.539X – 1.532	27.49	30.97 – 24.37	--
Thiamethoxam 25% WG	5	0.139	3.773X – 0.508	28.84	33.02 – 25.19	--
Acetamiprid 20% SP	5	0.348	3.989X – 0.535	24.41	28.10 – 21.18	--
Beauveria bassiana + Imidacloprid (1:1)	4	0.925	4.186X – 0.878	25.35	28.99 – 22.18	1.09
Metarhizium anisopliae + Imidacloprid (1:1)	4	0.089	4.700X – 1.761	27.29	30.71 – 24.25	1.01
Lecanicillium lecanii + Imidacloprid (1:1)	4	0.572	4.470X – 1.175	24.07	27.48 – 21.08	1.14
Beauveria bassiana + Thiamethoxam (1:1)	5	4.809	5.794X – 3.169	25.71	28.45 – 23.23	1.12
Metarhizium anisopliae + Thiamethoxam (1:1)	5	3.801	3.798X – 0.545	28.48	32.72 – 25.43	1.01
Lecanicillium lecanii + Thiamethoxam (1:1)	4	0.147	3.993X – 0.340	21.75	25.49 – 18.55	1.33
Beauveria bassiana + Acetamiprid (1:1)	5	1.388	5.061X – 1.869	22.76	25.79 – 20.09	1.07
Metarhizium anisopliae + Acetamiprid (1:1)	5	1.826	3.667X – 5.038	23.85	27.93 – 20.37	1.02
Lecanicillium lecanii + Acetamiprid (1:1)	4	0.385	3.387X – 0.413	22.61	27.11 – 18.85	1.08

$$\text{Co-toxicity coefficient} = \frac{LT_{50}\text{ value of Neonicotinoid insecticide alone}}{LT_{50}\text{ values of insecticide and EPF mixtures}}$$

Thiamethoxam and Acetamiprid) and biopesticides *viz*., *Beauveria bassiana*, *Metarhizium anisopliae*, *Lecanicillium lecanii* were tested at half of their recommended doses and found compatible. Combination of Acetamiprid and *L. lecanii* took the lowest median lethal time (22.61 hour) with Co-toxicity coefficient (CTC) value (1.08). A similar observation was also noted in case of Thiamethoxam where *L. lecanii* when mixed with Thiamethoxam at half of recommended doses took the lowest median lethal time (21.75 hour) and with highest CTC value (1.33) (Table 4). Similar observation was also noted combination of Imidacloprid and three EPF. Co-application of these EPF with sub-lethal concentration of neonicotinoids could not only be a green eco-friendly option against this sucking pest but also able to minimize the chemical insecticides load in the environment.

Molecular characterization of indigenous entomopathogenic nematode strains of Uttar Pradesh: Soil samples were collected from research farm, ICAR-Indian Institute of Vegetable Research and ICAR-Central Agroforestry Research Institute, Jhansi, Uttar Pradesh and baited with last instar of wax moth (*G. mellonella*) larvae. Indigenous EPNs strains were isolated by using white trap method. Further identity of indigenous EPN strains was confirmed through molecular characterization by using the ITS-rDNA region. The sequence of EPN strains such as IIVR EPN03 (MG976754), IIVR JNC01 (MH208855) and IIVR JNC02 (MH208856) showed maximum identity with *Steinernema siamkayai* (99%) (Table 5).

Table 5: Molecular characterization of indigenous entomopathogenic nematode strains of Uttar Pradesh

Isolate code	Nematode species	NCBI Accession Number	Geogra-phical origin	Habitat
IIVR EPN03	*Steinernema siamkayai*	MG976754	IIVR Varanasi	*Alstonia scholaris*
IIVR JNC01	*Steinernema siamkayai*	MH208855	CAFRI Jhansi	Neem
IIVR JNC02	*Steinernema siamkayai*	MH208856	CAFRI Jhansi	Teak

Virulence assay of entomopathogenic nematodes against major insect pests of vegetable crops: A laboratory experiment was conducted to determine the pathogenicity of indigenous entomopathogenic nematode strain, *Steinernema siamkayai* IIVR JNC01(MH208855) against third instar larvae of *Spoladea recurvalis*, *Spodoptera litura* and *Spilosoma obliqua* and compared their efficacy with ICAR-NBAIR entomopathogenic strain *Heterorhabditis indica* NBAIIH38 (KX950751). The assay revealed that, *S. siamkayai* (MH208855) cause mortality between 30-85%, 60-100% and 62.5- 100% on third instar larvae of *S. recurvalis, S. litura* and *S. obliqua* respectively, while *H. indica* caused mortality between 40-100%, 50-100% and 65- 100% in *S. recurvalis, S. litura* and *S. obliqua* respectively at 25-200 IJ's concentration (Fig. 10, 11 & 12). Among the nematode species, irrespective of IJ concentration, *H. indica* caused significantly ($P < 0.05$) greater mortality only in *Spoladea recurvalis* compared to *S. siamkayai*, however, there was no significant difference in mortality caused by both EPN species on third instar larvae of *S. litura* and *S. obliqua*. The calculated LC_{50} and LC_{90} values for *S. siamkayai* and *H. indica* on *S. recurvalis, S. litura* and *S. obliqua* are shown in table 6.

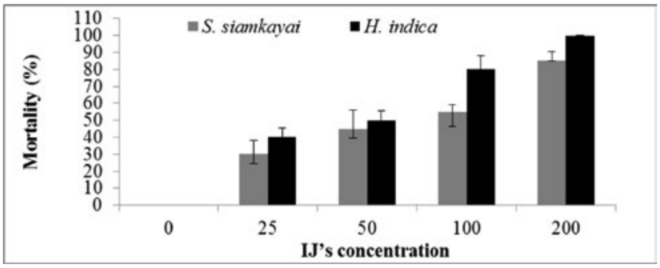


Fig. 10: Percent mortality of third instars larvae of *Spoladea recurvalis* at different infective juvenile concentrations of entomopathogenic nematodes, *Steinernema siamkayai* and *Heterorhabditis indica* at 48 HAT.

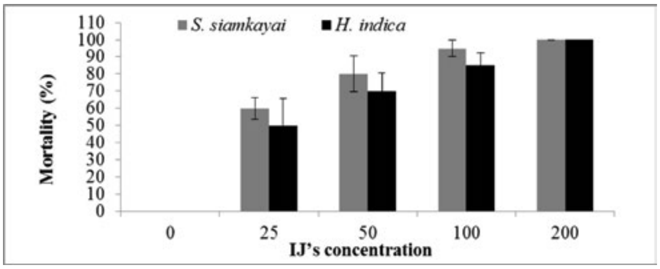


Fig. 11: Percent mortality of third instars larvae of *Spodoptera litura* at different infective juvenile concentrations of entomopathogenic nematodes, *Steinernema siamkayai* and *Heterorhabditis indica* at 48 HAT.

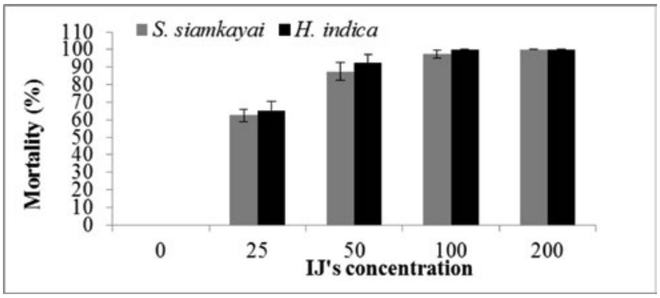


Fig. 12: Percent mortality of third instars larvae of *Spilosoma obliqua* at different infective juvenile concentrations of entomopathogenic nematodes, *Steinernema siamkayai* and *Heterorhabditis indica* at 48 HAT.

Table 6: The lethal concentration (LC_{50} and LC_{90}) of *Steinernema siamkayai* IIVR JNC01 (MH208855) and *Heterorhabditis indica* NBAIIH38 (KX950751) against major vegetable insect pests at 48 HAT

Insect pests	Entomopathogenic nematode species	LC_{50}	95% Fiducial limits	LC_{90}	95% Fiducial limits	Slope ± SE	X^2	*P* (<0.05)
Spoladea recurvalis (Lepidoptera: Crambidae)	*Steinernema siamkayai* IIVR JNC01 (MH208855)	60	33-95	385	188-3993	1.58 ± 0.45	12.07	0.0005
	Heterorhabditis indica NBAIIH38 (KX950751)	39	24-53	133	90-319	2.40 ± 0.56	18.26	<0.0001
Spodoptera litura (Lepidoptera: Noctuidae)	*Steinernema siamkayai* IIVR JNC01 (MH208855)	20	6-30	68	48-166	2.46 ± 0.75	10.80	0.0010
	Heterorhabditis indica NBAIIH38 (KX950751)	27	11-37	103	70-275	2.19 ± 0.59	13.49	0.0002
Spilosoma obliqua (Lepidoptera: Erebidae)	*Steinernema siamkayai* IIVR JNC01 (MH208855)	19	10-26	55	43-84	2.83 ± 0.65	18.49	<0.0001
	Heterorhabditis indica NBAIIH38 (KX950751)	20	11-25	43	35-65	3.83 ± 1.00	14.73	0.0001

Project 6.4: Management of important fungal diseases of vegetable crops

For the management of fungal diseases in tomato cv. Kashi Aman a experiment was conducted under field condition with seven management packages (treatments). The treatment details as follows.

T1-Chemical module: Seed treatment by carbendazim @ 0.2% and nursery drenching of pencycuron @ 5 l/m² of 0.1%; seedling drenching by fosetyl-Al @ 0.1% after 15 days of sowing; one spray of streptocycline @ 150 ppm on seedling after 20 days of sowing; seedling root dip in imidacloprid @ 0.04% for 30 minutes just before transplanting; one spray of copper oxychloride @ 0.3% after 25 DAT.

T2-Biological module: Seed treatment by *Trichoderma* sp. (BATF-43-1) @ 1%; Nursery application of talc based *Trichoderm*a sp. (BATF-43-1) @ 25 g/m²; seedling root dip in slurry BATF-43-1 10 g + 100 g FYM/ vermicompost + 250ml water; drenching by BATF-43-1 @ 1% thrice at 25 days interval started 25 DAT.

T3-Good Agricultural Practices (GAP) module: Soil application with neem cake @ 100 g/m² 10 days before sowing; seed soaking in cow urine for 60 minutes; nursery bed covering by 40 mesh nylon net; seedling root dipping in cow dung slurry; spot use of vermicompost @ 50g/plant thrice at 25 days intervall; foliar spray of micronutrient @ 0.2% twice at 25 days interval; Neem oil (Azatarachtin 0.03%) sprays @ 0.3% twice at 20 days interval.

T4-Integrated module: Seed treatment by *Trichoderma* sp. BATF-43-1**;** seedling drenching of talc based BATF-43-1@ 1% after 15 days after sowing; spray of streptocycline @ 150 ppm on seedling 20 days after sowing; seedling root dip in Imidacloprid @ 0.04% for 30 minutes followed by BATF-43-1 @ 1% for 10 minutes; spot application of (BATF-43-1) 10 g + vermicompost 50g/plant thrice at 25 days interval started 25 DAT; one spray of copper oxychloride @0.3% after 30 days of transplanting.

T5-Research gap module: Seed treatment by *T. asperllum* @ 0.5%+ *Bacillus subtilis* @ 0.5% -5a; seed treatment 0.25% and soil drenching by captan @ 5 l/m² of 0.25% -5b; seedlings dip by Imidacloprid @ 0.03% + *T. asperllum* @ 1%; drenching of CRB7 (*Bacillus subtilis* as antagonistic to *S. rolfsii*, *M. phaseolina* and RKN as well as IAA producer) @ 1% +TRB17 (*Stenotrophomonas)* as antagonistic to *F. oxysporum*) @ 1% thrice at 25 days interval started 25 DAT; foliar spray of BS2 @ 1% thrice at 25 days interval after 25 DAT.

T6-Control: Solarized nursery only.

T7-Botanical: Foliar spray thrice at 25 day interval @ 0.4%

Effect of modules on seed and nursery: *In vitro* seed germination after 7 days of plating varied from minimum 72% in control to maximum 90% in research gap biological module-T5 i.e. seed treatment by *T. asperllum* @ 0.5% + *Bacillus subtilis* BS2 @ 0.5%. There was no seed mycoflora appeared in any of the incubated seed till 10th days in T3-GAP, T4-IDM and T5a-reserearch gap biological module. After 25 days of seed sowing observation revealed that maximum seedling stand was 80% in T4-IDM (Table 7). Weed population mostly *Echinochloa* spp. was found 724/sq. feet in control bed while 12/sq.ft. in solarized bed after 14 days of polythene removal (Fig.13).

Fig. 13: Solarization effect on weed

Fig. 14: Plant status in T4 after 6th harvest

Fig. 15: Proportion in early blight fruits 15th harvest

Table 7: Effect of different IDM* package on diseases management of tomato nursery

Treatments	*In vitro* seed germination (%) 7 DAP	Seed mycoflora 18 DAP	Total seedling length (cm) 8 DAP	Seedling stand (%) 22 DAS
T1- Chemical module	85	+++	10.3	73.83
T2-Biological module	80	+++	10.73	57.83
T3- GAP module	84	-	11.0	58.0
T4-Integrated module	87	-	11.65	80.0
T5a- Research gap- biological	90	-	9.38	79.0
T5b- Research gap- chemical	80	++	12.1	69.3
T6- Control	72	++++	10.9	65.2

*Soil solarization of nursery bed was common in all the treatment modules.

Effect of modules on yield and fruit rots: Maximum marketable yield was recorded in T4-IDM (Fig. 14) is 80.74 t/ha followed by 78.63 t/ha in T1-chemical in comparison to control (65.53 t/ha). The unmarketable yield varied from 19.52 t/ha to 27.58 t/ha in different treatment modules which was about 24% of total yield. The unmarketable yield comprises of about 60% bird damage, 30% insect damage and 10% diseased. Minimum diseased fruits 3.7% were recorded in T3-GAP while maximum 9.9% in control. Among diseased fruits 85.9% to 100% was infected only by early blight with minimum in T1 chemical module. The other diseased fruits were identified as *Rhizoctonia solani, Alternaria alternata*, *Phythium apahindermatum*, *Myrothecium rorridum*, *Sclerotium rolfsii Geotrichum candidum*, *Phytophthora nicotianeae*, *Colletotrichum capsici*, *Xanthomonoas campestris* pv. *vesicatoria* and *Erwinia carotovora* pv. *carotovora*. The fruit rot by *Rhizoctonia solani* was recorded from first harvest to ninth harvest as second major pathogen after *Alternaria solani*. Early blight infected fruits were only recorded during 13th to 16th harvest in the month of March-April (Fig. 15).

Virus incidence was recorded in all the treatments with minimum 10.0% in T7-Botanical module and maximum 27.4% in control. Root knot nematode incidence reveled that minimum 13.2 galls /plant in T7 botanical module and maximum 32.4 galls/plant in T3. GAP module (Table 8).

Associated microorganisms in NADEP and vermicompost of IIVR: Two organic supplements i.e. NADEP and vermicompost were analyzed for presence of fungal populations. The fresh samples were collected in the first week of September and plated on Peptone dextrose Rose Bengal Agar medium at serial dilution 1:1000 and incubated for 10 days. All the colonies were purified by hyphal tip method on potato dextrose agar plate and microscopic observation were made to identify the pathogen (Fig. 16). The total fungal colony was more in vermicompost i.e. 14.33 x10^3 cfu/g of compost as compared to NADEP (11.5x10^3 cfu/g. Similarly pathogenic colony was more in vermicompost than NADEP (Table 9). Population of *Phythium* sp. was four times more 4.0 x 10^3 cfu/g in vermicompost than NADEP 1.0 x 10^3 cfu/g.

Incidence of Sclerotinia rot *sclerotiorum* on French bean and brinjal little leaf in organic block Observations on Sclerotinia rot in French bean revealed minimum disease incidence 20.0 % in treatment of 100% RDF inorganic fertilizer to all crops followed by 25.3% in 75% N NADEP + 25% neem cake + 25% N

Table 8: Effect of different IDM* package on disease management of tomato crop in main field.

Treatments	Marketable yield (t/ha)	Unmarketable yield (t/ha)	Diseased fruit yield (t/ha)	Early blighted fruits (t/ha)	Diseased fruits % of total un-marketable	Early blighted % of diseases	Virus infected plant/plot	RKN galls/ plant
T1-Chemical module	78.63	20.52	1.373	1.18	6.7	85.9	6.3	21.43
T2-Biological module	68.58	22.96	1.512	1.457	6.6	96.4	8.7	17.23
T3- GAP module	75.67	24.57	1.912	1.83	3.7	95.7	10.0	32.43
T4-Integrated module	80.74	27.58	2.48	2.26	9.0	91.3	8.7	21.97
T5-Research gap module	67.73	21.36	1.46	1.46	6.8	100	10.7	23.57
T6- Control	65.53	19.52	1.95	1.84	9.9	94.3	13.7	18.6
T7-Botanical Module	75.243	24.19	2.285	2.05	9.4	89.7	5.0	13.23

*Green manuring was common in all the treatment modules.

Table 9: Microorganisms in organic matter of IIVR

Microorganisms	NADEP (cfu/g)	Vermicompost (cfu/g)
Fusarium oxysporum	3.0×10^3	3.0×10^3
Aspergillus niger	1.5×10^3	2.0×10^3
A. flavus	1.0×10^3	0.5×10^3
A. ochraceous	1.66×10^3	2.5×10^3
Phyhtium sp.	1.0×10^3	4.0×10^3
Eurotium sp.	1.33×10^3	2.33×10^3
Pestalotia sp.	1.0×10^3	-
Rhizopus sp.	1.0×10^3	-
Total fungi	$\mathbf{11.5 \times 10^3}$	$\mathbf{14.33 \times 10^3}$
Pathogenic	$\mathbf{5.0 \times 10^3}$	$\mathbf{7.0 \times 10^3}$
Non-pathogenic	6.5×10^3	7.33×10^3

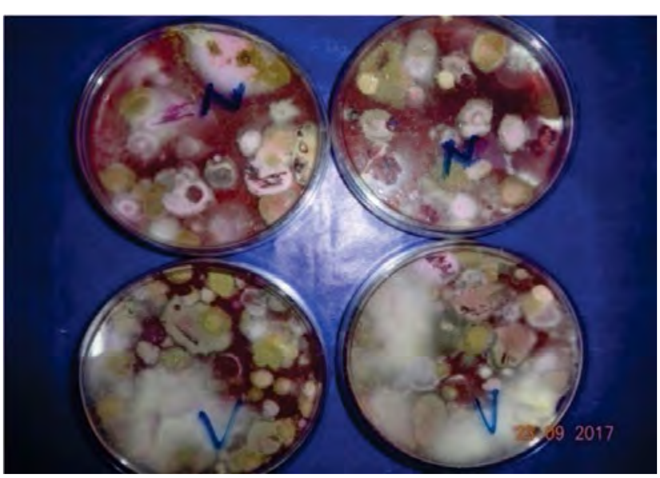


Fig. 16: Total fungi on PDRBA

Table 10: Incidence of *Sclerotinia sclerotiorum* on French bean and brinjal little leaf in organic block

Treatment details	White rot incidence (%)	Little leaf incidence (%)
100% RDF inorganic fertilizer to all crops	20.0	7.71
Control	29.4	8.4
100% N FYM + 25% N FYM to other crops	78.3	6.46
75% N FYM + 25% neem cake + 25% N FYM to other crops	81.7	5.53
100% N FYM	76.7	10.35
100% N VC	43.3	6.53
100N VC+25% N VC to other crops	46.7	4.56
75% N VC + 25% neem cake + 25% N VC to other crops	35.0	-
100% N NADEP + 25% N NADEP to other crops	28.3	-
75% N NADEP + 25% neem cake + 25% N NADEP to other crops	25.3	0
100% N NADEP	27.5	5.63
50%N NADEP + 50% N VC	55.0	17.92
50%N FYM + 50% N VC	30.0	6.89
50%N FYM + 50% N NADEP	42.7	5.29

NADEP to other crops. Maximum 81.7% Sclerotinia rot was observed in plot having 75% N FYM + 25% neem cake + 25% N FYM to other crops followed by 100% N FYM + 25% N FYM to other crops (Table 10). There was no little leaf incidence in brinjal where 75% N NADEP + 25% neem cake + 25% N NADEP to other crops applied while maximum 17.9% little leaf incidence was recorded in 50%N NADEP + 50% N vermicompost plot .

Project 6.5: Bioprospecting of microorganisms associated with vegetables against plant pathogens

Talcum based formulation of microbial bio-agents 1% WP viz. *P. fluorescens, Pb-3, Isaria farinosa, Serratia marcesens, Trichoderma asperellum, Stenotrophomonas maltophila, Alcaligenes* sp. and standard seed treating fungicide carbendazim 50% WP were evaluated as seed treatment and 3 subsequent foliar spraying at 15 days interval after 30 days of sowing in field on bottle gourd (cv. Kashi Ganga) to assess diseases incidence, polyphenol activities and yield. Among tested microbial bio control agents minimum Percent Diseaase Index (PDI) of cercospora leaf spot (47.77) and highest yield (55.43 t/ha) were recorded with seed treatment @ 10 g/kg and foliar spray @ 1% *T. asperellum* (T6) in compared to control. However highest peroxidase (PO) activity (2.01 g/min) was recorded in T1- carbendazim 50% WP after 75 days of sowing (Table11 and Fig. 17).

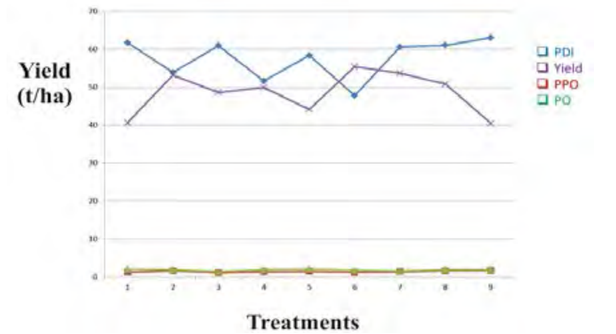


Fig. 17: Effect of treatments on Cercospora leaf spot, PO, PPO activities and yield

Determination of colony forming unit (cfu) in talcum formulation of microbial bioagents: The colony forming units (cfu) of bio agents were determined by serial dilution method on potato dextrose agar medium. Quantification of cfu/g in talcum based formulation was done by followed a standard formula: Viable cells/g of dry soil = Mean plate count × (1/ dilution factor) × (1/ Weight of sample). Quantified cfu data given in table 12.

Standardization of isolation protocol for *Actinomycetes* and their *in vitro* bio control efficacy against plant pathogens: Protocol for the isolation

Table 11: Effectiveness of bio-formulations on plant pathogens and yield in bottle gourd

Treatments	Cercospora leaf spot (PDI)	Poly phenol oxidase activity (75 DAS)	Peroxidase activity (75 DAS)	Yield (t/ha)
T1 - Seed treatment with carbendazim 50 WP@ 2g/ kg and spray* @ 0.1%	61.72	1.30	2.01	40.67
T2 - Seed treatment with *P. fluorescens* @ 10g/kg seed and spray @ 1%	53.90	1.63	1.85	53.16
T3- Seed treatment with *Pb-3* @ 10g/ kg seed and spray @ 1%	60.99	1.13	1.44	48.65
T4 - Seed treatment with *Isaria farinosa* @ 10g/kg seed and spray @ 1%	51.66	1.40	1.83	49.93
T5 - Seed treatment with *Serratia marcesens* @ 10g/kg seed and spray @ 1%	58.39	1.49	2.11	44.16
T6 - Seed treatment with *Trichoderma asperellum* @ 10g/kg seed and spray @ 1%	47.77	1.21	1.76	55.43
T7 - Seed treatment with *Stenotrophomonas maltophila* @ 10g/kg seed and spray @ 1%	60.63	1.36	1.55	53.70
T8 - Seed treatment with *Alcaligenes* sp. @ 10g/kg seed and spray @ 1%	61.05	1.63	1.89	50.83
T9 - Control	63.05	1.69	1.87	40.44
SE (M)	-	-	-	0.647
CD (5%)	-	-	-	1.956
CV	-	-	-	2.308

*No. of foliar sprays-3, first spray at 30 DAS and subsequent sprays at 15 days interval

Table 12: Quantification of cfu/g in formulation of microbial bio-agents

Microbial bio-agents	cfu/g
Trichoderma asperellum	2.0×10^{7}
Isaria farinosa	1.0×10^{8}
T. Viride (BATF-43-1)	4.5×10^{7}
Pseudomonas *fluorescens*	2.0×10^{8}
Pb-3	1.5×10^{8}
Serratia marcesens	2.5×10^{9}
Stenotrophomonas maltophila	2.6×10^{8}
Alcaligenes sp.	1.5×10^{9}
Bacillus subtilis (BS 2)	3.5×10^{11}

of *Actinomycetes* spp. from Vermicompost and NADEP samples obtained from IIVR research farm was standardized using different specific media like actinomycetes isolation agar, actinomyces agar, starch casein agar, and antifungal agent nystatin. A total 56 out of them 33 different morphotypes of actinomycetes were recovered from vermicompost, and 23 from NADEP sample. Colony farming unit count was comparably higher in vermicompost (23 X 10^4 cfu/g) compared with NADEP (13 X 10^4 cfu/g). Biocontrol potential of actinomycetes isolates were evaluated against three fungal plant pathogens namely *Macrophomina phaseolina*, *Sclerotium rolfsii* and *Scleritonia sclerotiorum* using dual plate culture. Strain N1.2, isolated from NADEP using actinomyces agar media showed promising result for *in vitro* inhibition of pathogen compared with the control.

Project 6.6: Management of Important Bacterial Diseases of Vegetable Crops

Survey and recording of prevalence of bacterial wilt, *Ralstonia solanacearum in* eastern Uttar Pradesh: Among bacterial diseases bacterial wilt caused by *Ralstonia solanacearum* was noticed most widely spread and economically important plant disease of solanaceous vegetable crops in the surveyed region of Sonebhadra and Mirzapur district of Eastern U.P. The bacterial wilt incidence were recorded 40% in chilli (*Capsicum annum* cv. VNR 305) fields at Kushi Dour village in Sonebhadra (Fig. 18). The disease was up to 75% in tomato (*Solanum lycopersicom cv* Namdhari 585) and brinjal (*Solanum melongena)* farmers' fields at Arazi line, Kiriyat and Chunar of Mirzapur during August to November 2017. Interestingly, 'VNR 305' the popular hybrid of chilli was found highly susceptible at Kushi Dour location in Sonabhadra district but chilli hybrid 'Josh' was unaffected at same location. Pure culture isolates of the bacterial pathogen were established by following standard procedure from chilli, tomato and brinjal. On the basis of colony characteristics on TZC medium and utilization of sugar (lactose, maltose, cellobiose) and sugar alcohol (sorbitol, manitol, dulcitol) pathogen isolates were identified as *Ralstonia solanacearum* (Fig. 19). On the basis of host range studies and oxidation of sugars and sugar alcohols all the tested isolates identified as race land biovar 3. Better understanding of variability in the pathogen is necessary for detailed and precise studies.

Fig. 18: Severe incidence of bacterial wilt on chilli at Kushi Dour, Sonebhadra.

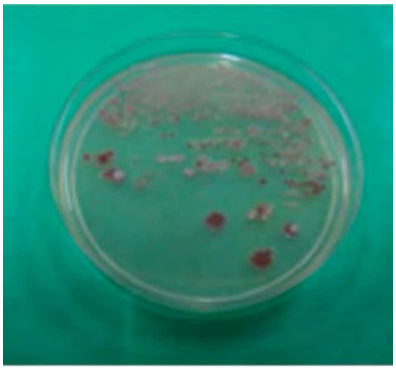


Fig. 19: Typical virulent colonies of *Ralstonia solanacearum* on TZC medium

Management of bacterial diseases on tomato: Pesticides viz. Azoxystrobin 23 SC, Copper oxychloride 50 WP, Copper hydroxide 53.8 DF, Streptocycline and bacterial bio agents namely *P. fluorescens* and *B. subtilis (BS 2)* were evaluated under field condition in tomato (cv. Kashi Amrit) against bacterial diseases as three subsequent foliar spraying at 15 days interval after 20 days of transplanting (DAT). The highest tomato yield (39.91 t/ha) recorded in Copper oxychloride 50 WP (T2) however all treatments were found free from bacterial spot (*Xanthomonas*) and bacterial speck (*Pseudomonas*) disease (Table 13).

Management of Black rot of cabbage: Different bioagents, bactericides and plant defense activator were evaluated under field in cabbage (cv. Golden Acre) against bacterial black rot caused by *Xanthomonas*

Table 13: Effectiveness of fungicides, bactericides and bacterial bio-agents against bacterial diseases on tomato and yield

Particulars	Yield (t/ha)
T1 - Azoxystrobin 23 SC spray @ 1ml/l	33.75
T2 - Copper oxychloride 50 WP sprays @ 2.5g/l	39.91
T3 -Copper hydroxide 53.8 DF sprays @ 2.0g/l	30.58
T4 – Streptocycline sprays @ 100 ppm	35.41
T5 - *P. fluorescens* WP sprays @ 10g/l	33.58
T6 - *B. subtilis (BS 2)* sprays @ 10g/l	32.66
T7 - Control	30.16
C.D.	1.227
SE(m)	0.394
C.V.	2.023

compestris pv *compestris* as foliar spraying after 20 and 35 days of transplanting. Among tested molecules highest yield (23 t/ha) recorded in cabbage with foliar sprays of treatment comprises *P. fluorescens* and Acibenzolar S- Methyl (ASM) sprays @ 0.5 g/l (T9) however all treatments were found free from incidence of black rot (Table 14).

Table 14: Effectiveness of bio-agents, fungicides and plant defense activators against black rot of cabbage and yield

Particulars	Yield (t/ha)
T1 - *P. fluorescens* WP sprays @ 5g/l	21.33
T2 - *B. subtilius* WP sprays @ 10g/l	22.00
T3 - T1+ Salicylic Acid sprays @1g/ l	21.83
T4 - T2+ Salicylic Acid sprays @ 1g/l	21.50
T5 - Salicylic Acid sprays @ 1g/l	22.83
T6 – Streptocycline sprays @ 100 ppm	21.50
T7- Copper oxychloride 50 WP sprays @ 2.5g/l	22.66
T8 - Copper hydroxide 53.8 DF sprays @ 2.5g/l	22.66
T9 - T1 + Acibenzolar S- Methyl (ASM) sprays @ 0.5gram/l	23.00
T10 - T2 + ASM sprays @ 0.5gram/l	22.00
T11 - Control	20.66
C.D.	1.224
SE(m)	0.412
C.V.	3.243

Project 6.7: Development of diagnostics kits for major viruses infecting vegetable crops

Characterization of begomovirus associated with mosaic disease of sponge gourd: The complete genome sequencing of begomovirus infecting sponge gourd revealed two DNA A (clone SP 1A and clone SP 22A) one DNA B (clone SP 7B) of Tomato leaf curl New Delhi virus (ToLCNDV) is associated. SP 1A and SP 22A are having 88.5% nucleotide identity among themselves with 2739 and 2736 nt, respectively. In BLAST analysis, clone SP 1A had 99% nucleotide identity with ToLCNDV isolate reported on sponge gourd from New Delhi, whereas clone SP 22A had 99% nucleotide identity with ToLCNDV isolate reported on bitter gourd from Varanasi. Both the DNA A components are comprising of two open reading frames (ORFs) [AV1, AV2] in virion sense and 5 ORFs [AC1, AC2, AC3, AC4 and AC5] in complementary-sense separated by an intergenic region (IR). In phylogenetic analysis, the complete nucleotide sequences of DNA A component of ToLCNDV infecting sponge gourd with other selected ToLCNDV indicates, SP 1A and SP 22A forms separate cluster with ToLCNDV isolate reported on sponge gourd from New Delhi and bitter gourd from Varanasi, respectively (Fig 20).

Similarly, clone SP 7B of DNA B had 96% identity with ToLCNDV isolate reported on snake gourd from Varanasi in BLAST analysis. Phylogenetic tree constructed for DNA B component with other ToLCNDV DNA B component reported earlier revealed clone SP 7B had form a separate cluster with other strains of ToLCNDV reported in Asia (India, Thailand, Pakistan) infecting tomato, potato, carrot and cucurbits (Fig 21).

In addition, ToLCNDV causing mosaic disease on sponge gourd is also found associated with α- and β-satellites. Complete genome sequencing of α-satellite showed that 90% similarity with Tomato leaf curl New Delhi alphasatellite (ToLCNDA) reported on tomato from New Delhi. In phylogenetic analysis, it is forming separate cluster with other α-satellites such as Cucurbit yellow mosaic alphasatellite, Papaya leaf curl alphasatellite and Croton yellow vein mosaic alphasatellite (Fig 22).

Similarly, complete genome sequencing of β-satellite showed that 94% similarity with Tobacco leaf curl Patna betasatellite (TbLCPB) reported on tobacco from New Delhi. In phylogenetic analysis, TbLCP is forming separate cluster with Tomato leaf curl Patna betasatellite (Fig 23).

Diagnostics of Begomovirus and its associated satellite on tomato: Totally 11 villages were surveyed and 34 samples were collected in 3 replications including

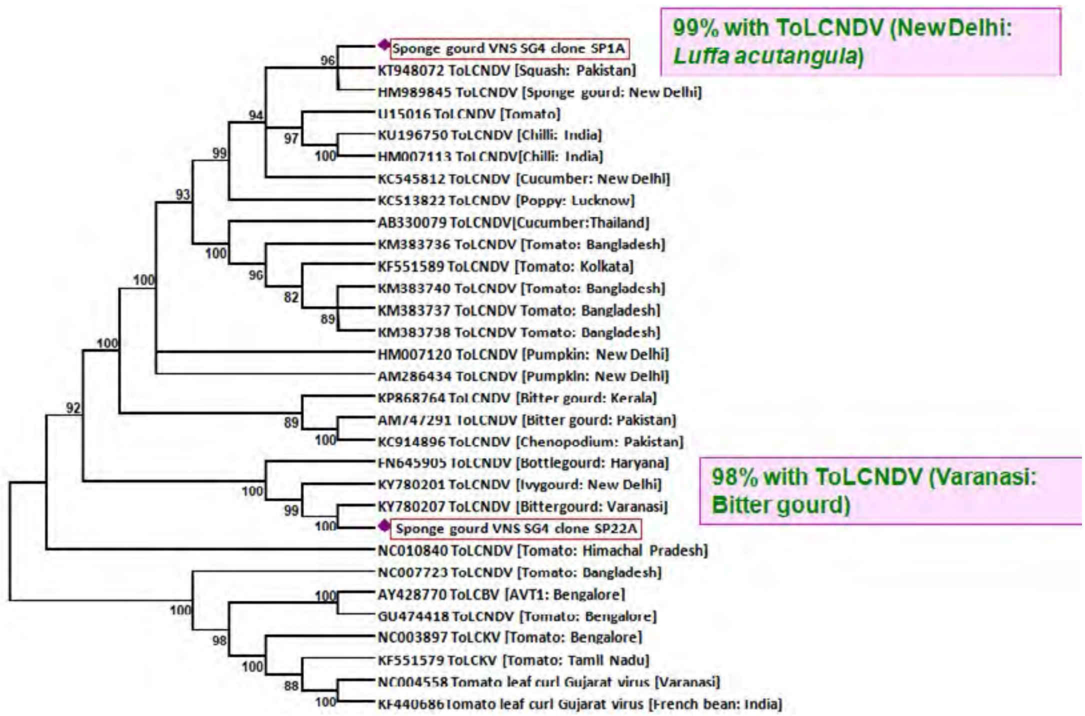


Fig. 20: Phylogenetic analysis of DNA A component of ToLCNDV infecting sponge gourd with other strains

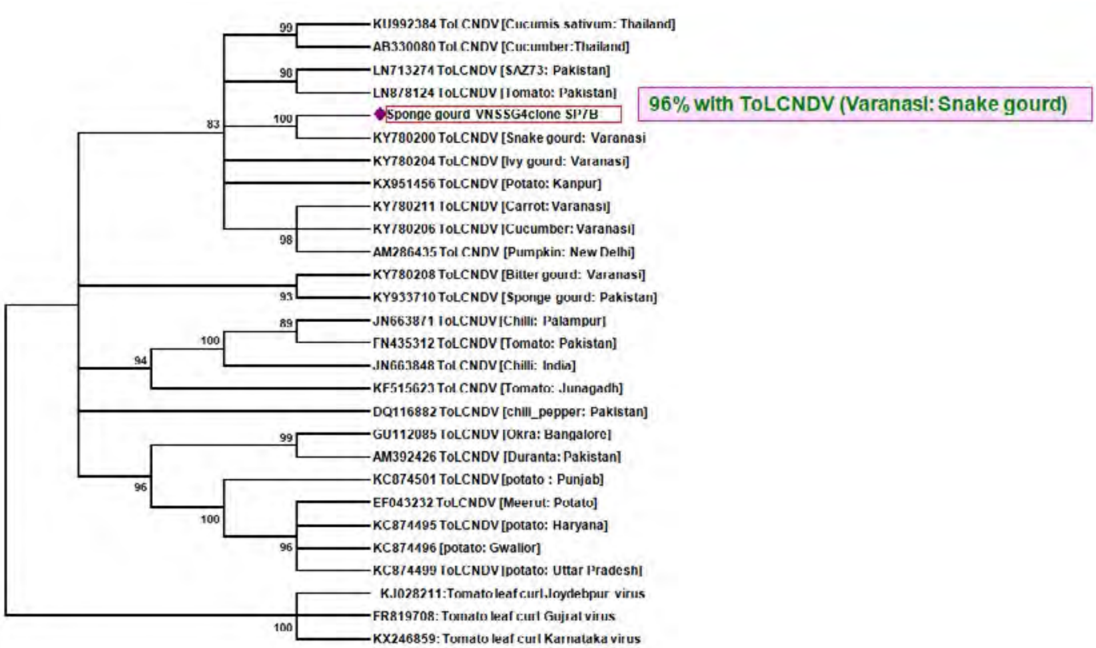


Fig 21: Phylogenetic analysis of DNA B component of ToLCNDV infecting sponge gourd with other strains

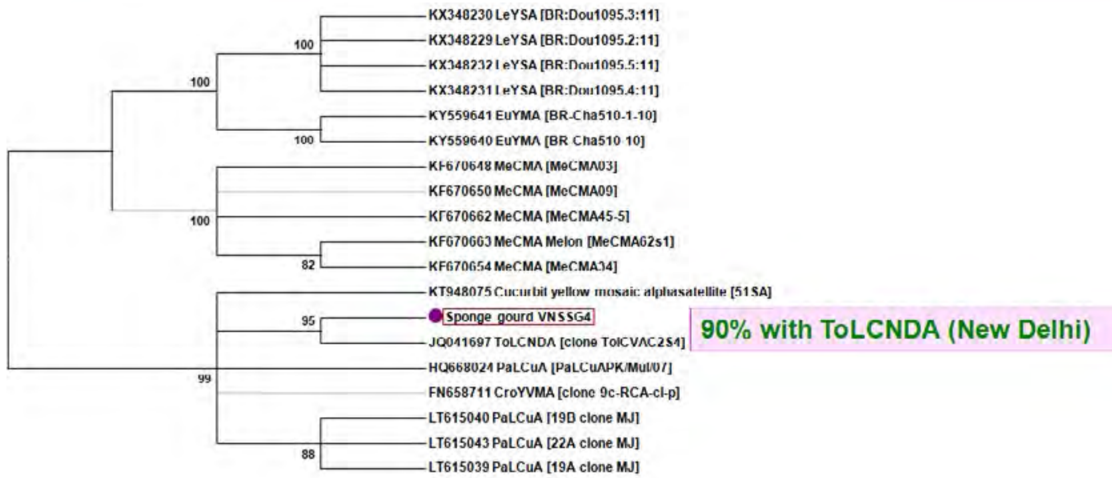


Fig 22: Phylogenetic analysis of α-satellite ToLCNDV infecting sponge gourd with other satellites

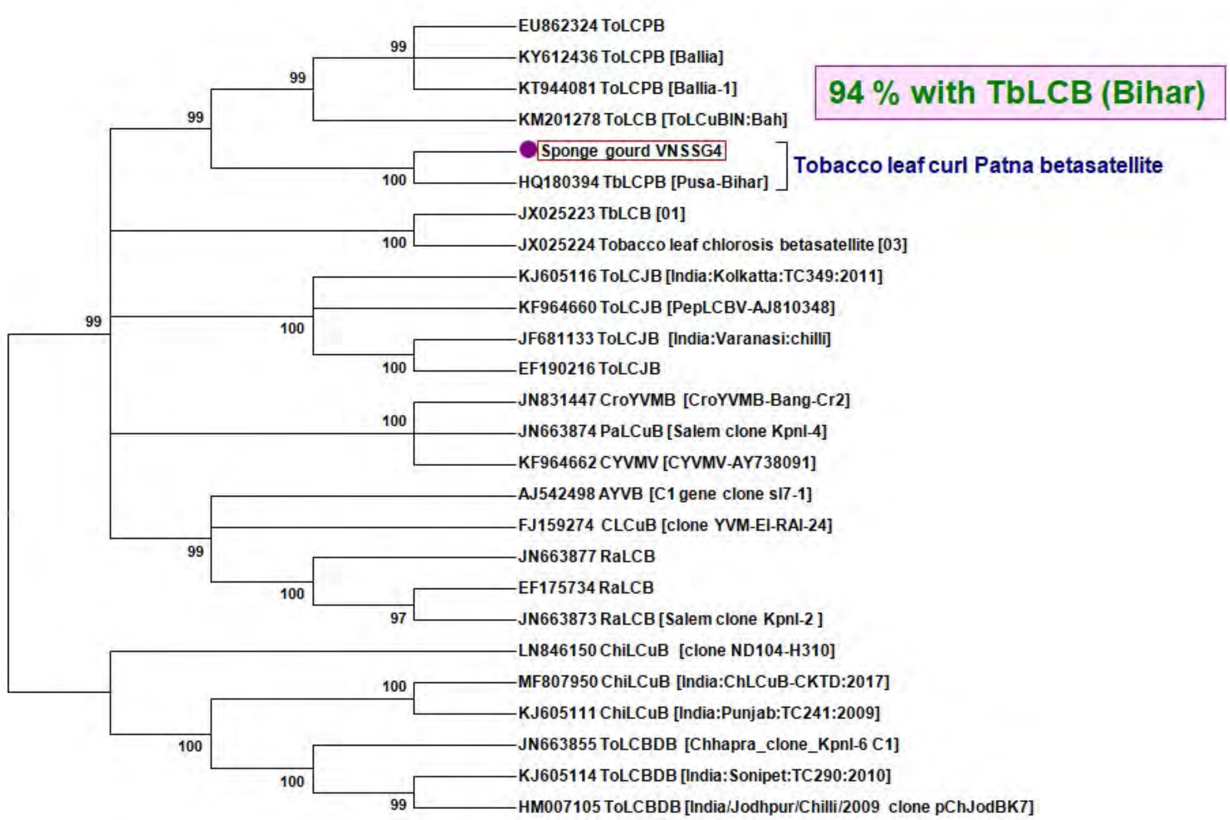


Fig 23: Phylogenetic analysis of β-satellites ToLCNDV infecting sponge gourd with other satellites

Mirzapur and Varanasi districts both at vegetative and fruiting stage of the crop. During the survey, tomato plants showing virus and virus-like symptoms (mosaic, yellowing, leaf curls, witches broom and bud necrosis) were collected along with the healthy samples to diagnose the begomovirus species associated. Total nucleic acid was extracted from the infected tomato samples using CTAB method (Doyle, 1990) and were tested for presence of begomovirus by PCR using begomovirus specific primers. Mean while, the samples were also tested with species specific primers of *Tomato leaf curl New Delhi virus* (ToLCNDV), *Tomato leaf curl*

Gujarat virus (ToLCGV), *Tomato leaf curl Palampur virus* (ToLCPalV), *Tomato leaf curl Bangalore virus* (ToLCBV) and *Tomato leaf curl Karnataka virus* (ToLCKV) to confirm the possible associated viruses with the tomato samples in Indo Gangetic plain.

Among 34 samples tested, 25 samples showing leaf curl showed positive for begomovirus. Among the 25 samples, 12 samples were found infected with ToLCNDV, 3 samples with ToLCKV and 1 sample with ToLCPalV and none of the samples were infected with ToLCBV. Interestingly, mixed of more than one begomoviruses combinations such as ToLCNDV+ToLCGV, ToLCNDV+ToLCKV, ToLCNDV+ToLCGV+ToLCKV and ToLCNDV+ToLCPalV+ToLCKV were also observed. Around 20 of 34 samples were associated with β-satellite and 27 of 34 samples associated with α-satellite. Among them, 17 of 34 samples were associated with both β-satellite and α-satellites (Fig 24).

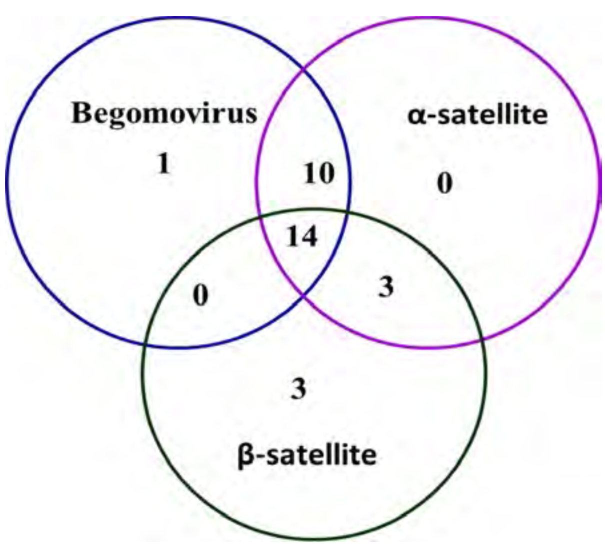


Fig 24: Diagnostics of Begomovirus and its associated satellites on tomato

Begomovirus-associated α-satellites are self-replicating whereas β-satellites are always associated with their specific helper component. Since the primers used in present study for the detection of begomoviruses in tomato are species specific, therefore some samples have negative to helper viruses and positive for satellite, may be due to evolution in primer bind site of the viruses or occurrence of new viruses in the samples.

Detection of viruses infecting on squash in Varanasi region: Virus like symptoms such as chlorosis, mosaic, vein banding, leaf thickening and leaf filiformity were observed on squash plants in the IIVR Research farm. Total RNA was extracted using QIAzol Lysis Reagent (Qiagen, Germany) and cDNA was prepared using iScript cDNA Kits (BioRad, USA). Samples were subjected to PCR analysis for the detection of viruses associated with the squash samples using specific primer pairs reported earlier. Among the 6 samples tested with different kinds of symptoms, crop is found infected with PRSV, CGMMV and Polerovirus. Samples were failed for CMV, ZYMV, Potexvirus, CABYMV and CCYV. For further confirmation, representative samples were send for sequencing. Interestingly, all samples were detected with more than one viruses (Table 15).

Molecular characterization of brinjal little leaf phytoplasma: Total DNA extracted from brinjal little leaf symptomatic sample collected from the IIVR farm. DNA was subjected to nested PCR assay using two primer pairs R16F/R followed by P1/P7. Amplified fragment of ~1300bp from the nested PCR was cloned and sequenced. In BLAST analysis, sequence had 99% identity with brinjal little leaf phytoplasma reported from Karnataka. In order to find the group of phytoplasma, phylogenetic analysis was performed with nucleotide sequences of previously reported group of phytoplasma from the GenBank database. Sequence revealed that, phytoplasma infecting brinjal in IIVR farm belongs to 16Sr VI group (Fig. 25). Further to identify the subgroup

Table 15: Detection of viruses using specific/universal primers

ID	CMV	Potyvirus			CABYV	Potex	CCYV	CGMMV	Polero
		Univ	ZYMV	PRSV					
Sq1	-	+	-	-	-	-	-	+	-
Sq2	-	+	-	-	-	-	-	+	+
Sq3	-	+	-	+	-	-	-	+	+
Sq4	-	+	-	-	-	-	-	+	+
Sq5	-	+	-	+	-	-	-	+	-
Sq6	-	+	-	-	-	-	-	+	-

of 16Sr VI, *in silico* virtual restriction analysis was performed with reference sequences. Results showed 16Sr VI-D is found infecting the brinjal having similar restriction pattern (Fig. 26).

Standardization of date of planting for the management of Brinjal little leaf disease: In order to standardize the optimum time of planting to obtain maximum yield with least disease incidence of brinjal little leaf disease, trial was conducted using four varieties with four different time of planting. Average disease incidences obtained at 4 different planting intervals were shown in the table 16. Crop planted on

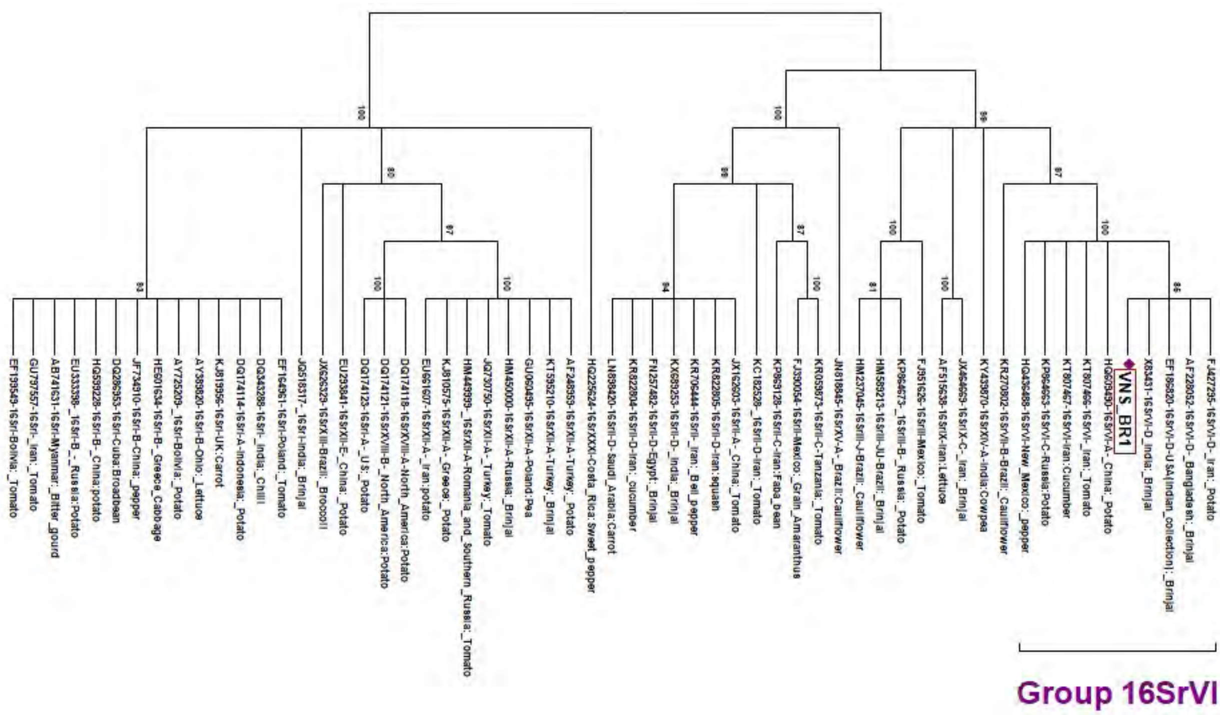


Fig. 25: Phylogenetic tree based on phytoplasma 16S rDNA showing the relationships among representative the phytoplasma strains

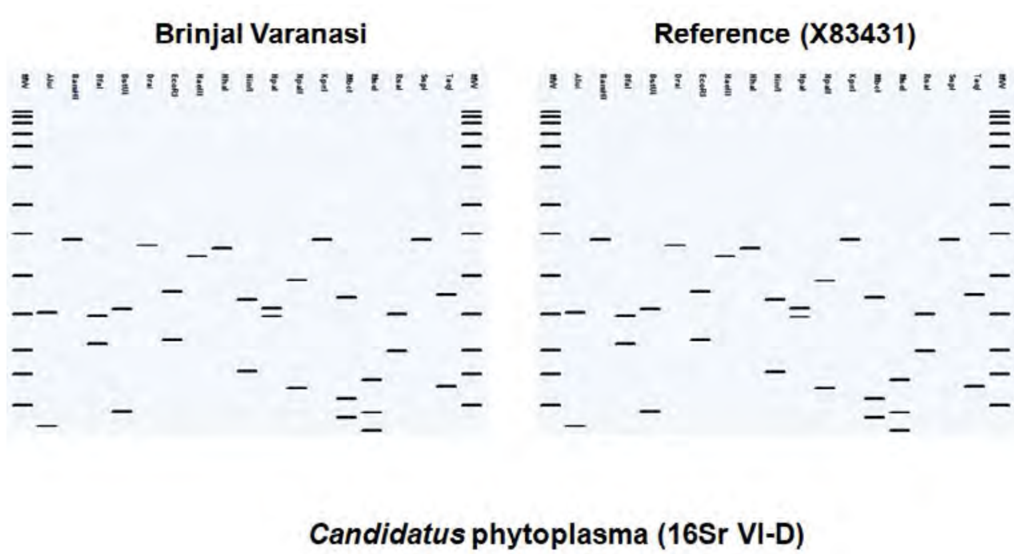


Fig 26: Virtual restriction analysis

29.09.2017 had recorded no disease incidence across all the varieties. For optimizing the time of sowing, yield has to be correlated. Recording of yield data is under progress.

Table 16: Incidence of brinjal little leaf disease in 4 cultivars with different date of planting

Variety	Date of planting			
	22.08.2017	05.09.2017	29.09.2017	5.10.2017
DBR-31	25.0	22.9	0	2.01
Kashi Taru	22.9	10.4	0	0
Br-14	41.67	16.67	0	0
Punjab Barsati	29.16	14.58	0	0

Project 6.9: Management of Nematodes Infesting Major Vegetable Crops

Molecular identification of Root knot nematode species: Root knot nematode species infecting vegetable crops in ICAR-IIVR, Varanasi research farm have been identified using molecular technique. DNA was isolated from newly hatched second stage infective juvenile and female using standard protocol described by Adam *et al.*, (2007). Further, identity of the root knot nematode species was confirmed through molecular characterization by using ITS 5S-18S ribosome region specific marker 194 (5 'TTAACTTGCCAGATCGGACG'3); 195 (5 'TCTAATGAGCCGTACGC' 3) (Blok *et al.*, 1997). The sequence of root knot nematode species showed maximum identity with *Meloidogyne incognita* (94%). In addition, the same root knot nematode species was again confirmed by using specific SCAR marker Inc-K14-F (5 'GGGATGTGTAAATGCTCCTG' 3); Inc-K14-R (5 'CCCGCTACACCCTCAACTTC' 3) (Randig *et al.*, 2002) (Fig. 27).

Fig 27: Molecular identification of *Meloidogyne incognita* using SCAR marker (INC-K-14F/INC-K-14R) infesting different vegetable crops at research farm

***In vitro* evaluation of PGPR isolates on egg hatching inhibition of root knot nematode *Meloidogyne incognita*:** Culture filtrates of three antinemic PGPR isolates such as *Bacillus marisflavi* (CRB2), *Bacillus subtilis* (CRB7) and *Bacillus subtilis* (CRB9) were tested for egg hatching inhibition of root knot nematode *Meloidogyne incognita* under *in-vitro* condition. The study revealed that, these three antinemic isolates were significantly inhibited (>90%) egg hatching compared to control (Fig. 28). Nevertheless, media control also inhibited egg hatching up to 27% after 120 h exposure period; however, there is no significant difference in egg hatching inhibition between three antinemic isolates.

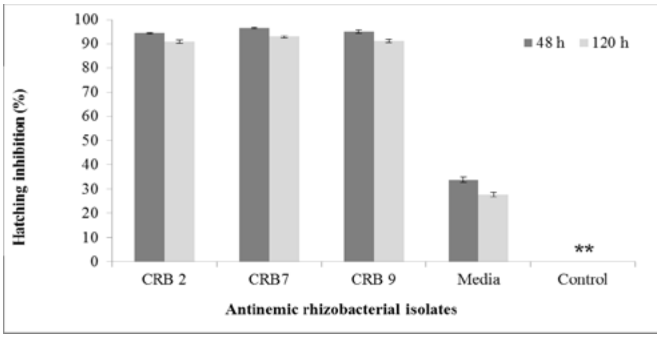


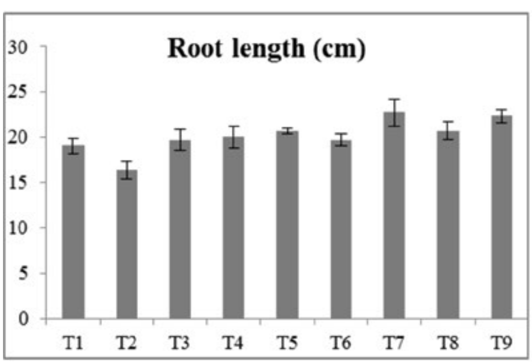
Fig. 28: Effect of antinemic rhizobacterial isolates on egg hatching inhibition of *M. incognita*

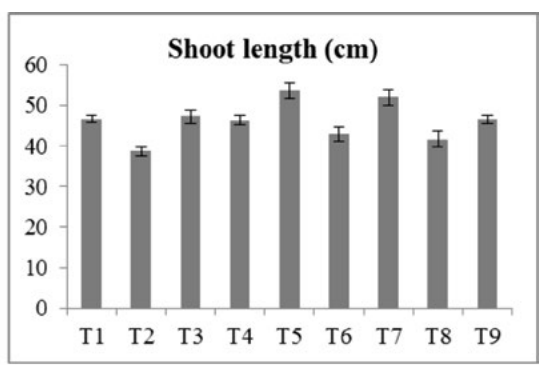
Evaluation of antinemic activity and plant growth promotion of PGPR agents against Root knot nematode, *Meloidogyne incognita* infecting okra under screen house pot condition: Antinemic activity of plant growth promoting rhizobacterial (PGPR) agents are evaluated against root knot nematode, *M. incognita* under pot condition in okra (cv Kashi Pragati). Three PGPR isolates such as *Bacillus marisflavi* (CRB2), *Bacillus subtilis* (CRB7) and *Bacillus subtilis* (CRB9) are tested at two potential dose (2.5g and 5.0 g) through soil drenching method under pot condition. The study revealed that irrespective of treatments, gall index, number of root galls, final soil population, egg mass per root system and reproductive factor were significantly reduced compared with inoculated control. Among them, the treatments T5 (CRB2 @ 5g/plant) and T7 (CRB7 @ 5g/plant) exhibited maximum antinemic activity against root knot nematode (*M. incognita*) by reducing gall index (1.67, 1.34), number of galls (83.3%, 86.8%), egg mass per root system (79.3%, 84.1%), final soil population (73.3%, 70.7%), and RF (0.68, 0.75) respectively compared to inoculated control. These treatments results were comparable with chemical check (T3) (Table 17). In addition, these treatments are also enhanced plant growth parameters including shoot length, root length, shoot weight, root weight and yield (Fig 29).

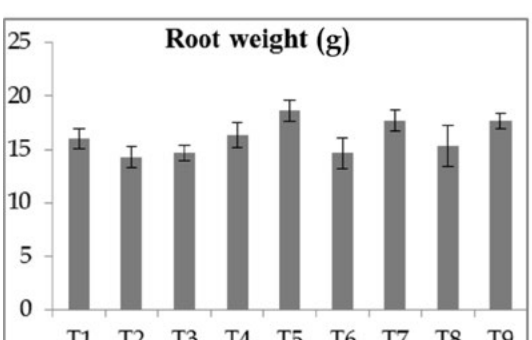
Table 17: Evaluation of antinemic activity and plant growth promotion of selected PGPR agents against Root knot nematode, *Meloidogyne incognita* infecting okra under screen house pot condition

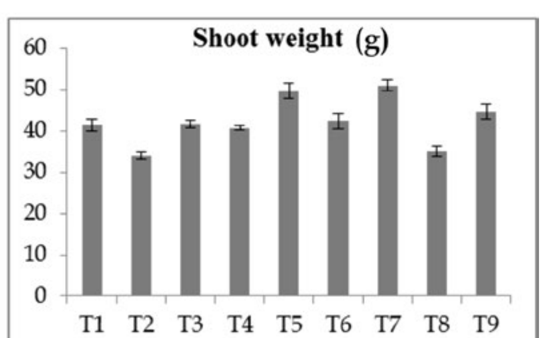
Treatments	GI (Scale) 0-5)	Galls/root system	Percent decrease over control	Egg mass/ root system	Percent decrease over control	FSP (250 CC)	Percent decrease over control	RF
T1-Healthy control	-	-		-		-	-	
T2-Inoculated control	4.8	415.3 (20.37)^{a}	-	69.3 (8.32)^{a}	-	644.3 (25.38)^{a}	-	2.58
T3-Carbofuran 3G (0.3 g a.i)	1.0	44.00 (6.63)^{f}	89.4	09.6 (3.10)^{e}	86.1	161 (12.68) ^{c}	75.0	0.64
T4-*Bacillus marisflavi* (CRB2) 2.5g	2.67	112.0 (10.58)^{c}	73.0	21.6 (4.65)^{c}	68.8	261 (16.15)^{bc}	59.5	1.04
T5-*Bacillus marisflavi* (CRB2) 5g	1.67	69.34 (8.32)^{e}	83.3	14.3 (3.78)^{d}	79.3	172 (13.11)^{c}	73.3	0.68
T6-*Bacillus subtilis* (CRB7) 2.5g	2.67	108.67 (10.42)^{c}	73.8	24.3 (4.93)^{c}	64.9	255.3 (15.97)^{bc}	60.4	1.04
T7-*Bacillus subtilis* (CRB7) 5.0 g	1.34	54.67 (7.39) ^{f}	86.8	11.0 (3.31)^{e}	84.1	188.6 (13.73)^{c}	70.7	0.75
T8-*Bacillus subtilis* (CRB9) 2.5g	4	200.0 (14.14)^{b}	51.8	32.3 (5.68)^{b}	53.3	311 (17.63)^{b}	51.7	1.24
T9- *Bacillus subtilis* (CRB9) 5.0 g	2	81.00 (9.0) ^{d}	80.5	17.6 (4.20)^{cd}	74.6	216.6 (14.71)^{c}	66.4	0.87
SE(m)		**0.31**		**0.24**		**0.68**		
CD		**0.93**		**0.75**		**2.07**		

Gall Index (0-5 scale) (Hussey and Janssen, 2002), FSP: Final soil population (250 CC), RF: Reproduction factor
Transformed values were used for statistical analysis, Means with the same letter are not significantly different
Talc formulation: *Bacillus marisflavi* (CRB2)(2.5x10^{11} cfu/g) *Bacillus subtilis* CRB7 (2.6x10^{11} cfu/g) *Bacillus subtilis* CRB9(2.5x10^{11} cfu/g)









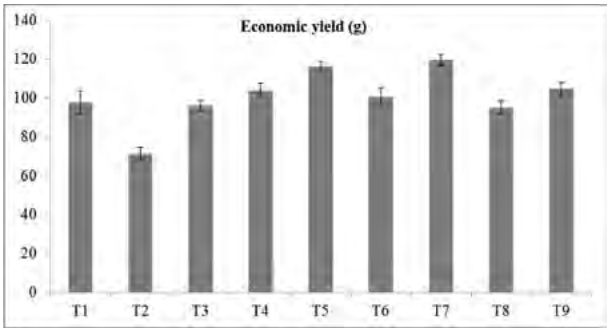


Fig 29. Effect of plant growth promoting rhizobacterial agents on okra root length, shoot length, root weight, shoot weight and economic yield.

Management of root-knot nematode (*Meloidogyne incognita*) in okra by using plant growth-promoting rhizobacteria: Antinemic talc based formulations of rhizobacterial agents *Bacillus marisflavi* (CRB2) and *Bacillus subtilis* (CRB7) are evaluated through different delivery mechanism including seed treatment, soil application (enriched with vermicompost) and soil drenching and their combination under naturally infested field condition for the management of root knot nematode *M. incognita* in okra cv. Kashi Pragati. The study revealed that, among the treatments the integrated module T7 (Seed treatment with *Bacillus*

Table 18: Management of root-knot nematode (*Meloidogyne incognita*) in okra by using plant growth-promoting rhizobacteria

Treatments	Gall index (0-5 scale)	Number of galls/plant	Percent decrease over control	Egg mass/ root system	FSP (250 CC soil sample)	Percent decrease over control	RF	Yield (Kg/plot)
T1	1.53	72.1 (8.49)[b]	30.4	29.3 (5.41)[b]	382.3 (19.55)[b]	24.6	1.0	5.5
T2	1.40	66.7 (8.16)[bc]	35.6	26.7 (5.16)[bc]	353.3 (18.79)[bc]	30.3	0.9	5.0
T3	1.17	43.5 (6.60)[d]	58.0	17.4 (4.17)[cd]	230.7 (15.18)[cd]	54.5	0.6	7.3
T4	1.20	48.2 (6.94)[c]	53.5	19.3 (4.39)[c]	255.5 (15.98)[c]	49.6	0.7	6.4
T5	1.27	55.1 (7.43)[bc]	46.8	23.6 (4.85)[bc]	292.2 (17.09)[bc]	42.4	0.8	7.1
T6	1.27	57.3 (7.57)[bc]	44.7	22.9 (4.78)[bc]	303.9 (17.43)[bc]	40.1	0.8	6.1
T7	0.97	30.5 (5.53)[d]	70.6	12.2 (3.49)[d]	161.8 (12.72)[d]	68.1	0.4	8.2
T8	1.07	35.0 (5.92)[cd]	66.2	14.0 (3.74)[d]	185.5 (13.61)[cd]	63.4	0.5	7.8
T9	1.80	83.6 (9.14)[ab]	19.3	33.4 (5.78)[ab]	443.1 (21.04)[ab]	12.6	1.2	4.8
T10	1.00	31.3 (5.60)[d]	69.8	12.5 (3.50)[d]	166.1 (12.88)[d]	67.2	0.4	6.1
T11	0.93	29.0 (5.39)[d]	72.0	11.6 (3.40)[d]	153.7 (12.39) [d]	69.7	0.4	7.8
T12	2.33	103.6 (10.18)[a]	-	38.3 (6.18)[a]	507.0 (22.51)[a]	-	1.4	5.6
SE(m)		**0.40**		**0.24**	**0.93**			
CD		**1.19**		**0.72**	**2.7**			

T1- Seed treatment with *Bacillus subtilis* (CRB7) @ 20g/kg seed; **T2** – Seed treatment with *Bacillus marisflavi* (CRB2) @ 20 g/ kg seed; **T3-** Soil application of *Bacillus subtilis* (CRB7) (5 kg/ha/2 ton of Vermi compost); **T4-** Soil application of *Bacillus marisflavi* (CRB2) (5 kg/ha/2 ton of Vermi compost); **T5** - Soil drenching *Bacillus subtilis* (CRB7) @ 1%; **T6** - Soil drenching *Bacillus marisflavi* (CRB2) @ 1%; **T7** - Seed treatment with *Bacillus subtilis* (CRB7) @ 20g/ kg seed+ Soil application of *Bacillus subtilis* (CRB7) (5 kg/ha/2 ton of Vermi compost) + Soil drenching *Bacillus subtilis* (CRB7) @ 1%; **T8** - Seed treatment with *Bacillus marisflavi* (CRB2) @ 20g/ kg seed+ Soil application of *Bacillus marisflavi* (CRB2) (5 kg/ ha/2 ton of Vermi compost) + Soil drenching *Bacillus marisflavi* (CRB2) @ 1%;**T9-** Vermicompost alone 2 ton/ha; **T10-** Carbofuran 3G@ 1kg a.i/ ha; **T11-** Vermicompost alone 2 ton/ha + Carbofuran 3G @ 1kg a.i/ha; **T12-** Control

Gall index (0-5), but it was based on the percentage of the root system with galls (Hussey and Janssen, 2002), where 0 = no galling; 1 = trace infection with a few small galls; 2 = < 25% roots galled; 3 = 26 to 50%; 4 = 51 to 75% and 5 = >75% roots galled, FSP: Final soil population (250 CC), RF: Reproduction factor; Transformed values were used for statistical analysis, Means with the same letter are not significantly different

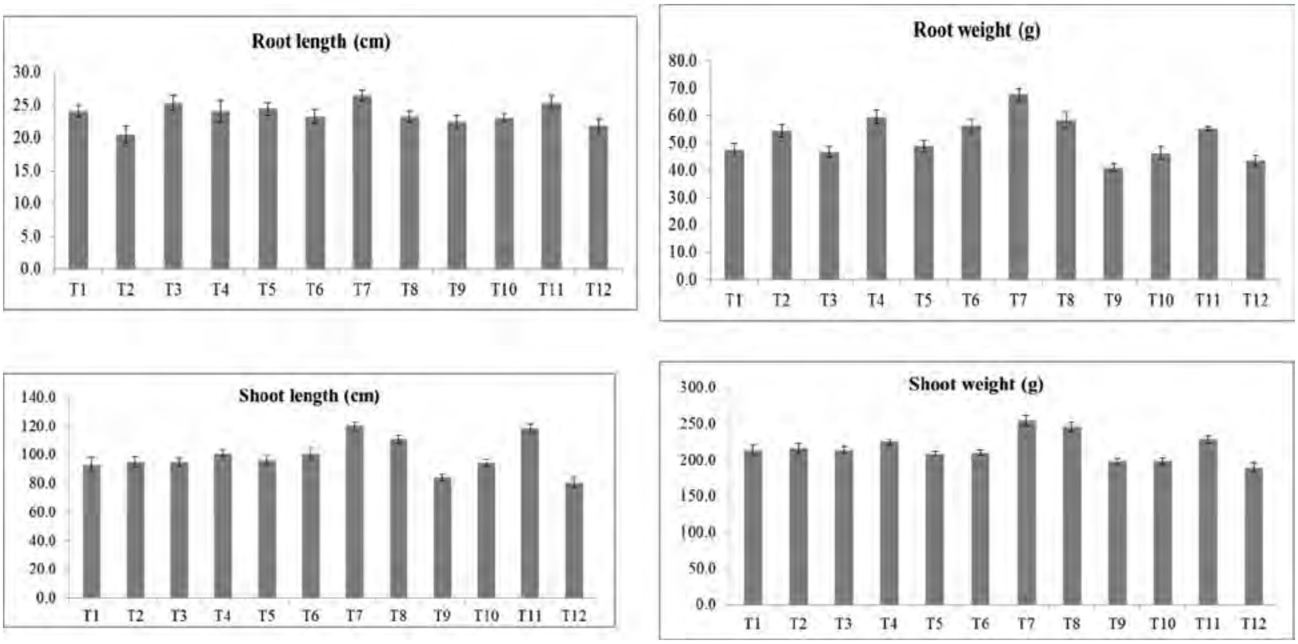


Fig.30. Effect of plant growth promoting rhizobacterial agents on okra root length, root weight, shoot length and shoot weight.

subtilis (CRB7) @ 20g/ kg seed; Soil application of *Bacillus subtilis* (CRB7) (5 kg/ha/2 tonnes of enriched vermi compost); Soil drenching *Bacillus subtilis* (CRB7) @ 1%) and T8 (Seed treatment with *Bacillus marisflavi* (CRB2) @ 20g/ kg seed; Soil application of *Bacillus marisflavi* (CRB2) (5 kg/ha/2 tonnes of enriched vermi compost); Soil drenching *Bacillus marisflavi* (CRB2) @ 1%) exhibited maximum antinemic activity against root knot nematode (*M. incognita*) by reducing gall index (0.97, 1.07), number of galls (70.6%, 66.2%), final soil population (68.1%, 63.4%) and RF (0.4, 0.5) respectively compared to control (Table 21). In addition, these treatments also enhanced plant growth by increasing shoot length, shoot weight, root length and root weight (Table 18, Fig. 30). Both the treatments were comparable with chemical check (T11).

Project 6.10: Dynamics of pest and diseases and development of forecasting models

Seasonal incidence of cucurbit fruit fly in and around Varanasi: The seasonal incidence of cucurbit fruit fly, *Bactrocera cucurbitae* infesting bottle gourd was observed almost throughout its growth period of during April, 2017 to March, 2018 in and around Varanasi, Uttar Pradesh, India. The fruit fly population was higher during Oct-Nov and April –June months. Highest number of fruit fly (119.33 per trap) was recorded during 16th SMW (3rd week of April, 2017) followed by 43rd SMW (4th Week of October, 2017) *i.e.*, 118/trap (Fig.31).

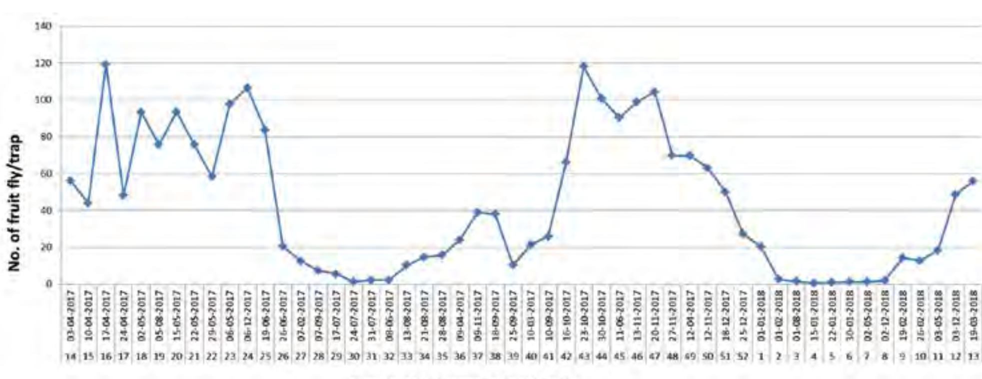


Fig. 31: Population dynamics of Cucurbit fruit fly, *B. cucurbitae*

Externally Funded Projects

EXTERNALLY FUNDED PROJECTS

Project 1: Genomics assisted selection of *Solanum chilense* introgression lines for enhancing drought tolerance in tomato

For *de novo* assembly of *S. chilense* genome, deep sequencing data was generated on the Illumina HiSeq2500 platform in Rapid Run Mode and on PacBio RS II system with a total of 18 Single Molecule Real Time (SMRT) Sequencing cells. This deep sequencing raw data was shared by the UK collaborators. Raw sequence data was quality checked using the program Fast QC.

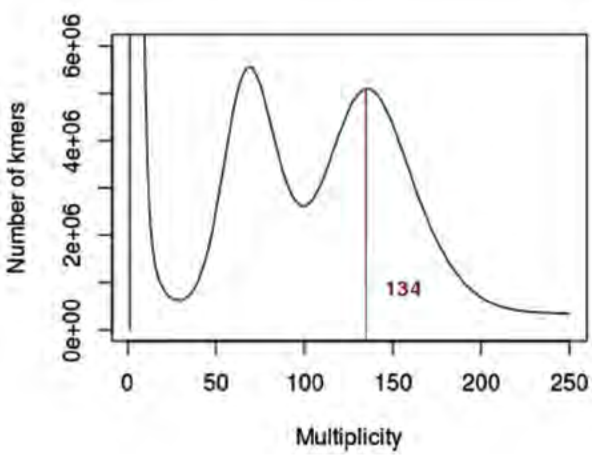


Fig. 1: Distribution of 19-mer frequency in raw sequencing reads

Estimation of the size of the *S. chilense* genome was performed through the k-mer abundance distribution using illumina sequence data. A k-mer value of 19 was chosen and the program Jellyfish was used for counting the k-mers and to make a histogram of the reported results. The k-mer distribution was analyzed using R. In order to estimate the genome size a read count was made counting the reads present for the lengths from 100-251bp.

The 19-mer abundance distribution graph yielded 2 peaks out of which the peak with the greater depth was the homozygous peak (Fig. 1). This depth of homozygous peak was ascertained from the graph and was found to be at 134x. The heterozygous peak depth (62x) was found to be exactly half of the homozygous peak (62x). The relations among the peak of 19-mer frequency (M), real sequencing depth (N), read length (L), and k-mer length (K) can be expressed in an experienced formula: $M = N * (L - K + 1) / L$. Then, we divided the total sequence length by the real sequencing depth and obtained an estimated genome size of 1.085 Gb.

The illumina reads were assembled using Discovar Denovo assembler using 48 processors and a memory usage of 1000GB. The assembly generated had the following statistics.

- Contig N50: 20,595
- Scaffold N50: 25,032
- Total bases in 1 kb+ scaffolds: 1,158,944,708
- Total bases in 10 kb+ scaffolds: 882,642,044

Hybrid assembly of *S. chilense* genome was carried out using illumina and PacBio deep sequencing data. This was achieved using the MaSuRCA hybrid assembler. The hybrid assembly resulted in a more contiguity and contig N50 improved by three fold. The overall statistics of the assembly is as follows.

- Contig N50: 62,650
- Total bases in 1 kb+ scaffolds: 1,183245172
- Total bases in 10 kb+ scaffolds: 1090153081

To create an introgression line population, various backcross generations have been developed for isolating introgression lines that include BC_2F_2, BC_3F_2 and BC_4F_1 generations. Backcross families exported to UK for advancement of generations were received and grown at ICAR-IIVR, Varanasi. These were backcrossed again to produce BC_5F_1 and BC_4F_1. This set was also progressed by self-pollination to constitute populations for isolating homozygous ILs. To develop an inbred backcross line (IBL) population with multiple introgressions, BC_2F_5, BC_2F_4 and BC_2F_3 families of Kashi Amrit and *S. chilense* interspecific cross were generated by repeated self-pollination.

Genomic DNA from the individuals of the multiple backcross generations and parents viz. VF36, Kashi Amrit and LA1972 accession were subjected to GBS analysis. GBS library preparation and sequencing was carried out at Center for Excellence in Genomics and System Biology, ICRISAT. GBS data analysis was performed using the GBS discovery pipeline of TASSEL version 5.0 software. The FASTQ and sample key files (containing the barcodes for each genotype) generated from raw sequence reads were used as input. The filtered, high-quality sequences from each sample were aligned to the latest version of the *S. lycopersicum* genome SL3.0 using Bowtie 2. A majority of the sequence tags (57.7%) were uniquely aligned to the tomato reference genome and were used for SNP discovery.

Tag alignment to the *S. lycopersicum* SL3.0 revealed

43213 unique SNP loci (pre-filtration SNPs). The proportion of missing loci was to the extent of 42.2% and proportion of heterozygous SNPs was 11.87%. The proportion of missing data was reduced by imputation using LD-KNNI implemented in the TASSEL plugin. After imputation, the proportion of missing SNPs was reduced to 15.14%. The ABH plugin of TASSEL removed SNPs with missing, ambiguous, or heterozygous parental genotypes. This resulted in final set of 14,266 SNPs across all the chromosomes of tomato genome (Table 1).

Table 1: Distribution SNPs discovered for *S. chilense* using Genotyping-by-sequencing

Chromosome	Number of SNPs
SL3.0Ch01	678
SL3.0Ch02	775
SL3.0Ch03	686
SL3.0Ch04	2195
SL3.0Ch05	404
SL3.0Ch06	2139
SL3.0Ch07	737
SL3.0Ch08	711
SL3.0Ch09	491
SL3.0Ch010	519
SL3.0Ch011	2393
SL3.0Ch012	2535
Total	**14,266**

Project 2: Introgression of begomovirus resistance genes in tomato (*Solanum lycopersicum* L.) using MAS and genomics approach

Pyramiding *Ty* genes: Two independent backcross programs were advanced further for pyramiding multiple *Ty* genes in the background of recurrent parents Kashi Vishesh and Kashi Aman. In the backcross program with Kashi Vishesh, tomato lines VRT 8-6-1 and VRT 2-2-3 were used as donors for *Ty-2* and *Ty-3* genes respectively. In the second backcross program, Kashi Aman is being used as recurrent parent. Tomato lines VRT 8-6-1 and FLA 456 are being used as donors for *Ty-2* and t*y-5/6* genes respectively (Fig. 2).

The BC_2F_1 generations developed for pyramiding *Ty-2* and *Ty-3* in the background of 'Kashi Vishesh' and pyramiding *Ty-2* and ty-5/ty-6 in the background of 'Kashi Aman' were grown in the field. Marker assays were performed to select the plants carrying target *Ty* genes using flanking markers linked to *Ty*-genes (*Ty*-2, *Ty*-3, *ty*-5 & *ty*-6). The plants that tested positive for markers linked to target genes and that resembled recurrent parent morphologically were selected. The selected plants were backcrossed to respective recurrent parents to generate BC_3F_1 populations.

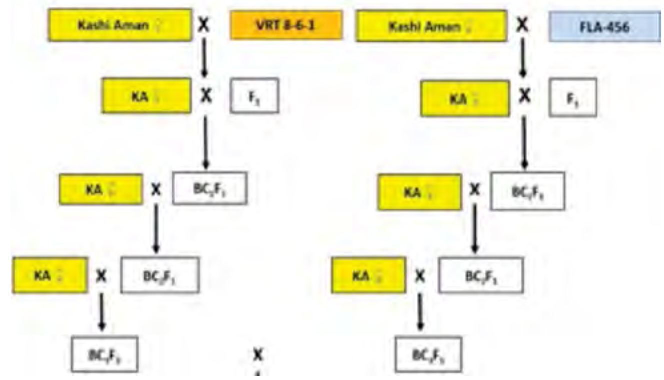


Fig. 2: Status of backcross scheme being followed to pyramid *Ty-2* and *Ty-5/6* genes in the background of 'Kashi Aman'.

Marker assays to test the presence of known *Ty* introgressions in tomato line VRT78-4: Set of tomato lines carrying known *Ty* genes (*Ty*-1: LA3473, *Ty*-2: C-8-6-1: *Ty*-3: D2-2-3 & *ty*-5+*ty*-6:IIHR2905), resistant tomato line and cultivars H-86, Kashi Aman and DVRT-1 were used to verify the presence of known *Ty* genes in tomato line VRT78-4. Genotyping assays were performed using the markers linked to known *Ty* genes on these set of lines. All the markers were monomorphic between resistant line VRT78-4 and susceptible cultivars except the marker SlNAC1 which is linked to *ty*-5. Interestingly, the marker SLM434 flanking the other end of *ty*-5 was monomorphic (Fig. 3). However, preliminary co-segregation tests performed to assess the association of polymorphic marker SlNAC1 with resistance showed no evidence of association in F_2 population.

QTL-Seq analysis: The F_2 generation derived from intraspecific cross viz. Kashi Vishesh X VRT-78-4 was phenotyped by agroinoculation usingTomato leaf curl

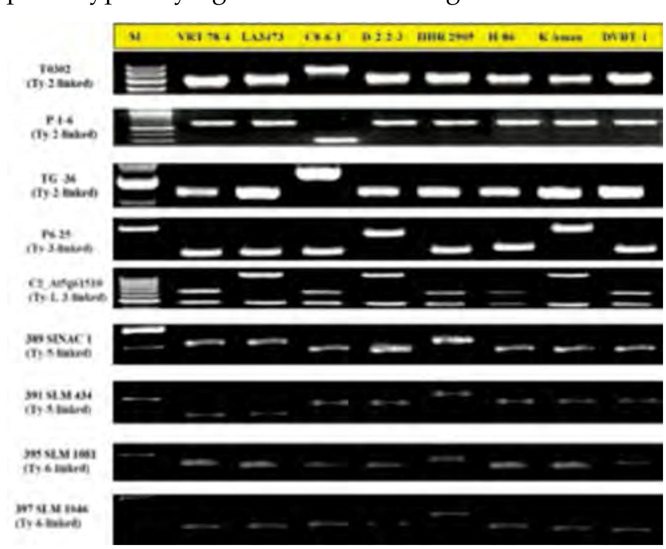


Fig. 3: Marker assays performed to test the presence of known *Ty* genes in the tomato line VRT78-4.

New Delhi virus (ToLCNDV) which was represented by DNA-A (GenBank accession HQ141673) and DNA-B (GenBank accession HQ141674) genome components of isolate ToLCNDV–IN [IN:Pune:JID:2010] and the associated Cotton leaf curl Multan betasatellite (CLCuMB–[IN:Sri:02]) (GenBank accession AY083590). Agroinoculation was performed on 4-5 weeks old tomato seedlings with *Agrobacterium tumefaciens* strain EHA105 bearing constructs of the associated components of ToLCNDV, described above.

The phenotyping data obtained from the agroinoculation experiment was used for construction of two F_2 bulks with extreme phenotypes namely resistant and susceptible bulks. Resistant parent (VRT78-4), resistant bulk and susceptible bulk were resequenced using Illumina HiSeq 2500 at Center for Excellence in Genomics and System Biology, ICRISAT.

QTL-Seq approach was used to identify the putative genomic regions using the resequencing data. The publicly available gold standard genome assembly of tomato genome version SL3.0 was used as reference. High-quality sequence reads generated from parental accession were mapped onto reference genome using BWA. After aligning sequence reads, the Coval software was used for post processing and filtering of the alignment files. The variants were called between resistant parent (VRT 78-4) and reference genome. Consequently, the reference genome was developed by replacing the detected SNPs with publicly available SL3.0 reference genome. After developing VRT78-4 assembly, the reads from both resistant and susceptible bulks were aligned onto VRT78-4 reference assembly. The variants (SNP index) were then called for both the bulks. Only those SNP positions which passed the criteria of having ΔSNP index of -1 are considered as the putative causal SNPs. ΔSNP index of -1 indicate that the allele called in resistant bulk was same as that of resistant parent while alternate SNP base in susceptible bulk. The preliminary analysis performed showed the presence of genomic region on chromosome 11 defined based on confidence of interval under null hypothesis of no QTLs ($P < 0.05$).

Project 3: National Innovations in Climate Resilient Agriculture (NICRA)

Development of F_1s for high temperature stress tolerance: A total of 165 F_1s were developed during 2017-18 by crossing 15 recipients (female parent), having traits of agronomic superiority with 11 heat tolerant lines as donors (male parent) *viz.* CLN-1621, CLN-2026, EC-538380, EC-620421, EC-620438, H-88-78-1, PR-161-L, PR-168, PR-180-3, PR-323-1 and Superbug, in line x tester mating design. F_1s were sown in last week of January, 2018 in order to evaluate the performance of the hybrids, particularly for fruit set during April-June, 2018.

Development of F_1s tolerant to moisture deficit condition: 152 F_1s were developed utilizing 8 donors, *viz.* C-9-1, D-3-1, E-1-1, EC-521047, EC-620401, EC-620402, EC-620422 and EC-620512, which were earlier identified for their better performance under moisture deficit condition. All the donors (as males) were crossed with 19 recipients (as females) having desirable agronomic traits, in line x tester mating design. Field evaluation of F_1s would be done during September, 2018–March, 2019.

Validation and evaluation of brinjal rootstocks for waterlogging tolerance: Scion of two cultivars of tomato *viz.* Kashi Aman & CLN-2026 were grafted on five eggplant rootstocks *viz.* IC-111056, IC-354557, BR-14, Kashi Sandesh and Surya. About 1100 tomato grafts were prepared by using side (splice) or apical top (cleft) grafting. Grafted plants, 40 days after transplanting were exposed to waterlogging stress for 48 and 72 h during September-October. Observations were recorded on physiological, biochemical and yield traits.

Graft combinations *viz.* BR-14 + CLN-2026 (R+S), IC-111056 + CLN-2026, Surya + CLN-2026, BR-14 + Kashi Aman, IC-111056 + Kashi Aman and IC-354557 + Kashi Aman survived 72 hours of waterlogging situation. Wilting was not prominent and also yellowing was not observed in surviving graft combinations. Optimum/enhanced enzymatic activities (SOD, APX and catalase) along with high osmo-protectant (proline) concentration were recorded in surviving graft combinations. Enhanced level of peroxide radical was observed among non-survivors i.e. non-grafted

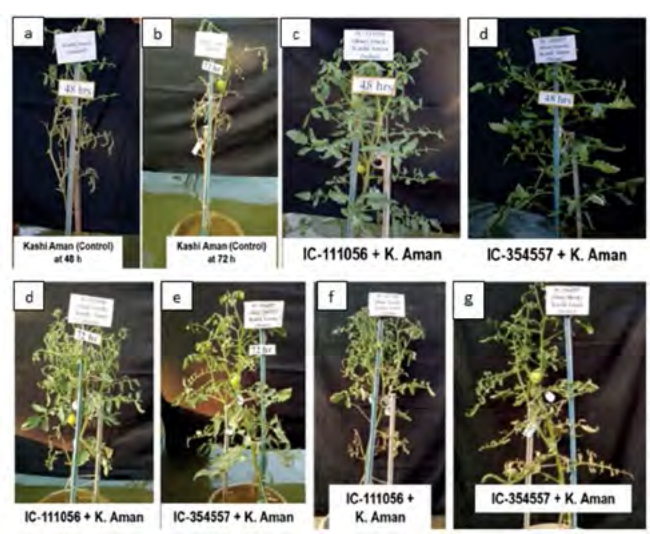


Fig. 4: Non-grafted (control) of Kashi Aman at 48 h and 72 h (a, b); graft combinations of Kashi Aman at 48 h (c, d), 72 h (d, e) and after recovery (f, g)

(control) and grafts combination of IC-354557 + CLN-2026 and Kashi Sandesh + Kashi Aman. In line with earlier findings, brinjal rootstocks IC-111056 and IC-354557 found promising in mitigating waterlogging stress in tomato (Fig. 4).

Stionic effect on yield, quality and physio-biochemical traits in tomato: There were 17 graft combinations prepared using 7 rootstocks viz. IC-354557, IC-111056, Surya, *Solanum torvum*, Kashi Sandesh, BR-14 and Kashi Taru and four scions viz. Kashi Aman, Kashi Chayan, CLN-2026 and CLN-1621. Observations were recorded on yield and related traits, fruit quality traits and physio-biochemical traits.

Highest yield was recorded in IC-354557 + Kashi Aman (6.13 kg/plant) followed by IC-354557 + Kashi Chayan (5.60 kg), IC-111056 + Kashi Aman (5.49 kg), Surya + Kashi Aman (5.41 kg) and IC-111056 + Kashi Chayan (5.40 kg). Highest yield improvement over control (66.67%) was observed when Kashi Chayan was grafted over IC-354557 followed by IC-111056 + Kashi Chayan (60.71%) and IC-354557 + Kashi Aman (38.37%). Differential interaction of same rootstock with different scions was observed for yield *per se*. e.g. IC-111056 and Surya results in yield reduction with scions CLN-2026 and CLN-1621 in contrary to their effect on other scions. A direct effect of grafting on yield was observed through its direct positive impact on fruit/plant, fruit size, fruit weight and fruit/cluster.

Among fruit quality traits level of ascorbic acid and titrable acidity were enhanced in majority of graft combinations, whereas TSS, lycopene and β-carotene were mostly unaffected or declined. Physio-biochemical traits (chlorophyll a, b, total chlorophyll, total carotenoids) were either unaffected or statistically at par or exhibited significantly lower levels in most of the graft combination over non-grafted (control).

Expression analysis of heat shock proteins in tomato high temperature tolerant genotypes: Expression analysis of these genes was carried out in heat tolerant tomato genotypes H88-78-1 and CLN-1621 and a susceptible variety Punjab Chhuhara. Heat stress was given to plants by transferring them to a growth chamber at 42 °C after 25 days of transplanting. The stress was given for a period of 16 h, 32 h and 48 h and leaf samples were collected after each stress treatment. Expression of twenty heat shock proteins was studied. HSF17 showed to be playing major role in heat stress tolerance as its expression was highly induced in all the genotypes, being highest in CLN-1621. Similarly, HSF2, HSF4, HSF16, HSF18, HSF19 and HSF20 were highly expressed in both H88-78-1 and CLN-1621 genotypes indicating their contribution in tolerance to heat stress. Other parameters such as relative water content (RWC), electrolyte leakage, photosynthetic pigments, lipid peroxidation, proline content, hydrogen peroxide (H_2O_2), catalase (CAT), superoxide dismutase (SOD) and ascorbate peroxidase (APX) activity were also studied and in all the aspects H88-78-1 and CLN-1621 genotypes were found superior to Punjab Chhuhara, which supports the heat tolerance trait of these genotypes.

Project 4: CRP on hybrid Technology (Tomato)

Re-evaluation of F_1s for ToLCV tolerance: Forty one F_1 along with 10 private sector developed hybrids and 04 open pollinated varieties were evaluated against ToLCV for confirmation. The F_1s *viz;* VRTH-16-4, VRTH-16-75, VRTH-16-86, VRTH-16-70, VRTH-16-74 and VRTH-16-5, VRTH-17-41, VRTH-17-83, VRTH-17-87, VRTH-17-89 and VRTH-16-14 showed tolerance to ToLCV with high yield.

Evaluation of F_1s under moisture deficit condition: On the basis of yield, physiological, and biochemical attributes, the F_1s VRTH-17-2, VRTH 17-68, VRTH-16-3 and VRTH-17-81 showed ability to give high yield under moisture deficient condition (Fig. 5).

Fig. 5: Performance of selected 4 F_1s in tomato under moisture deficit condition (80 days moisture deficit).

Evaluation of F_1s for salt tolerance: Twenty four F_1s developed using breeding lines EC-620543, EC-620533, EC-620435, EC-620533, Mountfevet, Agata 30, KT-8, EC-520078, EC-520076, Angoorlata, FLA-7171 and Sel-7 were evaluated under salt in two phases. Phase1: All the hybrids were subjected to physiological and biochemical screening for their ability to tolerate 200 mM and 400 mM of salt stress. Phase 2: The selected hybrids were subjected to 400 mM salt stress for confirmation and validation of tolerant hybrids. On the basis of yield traits, physiological and biochemical basis the hybrids VRTH-17-49, VRTH-17-51, VRTH 17-47, VRTH-17-44 and VRTH-17-55 were superior in their performance at 400 mM of salt stress (Fig. 6-7).

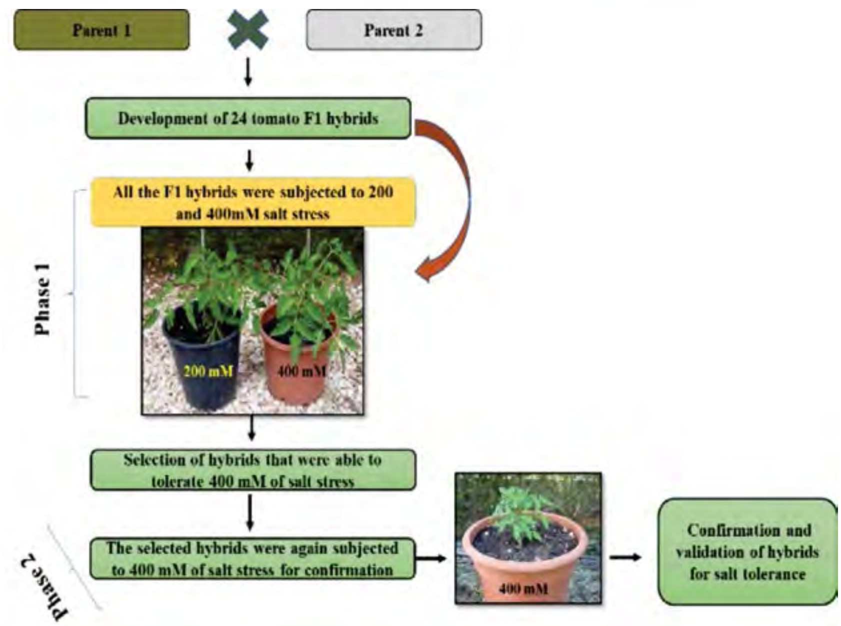


Fig. 6: Physiological and biochemical screening for salt tolerance

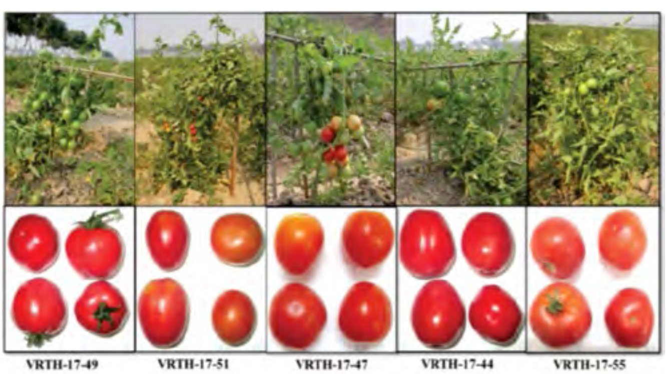


Fig. 7: Promising tomato hybrids for salt tolerance

Screening of F$_1$s against Root knot nematode (*Meloidogyne incognita*) resistance: Six F$_1$s along with susceptible check Kashi Adarsh were screened for root knot nematode (*Meloidogyne incognita)* resistance under artificial condition. The F$_1$s VRTH-17-82, VRTH-17-29, VRTH-17-23, VRTH-17-67, VRTH-17-66 and VRTH-17-01 showed resistance reactions against the nematode.

Project 5: Network Project on Transgenic Crops (NPTC)

Fruit and shoot borer resistant transgenic brinjal –*Cry1Aa3* gene*:* Homozygous T$_6$ generation plants of three c*ry1Aa3* transgenic brinjal (cv. Kashi Taru) events (A2, A3 and A7) developed earlier were grown in a glass house. To advance the generation, flowers of these events were self-pollinated, and T$_7$ generation seeds were harvested from the developed fruits.

Fruit and shoot borer resistant transgenic brinjal – *Cry1Ac* gene: Generation advancement of *Bt*-brinjal lines (Pant Rituraj, Uttara, Punjab Barsati, VR-14, IVBL-9, VR-5, EV-1 and EV-4) with high protein expression and similar to recurrent parent were selected and further selfing was repeated in this season. Plants were again raised for seed multiplication. *Bt*-brinjal seeds were sown in pots in containment proof insect house. After 20 days of germination, six successive kanamycin sprays (200 mg/l) were applied to find any escape of transgenic or low expression on the transgene. All the seedlings survived after kanamycin sprays showing optimum expression of the transgene. Further, the positive plants of each line were transplanted in net house. Selfing was performed on fully grows plants for multiplication and T$_{10}$ seeds of mature selfed fruits from all the six lines were harvested and stored.

Fruit borer resistant transgenic tomato – *Cry1Ac* gene*:* Eight events of transgenic tomato plants *cv*. Kashi Vishesh carrying *Cry1Ac* gene were advanced to T$_{10}$ generation. Seeds of the best events IVTT-5 and all other events were germinated in a glass house. After 30 days of germination, six successive sprays of kanamycin (200 mg/l) were applied to find any escape of transgenic or low expression of the transgene. All the seedlings survived after kanamycin spray showing optimum expression of the transgene. Ten seedlings of each event were transplanted in insect-proof net house, their flowers were self-pollinated and seeds from matured fruits were harvested for further multiplication.

Drought, salt and cold stress tolerance transgenic tomato- *AtDREB1A:* Transgenic tomato lines D41, D53, D76 and D86 expressing *AtDREB1A* gene were advanced to T$_8$ generation. The seeds of all events were germinated in a glass house. After 30 days of germination, screening was done with kanamycin sprays (200 mg/l) to find any escape of transgenic or low expression on the transgene. All the seedlings were survived after kanamycin spray showing optimum expression of the transgene. Eight seedlings of each event were transplanted in insect-proof net house, their flowers were self-pollinated and matured fruits were picked up. Seeds of such fruits were harvested for further multiplication.

Drought, salt and heat stress tolerance transgenic tomato- *BcZAT12:* Drought, salt and high temperature stress tolerance transgenic tomato lines ZT1, ZT5 and ZT6 expressing *BcZAT12* gene were advanced to T$_8$

generation. The seeds of all events were germinated in a glass house. After 30 days of germination, screening with kanamycin sprays (200 mg/l) was done to find any escape of transgenic or low expression on the transgene. All the seedlings were survived after kanamycin spray showing optimum expression of the transgene. Eight seedlings of each event were transplanted in insect-proof net house, their flowers were self-pollinated and matured fruits were picked up. Seeds of such fruits were harvested for further multiplication.

Pyramiding of *AtDREB1A and BcZAT12* transgenes for abiotic stresses: Pyramiding of *AtDREB1A* and *BcZAT12* transgenes were done by crossing both the transgenic lines in a reciprocal manner. The F_3 plants expressing *AtDREB1A* and *BcZAT12* transgenes gene were tested by PCR amplification for both *AtDREB1A* and *BcZAT12* specific primers and scored according to banding patterns. Progenies having both the transgenes were further used for morphological and physiological characterization for generation advancements. At the same time, both the transgenic lines, *AtDREB1A* and *BcZAT12* are being multiplied for generation advancement programme.

Evaluation of *AtDREB1A* transgenic tomatoes for cold stress tolerance: Transgenic tomato plants containing *AtDREB1A* gene under the control of drought specific promoter *rd29A* were evaluated for tolerance to cold stress. The plants were exposed to 4 °C for 5 days and samples were collected. Cold stress resulted in increased reactive oxygen species in both wild-type and transgenic plants. However, increase was less in transgenic plants. Moreover, a simultaneous increase in antioxidants like catalase, superoxide dismutase, reduced ascorbate and total ascorbate managed the oxidative stress caused by ROS as evident by the reduced amount of hydrogen peroxide and superoxide anions. Photosynthetic potential of transgenic plants was slightly affected whereas effect was more on wild-type plants. Transgenic plants showed more adaptability to stress by retaining more relative water content and electrolyte leakage compared to wild-type plants. Under cold stress, the osmotic balance was maintained by higher accumulation of osmoprotectants like cellular proline and total soluble sugars in transgenic crops. The higher expression of stress-responsive genes *P5CS, CAT, SOD* and *LPO* under cold stress supported the higher levels of proline and respective antioxidants. Further, subsequent cold stress at reproductive stage of plants exhibited higher expression of these cold-responsive genes compared to the plants that are exposed to cold stress for the first time. This indicates the signs of stress memory in tomato plants. The plants with stress memory of cold stress were found to respond quickly and more effectively to the recurrent cold stress showing efficient acquisition of stress memory.

Project 6: Evaluation of high yielding varieties/ hybrids of cucurbitaceous vegetables for river bed (*diara* land) cultivation and standardization of their agro-techniques

The experiment was conducted on nine cucurbitaceous vegetables including bottle gourd, bitter gourd, muskmelon, ridge gourd, long melon, sponge gourd, watermelon, cucumber, and pumpkin. The results of different year of experiments were assembled The different varieties were sown in the river bed between 27th October and 5th November. The seeds were soaked in water up to 12 hrs and after that removed from water. The sprouted seeds were sown in the pits. The recommended management practices followed for control the insects. The open pollinated variety Azad Harit and Arka Bahar of bottle gourd, Kalyanpur Baramasi of bitter gourd; Kashi Madhu of muskmelon; Prasad Komal of long melon, Pusa Sneha of sponge gourd; Arka Manik of watermelon; Kashi Harit of pumpkin and Swarna Manjari of ridge gourd were found promising under river bed condition. The hybrid Sarita of bottle gourd; Prachi of bitter gourd; Muskan of muskmelon; Alok of sponge gourd; Malini of cucumber, NS-750 of watermelon and NS 471 of ridge gourd were recommended for cultivation under river bed condition. The nutritional requirements of cucurbitaceous vegetable crops for river bed conditions were standardized. The application of FYM (1 kg) + NPK (25:50:50 g) in the form of urea, DAP per pits and MOP and spray of liquid fertilizers (19:19:19) 4 times @ 3-5 g/liter were gave maximum yield in all cucurbitaceous crops. The sowing of 2 seeds per pit was recommended instead of 6-8 seeds/pit as practiced by farmers. The transplanting of seedlings in the month of December and in January was found better for all crops and no significant yield difference was recorded as compare to direct sown crop. The transplanting of cucurbits in river bed condition needs watering at early stage (7-10) for better establishment. The sowing of 100-150 plants per hectare sunflower plants has been recommended. This practice attracts the pollinating agents results overall increase in yield. The infestation of nematode has been identified as major problem, which causes major yield loss. Use of neem cake 1 kg/ pit at was found most effective to manage the nematode in riverbed condition. One day training cum awareness programme was conducted on Riverbed Cultivation of Cucurbitaceous Vegetables on 26 July, 2017 in which 64 farmers participated.

Project 7: CRP on Agrobiodiversity

Okra: Screening of 532 okra accessions for YVMV and OELCV diseases were done in *Kharif* -2017. Two accessions IC009856-B and IC022237-X did not produce fruits, 6 genotypes (EC133336, EC169384, EC305648, IC140914, IC385770 and EC169147) were found moderately resistant to both YVMV and OELCV diseases, one genotype (IC-506056) was identified as round fruited; IC043796-X, IC344687-X, IC043742-X, IC551791 and IC331047 were red fruited and twenty nine accessions (IC306741-A, IC506119, IC510692, IC510758, IC026862, IC033344, IC026862, IC040929, IC090183, IC090216, IC090227, EC305672-2, EC305736-1,EC305745, EC305749-A, EC306731-P, EC306741-A, EC306741-B, IC018532, IC033350-A, IC264736, IC52273, IC588197, IC600844, IC601315, IC506035, IC506041, IC506042, IC506045) were identified as *Abelmoschus caillei*. Accessions IC112504, IC201143, and IC411880 were bush–type; IC027881, IC042524 were thin fruited and IC169511, IC169511, IC282241, IC325925, IC394158, IC604948, IC117015, IC427560, IC506120, IC117093 and IC427732beared fruits with eight ridges.

Project 8: Central Sector Scheme for Protection of Plant Varieties and Farmers' Rights Authority (DUS Testing of tomato, brinjal, okra, cauliflower, cabbage, vegetable pea, French bean, bottle gourd, bitter gourd, pumpkin and cucumber)

Maintenance of reference varieties: Reference of varieties of tomato, okra, brinjal, cauliflower, cabbage, vegetable pea, French bean, bottle gourd, bitter gourd, pumpkin, cucumber and pointed gourd were collected from different ICAR institute and SAUs and maintained. All the varieties of these crops were sown in randomized block design with 3 replications. The details of varieties of these crops and the number of morphological traits are presented in Table 2.

Table 2: Details of reference varieties and their morphological traits

Crops	Reference varieties	Morphological traits
Tomato	95	47
Cauliflower	05	28
Cabbage	04	28
Brinjal	77	47
Vegetable pea	42	20
French bean	27	22
Okra	42	31
Cucumber	28	35
Bitter gourd	25	31
Bottle gourd	31	31
Pumpkin	23	30
Pointed gourd	12	15
Total	**411**	

DUS Testing of vegetable crops: Nineteen okra, 58 brinjal, 4 cauliflower, 9 cabbage, 20 tomato, 8 bottle gourd, 2 bitter gourd, 15 cucumber and 12 pumpkin entries were evaluated under DUS Testing along with reference varieties (Table 3).

Table 3: DUS testing vegetable crops

Type of variety	New Entries					
	1st year	2nd year	VCK	FV	Total	Date of Monitoring
Okra	6	2	4	7	19	07.10.2017
Brinjal	-	7	-	51	58	06.01.2018
Cauliflower	-	4	-	-	4	06.01.2018
Cabbage	2	7	-	-	9	07.02.2018
Tomato	9	2	-	9	20	07.02.2018
Bottle gourd	-	-	1	7	8	09.05.2017
Bitter gourd	-	-	-	2	2	09.05.2017
Cucumber	-	7	-	8	15	09.05.2017
Pumpkin	-	-	-	12	12	09.05.2017
Total	**17**	**29**	**5**	**96**	**147**	

Project 9: Agri-Business Incubator (ABI) -IIVR, Varanasi

To facilitate technology commercialization, development of agri-entrepreneurships and to provide Human Resource Development support for empowering entrepreneurs through training for industry oriented vocations, an ABI unit has been established by the Council under NAIF at ICAR-IIVR, Varanasi. This unit was instrumental in showcasing the technologies of

Fig. 8: Showcasing the technologies of the Institute for commercialization in National Seminar at Sunbeam College for Women, Varanasi during 8-9 September, 2017

the Institute for commercialization (Fig. 8) at National Seminar on "Water and soil management for agriculture and livelihood security under climate change" organized at Sunbeam College for Women, Varanasi during 8-9 September, 2017. One day awareness-cum-training programme on Entrepreneurship Development in Vegetable Seeds was organized on 13th November, 2017 at the institute (Fig. 9), which was attended by 73 students pursuing graduation programme in agriculture at Government Degree College, Jakhini, Varanasi.

Two Entrepreneurship development program on Vegetables were also organized during the year. The first one was organized on 26th December, 2017 (Fig.10) for 38 professionals from Agri-business Agri-clinic of Ghazipur. Another EDP on Vegetables was organized on 23rd March, 2018 (Fig.11) for 39 Post-graduate students from Chandra Shekhar Azad University & Technology, Kanpur from various disciplines of agriculture.

Towards the commercialization of IIVR technologies, 14 technology commercialization license agreements were executed during this period as a result of the efforts undertaken in this direction by ABI unit. Overall, revenue of Rs. 19.61 lakhs was generated by the unit during this year.

Fig. 9: Awareness cum Training Programme on Entrepreneurship Development in Vegetable organized on November 13, 2017

Fig. 10: Entrepreneurship Development in Vegetables organized on December 26, 2017

Fig. 11: Entrepreneurship Development in Vegetables organized on March 23, 2018

Project 10: Zonal Technology Management Unit-IIVR, Varanasi

To help ITMUs of the zone in commercialization of technologies, showcasing of technologies, management of IP portfolio, helping in IPR related issues and to serve as a link between IPTM unit of the Council and ITMUs of the zone, a Zonal Technology Management Unit has been established by the Council under NAIF at ICAR-IIVR, Varanasi. The unit has eleven different ICAR Institutes under its umbrella *viz*. ICAR-Central Institute of Arid Horticulture, Bikaner; ICAR-Central Institute of Sub-Tropical Horticulture, Lucknow; ICAR-Central Institute of Temperate Horticulture, Srinagar; ICAR- Central Potato Research Institute, Shimla; ICAR-Directorate of Medicinal and Aromatic Plants Research, Anand; ICAR-Directorate of Mushroom Research, Solan; ICAR-National Research Centre for Litchi, Muzaffarpur; ICAR-National Research Centre on Orchids, Pakyong, Sikkim; ICAR-National Research Centre on Seed Spices, Ajmer, ICAR- Central Island Agricultural Research Institute, Port Blair and ICAR-Indian Institute of Vegetable Research, Varanasi.

The reports from all the ITMUs in domain on management of IP portfolio, commercialization of technologies, outreach activities, capacity building in IP Management and training/workshop/seminar etc. organized was compiled and sent to IPTM unit of the Council on a regular basis.

This unit was instrumental in organizing a market sensitization programme for okra varieties/ hybrids and promising lines on 7th October 2017 (Fig. 12) and for varieties/hybrids and promising lines of solanaceous vegetables on 6th Jan 2018 (Fig. 13) for promoting the commercialization of ICAR-IIVR technologies. The okra day program was attended by 39 representatives from 25 leading private sector seed companies. The participants thoroughly observed the promising genotypes and expressed their willingness to get the promising advanced breeding lines having high degree of resistance to YVMV and OELCV, high potential and good fruit quality. The delegates had a fruitful interaction with the breeders of the institute and

Fig. 12: Okra Field Day organized on 7th October, 2017 at ICAR-IIVR, Varanasi

provided critical and valuable feedback on the present market needs of okra.

The solanaceous day programme was attended by 25 representatives, which included breeders and marketing strategists from of 11 private sector seed companies dealing in vegetable seeds. The representatives from seed-companies visited the field of the Institute and appreciated the performance of different varieties, hybrids, advanced breeding lines and pre-breeding lines of solanaceous vegetables developed by the Institute.

Fig. 13 : Solanaceous Day organized on 6th January, 2018 at ICAR-IIVR, Varanasi

Project 11: Cowpea Golden Mosaic Disease (CPGMD) resistance: Agroinfectious clone development, screening, genetics of inheritance, molecular tagging and mapping for CPGMD resistant genes in cowpea by using linked markers

Agroinfectious clones development for mungbean yellow mosaic India virus causing cowpea golden mosaic disease: Total DNA extracted from the golden mosaic symptomatic samples of cowpea collected from ICAR-IIVR farm was subjected to PCR assay using universal begomovirus specific primer pair. Result showed the symptomatic samples were found associated with begomovirus Also they were tested for the association of α- and β- satellites using universal primer pairs. Results revealed that, there were no association of α- and β- satellites. Further, for the development of infectious clone, DNA was subjected to RCA and RFLP analysis. RCA restricted product resolved on gel showed ~2.7kb fragment with *Bam*HI, *Hind*III and *Kpn*I enzyme. Fragments of *Bam*HI and *Kpn*I were cloned in pBluescriptK+ vector. Based on restriction analysis selective clones were subjected for partial sequencing. Sequencing results revealed that, *Bam*HI and *Kpn*I clones are found to be DNA B of mungbean yellow mosaic India virus (MYMIV) and one clone of *Kpn*I is found to be DNA of MYMIV. Hence, selected clone of *Bam*HI and *Kpn*I were sequenced to obtain the complete sequence. Both DNA A and DNA B sequences had 99% sequence identity with the MYMIV isolate reported from the Varanasi region. In phylogenetic analysis, both DNA A and DNA B were grouped with the previously reported isolates of MYMIV from Varanasi region (Fig. 14-15). After sequencing for monomer construction of DNA B, *Hind*III site has been chosen and restricted product (*BamH*I+ *Hind*III) has been sub-cloned in the binary vector pCAMBIA. Around 50 colonies were screened for the insert of ~1.4kb and the one colony was selected for further tandem dimer construction. Around 40 colonies were screened for the dimer in which only one colony was positive for the tandem dimer based on the restriction analysis. Same colony has been subjected to plasmid isolation and extracted plasmids were transformed into Agrobcterium strain GV3101 and the infectious clone has been produced. Further, it was confirmed with the PCR analysis using universal primer pair and restriction analysis. Tandem dimer construction for DNA A is under progress with the *Kpn*I clone.

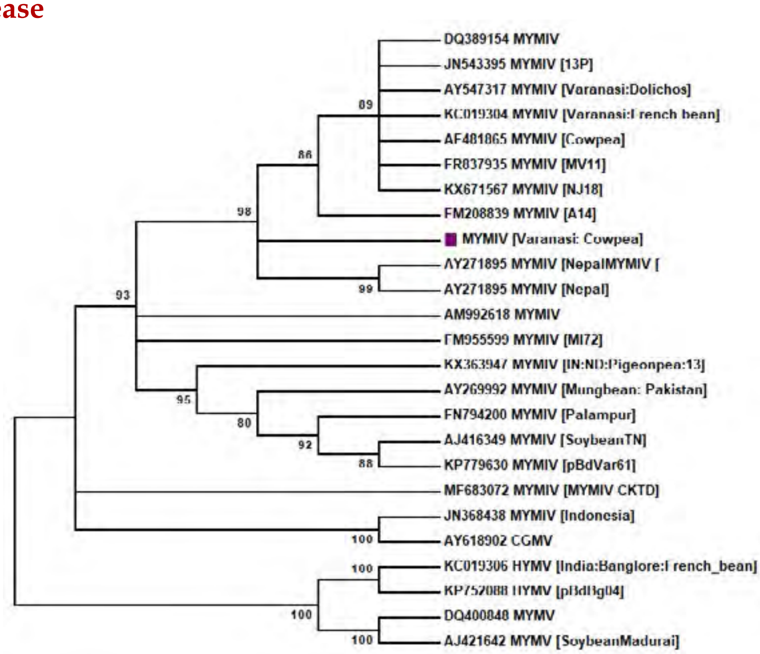


Fig. 14. Phylogenetic analysis of DNA A component of MYMIV infecting cowpea crop

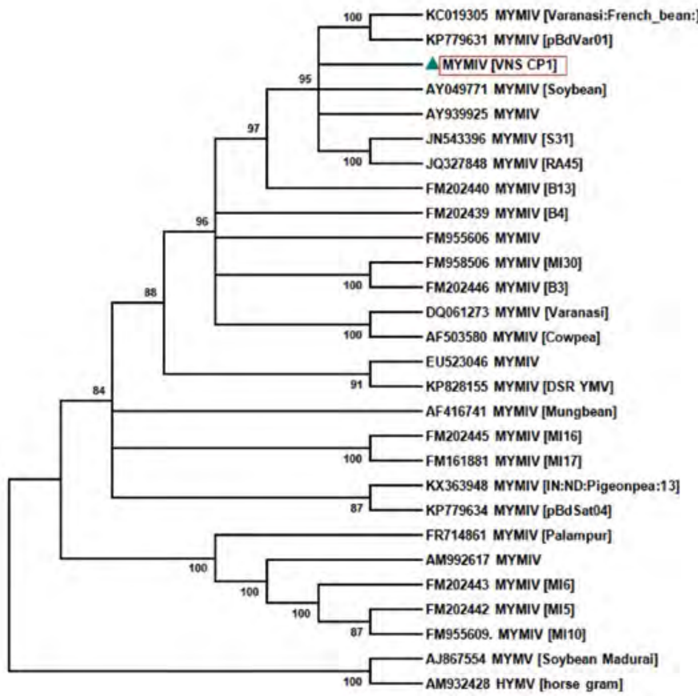


Fig. 15: Phylogenetic analysis of DNA B component of MYMIV infecting cowpea crop

Project 12: A total value chain on commercialization of value added convenience vegetable products

Development of instant protein rich tomato soup mix

Drying of tomato: The blanched whole tomatoes for 100°C for 2 min are sliced into thin slices. Tomato slices are placed in gelatinized starch for 10 min at 40°C. Subsequently tomato slices are dried in cabinet dryer at 50-55°C to reduce the final moisture to 1-2%.

Formulation of instant protein rich tomato soup mix: The formulation of instant protein rich tomato soup consisted of dry blending of 15-20% WPC-70, 10-15% corn flour, 15-20% modified starch, 2-5% onion powder, 0.5-2.0% garlic powder, 20-30% tomato powder, 4-6% table salt, 0.5-1.5% black pepper, 0.5-1.0% cumin powder, 4-6% flavour enhancer, 50-100 ppm red colour, 3-8% sugar, 4-8% black salt and 0.5-1.5% dextrin white (Fig. 16).

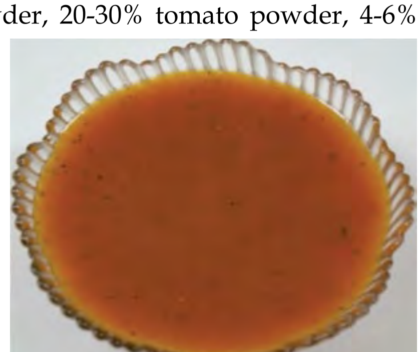


Fig. 16: Protein rich tomato soup mix

Physico-chemical properties of instant protein rich tomato soup powder: Moisture content in protein rich tomato soup mix varied from 3.5-4.53%. Loose and packed bulk density ranged 0.40-0.42 g/cc and 0.70-0.74 g/cc, respectively. Insolubility index and hydroxyl methyl furfural 30.5-35 ml and 37.90-42.12 µ mol/L, respectively in tomato soup mix. During storage, moisture, loose and packed bulk density, insolubility index, and hydroxyl methyl furfural increased during storage both at 10 and 25°C. However, the increase was more after months of storage at 25°C as compared to the storage of 6 months at 10°C.

Sensory score of instant protein rich tomato soup mix: Reconstituted tomato soup exhibited good consumer acceptability. Sensory score for flavour, colour and appearance, consistency and overall acceptability decreased during storage. Reconstituted tomato soup was acceptable to the judges after 6 months of storage at 10°C while the judges observed acceptability of tomato soup up to 4 months of storage at 25°C.

Project 13: Efficient Water Management in Horticultural Crops under Agri-CRP on Water

Studies were conducted on vegetable based cropping sequences and crop geometry under micro-irrigation and fertigation. Three cropping sequences, i) Cowpea-summer squash- amaranth, ii) Okra- sweet pepper- baby corn, iii) Okra-pea- bitter gourd were taken under the study. The studies on crop geometry for standardization under micro-irrigation were conducted with single, two, three, and four plants to be irrigated by each emitter of different laterals. Crops were sown on raised bed and applied with 60%, 80%, 100% and 120% recommended dose of NPK for each lateral row maintained with 1, 2, 3 and 4 plant geometry. Water was applied to the crops as per irrigation scheduling based on the crop water requirement.

Summer squash: Maximum summer squash yield of 42.53 t/ha was obtained with 3-plant crop geometry configuration and 120% NPK drip fertigation against minimum 13.7 t/ha under control. Maximum and minimum WUE of 2.04 t/ha-cm and 0.35 t/ha-cm were

Fig. 17: Summer squash crop with 3-plant geometry and 120% NPK fertigation

recorded in above, respectively. Yield enhanced 46-89% with 60-120% fertigation over control and 44-80% with 2, 3 & 4-plants geometry configuration over single plant geometry (Fig. 17).

Amaranths: Maximum yield of amaranths 20.14 t/ha was realized with 4-plant geometry configuration and 120% NPK fertigation, and minimum 7.5 t/ha under control. Yield and water use efficiency of amaranth enhanced by 24.4-56.3% and 24.4-45.4%, respectively with 2 to 4 times plant populations. Increase in yield and WUE were, respectively, 15.3-84.6% and >2 times over control with drip fertigation of 60-120% NPK

Cowpea: Maximum cowpea yield of 12.5 t/ha was realized with 4-plant geometry configuration and 120% NPK fertigation, and minimum 2.73 t/ha under control. Yield enhanced by 35.3-146.4% with 60-120% fertigation over control, and 54-118% with number of 2, 3 and 4-plants over single plant geometry configuration.

Sweet pepper: Sweet pepper with 3-plant geometry configuration at 120% NPK realized maximum yield (35.8 t/ha) and WUE (0.085 t/ha-cm), whereas the minimum yield (11.55 t/ha) and WUE (0.012 t/ha-cm) was reported under control. Yield enhanced by 42.9-90.6% with 2, 3 and 4-plants over single plant geometry and 34.7-95% with fertigation. WUE enhanced by 59.1-71.4% with 3-plants geometry configuration over control.

Baby corn: Maximum baby corn yield (24.7 t/ha) was achieved with 3-plant geometry and 120% NPK fertigation and minimum 8.83 t/ha under control. The yield was lower at 4-plant geometry configuration as compared to 2 and 3-plant geometry configuration. Increase in yield and water use efficiency was 4.6-79.5% and 26.1-61.9%, respectively with 2, 3 and 4-plants geometry over single-plant geometry.

Okra: The maximum yield of okra 15.3 t/ha was obtained with 3-plant geometry configuration and 120% NPK fertigation while minimum was 8.02 t/ha under control. Increase in yield and WUE was 8-52% and 1.36-2.52 times, respectively with fertigation over control. Increase in yield and WUE 8.1-36.3% and 25.3-36.3%, respectively with 2 to 4-plants geometry configuration over single plant geometry.

Pea: The maximum yield of pea 10.8 t/ha was recorded with 3-plant geometry configuration under 120% NPK fertigation over control (3.56 t/ha). Increase in yield to the tune of 16-79% over control was recorded under drip fertigation. Plant geometry configuration of 2-4 plants realized 21.8-76.5% increase in yield over one plant configuration. Maximum WUE of pea was found to be 0.597 t/ha with 3 plant configuration and 120% NPK against 0.111 t/ha with one plant geometry configuration under control (Fig. 18).

Fig. 18: Pea crop under 3-plant geometry

Bitter gourd: The yield of bitter gourd was found maximum (12.9 t/ha) with 2-plant geometry at 120% NPK and minimum (6.77 t/ha) with control. Increase in yield and WUE were 18.5-56.7% and 1.11-1.45 times with fertigation over control. The yield and WUE enhanced up to 38.4% and 21.6-38.4%, respectively with enhanced plants geometry configuration.

Equivalent yield: Cowpea equivalent yield of all three vegetable crop sequences was calculated based on the yield and the average selling price of the vegetables in the vicinity (Table 4). It is evident that crop sequence-II i.e. okra- sweet pepper- baby corn appears to be the superior as cowpea equivalent yield was maximum (145.2 t/ha) among all sequences, and was 3.8 times

Table 4: Cowpea equivalent yield of crop sequences

Crop Sequence	Name of crops	Duration (days)	Max Yield, t/ha	Control yield, t/ha	Rate, Rs/ kg
Crop sequence-I Cowpea - summer squash- amaranth	Summer squash	100	42.53	13.7	23
	Amaranth	90	20.14	7.5	8
	Cowpea	95	12.5	2.7	17
Cowpea equivalent yield for crop sequence -I		285	79.5	18.3	
Crop sequence-II Okra- sweet pepper - baby corn	Sweet pepper	145	35.8	11.55	34
	Baby corn	90	24.7	8.83	42
	Okra	110	15.3	8.02	14
Cowpea equivalent yield for crop sequence -II		345	145.2	38.1	
Crop sequence-III Okra-pea-bitter gourd	Pea	100	10.2	3.56	18
	Bitter gourd	115	12.9	6.67	22
	Okra	110	14.7	8.51	14
Cowpea equivalent yield for crop sequence -III		325	40.2	14.3	

higher than that of control in that sequence. Under this sequence the irrigation system and land was under maximum use for 345 days. The crop sequence-I i.e. cowpea-summer squash-amaranth was found to be the second best with equivalent yield of 79.5 t/ha and enhancement of maximum 4.3 times during total 285 days. The crop sequence-II was found the best in terms of water use efficiency 1.528 t/ha-cm followed by crop sequence-I (Fig. 19).

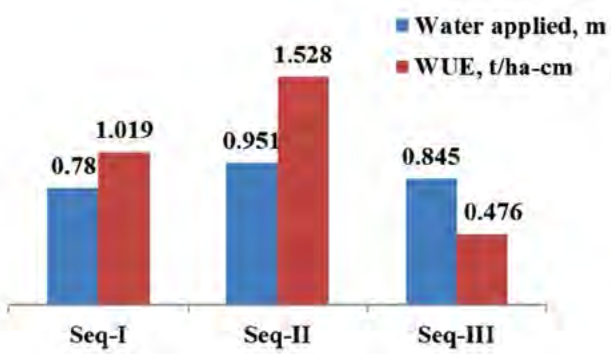


Fig. 19 Water use efficiency for different crop sequences

Project15: Farmer FIRST Program on "Intervention of Improved Agricultural Technologies for Livelihood and Nutritional Security Adhering Local Resources and Working Knowledge of the Farmers"

The present project is operated in 5 villages of Araziline block of Varanasi district of U.P. namely Upadhaypur, Baburam Ka Pura, Paniyara, Dhanapur and Laskariya with different interventions under Horticulture, Crop, IFS and Enterprise based modules.

Horticulture based Module

Under Horticulture based modules focus was given on promotion of improved vegetables varieties through demonstrations and comparing with existing cultivar in both quality and yield. YVMV and OLCV resistant okra variety Kashi Kranti was demonstrated at 186 farmers' field in an area of 26.7 ha fetched an average yield of 127.8 qt/ha which was 21.3% more than local practicing cultivar. In root crops, radish (Kashi Hans) and carrot (Kashi Arun) were demonstrated in an area of 5.0 ha each in Upadhaypur, Paniyara, Dhanapur and Laskariya villages of Araziline block of Varanasi district in Uttar Pradesh. These short duration varieties of radish and carrot with quality produced was heartily accepted by the growers as it fetched an average yield of more than 400 q/ha (Fig. 20 & 21).

Fig. 20: Ms. Roopkala Patel, young progressive grower from Dhanapur felicitated by Hon'ble Sh. Radha Mohan Singh, Union Minister of A&FW, GOI

Fig. 21: Demonstration of okra cv. Kashi Kranti in Laskariya village

Similarly, in Solaneceous crops, TLCV resistant tomato variety Kashi Aman was successfully demonstrated at 115 farmers' field in an area of 20.0 ha in Paniyara, Dhanapur, Babu Ram Ka Pura and Laskariya villages of Araziline block in Varanasi. The average fruit weight of this demonstrated variety was 89 g with a yield of 487.3 q/ha which is better than the hybrids grown by the farmers in the village as responded by the beneficiaries. High yielding brinjal variety Kashi Uttam was demonstrated at 190 farmers' field in an area of 20.0 ha fetched purple, medium size, round shape fruits with an average yield of 363.7 q/ha and maximum of 410 q/ha. Farmers of the villages said that fruit of Kashi Uttam is as good as hybrids they were practicing. Chilli variety Kashi Anmol was demonstrated in an area of 20.0 ha during kharif season fetched an average yield of 187.3 qt/ha within 130 days of transplanting with first picking after 65 days of transplanting. Later, in December farmers uprooted the chilli plant as advised by our FFP team and sowed wheat in the same plot which fetched them higher return as compared to sole chilli crop traditionally practiced by the villagers.

Further in cucurbits, improved open pollinated high yielding variety of bottle gourd cv. Kashi Ganga was demonstrated in an area of 16.7 ha in selected villages. The farmers harvested 18-20 fruits per plant with an average fruit weight of 863.2 gm. and a yield of 498 q/ha. The farmers were happy with the demonstrations and also 13 farmers produced the seeds of demonstrated variety Kashi Ganga. Pumpkin cv. Kashi Harit was demonstrated in an area of 10.0 ha which created a new interest among farmers for growing in larger areas due to its small fruit size with average fruit weight of 2.2 kg and high yield (average demonstrated yield 334.3 q/ha) in a short period of 65-70 days.

In legumenous crops, cowpea cv. Kashi Nidhi resistant to golden mosaic virus and Pseudocercospora cruenta was demonstrated at 141 farmers' field in an area of 20.0 ha area fetched 25-30 cm long green fruits pod with an average yield of 141.3 q/ha. Vegetable pea namely Kashi Nandini and Kashi Udai were demonstrated in an area of 9.4 ha and 20.0 ha respectively during rabi 2017. The sowing of crop was made during 25 October to 05 November and fetched an average yield of 94.7 q/ha (Kashi Nandini) and 102.1 q/ha (Kashi Udai). The farmers were very happy with the demonstrations as Kashi Nandini variety reached in the market by the 20th December fetching higher price upto Rs. 70-80 per kg. French bean was not grown by the farmers of selected villages, hence this crop was introduced and Kashi Rajhans variety was demonstrated at 30 farmers' field in Dhanapur and Paniyara village of Varanasi in an area of 0.4 ha. The crop was line sown by 30-31 October 2017. Plants were determinate (70 cm long) with round, dark green fleshy pod of average length 14.7 cm. Farmers were excited with the average yield of 121.3 q/ha within the cropping period of 4 months and a B-C ratio of 3.63 (Fig. 22).

Fig. 22: Demonstration of Cowpea cv. Kashi Nidhi in Dhanapur village

Crop based Module

Rice-wheat is a major cropping system followed by the farmers in the selected villages under FFP but, one of the major constraints that farmers are facing is quality seeds of paddy and wheat as majority of them are continuing with their own seeds. In this circumstances as discussed in the site committee meeting, medium duration of paddy (CSR 43 & HUR 917) and high yielding wheat (HD 2967) (Fig. 23) were demonstrated in an area of 50.0 ha each. The demonstrations of paddy were liked by the farmers in double transplant method as the average yield of CSR-43 was recorded 41.8 q/ha and in HUR-917 32.6 q/ha. Whereas, the demonstrations of wheat varieties showed a smile on farmers' face as they harvested the yield upto 51.7 q/ha with an average yield of 46.3 q/ha. Lentil variety HUL-57 were also demonstrated in an area of 6.5 ha with an average yield of 16.64 q/ha harvested by the farmers.

Fig. 23: Harvesting of demonstrated wheat cv. HD-2967 in Upadhyaypur village

IFS based Module

Some of the farmers in the selected villages use to protect nursery by putting green shade on bamboo placed at 3-4 feet height but in this process they felt difficulty in weeding and other agronomic practices. This technology was further modified by FFP team and 65 demonstrations of bamboo made shade-net house (50 sq meter area) were conducted which not only helped them in raising healthy vegetable nursery with all agronomic practices but also helped them in producing quality produce of capsicum, cucumber, indeterminate type tomato etc. Considering the importance of nutritional security in rural areas, promotion of kitchen garden by providing quality seeds of tomato, brinjal, chilli, sem, radish, palak, carrot, cowpea, okra, cucumber, bottle gourd, sponge gourd and pumpkin for an area of 150 sqm. were given to 1095 farm families in all selected villages under Farmers FIRST Programme (Fig. 24).

Fig. 24: Demonstration of capsicum in shade-net house at Dhanapur village

Enterprise based Module

Farmers in the selected villages use to store their vegetable seeds in simple clothes or jar which resulted in poor germination of seeds when used and hence, in every season farmers had to purchase vegetable seeds from retail shops of villages. In this direction FFP team after interaction with growers demonstrated low cost seed storage technologies ie., storing seeds with zeolite beads in air tight plastic container. This technology was given to 150 small farmers of Paniyara, Dhanapur and Laskariya villages where farmers started storing their seeds in the given container along with zeolite beads.

Apart from the demonstrations, 17 Farmers-Scientists interaction programme were organized in selected 5 villages namely Upadhaypur, Baburam Ka Pura, Paniyara, Dhanapur and Laskariya village of Araziline block of Varanasi district of Uttar Pradesh during different cropping season in which apart from farmers and farm women, Gram Pradhan of concerned villages had participated and discussed the needs and performance of demonstrated interventions under the objectives of FFP.

Project 16: NHB Project on "Promotion of Vegetables for Nutritional Security in Eastern Uttar Pradesh including East Champaran, Bihar"

Vegetables are not only rich in minerals and vitamins but also; contribute in a big way in maintaining health, overcoming hunger and malnutrition. Among the rural community, their consumption is very low due to lack of purchasing power, ignorance and other factors including unavailability. Promotion of kitchen garden by demonstrations of different vegetable crops for gardening in a systematic manner in small pieces of land available in households was one of the key objectives of this project for nutritional security. The efforts were made by providing 15000 kitchen garden packets consisting of quality seasonal vegetable seeds to rural households during different seasons in 40 villages of 06 districts in Uttar Pradesh and East Champaran in Bihar to ensure healthy diet with adequate macro and micronutrients at doorstep (Fig. 25).

Fig. 25 : Distribution of kitchen garden packets to farmers by Hon'ble Shri Radha Mohan Singh, Union minister of A & FW GOI.

Project 17: Scheduled Tribes Component (Earlier Tribal Sub Plan) for Tribals of Sonbhadra district in Uttar Pradesh (National Assignment by Department of Agricultural Research & Education, Ministry of Agriculture and Farmers Welfare, Govt. of India)

An integrated agricultural approach was followed to improve the livelihood and nutritional status of 1512 tribal households of Chopan block in Sonbhadra districts, adopted under Scheduled Tribes Component of DARE, Ministry of Agriculture and Farmers Welfare, Govt. of India. Considering vegetables as an important aspect for nutritional and economic security, demonstrations of cucurbits viz., bottle gourd (Kashi Ganga); sponge gourd (Kashi Divya); pumpkin (Kashi Harit); bitter gourd (Kalyanpur Baramasi) and Ash gourd (Kashi Dhawal) were conducted at 713 tribals' field in an area of 39.9 ha, which fetched an average increase of 41.3% quality yield compared to local cultivar. Demonstrations of tomato (Kashi Aman); brinjal (Kashi Uttam) and chilli (Kashi Anmol & Kashi Tej) were conducted at 544 tribals' field in an area of 21.2 ha, 20.0 ha and 22.5 ha, respectively which showed an increase in yield upto 48% compare to local cultivar with much superior fruit quality. Demonstrations of legume vegetables *viz.,* cowpea (Kashi Kanchan); sem

(Kashi Haritima); pea (Kashi Udai & Kashi Nandini) and french bean (Kashi Rajhans) were conducted at 503 tribals' field in an area of 30.7 ha which fetched an average increase of 34.8% quality yield compared to local cultivar. Demonstrations of root crop and leafy vegetables viz., radish (Kashi Hans); carrot (Kashi Arun) and Palak (All Green) were conducted at 431 tribals' field in an area of 8.3 ha which showed an increase of 44.7% quality yield in radish compared to local cultivar (Fig. 26).

Fig. 26 : Crop at farmers' field

Demonstrations of wheat (HD 2967) and lentil (HUL 57) were conducted at 824 tribals' field in an area of 52.0 ha which fetched an average increase of 37.4% quality yield compare to local cultivar in wheat. For ushering nutritional and livelihood security, 3200 kitchen garden packets containing seeds of tomato, brinjal, chilli, cowpea, radish, okra and cucurbits were provided to 1512 tribal households as well as neighboring tribal during different cropping season in 2017-18.

Further, 11 on/off campus training were organized on different aspects of agriculture, which were attended by 1463 tribal. Apart from exposure visit of 1200 tribal to North Zone Regional Agriculture Fair, Varanasi on 23-25 February 2018, a Tribal Farmers Fair cum Gosthi was organized during 28-29 March 2018 at Kota Gram Panchyat, Sonbhadra where more than 3000 tribal from selected villages as well as adjoining villages had participated. Knapsack sprayer and vermicompost making bag were also provided to 216 tribal households after adequate training (Fig. 27).

Fig. 27: Distribution of Knapsack sprayer and vermi compost making bags to farmers' by the Director, ICAR-IIVR, Varanasi

Project 18: Synthesis and validation and sustainable and adaptable IPM technologies for cucurbitaceous vegetables

The following IPM technology for bottle gourd were synthesized and validated in the selected villages of Varanasi, Deoria and Mirzapur districts of Uttar Pradesh for the management of its insect pests and diseases was as followed –seed treatment with *Trichoderma* @ 5g/kg of seed; need based spraying of neem @ 5 ml/l; installation of cue lure traps for fruit flies for wider area management @ 10/acre, cue lure trap @ 25-30/ha; raking of soil for exposing fruit fly to sunlight and predatory fauna; need based application of insecticides like *Bt* @ 2 g/l against white plume moth, *Sphenerches caffer* in bottle gourd; need based spraying of Imidacloprid @ 1 ml/3 l or Thiamethoxam @ 1 g/3 l of water against mirid bugs; need based Cymoxanil+Mancozeb against downy mildew and *Cercospora* leaf spot. In bottle gourd, IPM adopted fields suffered lowest fruit fly damage as compared to non-IPM fields where only farmers' practices were followed. Fruit damages by fruit fly were 5.57 per cent which were significantly lower than the non-IPM fields (14.18 %). Same trend was also observed in case of white plume moth. Only 4.42 *S. caffer* larvae per plant were recorded in IPM adopted fields where the corresponding value for non-IPM fields was 7.76 larvae per plant. IPM adopted fields also harboured minimum mirid bugs population (4.90/ fruit) than the non-IPM fields (12.75) (Table 5).

Table 5: Pest incidence in bottle gourd in IPM and non-IPM field (farmers' practice)

Crop	IPM field	Non-IPM field
Fruit fly (%)	5.57	14.18
Sphenarches caffer / plant	4.42	7.76
Mirid bugs / fruit	4.90	12.75

In contrast, natural enemy population recorded highest in number on IPM adopted fields as compared to farmers' practices. Number of spiders and braconid parasitoid (*Apanteles paludicole*) populations were higher (4.38, 4.05 numbers per plant, respectively) in IPM adopted fields than the non-IPM fields (1.13, 0.94) (Table 6).

Table 6: Natural enemies population (per plant) in IPM and non-IPM field

Predators	IPM field	Non-IPM field
Spiders	4.38	1.13
Apanteles paludicole	4.05	0.94

Two important diseases *viz*., Downy mildew and *Cercospora* leaf spot were recorded during the cropping season. In IPM plot, both the disease incidences were less compared to non-IPM plots. IPM plots recorded only 9.4% of downy mildew compared to non-IPM plots (27.5%) throughout the crop growth. Similarly, only 5.2% leaf spot disease incidence was recorded in IPM plots compared to non-IPM plots where the 18.5% disease incidence was noted. Crop duration also was more in IPM plots than non-IPM plots giving higher B:C ratio (1:2.41) in IPM practices than to the non-IPM practices (1:1.11) (Table 7 & Fig. 28).

Table 7: Occurrence of downy mildew and *Cercospora* leaf spot in bottle gourd (per cent disease index)

Disease	IPM field	Non-IPM field
Downy mildew	9.4	27.5
Cercospora leaf spot	5.2	18.5

Fig. 28: Demonstration of IPM at farmer's field at Varanasi and Mirzapur

Nematological studies revealed that the regions are mostly infected with root knot nematode and reniform nematode on cucurbitaceous vegetable crops.

Project 19: Development and validation of effective formulation(s) of plant growth promoting rhizobacteria (PGPR) having multicide mechanisms for pest management in vegetables

Sequencing and sequence analysis: The PCR products were sequenced from the M/s. SciGenome Labs Pvt. Ltd, Cochin, Kerala. The 16s rDNA homology searches were performed using the BLASTn program

Table 8. List of identified PGPR isolates along with their NCBI Accession number

Isolates	Organism	NCBI accession no.
Chilli isolates		
CRB1	*Bacillus marisflavi*	MH109282
CRB2	*Bacillus marisflavi*	MH109283
CRB3	*Bacillus subtilis*	MH108167
CRB4	*Bacillus pumilus*	MH109284
CRB5	*Leucobactor aridicolis*	MH109293
CRB6	*Alcaligenes faecalis*	MH108168
CRB7	*Bacillus subtilis*	MH108169
CRB8	*Bacillus subtilis*	MH109299
CRB9	*Bacillus subtilis*	MH109300
CRB10	*Alcaligenes faecalis*	MH108170
CRB11	*Bacillus subtilis*	MH108171
CRB12	*Alcaligenes faecalis*	MH108172
CRB13	*Leucobacter aridicollis*	MH108173
CRB14	*Bacillus subtilis*	MH108174
CRB15	*Bacillus subtilis*	MH108175
CRB17	*Bacillus subtilis*	MH109304
CRB18	*Bacillus subtilis*	MH108176
CRB19	*Microbacterium arborescens*	MH108177
CRB20	*Leucobacter aridicollis*	MH109301
CRB21	*Pseudomonas stutzeri*	MH108178
CRB22	*Alcaligenes faecalis*	MH108179
CRB23	*Alcaligenes faecalis*	MH108180
CRB26	*Bacillus safensis*	MH108181
Tomato Isolates		
TRB1	*Bacillus subtilis*	MH108184
TRB2	*Bacillus pumilus*	MH109288
TRB3	*Bacillus pumilus*	MH109289
TRB4	*Bacillus subtilis*	MH109292
TRB5	*Bacillus aerophilus*	MH109298
TRB6	*Bacillus pumilus*	MH109297
TRB7	*Alcaligenes faecalis*	MH109290
TRB8	*Bacillus aerophilus*	MH109296
TRB9	*Bacillus safensis*	MH109295
TRB10	*Bacillus pumilus*	MH108166
TRB11	*Agromyces mediolanus*	MH108183
TRB12	*Bacillus cereus*	MH108182
TRB14	*Bacillus oceanidediminis*	MH10928
TRB15	*Bacillus pumilus*	MH109286
TRB17	*Stenotrophomonas maltophilia*	MH109303
TRB18	*Arthrobacter pascens*	MH108165
TRB19	*Microbacterium arborescens*	MH109294
TRB20	*Rhodococcus equi*	MH108164
TRB21	*Arthrobacter pascens*	MH108163
TRB23	*Bacillus oceanidediminis*	MH109291
TRB24	*Alcaligenes faecalis*	MH109287

and sequences were compared with the GenBank database. Organisms were identified based on the top hit in BLAST analysis and sequences were submitted to the GenBank database (Table 8).

Among 44 PGPR strains characterized, 26 isolates belongs to genera *Bacillus*, 7 belongs to *Alcaligenes*, 3 belongs to *Leucobacter,* 2 belongs to *Arthobacter*, 2 belongs to *Microbacterium* and *Pseudomonas, Agromyces, Rhodococcus* and *Stenotrophomonas* represents one isolate each.

Phylogenetic analysis

Sequences of the PGPR strain identified were used for comparison with the nucleotide sequences from the GenBank database. The phylogenetic tree has been constructed using MEGA 6.0 with 1000 bootstrap replicates in Neighbour joining method (Fig 29). From the phylogeograph, it has been identified that, all the identified organism were polyphyletic in nature by froming group based on their species. This phylogentic analysis furthers confirms the identified organisms were same as that of BLASTn analysis.

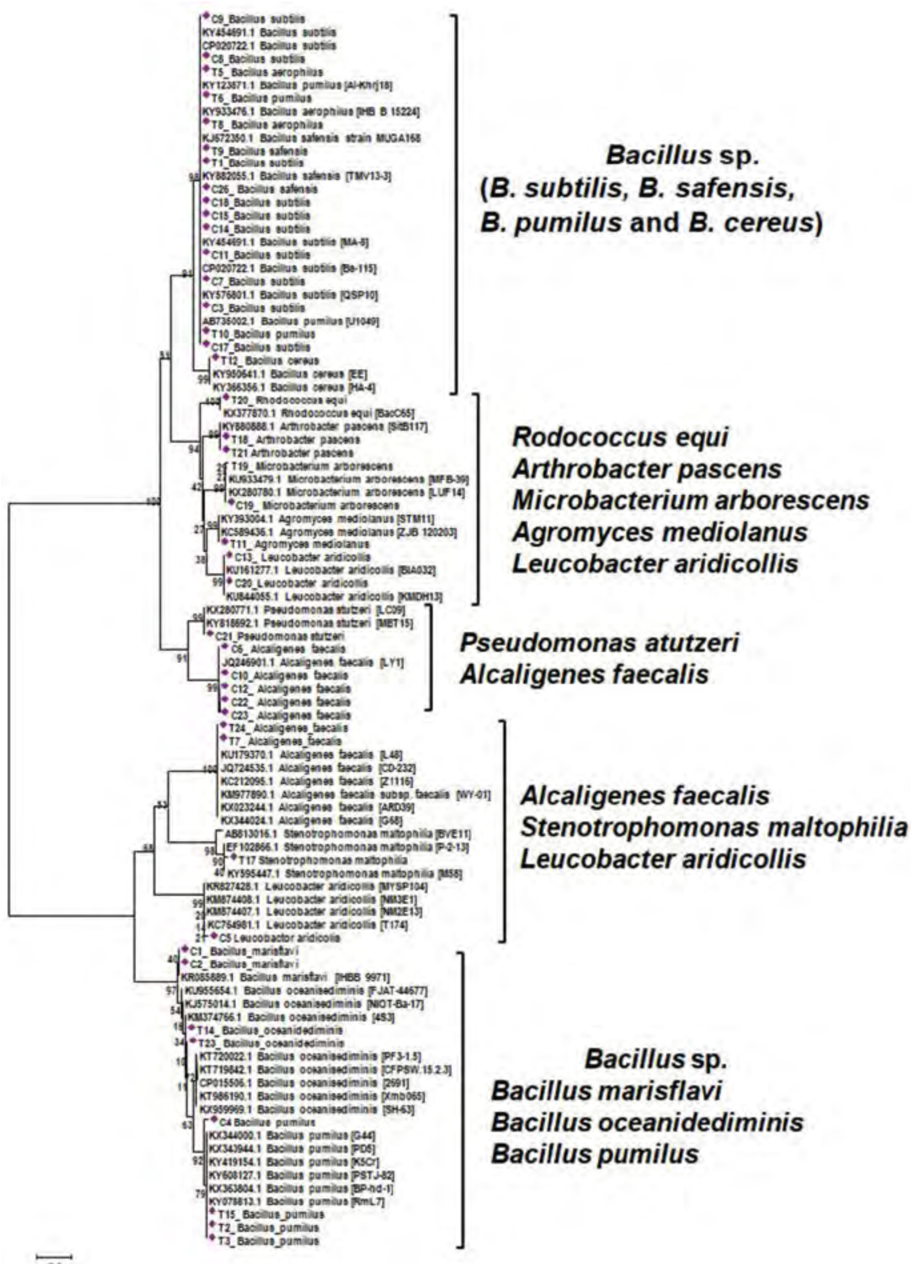


Fig. 29 : Phylogenetic analysis based on 16s rDNA region for the PGPR isolates from tomato and chilli rhizosphere with known organism available in GenBank database (NCBI) using Mega 6.0 with 1000 boostrap replicates.

Forty four bacterial isolates of PGPR are forming five major clusters or groups in phylogenetic analysis of 16s rDNA sequences. Interestingly *Bacillus* species formed two groups where in first group includes *Bacillus subtilis, B. pumilus, B.sefensis. B. cereus* whereas second group includes *B. marisflavi, B. oceanidedimiius, B. pumilus*. Third group includes diverse species like *Rhodococcus equi, Arthrobacter pascens, Microbacterium arborescens, Agromyces mediolanus* and *Leucobacter aridicollis. Pseudomonas stutzeri* and *Alcaligenes faecalis* forms separate cluster. *Alcaligenes faecalis* also formed a separate group with *Stenotrophomonas maltophilia* and *Leucobacter aridicollis*. The phylogenetic analysis suggests the variation among the few bacterial species like *B. pumilus*, *Alcaligenes faecalis* and *Leucobacter aridicollis* at strain level as the different strains are forming the more than one clusters with other species.

ISR analysis of tomato seedlings in pathogen (*Fusarium* and *Sclerotium*) inoculated soil under pot condition: Induced systematic resistance (ISR) study was performed with promising strains individually upon challenge inoculation with *Fusarium* and *Sclerotium* separately under pot conditions. The same has been studied in consortia treated tomato plants upon challenged with pathogens. Analysis for total phenol, peroxidase and polyphenol oxidase activity were performed by following the standard protocol.

Induction of defence enzymes and total phenol: Induction of defence enzymes such as peroxidase and polyphenol oxidase and total phenol content found to be highly variable with respect to pathogen concerned. Some isolates are inducing higher enzyme peroxidase activity upon challenge inoculation with *Fusarium* but not against *Sclerotium* and vice versa. Similarly same trend was observed in case of PPO and total phenol. However, many of the isolates induced higher defence activities than the healthy control and pathogen inoculated control (Fig 30, 31 & 32). However, when these strains of bacteria were applied in consortia, they

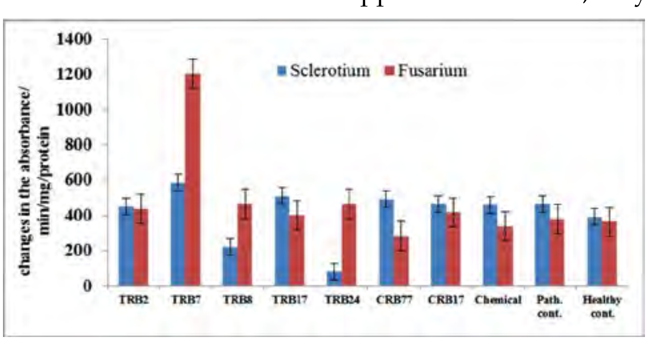


Fig. 30: Effect of different PGPR isolates on peroxidase activity in tomato plants challenged with *Fusarium oxysporum* f. sp. *Lycopersici* and *Sclerotium rolfsii*

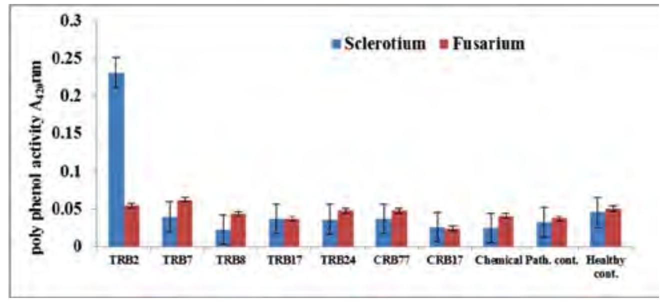


Fig. 31: Effect of different PGPR isolates on polyphenol oxidase activity in tomato plants challenged with *Fusarium oxysporum* f. sp. *Lycopersici* and *Sclerotium rolfsii*

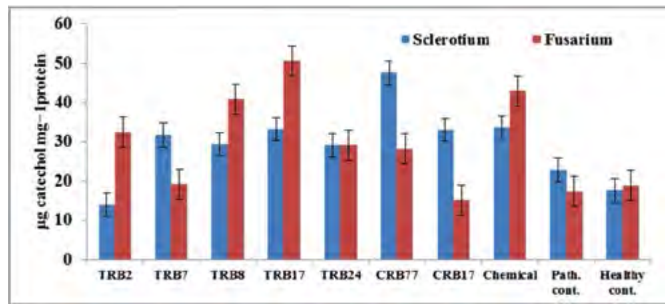


Fig. 32: Effect of different PGPR isolates on total phenol content in tomato plants challenged with *Fusarium oxysporum* f. sp. *lycopersici* and *Sclerotium rolfsii*

have induced higher activity. This may be due to the synergistic action by the strains in consortia.

Determination of shelf-life of bio-formulations: The shelf life was determined by enumerating the population bacterial antagonists by dilution plate method using nutrient agar medium. The dilution was prepared by taking 10 g of formulations in 90 ml sterilized water, shaken well on orbital shaker for half an hour and further diluted to get the final dilution of 10^{-6}. From these dilutions, 0.1 ml was spread in sterilized Petri plates and sterilized media was poured over it. The plates were then given a gentle swirl clockwise and anticlockwise to distribute the suspension uniformly

Table 9: Shelf-life study of talc based of selected PGPR formulations

Isolates	No. of days (cfu 10^6 x /g of formulation)						
	1	15	30	60	120	240	360
TRB7	158	92.5	62	61.5	22.5	1	0
TRB2	80	65	56.5	20	12.5	1.5	0
TRB8	47	32.5	33	9	6	0	0
TRB17	194	170	98	27	27.5	0	0
TRB24	144	61.5	38	19	12.5	0	0
CRB17	160	20	17.5	14.5	5	2	0
CRB7	105	29.5	22.5	11	17.5	0	0

in the medium. These plates were incubated at 28± 1°C and observed regularly for the appearance of microbial colonies. The population (cfu/g) was counted by average number of colonies developed in each treatment multiplied by dilution factor (10^{-6}) of sample. In the shelf-life study, it was established that upto 120 days formulation is having required population i.e., 5 × 10^6 cfu. All the formulation were showing steady decrease in the population other than CRB17 where it shows sudden decrease in the population with in 15days (Table 9).

Project 20: Agro infectious clones development for probing resistance to chilli leaf curl diseases caused by begomoviruses and devising integrated management strategy

Survey and disease incidence: Totally, 101 samples were collected from the chilli leaf curl disease infected field from 50 different locations. Among the states surveyed, maximum average disease incidence was recorded in Madhya Pradesh (56.6%) followed by Tamil Nadu (50.0%), Bihar (28.3%) and Uttar Pradesh (21.9%). Among the areas surveyed, field at Mettur (Tamil Nadu) recorded the maximum incidence of 85% and least recoded at Mirzapur (Uttar Pradesh) with 2%. Different types of symptoms such as mild to severe curling of leaves, stunting of plants, vein clearing, thickening of the leaves, brittleness of the leaves, puckering of leaves, etc., were observed on the chilli leaf curl disease infected chilli plants collected from different regions.

Detection of Begomovirus and its associated satellites: In PCR assay using universal begomovirus specific primer pair, 53 samples were found infected with begomovirus. Also they were tested for the association of α- and β- satellites using universal primer pairs. Results revealed that, association of β- satellites were found in 40 samples but there was no association of any α-satellites with any of the samples. Interestingly, some samples were detected with β- satellites without any helper begomovirus. This might be due to the evolution in the primer binding site of the universal begomovirus primer pair. Among the total samples tested, nearly 60.40% samples found associated with begomovirus. (Table 10).

Randomly samples from the Uttar Pradesh, Tamil Nadu and Andaman & Nicobar Islands were subjected to partial sequencing of PCR amplification to identify the begomovirus associated. Sequencing results shows that, Pepper leaf curl Bangladesh virus (96%) and Tomato leaf curl Joydebpur virus (97%) were causing leaf curl. Further, tomato leaf curl Joydebpur betasatellite was found associated with leaf curl diseased samples (Fig 33). In phylogenetic analysis based on complete genome sequencing of selective samples from UP are Pepper leaf curl Bangladesh virus and Tomato leaf curl Joydebpur virus; whereas from Madhya Pradesh is Chilli leaf curl virus associated with leaf curl disease of chilli (Fig 34).

Table 10: Detection of begomovirus association with chilli samples

State	No. of Samples	Begomovirus	β-satellite	Total samples infected	% of virus infection
Uttar Pradesh	33	28	19	31	93.9
Andra Pradesh	4	0	2	2	50.0
Tamil Nadu	8	5	5	6	75.0
Bihar	6	0	0	0	0
Andaman & Nicobar Is.	15	12	13	14	93.33
Madhya Pradesh	31	7	0	7	22.58
Rajasthan	4	1	1	1	25.0
Total	101	53	40	61	60.40

Development of infectious clone: Based on the sequencing data, Tomato leaf curl Joydebpur virus is found to be predominant in causing leaf curl disease on chilli. One sample from UP (C11) was chosen for the infectious clone development. Initially RCA restricted product resolved on gel showed ~2.7kb fragment with *Xba*I enzyme and has been cloned in pUC19 vector and sequenced. After sequencing for monomer construction, *Hind*III site has been chosen and restricted product (*Xba*I+ *Hind*III) has been sub-cloned in the binary vector pCAMBIA. Around 40 colonies were screened for the insert of ~2.4kb and the one colony was selected for further tandem dimer construction. Around 50 colonies were screened for the dimer in which only one colony was positive for the tandem dimer based on the restriction analysis (Fig 36). Same colony has been subjected to plasmid isolation and extracted plasmids were transformed into Agrobcterium strain GV3101 and the infectious clone has been produced. Further it was been confirmed with the PCR analysis using

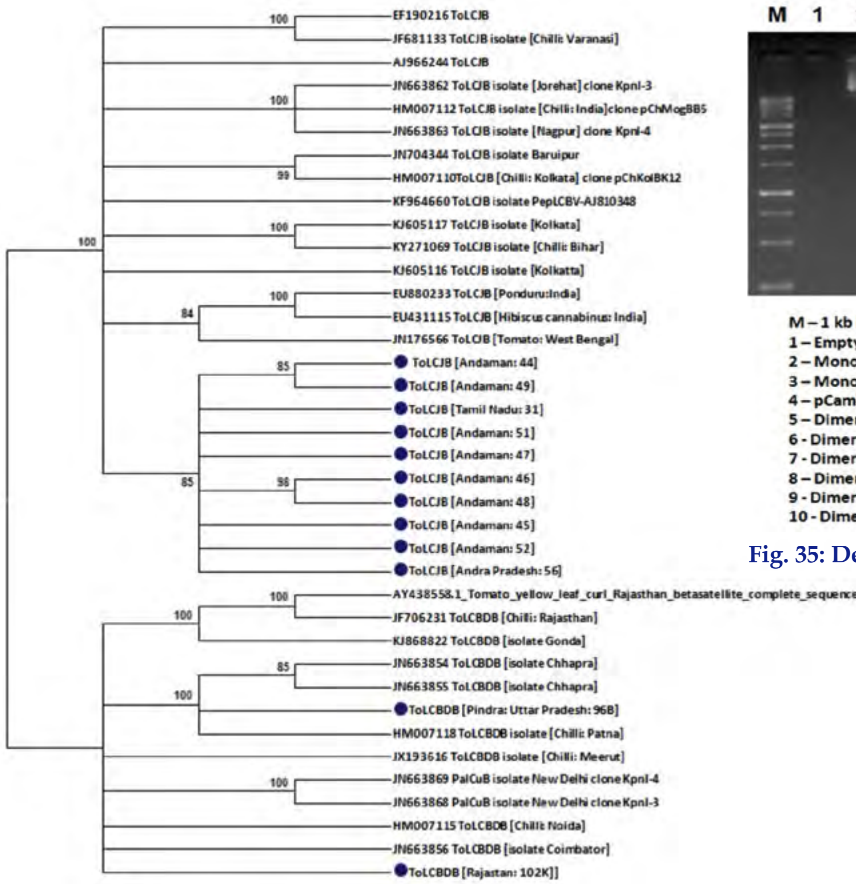


Fig. 33: Phylogenetic analysis of β-satellites associated with begomoviruses infecting chilli crop in India

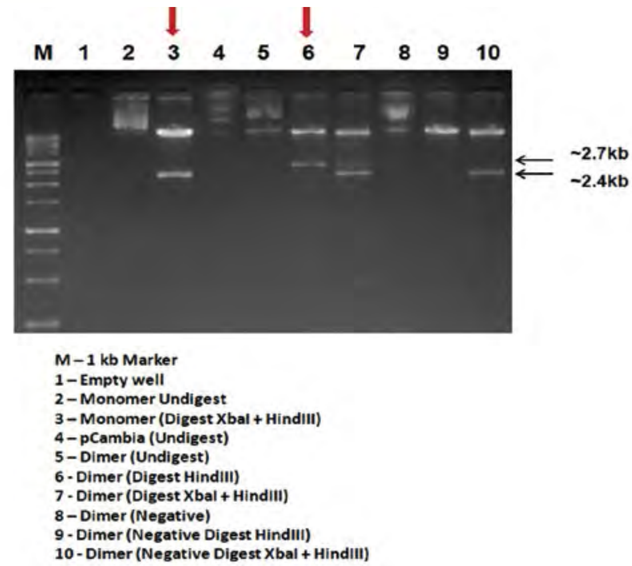


Fig. 35: Development of infectious clone

universal primer pair and restriction analysis. Agrobacterium culture having the construct has been inoculated on the *Nicotiana benthamiana* plants for testing the infectivity and the plants are under observation.

Leaf extract: In order to identify the leaf extract having antiviral properties against the leaf curl virus, selected 35 plant extracts were sprayed on chilli plants severely infected with leaf curl diseases under field conditions. Aqueous leaf extracts were prepared @ 10% concentration and applied on the infected plants twice @ 10 days interval. Plants were observed for the recovery of the symptoms upto 15 days after second spray. Results showed none of tested plant extracts reduced the severity of the leaf curl disease.

Field study: Integrated Disease Management (IDM) module has been devised for the management of leaf curl disease on chilli. Components used in the modules are as follows: Seed disinfection using Virkon S @5g/l for 20 mins; Seed treatment with imidacloprid; Covering of nursery seedlings with insect proof net; Soil application with bioconsortia formulation@5g/l; Seedling dip with imidacloprid @ 0.5ml/lit followed by Carbendazim + Mancozeb @ 2.5g/l for 20 mins each; Soil application of FYM enriched with bioconsortia formulation; Black silver mulching; Soil application of neem cake @5g/plant; Raising of

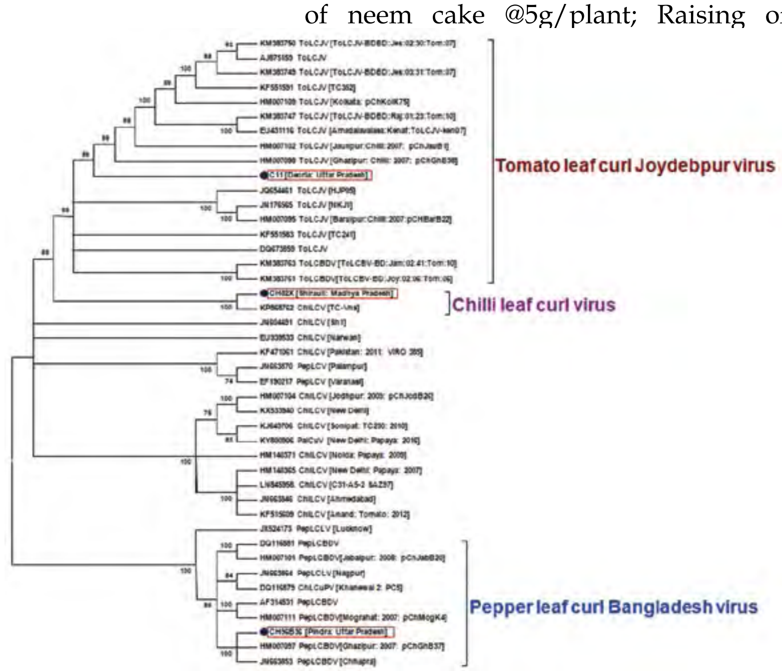


Fig. 34: Phylogenetic analysis of DNA A component of begomoviruses infecting chilli crop in India

bajra as border crop; Installation of yellow sticky traps @20nos./acre; Periodical spraying with micronutrient mixture and Salicylic acid (2mM); Soil drenching with humic acid @5ml/l; Need based application of insecticides such as neem oil, cyzpyr, Intrepid, flonicamid, chlorantraniliprole and flupyridifurone. Among the different module, integrated module with black silver mulching (Module 2) is performing better than other module in recording highest chilli yield. Due to minimum natural incidence of leaf curl disease, effect of different module on leaf curl disease could not be seen. But for yield point of view, module 2 performs better than other modules and yield data is in progress on susceptible cultivar Pusa Jwala.

Project 21: Establishment of Integrated Beekeeping Development Centre (IBDC)/Centre of Excellence (CoE) on Beekeeping at ICAR-IIVR, Varanasi

Development of IBDC/CoE on Beekeeping: Renovation works for establishing queen bee (*Apis melifera*) rearing unit, stock culture mentenance unit, Honey extraction & processing unit completed by August 2017 and inaugurated by Shri S.K. Pattanayak, Secretary, Department of Agriculture, Cooperation & Farmers' Welfare, Ministry of Agriculture and Farmers' welfare, Govt. of India inaugurated Integrated Beekeeping Development Centre/Centre of Excellence on 13th August, 2017. He emphasized that the Beekeeping would be an important component for doubling farmers' income. An extension folder on Integrated beekeeping at a Glance (in Hindi) was also released on this occasion for the benefit of stake holders (Fig. 36).

Fig. 36 : Integrated Beekeeping development centre (IBDC)

Diagnostic laboratory: For diagnosis of pathogens infecting honey bee, disease diagnostic laboratory is being established. Instruments used in the diagnosis such as thermal cycler, refrigerated high speed table top centrifuge and laminar air flow chamber have been procured. DNA and RNA has been extracted from dead honey bees collected near the bee hives were tested with the specific primers through PCR and RT-PCR assays for viruses infecting honey bees. But no viruses were detected. Death of honey bees might be due to other factors.

Demonstration unit: Live honey bee colony of *Apis melifera* and other related tools for handling bee colony and other hive products established and further maintenance and bee multiplication is in progress. Total 09 boxes with live colony as demonstration unit for the benefit of beekeepers/startups were installed in IIVR research farm near old laboratory complex with proper shade nets during August, 2017 for further multiplication and demonstration (Fig. 37).

Fig. 37 : Beekeeping accessories and apiary

Floral garden development: Apart from roadside area in institute's premises, a separate block in the institute farm was developed by planting 25 days seedlings of moringa at 4.6 m x 2.0 m (R x P) planting distance. The seeds were sown in plastic bags on 12.7.2017 and transplanting completed on 7.8.2017. A total of 2081 saplings were planted in surrounding areas of ICAR-IIVR farm at 3 different locations. Heavy rains killed several juvenile seedlings and thus gap filling of seedlings were done during August and September

Table 11 : Establishment and maintenance of bee flora

Crop	Site	Date of Transplanting /Sowing	No. of plants	Flowering Period
Morionga	5 Acre	07/08/2017	406	Oct.-Dec. 2017, February-May, 2018
	IIVR farm	03/08/2017	1547	Oct.-Dec. 2017, February-May, 2018
	IBDC	12/08/2017	86	Oct.-Dec. 2017, February-May, 2018
	IIVR Farm old trees	-	268	Oct.-Dec. 2017, February-May, 2018
		Total	2307	
Mustard	5 Acre	08/11/2017	-	Dec.2017-Feb., 2018
Sunflower	5 Acre and IIVR farm	March, 2018	-	May-June, 2018

Fig. 38: Bee flora maintained during different seassons for honey bee A. Moringa, B. Mustard C. Sunflower D. Honey bee visiting Moringa, E. Honey bee visiting Mustard and F. Honey bee visiting Sunflower

2017. Drenching of mancozeb @ 0.25% was carried out on 10.8.2017 to prevent root rot of moringa plantlets followed by spray of carbendazim + mancozeb @ 0.2% on 5.9.2017 to manage Colletotrichum leaf blight. Spray of chlorphyriphos @ 2 ml/l was given to manage leaf defoliators of growing moringa plantlets. Besides, 268 old established moringa plants were also available in the farm area. These tree plants served as bee flora in Oct.- Dec.2017-18 and March-May, 2018. Apart from moringa, this floral park is also having well established plants of mango, awala, guava, teak and banana which will serve as food for honey bees. Mustard was also grown in 4000 sq. m area at 5 Acre area of the institute during Nov., 2017 which served as bee flora from Dec.2017 –Feb., 2018. Staggered sowing of sunflower was also followed in and around IIVR farm at weekly interval to ensure food availability to honeybee from May-June, 2018 (Table 11 & Fig. 38)

Floristic studies in relation to honey bee, *Apis mellifera:* Honey bee visits various groups of plant viz., vegetables, cereals, fruit trees, plantation trees, ornamental plants etc., to obtain nectar, pollen or both for multiplication and survival. So continuous availability of bee flora is the necessity for bee keeping. Flowering duration of different plants differs from one place to other due to variation in topography, climate and other cultural and farming practices. The knowledge of bee flora of a Jakhini region enable us to utilize bee flora at the maximum level, so that we can harvest a good yield of honey and other bee products in addition to effective pollination, which enhances crop yields. This region has its own honey flow and floral dearth periods of short and long duration. Such kind of knowledge on bee flora help in the effective management of bee colonies. Different vegetables grown in the research farm including other crops available in adjoining area of the institute were periodically observed for flowering, duration of flowering and visit of honey bee for nectar and pollen. The observations revealed that among the different vegetable crops, Bottle guard, Bitter guard, okra, sponge guard, pumpkin, cucumber served as bee flora from august – november while brinjal and Indian bean were found suitable bee flora from september-march. Pea, tomato and summer squash served as bee flora from december- march. In case of cereals maize, bajra and sorghum were found suitable as bee flora from september to october. Among the oilseeds, mustard served as bee flora from december-march. In fruit crops, banana, guava, papaya and wood apple were found useful and visited by honey bees. Among the ornamental plants, *Hemelia* and *Ixora* were not visited by honey bees. In ornamentals, rose, calendula, marigold and *Tecoma* were preferred by honey bees as a bee flora. Amongst the plantation trees, moringa was highly preferred tree by honey bees followed by babul, kadam and peepal (Table 12).

Table 12: Floristic studies of *A. mellifera* in Varanasi region

Group	Common name/ vernacular name	Scientific name	Family	Availability of Flora		Flowering season	Utility to honey bees
				IIVR Field	Farmers field		
Vegetable	Bottle gourd	*Lagenaria siceraria*	Cucurbitaceae	Y	Y	Sept-Jan	Y
Vegetable	Bitter gourd	*Momordica charantia*	Cucurbitaceae	Y	Y	Sept-Nov	Y
Vegetable	Okra	*Abelmoschus esculentus*	Malvaceae	Y	Y	Sept-Nov	Y
Vegetable	Sponge gourd	*Luffa cylindrica*	Cucurbitaceae	Y	Y	Sept-Nov	Y
Vegetable	Chilli	*Capsicum* spp.	Solanaceae	Y	Y	Sept-Feb	N
Vegetable	Pumpkin	*Cucurbita pepo L.*	Cucurbitaceae	Y	Y	Aug-Oct	Y
Vegetable	Ash Gourd	*Benincasa hispida*	Cucurbitaceae	Y	N	Aug-Oct	Y
Vegetable	Cucumber	*Cucumis sativus*	Cucurbitaceae	Y	Y	Aug-Oct	Y
Vegetable	Amaranthus	*Amaranthus* spp.	Amaranthaceae	Y	Y	Oct-Nov	N
Vegetable	Ridge gourd	*Lufa acutangula*	Cucurbitaceae	Y	Y	Aug.-Sept.	Y
Vegetable	Brinjal	*Solanum melongena L*	Solanaceae	Y	Y	Sep-March	Y
Vegetable	Sem (Field bean)	*Lablab purpureus*	Fabaceae	Y	Y	Oct-March	Y
Vegetable	Cow pea	*Vigna unguiculata*	Fabaceae	Y	Y	Sep-Jan.	Y
Vegetable	Soybean	*Glycine max*	Fabaceae	Y	N	Aug.-Sept.	Y
Vegetable	French bean	*Phaseolus vulgaris*	Fabaceae	Y	N	Nov-March	Y
Vegetable	Pointed gourd	*Trichosanthes dioica*	Cucurbitaceae	Y	N	Sept-Dec	Y
Vegetable	Pea	*Pisum sativum*	Fabaceae	Y	Y	Dec-Feb	Y
Vegetable	Tomato	*Solanum lycopersicum*	Solanaceae	Y	Y	Dec-March	Y
Vegetable	Winged Bean	*Psophocarpus tetragonolobus*	Fabaceae	Y	Y	Sept.-Feb.	N
Vegetable	Chappan kaddu	*Cucurbita pepo*	Cucurbitaceae	Y	N	Dec- Mar	Y
Cereal crop	Bajra	*Pennisetum glaucum*	Poaceae			Sep-Oct	N
Cereal crop	Paddy	*Oryza sativa*	Poaceae	Y	Y	Sep-Oct	N
Cereal crop	Maize	*Zea mays*	Poaceae		Y	Sep-Oct	N
Cereal crop	Sorghum	*Sorghum bicolor*	Poaceae	Y	N	Sep-Nov	N
Oilseed crop	Mustard	*Brassica nigra*	Brassicaceae	Y	Y	Dec-Feb	Y
Oilseed crop	Castor (Arandi)	*Ricinus communis*	Euphorbiaceae	N	Y	Oct-March	N
Fruit	Ber	*Ziziphus mauritiana*	Rhamnaceae	N	Y	Oct-Dec	N
Fruit	Banana	*Musa acuminata*	Musaceae	Y	Y	Oct-Jan, April-June	Y
Fruit	Guava	*Psidium guajava*	Myrtaceae	Y	Y	Nov-Jan, April-May	Y
Fruit	Papaya	*Carica papaya*	Caricaceae	Y	Y	Nov-June	Y

Fruit	Wood apple (Bel)	*Aegle marmelos*	Rutaceae	Y	Y	Dec-Jan, May-June	Y
Fruit	Mango	*Mangifera indica*	Anacardiaceae	Y	Y	Feb-March	Y
Ornamental	Hibiscus	*Hibiscus rosa-sinensis*	Malvaceae	Y	Y	Oct-May	Y
Ornamental	Rose	*Rosa*	Rosaceae	Y	Y	Oct-May	Y
Ornamental	Tecoma	*Tecoma stans*	Bignoniaceae	Y	N	Sep-May	Y
Ornamental	Hamelia	*Hamelia patens*	Rubiaceae	Y	N	Sep-May	N
Ornamental	Ixora	*Ixora coccinea*	Rubiaceae	Y	N	Sep-May	N
Ornamental	Marigold	*Tagetes* spp.	Asteraceae	Y	Y	Oct-March	Y
Ornamental	Calendula	*Calendula officinalis*	Asteraceae	Y	N	Dec-Mar	Y
Plantation tree	Drumstick	*Moringa oleifera*	Moringaceae	Y	N	Oct.-Dec.; March-May	Y
Plantation tree	Teak	*Tectona grandis*	Lamiaceae	Y	Y	Sep-Dec	N
Plantation tree	Babul	*Acacia nilotica*	Fabaceae	Y	Y	Oct-Dec	Y
Plantation tree	Kadam	*Neolamarckia cadamba*	Rubiaceae	N	Y	Oct-Dec	Y
Plantation tree	Peepal	*Ficus religiosa*	Moraceae	N	Y	Oct-Dec	Y
Fiber crops	Sunhemp	*Crotalaria juncea*	Fabaceae	Y	Y	Sep-Nov	Y

Trainings to farmers and other stakeholders on bee keeping

Training cum Awareness Programme on Beekeeping: With an aim of promoting Beekeeping in Eastern Uttar Pradesh, the institute organized 02 days training cum awareness programme on "Motivation and Technology Transfer on Beekeeping" during 20-21 March 2017 with the help of Hi-Tech Natural Products (India) Ltd., Saharanpur (approved bee breader and training institute by NBB). This training cum awareness programme was attended by 153 beekeepers/farmers/ tribal from Varanasi, Mirzapur, Chandauli and Sonbhadra districts of Uttar Pradesh. The participants were acquainted with benefits and advanced technologies of beekeeping with both theoretical and visuals. On this occasion, Dr. Janardhan Singh, Former Head, Department of Entomology, Institute of Agriculture Science, BHU, Varanasi discussed the importance and scope of beekeeping in Eastern Uttar Pradesh with the participants.

Besides this, farmers and other stakeholders coming from different places for scheduled training programmes on vegetable production were given one day hands on training to promote scientific bee keeping in the region. We dealt with issues in a scientific manner and provided technical skill to help them to initiate beekeeping enterprise for earning their livelihood and enhancing their income. During the training, hands-on experience was provided to the trainees by practically showing them on various aspects of beekeeping especially for European bee (*Apis mellifera*) such as management of honey bee colonies in different seasons, routine apiary

Table 13: Trainings organised at IBDC, IIVR, Varanasi

Training Date	Training Name & Address	No. of participants
5/1/2018	BPIU, Gurua Gaya Bihar	29
10/1/2018	Baraghatti Gaya Bihar	26
20/01/2018	Rajnagar Madhubani Bihar	31
31/01/2018	Jeevika-Khajauli Madhubani Bihar	29
1/2/2018	ATMA-Jabalpur M.P	28
13/03/2018	NABARD-Lucknow - Scientific and Advanced technology for enhancing Vegetable production	30
15/03/2018	ATMA-Bihar	10
17/03/2018	Krishi Unnati Mela	29
23/03/2018	ATMA-Darbhanga Bihar Integrated Production & Management of Vegetable Crops	17
		229

management, colonies multiplication, management of bee enemies and diseases and extraction, processing and preservation of honey, etc. . Lab and apiary visits were organised for participant trainees. The trainees were also apprised of the finance and subsidy schemes available for beekeepers by the technical experts from banking sector and State Department of Horticulture.

Fig. 39: Trainees visiting IBDC

During the training, study material eg. Technical bulletin *'Madhumakhi palan-Digdarshika'* and extension folders *viz., Samekit Madhumakhi palan-Ek Drishtikon, Madhumakhi palan evm Prabhandhan, Madhumakhi palan mein rogon evm keeton ka surakshit Prabhandhan* and *Madhumakhi palan ek Labhprad Vyaysay'* in Hindi were distributed to the trainees. Altogether 229 farmers including farm women, youth, self-help groups and other stakeholders from 12 different districts of 3 states *viz.,* Uttar Pradesh, Madhya Pradesh and Bihar were trained on different aspects of scientific beekeeping. (Table 13 & Fig. 39). In addition to this, a technical and practical session was also arranged on 'Beekeeping: A Vital Input for Sustainable Agriculture' for 21 trainees from Afro-Asian countries on 16th February, 2018.

Production of honey and other bee products: Initially 50 honey bee boxes were procured during December 2017 and kept at 4 different sites in ICAR-IIVR, Varanasi and raised to 72 boxes within 3 months from Jan.2017- to March; 2018. Total yield of honey obtained from these boxes was 347.1 kg. About 5 kg wax, 75 g royal jelly were also obtained (Table 14 & Fig. 40, 41, 42).

Table 14: Honey production at IBDC

Location/Site	No. of bee boxes (Initial)	No. of bee boxes (Final)	No. of Frames used for honey extraction	Honey Extracted (Kg)
5 Acre	5	17	148	71.5
IBDC	8	14	168	62.9
IIVR farm pond area	25	29	250	107.2
IIVR Organic Farm area	12	12	274	105.5
Total	**50**	**72**	**840**	**347.1**

Fig. 40: Extraction of honey

Fig. 41: Processed, bottled and labelled honey at IBDC

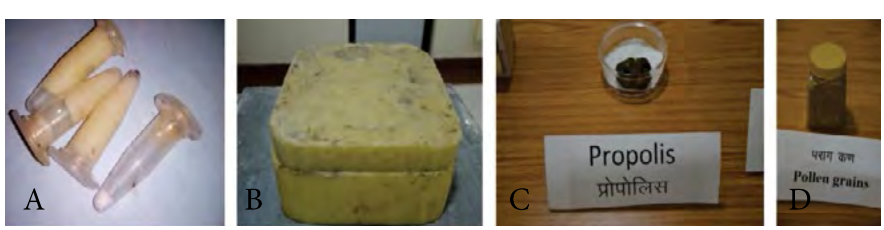


Fig. 42: Bee by-products produced at IBDC, IIVR A. Royal jelly B. Wax C. Propolis, D. Pollen grains

All India Coordinated Research Project (Vegetable Crops)

ACHIEVEMENTS OF ALL INDIA COORDINATED RESEARCH PROJECT ON VEGETABLE CROPS

During the year 2017-18, 1629 trials were conducted at 36 regular centres and 24 voluntary centres of AICRP on Vegetable Crops (Table-1)

The following recommendations under Crop improvement, Crop production and Crop protection were made during 35th Group Meeting of AICRP (VC) held at ICAR-IIHR, Bengaluru from 24-27 June, 2017 (Table-2, 3 & 4).

Crop Improvement

Variety evaluation trial: Three entries of 3 crops were identified for release and notification for different agro-climatic zones of the country.

Hybrid Evaluation Trial : Three entries of 3 crops were identified for release and notification for different agro-climatic zones of the country:

Table-1: Details of the trials conducted during 2016-17 through AICRP (VC)

Trials		No. of Trials	No. of trials conducted by the centre
Crop Improvement	Plant Genetic Resources	27	96
	Varietal trials	41	770
	Hybrid trials	24	340
	Resistant varietal trials	6	106
Crop Production	Vegetable production trials	18	58
	Protected cultivation	8	28
	Seed production trials	24	56
	Physiology & biochemistry trials	5	13
Crop Protection	Integrated pest management	16	69
	Integrated Disease management	8	93
Total		**177**	**1629**

Table -2: Varieties identified for release and notification

Sl. No.	Crop	Name of the entry	Original name	Name of the developing centre	Zone
1.	Brinjal Long	2013/BRLVAR-03	JBL-08-8	JAU, Junagadh	**VI** (Rajasthan, Gujarat, Haryana and Delhi)
2.	Dolichos bean (Pole type)	2013/DOLPVAR-03	IIHR-162-2	ICAR-IIHR, Bengaluru	**VII** (Madhya Pradesh, Mahrastra & Goa) and **VIII** (Karnataka, Tamil Nadu & Kerala)
3	Bottle gourd	2013/BOGVAR-04	BBOG-3-1	OAUT, Bhubaneswar	**VI** (Rajasthan, Gujarat, Haryana and Delhi)

Table -3: Hybrids identified for release and notification

Sl. No.	Crop	Name of the entry	Original name	Name of the developing centre	Zone
1	Tomato Hybrid (Determinate)	2013/TODHYB-05	-	-	**IV** (Punjab, Uttar Pradesh, Bihar and Jharkhand)
2	Brinjal Hybrid Round	2013/BRRHYB-05	PBHR-41	PAU, Ludhiana	**IV** (Punjab, Uttar Pradesh, Bihar and Jharkhand)
3	Ridge gourd	2013/RIGHYB-02	-	-	**VIII** (Karnataka, Tamil Nadu,Kerala & Pondicherry)

Table -4: Resistant Line identified for breeding purpose

Sl. No.	Crop	Name of the entry	Original name	Name of the developing centre	Zone
1	Brinjal Bacterial Wilt	GB 09-05-3	2013/BRBWRES-02	AAU, Jorhat	All

Resistant Varietal Trials: One entry of brinjal was identified as bacterial wilt resistant line for breeding purpose.

Vegetable Production

Micro nutrient

- **At Ludhiana and Bangalore**, the foliar application of six micronutrients(Boric acid (100 ppm), Zinc sulphate (100 ppm), Ammonium molybdate (50 ppm), Copper sulphate (100 ppm), Ferrous sulphate (100 ppm), Manganese sulphate (100 ppm) was found suitable for getting optimum fruit yield with highest B:C ratio in tomato. Hence, it is recommended for agro-climatic condition of Zone I and VIII.

Organic and Drip irrigation

- **At Kalyanpur**, the organic package for coriander-radish sequence consists of growing coriander cv. Pant Haritima and radish cv. Japanese White with the application of 100% recommended dose of nitrogen through vermicompost + IIHR microbial consortium @ 12.5 kg/ha was found suitable for optimum yield and highest B:C ratio. Hence, it is recommended for agro-climatic condition of Zone IV.

- **At Vellanikkara** in pickling cucumber, drip irrigation at 0.5 bar coupled with black polythene mulch was found suitable for getting optimum yield with highest B:C. Hence, it is recommended for agro-climatic condition of Zone VIII.

- **At Bangalore** in hybrid chilli (Arka Meghna), the NK fertigation with water soluble fertilizer or NPK with normal fertilizer was found suitable for getting maximum yield with highest B:C ratio. Hence, it is recommended for agro-climatic condition of Zone VIII.

Weed Management

- **At Jabalpur**, mulching with black polythene in chilli was found suitable for maximum yield and B:C ratio followed by soybean straw mulch @ 12 t/ha. Hence, it is recommended for agro-climatic condition of Zone VII.

- **At Coimbatore** in chilli, use of organic mulch @ 12 t/ha was found suitable for maximum green fruit yield and dry fruit yield with highest benefit cost ratio followed by black/white polythene (double coated 30 micron). Hence, it is for recommended agro-climatic condition of Zone VIII.

- **At Durgapura and Vellanikkara**, during kharif in cowpea mulching with black polythene was found suitable for obtaining maximum pod yield with highest benefit cost ratio. Hence, it is recommended for agro-climatic condition of Zone VI and VIII.

- **At Coimbatore** in okra pre-emergence application of pendimethal in @ 6 ml/lit + one hand weeding after 25 days of sowing was found suitable for maximum yield with highest B:C ratio. Hence, it is recommended for agro-climatic condition of Zone VIII.

Protected Cultivation

Low tunnel production technology:

- At **Hisar,** maximum yield (205q/ha) and B:C ratio (2.42) in capsicum cv. Pusa Deepti was recorded when the crop was transplanted on 15th October with black polythene mulch under low tunnel. Hence, it is recommended for Hisar condition of agro climatic Zone - VI

- At **Ludhaina**, 15th September transplanting with black polythene mulch recorded maximum yield of 375q/ha and B:C ratio of 3.17 in capsicum cultivar Indra under low-tunnel. Capsicum plants have to be covered with 40-mesh agro-net from mid September to Mid November and later with 50 micron polyethylene sheet till mid February. Hence, low tunnel technology is recommended for Ludhana condition of agro climatic Zone - IV.

- At **Durgapur,** transplanting of capsicum on 15th September under low tunnel with polyethylene mulch was found superior to all other treatments with a yield of 336 q/ha and a B:C ratio of 2.72.

Hence, it is recommended for Durgapur condition of agro climatic Zone - VI.

- At **Jabalpur**, on the basis of three years yield and economic data for bottle gourd and bitter gourd, it is concluded that, sowing of bottle gourd (Hy. NS 443) and bitter gourd (Hy. NS 454) on15th February under open field condition recorded highest yield of 172.4 and 76.4 q/ha along with of B:C ratio 4.93 and 5.46, respectively. Hence, it is recommended for Satpura hills and Kaymore Plateau of M.P. of agro climatic Zone - VII.

Production of Cherry tomato under controlled environment

- At **Hisar,** cherry tomato variety Olay transplanted at a spacing of 100 x 45 cm with pinching + stacking treatment gave maximum yield of 385.4 q/ha with B: C ratio of 3.84 under polyhouse condition . Hence, it is recommended for Hisar condition of agro climatic Zone – VI.

Production technology under rain-shelter conditions

- **At IIVR, growing of tomato in** rain shelter with transplanting at a spacing of 100 x 75 cm, recorded highest yield of 435q/ha with a B:C ratio of 3.71. Hence, it is recommended for Varanasi condition of agro climatic Zone – IV.

Seed Production

- Planting ratio of 2:1 was most suitable for maximizing hybrid seed yield (6.29q/ha) in cabbage (Pusa cabbage hybrid-1) using Self-Incompatibity system under Katrain condition.
- At Raipur, soil application of poultry manure @ 3.5 t/ha + vermicompost @ 3 t/ha recorded maximum plant growth and seed yield (12.19q/ha) with cost benefit ratio of 3.61 in garden pea cv. Arkel.
- Sowing in second fortnight of October with single cuttings gave highest seed yield (28.13q/ha) and quality with cost benefit ratio of 3.66 in palak cv. Shalimar Green under Srinagar.
- Bottle gourd cv. Pusa Naveen seeds treated with Potassium dihydrogen phosphate 10^{-1} M for 24 h recoded less germination time, more germination (77.36%), and vigour indices at Raipur.
- Bitter gourd cv. Preethi seeds treated with GA_3 500ppm for 24 h recoded highest gemination (89.7%), and vigour indices at Vellanikara.
- Seed coating with carbendazim @ 2 g/kg seed + Imidacloprid @ 2ml/kg seed + micro nutrient mixture @ 20 g/kg seed in tomato cv. T-6 recorded the maximum seed gemination (92%), and vigour under Kalyanpur condition. CD and CV of pooled data is 1.05 (at 5%) & 5.78.
- Seeds of dolichos bean coated with carbendazim @ 2 g/kg seed + imidacloprid @ 2 ml/kg seed + micronutrient mixture @ 20 g/kg of seed recorded maximum germination (80.5%) and vigour at Rahuri.
- Seed priming with K_2HPO4 (0.01M) or GA_3 (100ppm) for 24h improved germination (71%) and vigour of less viable brinjal cv. Kashi Taru seed at ICAR-IIVR, Varanasi (6.74).
- Seed priming with at K_2HPO4 (0.01M) for 24hrs is recommended for better germination (74%) and vigour in less viable seeds of okra cv. Arka Anamika at ICAR-IIHR, Bengaluru.
- Seed priming with at K_2HPO4 (0.01M) for 6hrs is recommended for better germination (80.3%) and vigour in less viable seeds of French bean cv. Arka Komal at ICAR-IIHR, Bengaluru.
- At Solan cauliflower cv. PSBK 1 transplanted in the last week of October at spacing of 60 x 60 cm gave higher quality seed yield (3.42q/ha).
- Mulching with black polythene effectively controls weeds and gives higher seed yield (3.39q/ha) with cost benefit ratio (1:3.54) in brinjal cv. Punjab Nagina seed crop under Ludhiana condition.
- Soil mulch with black polythene effectively controls weeds and gives higher seed yield (0.98q/ha) with cost benefit ratio (1:3.1) in capsicum cv. Solan Bharpur seed crop under Solan condition.
- Three hand weeding at 20, 40 and 60 days after sowing in vegetable pea cv. Azad Pea-3 recorded highest seed weight, seed yield (15.95q/ha) and quality attributes with cost benefit ratio of 1:2.48 under Kanpur condition.
- At Solan carrot cv. Solan Rachna stecklings transplanted at spacing of 30 x 30 cm produced maximum seed yield (6.52 q/ha) and quality.

- To get maximum seed weight (14.19g), germination (94%) and vigour indices the fruits of bottle gourd cv. Punjab Komal should be harvested 45 days after anthesis and prestored for at least 10 days before seed extraction under Ludhiana condition.

- Foliar application of NAA 40 ppm at 40 days after sowing in okra cv. Utkal Gaurav recorded higher seed yield (7.65q/ha) and cost benefit ratio (1:6.45) under Bhubaneshwer condition.

- To raise the muskmelon cv. Punjab Sunehri seeds under sub optimal conditions, should be treated with GA_3 @ 100ppm or ethrel@ 100 ppm for 24 hours followed by covering in moist gunny bags for 24 hours under Ludhiana condition.

Insect Pest Management

- In brinjal, cassava based biopesticide (Nanma) @ 10 ml/Lwas found to be the most effective in reducing the shoot (7%) as well as fruit damage (10.11%) by *L. orbonalis* with highest fruit yield (341.67 q/ha) at Sabour conditions.

- Growing round shaped brinjal under protected cultivation reduced the fruit damage infestation by fruit borer *L. orbonlais* to the tune of 4.55, 3.93 and 1.08 % at Sabour, Varanasi and Bengaluru conditions, respectively with maximum marketable fruit yield as compared to the brinjal grown in open field conditions.

- Insecticide resistance management (IRM) strategy comprising of rotation of rynaxpyr 18.5 SC @ 0.4ml/L followed by E. benzoate 25 WG @ 0.5gm/L, spinosad 45 SC @ 0.5ml/L, chloropyriphos 20 EC @ 2ml/L and cypermethrin 25 EC @ 0.5ml/L at 10 days interval gave lowest shoot and fruit damage by *L. orbonalis* in brinjal, with highest marketable fruit yield at Varanasi, Sabour and Rahuri conditions. Highest B:C ratio of 10.55 and 11.37 was obtained at Rahuri and Varanasi centres, respectively. This strategy reduces number of pesticide sprays in brinjal by 50 % as compared to farmers practice and it can be used as a pest and resistance management strategy for shoot & fruit borer in brinjal.

- Integrated pest management module consisting of seedling root dip with imidacloprid 17.8 SL @ 0.5ml/L and subsequent spray of buprofezin 25 SC @ 1ml/L followed by fipronil 5 SC @ 1.5 ml/L, *Verticellium lecanii* (1 x 108 CFU/g) @ 5.0 gm/L, chlorfenapyr 10 SC @ 1.0 ml/L and neem oil @ 10ml/l at an interval of 10 days starting from 25 days after transplanting was found most effective with 76.57 and 70.80 % reduction in chilli thrips and mites, respectively and gave 53.70 % increase in marketable fruit yield with highest B:C ratio of 5.83 at Rahuri conditions.

- Pest management module consisting of erection of yellow sticky traps @ 2 traps/50-100 m^2, foliar spray of imidaclorpid 200 SL @ 0.5 ml/L at 20 and 30 days after transplanting, spray of chlorantranliprole 18.5 SC @ 0.5 ml/L at 15 days interval at the initiation of flowering and spray of fenzaquin 10 EC @ 2.5 ml/L was found most effective with 78.06, 66.32 and 72.41 % reduction in whitefly, leafminer and fruit borer, respectively in tomato and gave 59.77% increase in yield and highest B:C ratio of 22.68 at Rahuri conditions.

Disease Management

- Seed treatment @4g/kg + soil application @10g/m^2 + soil drenching @5% of *Trichoderma viride-2* (IIVR-2) was most effective in reducing the damping off incidence and improving germination and seedling vigour index in chilli, tomato and brinjal with 1:4.45, 1:2.86 and 1:3.48 CB ration respectively at Lam.

Breeder Seed Production

During the year 2016-17 a total of 35669.243 kg Breeder Seeds produced against the indent of 14955.94 kg for 119 varieties of 33 vegetable crops by 19 coordinating centres. During the year 2017-18, an indent of 20615.99 kg breeder seed for 193 varieties of 36 vegetable crops have been received from the Deputy Commissioner (Seed) DAC,GOI, New Delhi and the same have been allotted to 21 coordinating centres for under taking the production. A total of 34426.840 kg of Breeder Seeds has been produced against the indents. However, the final production figures are awaited from many centres for some vegetable crops.

Krishi Vigyan Kendras

ICAR-KRISHI VIGYAN KENDRA, BHADOHI

Training Programme: KVK-Bhadohi conducted 57 training programme to farmers, rural youths and extension personnel to orient them in the frontier areas of technology development under cereals, oilseeds, pulses, vegetables, fruits, livestock and home science covering a total of 1275 beneficiaries including 940 male and 335 female participants (Table-1 & Fig. 1&2).

Fig. 1: On Campus Training on candle making

Fig. 2: Off Campus training to the farmers

Table 1: Training programmes organized

Clientele	No. of Courses	Male	Female	Total participants
Farmers & farm women	52	889	303	1192
Rural youths	04	30	32	62
Extension functionaries	01	21	0	21
Total	**57**	**940**	**335**	**1275**

Fig. 3: FLD on Wheat (KRL-213)

Fig. 4: FLD on Pigeon pea (NDA-2)

Front Line Demonstration on agricultural discipline: A total of 455 front line demonstrations (FLDs) on pulses, oilseeds, paddy, wheat, vegetables and fodder crops were conducted in 79.96 ha area in order to establish the production potential of improved technologies at the farmers' fields (Table 2, Fig. 3 & 4).

Table 2:Front Line Demonstration on improved variety and sowing time

Crop	Variety	No. of farmers	Area (ha)	Yield (q/ha)				% Increase in yield
				Demo			Check	
				High	Low	Average		
Oilseed								
Mustard	RH-749	66	26.51	33.34	21.35	25.4	18.6	36.56
Pulses								
Pigeonpea	NDA-2	34	10.0	24.5	16.10	20.20	19.40	32.02
Pigeonpea	P-992	03	0.5	10.25	6.30	8.35	7.7	8.44
Lentil	L-4076	04	0.2	12.5	8.2	10.75	7.5	43.33
Blackgram	Pant Urd-31	29	10.0	13.6	8.5	12.30	8.40	46.42
Other Crops								
Paddy	CSR-36	13	2.5	608	46.0	52.90	42.30	25.06
Paddy	CSR-43	27	7.5	54.0	40.0	50.10	42.30	18.44
Paddy	P-2511	12	2.5	59.25	42.5	56.2	32.3	73.99
Paddy	P-1592	05	1.0	56.0	41.8	52.10	32.3	61.61
Wheat	KRL-210	13	1.66	30.0	22.0	27.50	20.25	35.80
Wheat	KRL-213	20	3.34	32.5	24.0	29.75	20.25	46.91
Wheat	HD-2967	21	3.0	59.5	48.3	57.2	46.4	23.28
Bajra	NBH-4903	09	2.250	26.10	15.70	21.71	15.660	39.17
Sponge gourd	Kashi Divya	35	1.5	238.8	182.1	211.6	167.6	26.25
Tomato	Kashi Aman	26	1.0	473.8	396.1	428.6	272.3	57.40
Vegetable pea	Kashi Mukti	29	1.0	157.8	126.3	144.9	131.5	10.19
Okra	Kashi Kranti	20	0.75	113.6	93.4	102.8	84.5	21.65
Onion Kharif –Bulblet	AFDR	12	0.50	252.6	189.2	218.2	-	-
Onion Kharif –Seed	AFDR	32	1.25	210.7	172.4	197.6	-	-
Berseem	Vardan	45	2	1110	780	952	691	37.97
Total		**455**	**79.96**					

Front Line Demonstration on Livestock and Poultry: Under livestock production, 03 demonstrations were conducted on disease management in dairy animals and sheep & goat (Table-3 and Fig. 5 & 6).

Table 3: Front Line Demonstration on livestock for disease management

Category	Name of the technology demonstrated	No. of farmers	No. of units (Animal/ Poultry/ Birds, etc)	Major parameters		% change in major parameter
				Demo	Check	
Cattle	Control of Liver Fluke infestation in Cattle	48	131	Survivality, Milk Production and Growth parameter		99% animal are disease free
Buffalo	Control of Liver Fluke infestation in Buffalo	34	34	Survivality, Milk Production and Growth parameter		99% animal were disease free
Sheep & Goat	Control of Liver Fluke infestation in Sheep/ Goat	58	3252	Survivality and Growth parameter		99% animal were disease free

Fig. 5: FLD on Sheep & Goat (Animal Health Camp)

Fig.6: Animal Health Camp on Infertility

FLD on other enterprise: A total of 03 demonstrations on kitchen gardening conducted at 30 farmers field (Table 4).

Table 4: FLD on other enterprise (Nutritional Security)

Season	No. of Farmers	No. of Units	Yield (Kg)		% change in yield
			Demons-tration	Check	
Zaid	10	10	686.1	389.3	76.23
Kharif	10	10	732.0	467.8	56.47
Rabi	10	10	1237.8	667.9	85.32

Technology Assessment and Refinement: A total of 07 on farm trials (OFTs) were conducted in different villages of KVK Bhadohi for assessment of selected technologies in agriculture & allied subjects.

- **Low productivity of marigold due to local cultivar:** An OFT was conducted at 03 farmers fields on marigold variety Pusa Basanti, Pusa Narangi and Local. Among all the varieties, Pusa Narangi gave highest yield (198.4 q/ha), net income (Rs. 93730 per ha.) and B.C. ratio (3.08) followed by Pusa Basanti & Local.

- **Low production / failure of kharif crops due to aberrant weather condition:** A mixed crop of pigeon pea, jowar & urd bean in alternate rows was under taken as trial to cover the risk of aberrant weather conditions as against farmers practice only i.e. pigeon pea & jowar. The treatment pigeonpea + jowar + urd bean gave an yield of 15.20, 8.30 & 7.70 q/ha. as against farmers practice Pigeonpea + Jowar 16.5 & 6.0 q/ha respectively. The net income (Rs. 101411 / ha.) and B.C ratio (3.30) was more in this treatments than farmers practice (Rs. 64987 / ha.) and 2.49 respectively.

- **High incidence of leaf curl and low productivity of tomato due to infestation of fruit borer :** Effect of biointensive & chemointensive IPM was studied to manage leaf curl & fruit borer in tomato variety Kashi Aman and Namdhari 585. Chemointensive IPM (planting of 40 days old seedling of marigold around the field + installation of yellow sticky trap @15 per ha + removal and destruction of infested plant/fruit + application of chlorantranilprole 18.5 SC @ 0.35 ml/l and flubendiamide 20 WG @ 0.25 g/l in alternate manner) reduced the leaf curl incidence from 28.60 to 0.62 and fruit borer infestation from 25.73 to 5.0 percent and reduced the number of spraying from 6 to 3.

- **High incidence of anestrus in dairy animals due to maceration and mummification:** In dariy cattle due to incidence of mummification and maceration, there are the chances of corpus lutcum in ovary. As a result, the animals does not come into heat & show the estrus, which is a great loss for the dairy owners. Keeping in view, PGF2α analogues were used to disrupt anestrus phase and start cycle again with normal physiology. Due to trial 60% animals come into heat & conceived as against zero percent in farmers practices.

- **Low milk yield and anestrus in dairy animals due to micro nutrient deficiency & endoparasitic infestation:** In trial deworming with broad spectrum anthelmintic drugs and supplementation of urea mineral molasses block were undertaken for a period of 03 months. As a result 63.63 % animals conceived within 03-04 month and milk production enhanced 0.87 lit/day with an additional cost of Rs. 2.2 to 2.5 per head/day as against farmers practice.

- **Assessment of drudgery reduction and enhancing work efficiency of farm women during harvesting of paddy through serrated sickle :** Five trials were conducted on women in agriculture, during harvesting of paddy with serrated and non-serrated sickle. It was observed that using serrated sickle is more effecting in reducing drudgery, saving time and safety in use. The harvesting rate was 18.04 m^2/hour with serrated sickle and 15.34 m^2/hour with non-serrated sickle.

- **Assessment and enhancement in work efficiency and high drudgery of farm women involved in okra plucking:** Five trials was conducted on women during plucking of okra with and without plucker. It was observed that plucking of okra with plucker reduced drudgery, save time and easy to work. The working efficiency was 4.58 kg/hrs with plucker and 2.14 kg/hrs without plucker. (Fig. 7 & 8)

Fig. 7: Drudgery reduction by serrated sickle

Fig. 8: Drudgery reduction by okra plucker

Extension programmes: Extension programmes were conducted to disseminate and popularize improved agricultural technology for the benefit of the stakeholders of the farming community. During the period under report, KVK has organized 07 special days like International Yoga Diwas, Sadbhawana Diwas, Parthenium Control, World Honey Bees Day, Mahila Diwas, Agriculture Education Day, etc., involving 326 beneficiaries. For the dissemination of the technology at rapid mass KVK has organized 07 field days on Marigold, Pigeon pea, Vegetable Pea, Tomato and Wheat, where 252 farmers & farm women participated.

Table 5: Other extension activities

Activities	Date/No.	Beneficiaries
International Yoga Diwas	21.06.2017	20
Sadbhawana Divas	18.08.2017	25
Parthenium Control	18.08.2017	25
World Honey Bees Day	19.08.2017	48
Mahila Diwas	15.10.2017	51
National Agriculture Education Day	03.12.2017	57
World Soil Health Day	05.12.2017	100
Rastriya Krishi Unnat Mela	17.03.2017	115
Field Days	07	252
Sankalp Se Siddhi [New India Manthan]	25.08.2017	445
PPV& FRA	23.03.2018	175
Exposure Visit	23-25.02. 2018	275

Besides these, KVK organized Krishi Unnati Mela, Sankalp Se Siddhi, PPV&FRA Training programme and exposure visit involving 1010 participants (Table 5 and Fig. 9 & 10).

Fig. 9: Field Day on Mustard (RH-749)

Fig. 10: Sankalp Se Siddhi- New India Manthan

ICAR- KRISHI VIGYAN KENDRA, DEORIA

Training programmes: After the survey and assessment of farmer`s need, the priority based training, frontline demonstrations and other extension programmes were conducted on promotion of HYV in cereal, oilseed, pulses, vegetable & fruit crops promotion of farming system approach for sustainable agriculture entrepreneurship development, promotion of self help groups, integrated pest management, integrated nutrients management, direct sowing of rice, zero till sowing of wheat, beekeeping, importance of mulching, inter cropping, zero energy cool chamber, artificial flower making, designing of cloth by tie and dye method, mushroom cultivation, value addition, drudgery reduction, resource conservation technology, maintenance of farm machinery, equipments & storage of grains (Table 6 and Fig. 11).

Table 6: Training programmes organized

Training	No of courses	No. of participants
Practicing Farmers	80	2107
RY/ school dropouts	18	0476
Extension functionaries	06	0188

Fig. 11: Training programme to farm women

To improve economy of farmers 80 training courses in different thematic areas were conducted in which 2107 farmers & farm women participated. In addition, 18 vocational training programme on income generations for rural youth / school dropout were organized at on and off campus in which 476 rural youth participated. Six training courses on different topics were organized for extension functionaries of the district in which 188 extension functionaries were participated.

OFT (On Farm Trials): On the basis of problem diagnosed a total of 5 OFT on integrated disease management, value addition, drudgery reduction, livestock enterprises conducted at 33 farmers field for assessment/refinement of technologies (Table 7).

Table 7: Details of OFT

Category	No. of technology assessed & refined	No. of trials	No. of farmers
Technology assessed			
Crops	3	23	23
Livestock	02	10	10
Total	**5**	**33**	**33**

Front Line Demonstration: A total of 488 FLDs were the conducted in 48.2 ha area at farmers fields under oilseed, pulses, cereals, vegetables, livestock and other enterprises during the year of report (Table 8, 9, 10, & 11 and Fig. 12, 13, 14 & 15).

Fig. 12: FLD on Azolla Production

Fig. 13: FLD on Bottle Gourd

Fig. 14: FLD on Kitchen Garden

Table 8: Details of Front Line Demonstration

Crop	Thematic Area	Technology demonstrated	Variety	No. of farmers	Area (ha)	Yield (q/ha)				% Increase in yield
						Demo			Check	
						High	Low	Average		
Oilseed										
Mustard	Varietal Evaluation	Introduction of HYV	Pusa Tarak	38	12	21.8	16.2	20.1	15	34
Mustard (CFLD)	Varietal Evaluation	Introduction of HYV	Pusa Tarak	50	20	21.0	16.8	20.5	15.0	36.66
Pulses										
Chick pea	Integrated pest management	Technology	-	14	1.5	18.0	12.0	15.0	12.9	15.30
Cereals										
Paddy	Resource Conservation	DSR	HUBR 10-9	7	2.8	55.2	49.1	51.4	49.8	2.4
	Resource Conservation	DSR	BPT 5204	6	2.4	56.9	51.8	55.6	52.9	5.1
Scented Rice	Varietal Evaluation	HYV	P-1612	2	0.8	40.6	37.2	38.9	36.8	5.7
Vegetables										
Bottle gourd (Kharif)	Integrated pest management	Technology		05	0.4	140	110	125	110	13.6
	Varietal Evaluation	HYV	Narendra Rashmi	23	1.0	248.1	215.3	226.28	188.95	19.75
Bitter gourd	Integrated pest management	Technology		11	1.0	110	90	100	86	16.27
Tomato	Integrated Disease management	Technology		11	1.0	320	280	300	240	25.0
Chilli	Integrated Disease management	Technology		14	1.0	153	143	148	112	32.14
Brinjal	Varietal Evaluation	Introduction of HYV	Kashi Sandesh	21	2.0	631.44	424.27	582.68	408.45	42.65
Onion	Varietal Evaluation	Performance of HYV (Rabi 2016-17)	ALR, NHRDF Red 3	25	1.0	370.5	311.0	325.5	230.2	41.3
	Varietal Evaluation	Performance of HYV (Kharif 2017)	Nashik Red	18	1.3	297.2	243.0	275.80	232.60	18.5
			Total	**488**	**48.2**					

Table 9: FLD on livestock

Category	Thematic area	Name of the technology demonstrated	No. of farmers	No.of units (Animal/ Poultry/ Birds, etc)	Major parameters		% change in major parameter
					Demo	Check	
Cattle	Disease Management	Control of endoparasite	112	128	5.45	4.30	26.74
	Breeding Management	Improvement in conception percentage	10	10	06	03	80.00
	Feed Management	Feeding of Azolla in lactating cattle	04	04	6.1	4.8	27.08
Buffalo	Breeding Management	Improvement in conception percentage	08	08	05	02	150.00
Pig	Management	Pig farming through improved cross breed	01	01	88	52	69.23
Goat	Disease management	Endoparasite control	61	134	19.4	13.6	42.64

Table 10: FLD on Mushroom Production

Category	Name of the technology demonstrated	No. of Farmer	No. of units	Major parameters		% change in major parameter	Other parameter	
				Demo	Check		Demo	Check
Oyster Mushroom	Production of oyster mushroom (*Pleurotus florida*)	11	11	75 Kg/100 Kg wheat straw	-	-	1.5kg/ bag (of 2 kg)	-

Table 11: FLD on nutritional food security through kitchen gardening

Category and Crop	No. of farmers	No. of units	Yield (Kg)		% change in yield
			Demon-sration	Check	
Seasonal Vegetables (Zaid, 17)	40	40	197.50	157.81	25.15
Seasonal Vegetables (Kharif, 17)	25	25	183.28	149.81	22.34
Seasonal Vegetables (Rabi, 17-18)	6	6	64.15	58.25	10.12

Fig. 15: FLD on Oyester Mushroom

Extension Activities- A total of 453 programmes of extension activities viz., Advisory Services, Diagnostic visits, Kisan Ghosthi, Kisan Mela (Sankalp Se Sidhhi), Exhibition, Scientists' visit to farmers field, Celebration of important days, Special day celebration and Exposure visits were organized in which 13318 persons including 93 extension persons participated (Table 12 & Fig. 16 & 17).

Fig. 16: Vermicomposting unit visit by Hon`ble Agriculture Minister

Fig. 17: Celebration of women day

Table 12: Extension activities

Activities	No. of progra-mmes	No. of farmers	No. of Extension Personnel	TOTAL
Advisory Services	29	206	17	223
Diagnostic visits	165	713	-	713
Kisan Ghosthi	6	414	19	433
Kisan Mela (Sankalp Se Sidhhi)	1	760	20	780
Exhibition	11	8897	27	8924
Scientists' visit to farmers field	227	334	0	334
Celebration of important days	7	811	10	821
Special day celebration	5	1000		1000
Exposure visits	2	90		90
Total	453	-	-	-
Extension Literature	6	-	-	-
News paper coverage	81	-	-	-
Popular articles	24	-	-	-
TV Talks	3	-	-	-
Total	**477**	**13225**	**93**	**13318**

Table 13: Quantity of seeds plating materials produced and distributed

Crop	Name of the crop	Name of the variety	Quantity of seed (q)	Value (Rs)	Number of farmers
Cereals	Paddy	BPT 5204	0.84	2316	2
	Paady	HUR 10-9	4.35	12180	19
	Wheat	HD 2967	10.0	29000	2
Pulses	Pigeon Pea	NA 2	0.45	5400	1
Vegetables	Bottlegourd	Narendra Rashmi	0.0905	9050	11
	Cow Pea	Kashi Kanchan	0.147	5880	49
	Potato	K. Sinduri	32.0	64000	3
	Pumpkin	Kashi Harit	0.012	1435	20
	Spong Gourd	Kashi Divya	0.01185	1185	25
Fodder crop seeds	Azolla	Azolla (Animal Feed)	1.905	95120	3
Total			49.80635	225566	135

Table 14: Production of planting materials by the KVKs

Crop	Name of the crop	Name of the variety	Number	Value (Rs.)	Number of farmers
Vegetable seedlings	Brinjal	Kashi Uttam	1102	551	19
	Chilli	Kashi Anmol	942	471	18
	Onion	ALR	500000	10410	59
	Tomato	Kashi Aman	542	271	9
Total			502586	11703	105

Farm Produce: During the year under the seed production programme about 50 q. seed of cereals, Pulses, vegetables and about 5 lakhs vegetable plating materials were produce and distributed to the farmers. (Table 13 & 14).

ICAR - KRISHI VIGYAN KENDRA, KUSHINAGAR

Training Programmes: Krishi Vigyan Kendra, Kushinagar organized 88 need based on and off–campus training programmes under human resource development comprising diverse aspects of production technologies of cereals, oilseeds, pulses, vegetables, livestock, soil health management, value addition, household food security, rural craft and women empowerment benefitting a total of 2025 participants comprising 337 female and 1778 male farmers, rural youth and extension functionaries (Table 15 & Fig. 18).

Table 15: Training programmes organized

Clientele	No. of Courses	Male	Female	Total participants
Farmers & Farm Women	77	1504	205	1709
Rural Youths	3	47	15	62
Extension Functionaries	3	75	-	75
Sponsored Training	3	132	27	159
Vocational Training	2	20	-	20
Total	**88**	**1778**	**247**	**2025**

Fig. 18: On Campus Training on Bee Keeping

Frontline demonstration: Front line demonstration were conducted in 108.26 ha area at 848 farmers field on Paddy, Wheat, Mustard, Lentil, Sesmum, Soybean, Cauliflower, Onion, Green Gram, Pigeon Pea, Vegetable Pea, Marigold, Maize sheller, Kitchen Garden, Mushroom and Drumstick plant (Table 16 & Fig. 19 & 20).

Technology Assessment and Refinement

Inter cropping of potato in sugarcane: Krishi Vigyan Kendra conducted On Farm Trial on intercropping of potato in sugarcane sown by trench

Table 16: Frontline Demonstration at KVK, Kushinagar

Crop	Technology demonstrated	Horizontal spread of technology				
		No. of farmers	Area in ha	Demo Yield q/ha	Check Yield q/ha	Yield increase %
Paddy	Drum Seeder BPT 5204	12	4.22	42.33	30.66	27.57
Paddy	Drum Seeder Sarju-52	6	4.27	45.55	30.25	26.26
Paddy	Swarna sub 1	6	2.0	48.35	35.75	26.05
Paddy	P-1612	05	2.0	48.35	35.75	26.05
Paddy	Line Sowing in water logging	12	5.0	45.55	30.25	33.59
Wheat	Zero Tillage HD 2967	12	3.0	44.3	33.2	33.43
Wheat	Zero Tillage PBW 373	3	1.6	40.5	33.5	20.89
Vegetable Pea	Varietal performance Var-Kashi Udai	8	0.5	92.5	77.5	19.35
Cauliflower	Sabour Agrim	25	2.0	215.8	145.63	32.52
Potato	Intercropping of Potato Var. Kufri Sindhuri	5	0.4	74.0	-	
Sesamum	Line Sowing Var.Virat	15	10.0	3.93	0.75	80.92
Mustard	Line Sowing Var.PM 26	34	10.0	14.5	10.6	36.79
Green gram	Green Manuring IPM 02-03	215	50.0	9.30	7.10	23.66
Lentil	Seed Production Var.PL 8	49	6.5	16.6	11.8	40.68
Mushroom production	Oyster	06	020	Yield/Bag-3.6 kg. Yield /unit-21.6 kg		
Drudgery Reduction	Use of Maize Sheller	16	1.59	18 Kg./hr shelling	5.7 Kg. / hr Shelling	215 %
Nutritional security	Drumstick (Var.- PKM-1)	359	0.8 ha	Consumption of leaves per head-25 g/day, Saving in money spent on purchasing of green leafy vegetable-25%		
Nutritional garden	Vegetables	10	10 (150 m^2)	401	298	34.6
Nutritional garden	Vegetables	10	10 (150 m^2)	439	230	90
Nutritional garden	Vegetables	13	13 (150 m^2)	237	145	63.4
		848	**108.08**			

Fig. 19: FLD on Paddy (Sarju-52)

Fig. 20: FLD on Wheat (HD-2967)

method at five farmer's field. In the first treatment (T1) farmer sown the sugarcane with trench method where as in the second treatment (T2) farmer sown sugarcane as main crop and potato as intercrop in which 135 q / ha yield of potato obtained while main crop is yet not harvested. The data showed that farmers got additional income of Rs. 87000/- from potato as intercrop.

Weed management in rabi maize

The another On Farm Trial on weed management was conducted on rabi maize hybrid DKC 9081at 08 farmer's field. Result showed that effective weed control was observed with the use of Atrazine @ 1.0 kg.a.i./ha + Halosulfuron 90 g/ ha as post emergence.

Effect of water soluble fertilizer on yield and quality of banana.

An on Farm Trial was conducted on effect of water soluble fertilizer on yield and quality of banana at six farmers field. The trial was conducted in a plot size of 600 m^2 per treatment and total three treatments were carried out. Three treatments i.e. chemical fertilizers in root zone i.e., T_1- N:P:K:-250: 50:650 kg/ha, T_2- N:P:K:-800: 280:1200 kg/ha as recommended dose of fertilizers T_3- N:P:K:-600: 210:900 kg/ha (75% of recommended dose of fertilizers) + soluble fertilizers N:P:K:-18: 18:18 (10 g/l water) applied. The result indicated that 75% of recommended dose of fertilizers i.e. N:P:K: 600: 210:900 kg/ha + soluble fertilizers N:P:K:-18: 18:18 (10 g/l water) at 30 days interval recorded highest yield of banana per hectare i.e., 706.16 q/ha followed by 595.83 q/ha under T_2 and 372.67 q/ha under T_1-farmers practice (Fig. 21).

Fig. 21: OFT on Banana

Performance of timely sown wheat variety HD 2967 on different date of sowing

An on Farm Trial on performance of timely sown wheat variety HD 2967 was conducted at four farmers field. The farmer sown wheat during 15 November to 31 December In T_1 sowing of wheat was done from 1st November to 10th November for reducing the heat stress during the month of March and April, In the treatment T_2 the sowing of wheat was done from 11th November to 20th November and in the treatment T_3 sowing of wheat was done from 21st to 30th November. The result showed that an increase of 52.52 % in yield was recorded with early sowing of wheat as compared to farmer practice. Thus, farmers got Rs. 24225.00 as extra income. Early sowing of wheat (1st to 10th Nov.) also recorded highest yield i.e. 51.25 q/ha.

Effect of paddy drum seeder on yield of rice

An on farm trial was conducted on effect of paddy drum seeder at 5 farmer's field. In the treatment first i.e., T_1 transplanting of 21 days old seedling (NPK 120:60:40 manual weeding). In treatment two i.e. T_2 Sowing of pre-germinated seeds with drum seeder (NPK 120:60:40 and spraying of post emergence of herbicides Bispyribac Sodium 10% SC 250ml/ha.) and in the treatment three i.e. T_3 Sowing of pre-germinated seeds with drumseeder (NPK 120:60:40 and spraying of pre emergence of Pyrazosulfuran @200g/ha followed by post emergence of herbicides Bispyribac Sodium 10% SC 200ml/ha.) after 20 days transplanting. The yield was found highest (64.66q/ha) followed by T_2 i.e 58.81q/ha.

Availability of fruit & vegetable through nutritional garden

An on farm trial was conducted on nutritional garden at five farmer's field. The OFT consisted of three treatments, In the first treatment (T_1) farmer grown bottle gourd in Kharif 2017,cauliflower in Rabi 2017-18 and sponge gourd in Zaid 2018 and got the 755 kg total

Fig. 22: OFT on Nutrition Garden

production while in treatment second (T_2) farmer grown leafy vegetables like spinach and coriander & root and tuber vegetable like radish and beetroot and got the 827 kg total yield. In case of treatment three (T_3) farmer established a systematic nutritional garden in which leafy vegetables, root & tubers and other vegetables were grown throughout the year (Kharif 2017- Rabi 2017-18 and Zaid 2018) and farmers got 1389.2 kg total yield. In case of treatments three (T_3) farmer obtained 59.53 % more yield over T_1 (Fig. 22).

Extension Activities

To expedite the process of transfer of technology programme the KVK, organized 2 kisan gosthis where in 181 farmers participated . Four field days were organized covering 170 farmers for demonstration of technologies. One kisan mela was organized covering 1059 farmers. KVK participated in 7 exhibitions for awareness creation of farmers benefitting a total of 6453 farmers. A total 301 scientific visits were made to farmer's field visits by KVK officials and 82 diagnostic visits were made by the KVK scientists and S.M.S. for the benefit of 2753 farmers. One soil health camp was organized to the ultimate benefit of 85 farmers. 53 lectures were delivered as resource person benefitting more than 4789 farmers of kushinagar and adjoining districts. 3289 farmers visited KVK during 2017-18 (Table 17, 18, 19 & Fig. 23, 24, 25, 26).

Table 17: Extension Programmes

Activities	No. of programmes	No. of farmers	No. of Extension Personnel	TOTAL
Advisory Services	1945	5082	32	5114
Diagnostic visits	82	230	5	235
Field Day	04	167	3	170
Group discussions	3	65	7	72
Kisan Gosthi	2	166	15	181
Film Show	4	1209	8	1217
Self –help groups	2	35	2	37
Exhibition	7	6203	253	6453
Scientists' visit to farmers field	301	2746	7	2753
Farmers visit to KVK	3204	3204	85	3289
Kisan Mela	1	1000	58	1059
Plant/animal health camps	-	-	-	-
Lecture delivered	53	26	7	33
Method Demonstrations	18	290	13	303
Soil Health Card Distributed	15	250	05	255
Soil Health Camp	01	85		85
Plantation Programme	2	65	2	67
Swachhata Hi SewaFortnight	1	456	9	464
Celebration of important days				
Important Day -Celebration World Soil Day 05.12.18	1	136	8	144
Kisan Diwas 23.12.17	1	100	11	111
International women Day 08.03.18	1	42	03	45
Training on PPVFRA	1	137	11	148
Special Day Celebration Mahila Kisan Diwas	1	76	5	81
Meeting	1	10	24	34
Jai Kisan Jai Vigyan week	1	281	16	297
Total	**5652**	**22061**	**589**	**22647**

Table 18: Soil & water Analysis

Samples	No. of samples	No. of farmers
Soil	3750	3750
Water	500	500
Total	**4250**	**4250**

Table 19: Seed & planting material production

Samples	Quintal/Number	Farmers
Seed (q)	2866.945	2527901
Planting materials (No.)	22347	15560
Bio-Products (q)	25.0	-

Fig. 23: Soil Health Card distributionBy Shri Ganga Singh Kushwaha, M.L.A., Fazil Nagar

Fig. 24: Field Day on vegetable pea (Kashi Udai)

Fig. 25: Celebration of Mahila Kisan Diwas

Fig. 26: Field Visit By K. Raju, Financial Advisor of Chief Minister, U.P

Institutional Activities

TRAINING PROGRAMME AND OTHER ACTIVITIES

Doubling Farmers Income through Vegetable Production under *Mera Gaon Mera Gaurav* Praogramme

ICAR-Indian Institute of Vegetable Research, Varanasi organized training on 'Doubling Farmers Income through Vegetable Production' at KVK, Gazipur on 20th May 2017 for the benefit of the farmers from five adopted villages under '*Mera Gaon Mera Gaurav*' programme. The main theme of the training was to reduce the cost of vegetable cultivation with the help of improved and scientific technologies so that income of the farmers can be doubled in coming days. Scientists from ICAR-IIVR and KVK, Gazipur delivered lectures followed by discussion with farmers and finally improved varieties of vegetable seeds were provided to 413 vegetable growers for demonstrations. Later, the team visited the farmers' field to see the demonstrations conducted earlier and discussed the problems encountered by the farmers in vegetable cultivation.

Kisan Gosthi was organized at *Sansad Adarsh Gaon* 'Siow' of District Varanasi

ICAR-Indian Institute of Vegetable Research, Varanasi has adopted *Sansad Adarsh Gaon* 'Siow' during last- three years to harness nutritional security through Hi-tech vegetable farming under *'Mera Gaon Mera Gaurav'* programme. In this line, a one-day *Kisan Gosthi* was organized on 16th June 2017 in the village to sensitize the farmers about improved vegetable farming and importance of establishment of kitchen garden in every farm household. On this occasion, 256 vegetable kitchen garden packets were provided to the farmers of the village for growing vegetables at their door steps.

Training and Awareness programme on Riverbed Cultivation of Cucurbitaceous Vegetables

One day training and awareness programme on riverbed cultivation of cucurbitaceous vegetables was organized by ICAR-IIVR on 26th July 2017 in which more than 60 farmers participated from different villages of Varanasi, Mirzapur and Gonda district residing near the riverbed areas of the Ganga and Ghaghara. Dr. B. Singh, Director of the institute inaugurated the programme and urged the farmers to utilize modern practices for riverbed cultivation to get better yield. The training programme was sponsored by UP Council of Agricultural Research, Lucknow under the project "Evaluation of high yielding varieties and hybrids of cucurbit vegetables and standardization of their agro-techniques".

Visit of Director General, International Rice Research Institute, Philippines

Dr. Matthew Morell, Director General, International Rice Research Institute, Philippines visited ICAR-Indian Institute of Vegetable Research, Varanasi on 05th August, 2017 during his course of visit for establishment of South Asia Regional Centre of IRRI at NSRTC, Varanasi. Dr. Bijendra Singh, Director, ICAR-IIVR, Varanasi briefed about the ongoing activities and achievements of the institute, its impact on farmers and about the Institute future road map of research. He also expressed for collaborative works with IRRI for diversification of vegetables under rice wheat cropping system. During interaction meeting with scientists of the institute, Dr. Morell showed his keen interest and agreed for collaboration with IIVR for utilization of IIVR developed varieties under vegetable based Rice-wheat cropping system to enhance the farmers' income. He also briefed about the goals of the Regional Centre of IRRI for south Asia established at Varanasi for enhancing nutritional and livelihood security of the farmers' along with employment generation by developing value added rice and promoting quality rice.

Inauguration of Integrated Beekeeping Development Centre/Centre of Excellence Building

Shri S.K. Pattanayak, Secretary, Department of Agriculture, Cooperation & Farmers' welfare, Ministry of Agriculture and Farmers' welfare, Govt. of India visited ICAR-Indian Institute of Vegetable Research, Varanasi on 13th August, 2017. He inaugurated the "Integrated Beekeeping Development Centre/ Centre of Excellence" sponsored by National Bee Board, Department of Agricultural Cooperation & Family Welfare, Govt. of India. He highlighted the significance of bee keeping not only for enhancing the overall agricultural production and productivity of the country but also for rural employment generation through honey and other bee hive products. Dr. Bijendra Singh, Director, ICAR-IIVR, Varanasi briefed about the ongoing activities and achievements of the institute, and its impact on farmers.

Sadbhavana Divas

Sadbhavana Diwas had been observed in ICAR-Indian Institute of Vegetable Research, Varanasi on 18th August 2017. During the programme Dr. Bijendra Singh, Director of the institute administered the Sadbhavana Pledge to all the staff of the institute.

World Honey Bee Day

World Honey Bee Day was celebrated at ICAR-IIVR Varanasi on 19th August 2017. Hon'ble Shri Kedar Nath Singh, Member of Legislative Council, Varanasi, Government of Uttar Pradesh was the Chief Guest of the function. Shri Singh highlighted several programmes being operated by Government of India, like soil health card, beekeeping, crop insurance scheme, etc. for the benefit of farmers' community and doubling farmers' income. Shri Surendra Singh, Ex-Member of Legislative Assembly, Jaunpur encouraged the farmers to harness the benefits of different scheme launched by Government of India in the interest of farming community. He also highlighted the medicinal importance of honey and other bee hive products. Dr Ramashrit Singh, Ex-professor, RPCAU, Pusa, Samastipur advocated to follow the scientific methods of beekeeping to harness the various direct and indirect benefit. More than 217 farmers and bee keepers from Farmer FIRST Programme and other villages of Varanasi and Mirzapur district had participated in this programme. All the participants also took oath for New India Movement (2017-2022) under *Sankalp Se Siddhi*. On this occasion, extension folder on Integrated Beekeeping: At a Glance in Hindi was released by the chief guest and distributed among participants.

Parthenium awareness week

The world parthenium awareness week was observed during 16-22 August 2017 at ICAR-IIVR, Varanasi and the three KVK'S functioning under its administrative control, to make farmers and general public aware about the menace of parthenium, which is responsible for causing health problem in human

beings and animals beside deteriorating environment, loss of productivity and biodiversity. The activities such as parthenium uprooting, spraying herbicides, composting of uprooted biomass, farmers training, displaying exhibition materials etc. were organized at the institute and the KVKs campus as well as in the neighboring villages with involvement of scientists, farmers and other stake holders. Dr. Brijendra Singh, Director of the Institute, informed that the research farm of IIVR is Parthenium free.

Sankalp Se Siddhi Programme at KVKs Deoria, Kushinagar & Bhadohi

Sankalp Se Siddhi, New India Manthan programme had been organized at all the three Krishi Vigyan Kendra functioning under the administrative control of ICAR-IIVR on 25th August 2017 at Bhadohi, 29th August 2017 at Kushinagar and on 30th August, 2017 at Deoria. The programme comprised of video film having message of Hon`ble Prime Minister for doubling the farmer's income followed by the Sankalp Se Siddhi pledge administered to the gatherings by the Chief guest, technical sessions on improved technologies of agriculture & livestock and farmers-scientists interaction programme.

On this occasion Chief Guest at KVK, Deoria, Hon'ble Shri Surya Pratap Shahi, Minister of Agriculture & Farmers Welfare, Govt. of Uttar Pradesh highlighted the importance of soil health card to know the nutrient status of soil so as to reduce the cost of cultivation by minimizing the excessive use of fertilizers. The Guest of Honour, Shri Ravinder Kushwaha, Hon'ble M.P., Salempur applauded the activities of KVK, Deoria. Dr. Bijendra Singh, Director, ICAR-IIVR, Varanasi welcoming the guests and farmers highlighted the efforts being made by the KVK and institute for the benefits of farmers in whole Eastern Uttar Pradesh. Similarly, at KVK, Bhadohi, Chief Guest Hon'ble Shri Virendra Singh, Member of Parliament (Bhadohi) asked the farmers to adopt latest agricultural technologies along with limited use of natural resources to maintain the ecological balance. Here, Dr. Bijendra Singh, Director, ICAR-Indian Institute of Vegetable Research, Varanasi apart from the institute's activities for farmers, discussed the seven formulae suggested by Hon'ble Prime Minister to make strong and vibrant New India. In KVK, Kushinagar Chief Guest Hon'ble Member of Parliament, Shri Kalraj Mishra inaugurated the event and addressed the farmers focusing on the different agriculture schemes given by Government of India.

On this occasion, various stalls from public and private sectors were displayed at all the three KVKs for providing hands-on information to the farmers and other stakeholders. A total of 760 participants at Deoria, 520 at Bhadohi and 866 at Kushinagar including farmers, farm women, student, extension functionaries and officer from development department of attended the programme.

ISO 9001:2015 Certificate awarded to ICAR-Indian Institute of Vegetable Research

Keeping its pace on committed research, timely, efficient and effective service delivery to all the stakeholders, ICAR-IIVR, Varanasi achieved yet another milestone, certifying its quality services, by receipt of

ISO 9001:2015 Certification. Dr. Bijendra Singh, Director, ICAR-IIVR, Varanasi received the award on behalf of the Institute on 24th August 2017.

Training programme on Principles and Production Techniques of Hybrid Seeds in Vegetables sponsored by ICAR

ICAR sponsored 12 days training programme on *Principles and Production Techniques of Hybrid Seeds in Vegetables* was successfully organized for technical working in ICAR institutes/SAUs/ KVKs during September 12-23, 2017. Dr. AK Vyas, ADG (HRM), ICAR, New Delhi while inaugurating this training programme at ICAR-Indian Institute of Vegetable Research, Varanasi on 12th September 2017, emphasized that HRM unit at ICAR has been established for guiding and supporting the institutes from time to time for effective and timely

implementation of training programme for all categories of staff including technical and supporting staff. He said that to bridge the gaps in monitoring, organizing, coordination and implementation of training related activities ICAR has deputed nodal officer (HRD) in all its institutes. On this occasion, a training manual was also released by ADG (HRM), ICAR, covering different aspects of hybridization techniques in vegetables and distributed among the participants for their future reference.

A total of 18 technical personnel working in different ICAR Institutes/ SAUs/ CAUs/ KVKs from 08 different states which includes Uttar Pradesh, Madhya Pradesh, Uttrakhand, Assam, Odisha, Jummu & Kashmir, New Delhi and Andhra Pradesh had participated in this training programme. A series of lectures (29) and practical sessions (13) on various aspects such as development and maintenance of hybrids, principle and practices of hybrid seed production in various vegetables, seed processing, seed storage, role of public and private sector in hybrids etc. were covered during this 12 days training programme. At the end of day, Director emphasized on importance of hybrid seeds for enhancing the vegetable productivity.

Massive Cleaning Drive for Sarwatra Swachhta

In continuation to the various cleanliness activities being undertaken under the banner of *Swachhata Hi Seva* campaign launched on 15th September, 2017, a massive cleaning drive for *Sarwatra Swachhta* was undertaken by the staff of ICAR – Indian Institute of Vegetable Research, Varanasi, in the Adalpura/ Arazilines area, under the dynamic leadership of the Director of the Institute. Rigorously this cleaning campaign continued for 15 days.

Visit of Joint Secretary (Seeds), Ministry of Agriculture and Farmers' welfare, Govt. of India

The Joint Secretary (Seeds), Ministry of Agriculture & Farmers Welfare, Govt. of India, Dr. B. Rajendra visited ICAR-IIVR, Varanasi on 29th September, 2017. After brief discussion with Director of the Institute Dr. B. Singh and all the Heads of Divisions about various activities and achievements of the Institute, he visited the experimental area and expressed satisfaction the way experiments were being maintained in the field. He also visited the low energy seed storage facility and the Integrated Bee Development Centre (IBDC) of the Institute and appreciated the activities of the Institute.

Hindi Chetna Maas

Institute celebrated *Hindi Chetna Maas* from 14th September 2017 to 14 October 2017 in which competitions like debate, poster making, quiz and essay writing were organized and the awardees were felicitated at the valedictory function. This event was inaugurated by Dr. Ram Sudhar Singh, Former HOD (Hindi), U.P. College, Varanasi on 14th September 2017. In her inaugural address Dr. Singh expressed that all researches should be published in Hindi apart from

other technical bulletin and reference book, so that it may be used by maximum number of stakeholders which ultimately help in achieving the target of *Double farmers' income by 2022*. On this occasion, Dr. B. Singh, Director of the institute appreciated the work done by the institute in Hindi and also promoting it through various events during *Hindi Chetana Maas*. He appealed the scientists to document and publishes their work in Hindi also along with English. All the members of vegetable family in this institute actively participated in this one month Hindi Parv.

Union Minister of Agriculture and Farmers Welfare, Government of India Inaugurated Trainers Training on Vegetables Production

ICAR-IIVR organized Trainers Training on *Improved Production and Protection Technologies* in Vegetables at KVK, Piprakothi, East Champaran, Bihar which was inaugurated by Hon'ble Shri Radha Mohan Singh, Union Minister of Agriculture and Farmers Welfare, Government of India on 1st October 2017 which was attended by 40 officials of FPOs functioning with 9000 farmers at different block level in the district East Champaran of Bihar. On this occasion, Hon'ble Minister asked the FPO officials to organize series of training and exposure visits of farmers under their respective FPO with the help of ICAR-IIVR. He said that processing and quality seeds are an important factor in vegetables farming and farmers should produce their own seeds along with value addition for in vegetables for doubling their income. During the technical session scientists of this institute delivered their talk in interactive mode regarding the advancement in vegetable production technologies. Director of this institute, Dr. B. Singh focusing the efforts made by the institute for livelihood security of farmers asked the officials to educate farmers and other stakeholders of their FPO through learnt vegetables technologies and help them in improving their farming. At the end, 10000 kitchen garden packets of quality vegetable seeds were given to farmers through different FPOs.

IVth Interactive Meeting of Finance Officers of North Zone of Indian Council of Agricultural Research, New Delhi

Fourth Interactive Meeting of Finance Officers of North Zone of ICAR was held on 03rd November, 2017 at ICAR-IIVR, Varanasi. Shri SN Tripathi, Additional Secretary & Financial Advisor (DARE)/ICAR, New Delhi chaired the meeting and emphasized that all old pending paras of external audit reports and outstanding advance of CPWD/Government Departments/Officials of concerned Institutes may be pursued and got settled at the earliest.

In the meeting, Director (Finance), ICAR, New Delhi discussed the Agenda item wise (unsettled paras, outstanding advances, Bank Reconciliation Statement and other financial matters) with Finance officers of different ICAR Institutes of North zone. The position

of budget release and utilization of funds was also discussed by the Director (Finance) from the respective Institutes of ICAR. Dr. A.K. Vashishth, ADG (PIM), ICAR, New Delhi emphasized that Institute should follow the guidelines mentioned in the approved EFC/SFC for financial utilization. The meeting was attended by the Finance Officers of 35 Institutes of North Zone of ICAR, New Delhi.

IVth Interactive Meeting of NABARD

ICAR-IIVR organized IV Interactive Meeting of NABARD on 24th November 2017 in which 60 DDMs & other officials from different district of Uttar Pradesh had participated. This prestigious meeting was inaugurated by Chief General Manager, NABARD, Shri A.K. Panda in presence of General Manager, NABARD, Smt. Tulika Pankaj and Shri M. Mukherjee and Director of this institute. At the outset, Director, Dr. Bijendra Singh welcomes the guests and participants and briefly highlighted the achievements of the institute and also the efforts made by the institute to disseminate these technologies to farmers and other stakeholders for adoption and improving vegetable farming. On this occasion Chief Guest, Shri. A.K. Panda focused the vision of NABARD for linking research organizations with farmers and other stakeholders for doubling farm income. He also asked their officials to link farmers group of NABARD with ICAR-IIVR for adoption of improved technologies developed by this institute. During technical session, scientists of this institute discussed some of the important technologies developed by this institute which is ready for transfer in the farmers' field.

National Unity Day

In order to commemorate the birth anniversary of Sardar Vallabhbhai Patel, *Rashtriya Ekta Diwas* (National Unity Day) was observed in ICAR-Indian Institute of Vegetable Research, Varanasi on 31st October, 2017 under the dynamic leadership of Dr. B Singh, Director of the Institute. The Director emphasized the significance of unity and integrity among the people for peace as well as progress of the country. He further administered the *Rashtriya Ekta Diwas* Pledge to all the scientific and administrative staff of the institute.

Vigilance Awareness Week

ICAR-Indian Institute of Vegetable Research, Varanasi observed Vigilance Awareness Week (30th October 2017 to 4th November 2017). The vigilance awareness week commenced with administration of an integrity pledge on 30th October, 2017 by Dr. B Singh, Director, ICAR- Indian Institute of Vegetable Research, Varanasi to all the scientific and administrative staff of the institute.

Certified Farm Advisor Programme on Advanced Vegetable Production Technologies for Enhancing Productivity and Nutritional Security

ICAR-IIVR organized MANAGE Sponsored Certified Farm Advisor Programme on Advanced Vegetable Production Technologies for Enhancing Productivity and Nutritional Security during November 14-28, 2017 in which 14 Agriculture/Horticulture officials from Himachal Pradesh, Telangana, Tamilnadu, Haryana and Karnataka had participated. in this 14 days Training Progarmme a series of lectures and wide practical sessions to trainees were organized on various aspects such as origin, taxonomy, botany, physiology, production, protection, harvest and postharvest handling, organic farming and marketing of different major and minor vegetable crops. At the end of training programme, vegetable seeds sample and training manual were provided to the participants. As the result of this training programme an average increase of 21.14% gain in knowledge was recorded from the participants. Lastly, all the participants were assigned one mentor for this institute for monitoring and helping the participants to work in their actual work place in transferring the knowledge gained during training programme.

Agriculture Education Day

Agriculture Education Day was celebrated by the institute on 03rd December 2017 under the chairmanship of Director, Dr. B. Singh. This important event was attended by Principals and about 217 students from 10 different schools along with 115 rural youth from nearby villages. On this occasion, students had participated in debate, quiz and painting competition organized by institute for which they were also awarded by Director of this institute. Considering the importance of agriculture in todays' education, Dr. B. Singh, Director, ICAR-IIVR sensitized the participants about perspective of agriculture and asked them to promote the agriculture

technologies in their locality for livelihood and nutritional security. On this occasion, Chief Guest, Prof. B.K. Tripathi, Director, IUCTI, Varanasi emphasized the importance of agriculture education in present national scenario and doubling farm income by 2022. During technical session, scientists of this institute discussed the advancement made in agriculture R&D to the students and rural youth.

National Conference Organized on Food and Nutritional Security through Vegetable Crops in Relation to Climate Change

National Conference on *Food and Nutritional Security through Vegetable Crops in Relation to Climate Change* was organized during 9-11 December, 2017 at ICAR- Indian Institute of Vegetable Research, Varanasi. The conference was inaugurated by Hon'ble Smt. Krishna Raj, Minister of State for Agriculture and Farmers Welfare, Govt. of India. The occasion was graced by the presence of Dr Kirti Singh, Former Chairman ASRB; Dr G Kalloo, Former DDG (Hort) ICAR; Dr AK Singh, DDG (Hort. Sci.), ICAR and Dr K V Peter, Former VC, KAU, Kerala. On this occasion several awards of ISVS were distributed by Hon'rable Smt. Krishna Raj to distinguished scientists and progressive farmer. Smt Krishna Raj also released the souvenir and abstract book of the conference along with technical bulletins and processed products like green chilli powder and Moringa soup powder. Dr. AK Singh emphasized on effective marketing of vegetables and value addition for better remuneration to farmers. Dr B Singh, the Director of the institute, welcomed the chief guest, and all the delegates of the conference. Dr G Kalloo advocated the adoption of modern breeding and biotechnological tools to develop the climate resilient crops. Dr Kirti Singh raised the issue of fluctuations in market price of vegetables and need of tackling the same to double the income of farmers.

Agriculture Awareness Campaign under Farmer FIRST Program

ICAR-IIVR organized *Agriculture Awareness Campaign* in Dhanapur village of Arazilines block, Varanasi adopted under Farmer FIRST Programme on 19th December 2017 in collaboration with Regional Directorate of Publicity, Ministry of Information and Broadcasting and Ministry of Agriculture & FW, Government of India. On this occasion, Director of this institute, Dr. B Singh provided agriculture slogan written T-Shirts to the students of Krishna Government School and showed the flag for agriculture awareness rally which after passing from different villages later turns into a farmers' interaction programme in the village Dhanapur. During interaction programme with students, farmers and farm women, Director of the institute, Dr. B Singh appealed the mass to work in small self-help group and focused on different agriculture based enterprises like honeybee keeping, mushroom cultivation, nursery management, net-house farming, vegetables & flower cultivation etc. for better return. He also felicitated students and rural youth for their participation in quiz programme organized on this occasion and making significant contribution in promoting agriculture in their village.

International Training Programme organized on Production, Processing and Marketing of Organic Vegetables

ICAR-Indian Institute of Vegetable Research had successfully organized 15 days International Training Program under FTF ITT on *Production, Processing and Marketing of Organic Vegetables* during 06-20 February 2018 for 21 Executives from 11 Afro-Asian countries viz., Afghanistan, Botswana, Republic of Congo, Ghana, Kenya, Liberia, Malawi, Mongolia, Myanmar,

Sudan and Uganda. There were 11 male and 10 female participants out of whom 14 participants were from Agriculture Department, 05 from Agriculture Marketing & Horticulture Department and 02 from Private Sector. Feed The Future India Triangular Training (FTF ITT) is a Joint Programme of US government represented by USAID and India government (Ministry of Agriculture & Farmers Welfare) represented by MANAGE, Hyderabad launched on 25th July, 2016 at New Delhi. FTF ITT aims to address Human and Institutional Capacity Gaps in Food & Nutritional Security in selected African and Asian Countries by providing training to Extension Practitioners.

Prof. Panjab Singh, presently Chancellor of RLB Central Agricultural University, Jhansi and President, NAAS, New Delhi was the Chief Guest during the inaugural function of this international event on 7th February 2018. While addressing the delegates, he shared his experience of working in different African and Asian countries and emphasized the participants to learn and implement the technologies of organic vegetables farming and processing in their respective countries as this is the need of hour in global perspective for income and employment generation along with nutritional security. On this occasion, Guest of Honour Dr. B.K. Paty, Director (OSPM), MANAGE, Hyderabad,

emphasized on agricultural marketing extension and supply chain management. Dr. Bijendra Singh, Director, ICAR-Indian Institute of Vegetable Research, Varanasi welcomed the delegates and guests on this occasion. While focusing on the achievements of the institute, he emphasized on the opportunities of organic vegetable production in Afro-Asian countries. On this occasion, a training manual prepared for the purpose was released and given to the participants.

During this international training program, apart from series of practical orientations and educational visit, a total of 40 classroom lectures were designed which were delivered by highly qualified experts from the institute as well as other organizations like BHU, Varanasi; APEDA, New Delhi; CH&AF, Passighat (Arunachal Pradesh); IIPR, Kanpur and IISR, Mau. This international training program showed a great impact on the knowledge level of the participants

which can be viewed by the average increase of 92.4% in pre and post score achieved by the 21 participants. This increase in score varied from 21.4% fetched by Ms. Igecha Anne Njeri of Kenya to 280.0% by Ms. Manal Ismael Mohammed Ahmed of Sudan. On the basis of this learnt experience in organic vegetables farming, the participants developed their Back at Work Plan and presented individually during sessions which was followed by healthy discussions among the experts and participants.

Chief Guest of valedictory session, Dr. Prithvish Nag, Vice-Chancellor, MG Kashi Vidyapeeth, Varanasi while addressing the participants, expressed the needs of organic farming seeing the hazardous effect of chemical residues in vegetables. Director, ICAR-IIVR, Dr. Bijendra Singh thanks the participants for their active participation and expects to implement the learnt experiences in their work place. Earlier, the Course Coordinator presented the training report to the house followed by feedback given by the participants. At the end, Certificate was given to the participants by the Chief Guest for their successful completion of training program.

Hon'ble Union Minister of Agriculture & FW Inaugurated North Zone Regional Farmers Fair

Hon'ble Union Minister for agriculture and farmers welfare, Shri Radha Mohan Singh Ji inaugurated the

North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi during 23-25 February 2018 in the presence of Shri Surya Pratap Shahi, Minister of Agriculture & FW, Government of Uttar Pradesh and other dignitaries from central and state departments. This Regional Farmers Fair was organized by ICAR-Indian Institute of Vegetable Research, Varanasi in collaboration with Department of Agriculture and Co-operation, Ministry of Agriculture and farmer welfare, Govt. of India, New Delhi; PPV&FRA, New Delhi; Department of Agriculture, Govt. of Uttar Pradesh and APIV, Varanasi.

Shri Radha Mohan Singh ji, while addressing the farmers briefed about the various schemes launched by the Govt. to achieve the target of doubling the farmers' income by 2022. He appreciated the efforts made by the scientists, farmers and other stakeholders engaged in agriculture for safeguarding the food security through green revolution by developing advanced technologies

and high yielding good quality seeds. He advocated the farmers to prepare the soil health card for proper management of soil health and to reduce the production cost by 8% to 10%. He also highlighted the schemes like *Pradhan Mantri Fasal Bima Yojna, Operation Green, e-NAM, Gokul Gram Nirman* and other schemes related to fishery, beekeeping, horticulture, watershed management, etc. He also motivated the farmers to create the Farmers Producer Organization to get maximum facilities of Government Scheme and harness better price of their produce.

Shri Surya Pratap Shahi ji, Minister of Agriculture & FW, Government of Uttar Pradesh, informed that during Kharif season of this year, total production of cereals were about 192 lakh metric tonnes, which is 12 lakh metric tonnes more, compared to last year. He also said that Banana in 8 district and Chilli in 7 district of UP has been covered under Pradhan Mantri Fasal Bima Yojna. Dr. Bijendra Singh, Director, ICAR-IIVR in his welcome address highlighted the importance of horticultural crops in doubling the farmers' income. He also said that for sustainable agriculture, diversified farming including dairy, beekeeping, poultry, fishery, etc. is the need of the hour.

'**Sabji Gyan**' a mobile APP developed by ICAR-IIVR in collaboration with TCS has been launched by Hon'ble Chief Guest, Shri Radha Mohan Singh Ji on this occasion. The APP will be useful to vegetables growing farmers where they can simply ask their queries by voice messages. 70,000 farmers have already been connected to this APP. This APP can be downloaded from http://iivr.tcsmkrishi.com/mKRISHI_IIVR/.

On this occasion, progressive farmers at national and state level were awarded with medals for innovative farming and setting the role model for other farmers. Shri Pradumn Singh from Punjab bagged gold medal at national level while silver medal was given to Shri Gauhar Ahmad Gani from Jammu & Kashmir and Shri Hariman Sharma from Himachal Pradesh. The Bronze medal was awarded to Shri Chaman Saini from Rajasthan and Shri Ramesh Zode from Maharashtra. At state level, Gold medal was awarded to Shri Rakesh

Singh from Kushinagar, Silver medal to Shri Naresh Sirahi from Meerut and Bronze medal to Sri Ramlal Verma of Varanasi. A special award, Pandit Deen Dayal Upadhyay Antyodaya Krishak Puraskar was awarded to Miss Rupkala Devi from Farmer FIRST Programme, Varanasi for her enormous contribution in family farming. One progressive farmer from Jammu & Kashmir, Uttarakhand, Haryana, Himachal Pradesh and Punjab were also awarded with a regional level award.

On this occasion Shri Anil Rajbhar, Minister for Food Processing, Govt. of UP, Shri Ravindra Jaiswal, MLA, Shri Surendra Singh, MLA, Shri Chet Narayan Singh, MLC, Shri KedarNath Singh, MLC, Smt Aparajita Sonkar, Zila Panchayat Adhyaksh and Dr KV Prabhu, Chairman, PPVFRA, New Delhi were present.

Farmers Witnessed the Address of Hon'ble Prime Minister of India during Awareness Programme on Honeybee Management

Honeybees play a major role as crop pollinator along with providing income and employment with its bye-products like honey, wax, royal jelly etc. Considering this importance and need for mass promotion, an awareness programme on honeybee management was organized by this institute under Farmer FIRST Program on 17th March 2018 in which more than 350 farmers and farm women from adopted villages had participated. On the same day, Hon'ble Prime Minister of India was addressing the farmers of nation live from ICAR-IARI, New Delhi on the inaugural function of Krishi Unnati

Mela which was witnessed by the participants. During his address, Hon'ble PM focused on enhancing farmers' income by promoting organic farming and other agriculture enterprises including honeybee keeping. He advocated four concepts to the farmers for enhancing their income *ie.*, first reduce cost of cultivation by adopting improved technologies, second take steps for getting appropriate price of produce, third reduce post-harvest losses from farm to market and fourth adopt new agriculture based enterprise for extra return. In his address Hon'ble PM appealed the farmers to join their hands with government programme for the benefit of farming and livelihood security. On the occasion of this awareness programme in the institute, Chief Guest, Hon'ble Member of Parliament, Shri Surindra Narayan Singh, address the participants and asked them to follow the appealed made by Hon'ble PM of India. He also focused on the government programme operating the district for the benefit of farmers.

AWARDS, HONOURS AND RECOGNITION

1. Anant Bahadur, Principal Scientist conferred Dr. Harbhajan Singh Gold Medal for the best research paper published in Vegetable Science 2016, 43(2): 208-215 entitled "Tomato genotypes grafted on eggplant: Physiological and biochemical tolerance under waterlogged condition" during XXXV Group Meeting of AICRP(VC) held in June 24-27, 2017 at ICAR-IIHR, Bangalore.
2. B.Singh conferred Dr. Kirti Singh Life Time Achievement Award-2017 for outstanding contribution and leadership in the area of Vegetable Science by Indian Society of Vegetable Sciences.
3. B.Singh conferred Kalayya Krishnamurthy National Award 2016-17 for outstanding contribution in Agriculture Research by University of Agriculture Sciences, Bengaluru on 04th October, 2017.
4. B.Singh conferred Mahamana Vidya Ratan Award-2017 by Devashram Trust (an International Charity for Global Peace) for outstanding contribution in the field of Agriculture, Environment and Good Governance and Discipline Innovations and Creativity on 25th November, 2017.
5. BR Meena, A Singh, M Manjunath, K Nagendran and AB Rai received Best poster award on "Characterization and evaluation of potential PGPR isolates as biocontrol agent against Fusarium wilt and collar rot pathogens in tomato" in National Conference on Food and nutritional security through vegetable crops in relation to climate change" (NCVEG17) at ICAR-IIVR Varanasi during 9-11 December, 2017.
6. C.Manimurugan, R. Kumar, P.M. Singh, M. Loganathan, S. Saha and B.Singh received best poster award on "Effect of botanical extracts on internally seed borne pathogen (*Macrophomina phaseolina*) of vegetable cowpea". Poster presented in Session on Quality Seed & Planting Materials in Natl. Conf. on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)' at ICAR-IIVR, Varanasi during 9-11 December, 2017
7. G.K. Choudhary conferred Best Ph. D. Student of the year award at Govind Ballab Pant University of Agriculture & Technology, U. S. Nagar, Pantnagar during November 2017.
8. G.K. Choudhary conferred Excellence in Extension award in national conference on Livelihood and Food Security (LFS) organized by Society for Agriculture Innovation, Ranchi and Development and BVC, Patna from 27-28 February 2018.
9. N. Gupta, C. Manimurugan, P. Chaukhande, J. Halder, P.M. Singh, R. Kumar, L. Mishra, S.K. Singh and T. Chaubey 2017 conferred best poster presentation award on "Effect of pollinator attractant with nano particles spray on pollinator activities seed yield and quality of bottle gourd and sponge gourd" in National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17) at ICAR-IIVR Varanasi during 9-11 December, 2017.
10. R. Singh conferred Excellence in Extension award in national conference on Livelihood and Food Security (LFS) organized by Society for Agriculture Innovation, Ranchi and Development and BVC, Patna from 27-28 February 2018.
11. R.P.Chaudhary conferred as Excellence in Extension award in International Conference on Advances in Agriculture & Applied Sciences for Promoting Food Security at Hotel Mirage Lord Inn Battishputli, Kathmandu, Nepal organized by Society of Agriculture innovation and Development, Ranchi from 13-15 May, 2017.
12. Rajesh Kumar, A.C. Rai, A. Rai, M. Singh and B. Singh conferred best poster award on "Transcriptome Analysis and Gene Expression Studies for Anthracnose Disease in Chilli (*Capsicum annuum* L.)" in National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17) at ICAR-IIVR Varanasi during 9-11 December, 2017.
13. Rajesh Kumar, P.M. Singh, C. Manimurugan, Nakul Gupta, S.K. Singh and B.Singh conferred best poster award on "Effect of Storage Temperature on Pollen Viability of Cucurbits". in National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17) at ICAR-IIVR Varanasi during 9-11 December, 2017.
14. Shweta Kumari, K Nagendran, C Manimurugan, AB Rai and B Singh received best poster presentation award on "Natural mixed infection of two orthotospoviruses on bitter gourd (*Momordica charantia* L.) in India" in National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17) at ICAR-IIVR Varanasi during 9-11 December, 2017.

15. Singh, R., Prasad, R.N., Singh, S.K., Chaukhande, P., Roy, S. and Singh, B. conferred best poster award on "Weed control with pre-& post emergence herbicides for increasing pod yield of french bean" in National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17) at ICAR-IIVR Varanasi during 9-11 December, 2017.

16. T. Chaubey, B. Singh, S. Pandey, R.K. Singh, D.K. Upadhyay, S. Choubey and J.K. Gupta conferred best poster award on "Morpho-phenological characterization based on distinctiveness, uniformity and stability test among the extant cultivars of tomato and view of phytochemical properties". in National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17) at ICAR-IIVR Varanasi during 9-11 December, 2017.

17. V. Singh conferred Scientist of the year award by Society of Biological Sciences and Rural Development, Allahabad, Uttar Pradesh he during 10-11 November, 2017.

18. B. Singh awareded Certificate of Recognition-2017 from Protection of Plant Varieties and Farmer's Right Authority (PPV&FRA) for properly implementing the IPR law, New Delhi on 27th Feb., 2017.

19. B. Singh conferred best article award for article entitled Subjiyon me bahumulya aanuvanshaik sansadhan, Phal Phool, ICAR-DKMA, New Delhi. A hindi article published in Phal Phool magazine in the vol. (Jan-Feb., 2017) by ICAR-DKMA, New Delhi.

20. Jagdish Singh, Principal Scientist & HOD (I/C) nominated by Hon'ble Governor of UP as Member, General Body of U. P. State Council of Science & Technology, Lucknow *w.e.f.* July 2016 - 2019.

21. Jagdish Singh, Principal Scientist & HOD (I/C) served as an Expert panel member (Pulse Crops) of FSKAN (Food Safety Knowledge Assimilation Network), a framework for scientific cooperation in the area of food safety and nutrition, {FSSAI}.

22. Sudhir Singh served as Expert panel member on Food Additives, Flavouring, and Processing aids, Materials in contact with food since 2017 at Food Safety and Standards Authority of India, New Delhi.

HUMAN RESOURCE DEVELOPMENT

Training and Capacity Building

Training

Name of Scientist	Title of training	Duration	Held at
Y. Bijen Kumar	Training programme on Pesticide Residue Analysis	17 April - 16 May, 2017	NIPHM, Hyderabad, Telangana
Vidyasagar	One month orientation training	18 April, 2017 - 17 May, 2017	ICAR-IIVR, Varanasi
	Three months attachment training	22 May, 2017 - 21 August, 2017	ICAR-IIHR, Bengaluru
	Management of plant genetic resources	06-19 March, 2018	NBPGR, New Delhi
A.N. Tripathi	Production protocol for biocontrol agents & microbials bio-pesticide	01-21 June, 2017	NIPHM, Hyderabad
	21 days Training programme on Production Protocol for bio-control agents microbial bio-pesticides and quality analysis of microbial bio pesticides	01- 21 September, 2017	NIPHM, Hyderabad, Telangana
B.R. Reddy	CAFT training on next generation sequencing of its applications in crop sciences	01-22 September, 2017	NRCPB, New Delhi
B.R. Meena	Current techniques & advances in mass culturing of microbials for the production of biopesticide	05-25 September, 2017	NBAIR, Bengaluru
C. Manimurugan	Advances in quality seed production of vegetable crops	06-26 September, 2017	Dr. Y.S. Parmar University of Horticulture & Forestry, Solan
Rameshwar Singh	Principal & practices for hybrid seed production in vegetables	12-23 September, 2017	ICAR-IIVR, Varanasi
	Enterprenureship development of sustainability in vegetable processing industries	01-21 December, 2017	ICAR-IIVR, Varanasi
Sunil Gupta	Principal & practices for hybrid seed production in vegetables	12-23 September, 2017	ICAR-IIVR, Varanasi
Viswanath	Principal & practices for hybrid seed production in vegetables	12-23 September, 2017	ICAR-IIVR, Varanasi
Pankaj Kumar Singh	Principal & practices for hybrid seed production in vegetables	12-23 September, 2017	ICAR-IIVR, Varanasi
P.C. Tripathi	Principal & practices for hybrid seed production in vegetables	12-23 September, 2017	ICAR-IIVR, Varanasi
Sanjay Kumar	Regular Driver Training	19-23 Septmber, 2017	ICAR-CIAE, Bhopal
Rani A. T.	One month orientation training	23 October - 22 November, 2017	ICAR-IIVR, Varanasi
	Three month attachment training	27 November, 2017 - 26 February, 2018	ICAR-IARI, New Delhi

K. Nagendran	CAFT training on Use of biotechnological & conventional tool in understanding virus-host interaction	07-27 November, 2017	ICAR-IARI, New Delhi
Rekha Singh	Enterprenureship development of sustainability in vegetable processing industries	01-21 December, 2017	ICAR-IIVR, Varanasi
Manjunatha Gowda Thondihalu	21 days training programme on Bio-pesticides for Crop Protection and Improvement: Emerging Technology to Benefit Farmers	02-22 February, 2018	GBPUAT Pantnagar, Uttarkhand

Training and Skill Development of Farmers, Field Functionaries & Entrepreneur

Name of the Programme	Date	Sponsored by	Number & Nature of Participants
Integrated Vegetable Production Technologies	7-9 June, 2017	Sadachar Samiti, Allahabad	45 Farmer
Integrated Vegetable Production Technologies	14-15 July, 2017	Agriculture Department, Pratapgarh	40 Farmer
Improved Vegetable Production Technologies	21-22 July, 2017	Department of Agrilculture, Mau	30 Farmer
Improved Vegetable Production Technologies	25-26 July, 2017	Durgavati Gramin Vikash Samiti, Gonda	32 Farmer
Improved Vegetable Production Technologies	18-19 August, 2017	PROW, NABARD, Varanasi	30 Farmer
Beekeeping management	19 August, 2017	Farmers FIRST, ICAR-IIVR, Varanasi	200 Farmer
Organic farming in vegetables	05-09 September, 2017	ATMA, Patna	30 Farmer
Principles and Production Techniques of Hybrid Seeds in Vegetables	12-23 September, 2017	ICAR-IIVR-HRD	19 Farmer
Integrated production and management of vegetable production	25-27 September, 2017	Sewa International, Ramnagar, Varanasi	20 Farmer
Kitchen garden for nutritional security	01 October, 2017	NHB, ICAR-IIVR, Varanasi	50 Farmer
Integrated production and management of vegetable	04-06 October, 2017	ATMA, Bhagalpur	20 Farmer
Seed Production Techniques in Vegetables	09-15 October, 2017	ATMA, Muzaffarpur	30 Farmer
Vegetables Production and Marketing	07-10 November, 2017	ATMA, Samastipur	20 Farmer
Improved Vegetables Production technologies and Marketing	18-20 December, 2017	ATMA, Motihari	35 Farmer

Improved Vegetables Production technologies and Marketing	27-29 December, 2017	NABARD, Mirzapur	20 Farmer
Sustainable Vegetable Production Technologies under Changing Climate Scenario	03-06 January, 2018	JEEVIKA, Bihar	30 Farmer
Sustainable Vegetable Production Technologies under Changing Climate Scenario	09-12 January, 2018	JEEVIKA, Bihar	30 Farmer
Sustainable Vegetable Production Technologies under Changing Climate Scenario	17-20 January, 2018	JEEVIKA, Bihar	30 Farmer
Sustainable Vegetable Production Technologies under Changing Climate Scenario	1-4 February, 2018	JEEVIKA, Bihar	30 Farmer
Integrated Vegetable Production Technologies	12-13 March, 2018	Jan Kalyan Sanstha, Meerut	31 Farmer
Integrated Vegetable Production Technologies	14-15 March, 2018	ATMA, Supol, Bihar	12 Farmer
Integrated production and management of vegetable	20-24 March, 2018	ATMA, Darbhanga	20 Farmer
Integrated production and management of vegetable	20-24 March, 2018	DHO, Darbhanga	10 Farmer

Training and Skill Development of ICAR/SAUs/State/KVKs Officials

Name of the Programme	Date	Sponsored by	Number & Nature of Participants
Entrepreneurship Development in Vegetables	13 November, 2017	ABI, ICAR-IIVR, Varanasi	70
Module II Training on Advanced Vegetable Production Technologies for Enhancing Productivity and Nutritional Security	14-28 November, 2017	MANAGE, Hyderabad	14
Entrepreneurship development for sustainability in vegetable processing industries	01-21 December, 2017	ICAR Winter School	20
Entrepreneurship development for sustainable vegetable seed production and marketing	26 December, 2017	MANAGE, Gazipur & ABI, ICAR-IIVR, Varanasi	40
Feed The Future India Triangular Training (FTF ITT) program on production, processing and arketing of organic vegetables	6-20 February, 2018	USAID & MANAGE, Hyderabad	21
Recent advancement in integrated management of pest and disease in vegetables	13-15 February, 2018	BAYER, Lucknow	10

Seminar/symposium/conference/workshop attended

Name of Scientist	Title of seminar/ symposium/ conference/ workshop	Duration	Held at
Nagendran K.	XXXV group meeting of AICRP (VC)	24-27 June, 2017	ICAR-IIHR, Bengaluru
	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
	National Workshop on Revisiting FOCARS: Reflections and feedback of Trained Scientist	15-16 March, 2018	ICAR-NAARM, Hyderabad
B.Singh	Attended Intenational Symposium on Horticuture: Priorities and Emerging trends; Co-Chaired the technical session II	5-8 September, 2017	ICAR-IIHR ,Bengaluru
	Participated in XXIV annual workshop on KVKs in U.P and Uttarakhand	8 June, 2017	ICAR-IISR, Lucknow
B R Meena	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR Varanasi
Pratap A. Divekar	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
A. Chaurasia	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
Y. Bijen Kumar	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
	National Workshop on Revisiting FOCARS: Reflections and feedback of Trained Scientist	15-16 March, 2018	ICAR-NAARM, Hyderabad
A.N. Tripathi	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
	National Seminar on Water and soil management for agriculture & livelihood security under climate change	8-9 September, 2017	Sunbeam College for Women, Varanasi
Shweta Kumari	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
	National Workshop on Revisiting FOCARS: Reflections and feedback of Trained Scientist	15-16 March, 2018	ICAR-NAARM, Hyderabad
Manjunatha Gowda Thondihalu	XXXV group meeting of AICRP (VC)	24-27 June, 2017	ICAR-IIHR Bengaluru
	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi

K.K. Pandey	XXXV group meeting of AICRP (VC)	24-27 June, 2017	ICAR-IIHR, Bengaluru
	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
Jaydeep Halder	National Conference on Food and nutritional security through vegetable crops in relation to climate change (NCVEG17)	9-11 December, 2017	ICAR- IIVR, Varanasi
	National Seminar on Water and soil management for agriculture & livelihood security under climate change	8-9 September, 2017	Sunbeam College for Women, Varanasi
	XXVI Annual Group meet of AICRP on Biological Control of Crop Pests	16-17 May, 2017	Dr.YSPUHF, Solan, Himachal Pradesh
D.R. Bhardwaj	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	9-11 December, 2017	ICAR-IIVR, Varanasi
	XXXV group meeting of AICRP (VC)	24-27 June, 2017	ICAR-IIHR, Bengaluru
	Advance Vegetable production technologies for enhancing productivity and nutritional security	14-28 November, 2017	ICAR-IIVR, Varanasi
	Feed The Future India Triangular Training (FTFITT) International Training programme on production, processing and marketing of organic vegetables	6-20 February, 2018	MANAGE, Hyderabad at ICAR-IIVR, Varanasi
N. Rai	XXXV group meeting of AICRP(VC)	24-27 June 2017	ICAR-IIHR, Bangalore
	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
	NICRA Annual group meeting	24 March, 2018	ICAR-IIHR, Bangalore
Rajesh Kumar	XXXV group meeting of AICRP(VC)	24-27 June, 2017	ICAR-IIHR, Bengaluru
	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
Vikas Singh	Nutritional Security, Environmental Protection: Present Scenario and Future Prospects	10-11 November, 2017	Vigyan Parishad, Main Hall, Allahabad
	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
	Novel Applications of Biotechnology in Agricultural Sectors: Towards Achieving Sustainable Development Goal (INABASDG-2018)	20-21 March, 2018	BHU, Varanasi
	XXXV group meeting of AICRP(VC)	24-27 June, 2017	ICAR-IIHR, Bengaluru

B.K. Singh	XXXV Annual Group Meeting of AICRP-VC	24-27 June, 2017	ICAR-IIHR, Bengaluru
	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
S.G. Karkute	Developing a roadmap for agricultural knowledge management in India	07-28 September, 2017	ICAR, DKMU, New Delhi
	National Workshop on Revisiting FOCARS: Reflections and Feedback of Trained Scientists	15-16 March, 2018	ICAR-NAARM, Hyderabad
	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
	5th Annual South Asia Biosafety Conference	11-13 September, 2017	BCIL, Bangalore
B.R. Reddy	XXXV group meeting on AICRP (VC)	24-27 June, 2017	ICAR- IIHR, Bangalore
	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
T. Chaubey	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
S.K. Tiwari	XXXV group meeting on AICRP (VC)	24-27 June, 2017	ICAR- IIHR, Bangalore
	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
	National Seminar on Water and soil management for agriculture & livelihood security under climate change	8-9 September, 2017	Sunbeam College for Women, Varanasi
Neeraj Singh	National Conference on Food & Nutritional Security through Vegetable Crops in relation Climate Change (NCVEG-17)	09-11 December, 2017	ICAR-IIVR, Varanasi
	National Seminar on Water and soil management for agriculture & livelihood security under climate change	8-9 September, 2017	Sunbeam College for Women, Varanasi

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2. Singh B. 2017. Presented a lead talk on "Resistance breeding in Okra" during International symposium on "Horticulture: Priorities & Emerging Trends" on 5-8 September, held at ICAR-IIHR, Bengaluru.
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59. Singh N, Roy S and Singh Jagdish. 2017. ICT tools for technology transfer in agriculture. *In: National Conference on Food and Nutritional Security through Vegetable Crops in Relation to Climate Change,* pp: 212-215.

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News letter

1. Vegetable news letter, vol. 4(1) January-June, 2017, ICAR-IIVR

2. Vegetable news letter, vol. 4(2) July-December, 2017, ICAR-IIVR

Extension Folder

1. Bahadur A, Prasad RN, Chaurasia SNS, Singh Jagdish, Singh DK, Roy S, Yadav RB, Gouda Manjunath, Singh YP and Singh B. *Niyantrit watawaran me sabjiyon ki kheti (In Hindi).* Extension Folder No. 33/2018.
2. Bahadur A, Singh DK, Singh Jagdish, Prasad RN, Yadav RB, Roy S, Singh PK and Singh B. *Sabjiyon me sinchai prabandhan (In Hindi).* Extension Folder No. 30/2018.
3. Bhardwaj DR, Pandey Sudhakar, Gautam Keshav Kumar, Rai AB and Singh Ashok Kumar. *Karela ki vaigyanik kheti (In Hindi).* Extension Folder No. 36/2018.
4. Chaubey T, Baradwaj DR, Singh B, Singh PM, Halder J, Nagendran K, Prasanna HC, Roy S, Gupta N and Goswami A. *Nasdar torai ki vaigyanik kheti (In Hindi).* Extension Folder No. 11/2018.
5. Chaubey T, Bhardwaj DR, Singh B, Singh PM, Halder J, Prasanna HC, Roy S, Meena BR, Nagendran K and Manimurugan C. *Chikani torai ki vaigyanik kheti (In Hindi).* Extension Folder No. 10/2018.
6. Chaudhary R P, Choudhary GK, Chaturvedi AK and Singh PC. *Sarson se adhik paidawar hetu unnat utpadan takniki (In Hindi).* Extension Folder No. 01/2018, KVK, Bhadohi.
7. Chaurasia SNS, Prasad RN, Singh J, Singh DK, Singh SK, Yadav RB, Manjunatha Gowda and Singh YP. *Tamatar me ekikrit poshak tatva prabhandan (In Hindi).* Extension Folder No. 32/2018.
8. Chaurasia Surya Nath, Prasad RN, Bahadur Anant, SM Vanitha, Tripathi PC and Kumari Shweta. *Sabjiyon ki swastha paudh taiaar karna (In Hindi).* Extension Folder No. 26/2018.
9. Dubey Rakesh Kumar, Devi Jyoti, Singh B, Singh PM, Singh Neeraj, Halder Jaydeep, Kumar Yengkhom Bijen and Chandra Subhash. *Sabji matar ki vaigyanik kheti (In Hindi).* Extension Folder No. 21/2018.
10. Gautam KK, Pandey S, Singh PM, Divekar PA and Singh AK. *Tarbuj ki vaigyanik kheti (In Hindi).* Extension Folder No. 08/2018.
11. Karmakar P, Bhardwaj DR, Halder J, Chaurasia A, Singh R, Roy S, Viswanath and Singh B. *Parwal ki vaigyanik kheti (In Hindi).* Extension Folder No. 05/2018.
12. Karmakar P, Bhardwaj DR, Halder J, Chaurasia A, Singh R, Roy S, Viswanath and Singh B. *Kumbhara ki vaigyanik kheti (In Hindi).* Extension Folder No. 06/2018.
13. Karmakar Pradeep, Singh B, Vidyasagar, Halder Jaydeep, Singh PM and Chandra Subhash. *Bhindi ki vaigyanik kheti (In Hindi).* Extension Folder No. 15/2018.
14. Lal H, Reddy BR, Singh PM, Halder J, Kumar YB, Singh N and Vishwanath. *Lobiya ki vaigyanik keti (In Hindi).* Extension Folder No. 20/2018.
15. Pandey S, Karmakar P, Pandey KK, Roy S, Singh YP and Singh B. *Kharbuja ki vaigyanik kheti (In Hindi).* Extension Folder No. 07/2018.
16. Pandey S, Rai AB, Roy S, Singh AK and Tripathi AN. *Kheera ki vaigyanik kheti (In Hindi).* Extension Folder No. 4/2018.
17. Pandey Sudhakar, Singh PM, Singh Neeraj, Pandey KK, Halder Jaydeep and Singh AP. *Petha ki vaigyanik kheti (In Hindi).* Extension Folder No. 35/2018.
18. Rai AB, Tripathi AN, Singh N. Pandey KK, Gupta S and Singh B. *Samekit Madhumakhi palan ek drishtikon (In Hindi).* Extension Folder No. 02/2017.
19. Rai N, Rai AB, Prasanna HC, Kumar R, Halder J, Tripathi AN, Reddy YS, Kurkutke S and Singh A. *Tamatar kee vaigyanik kheti (In Hindi).* Extension Folder No. 12/2018.
20. Rai N, Devi J, Heera lal, Singh BK, Halder J, Reddy R, Divekar PA and Rai RM. *Farash bean (French bean) ki vaigyanik kheti (In Hindi).* Extension Folder No.23/2018.
21. Rai N, Devi J, Rai AB, Singh N and Rai RM. *Sem kee vaigyanik kheti (In Hindi).* Extension Folder No. 24/2018.
22. Sagar V, Chandra R, Verma SK, Bhadur A, Roy S, Singh R, Singh YP, Divekar PA and Tripathi PC. *Ek varshiya sahjan ki unnat kheti (In Hindi).* Extension Folder No. 31/2018.
23. Singh BK, Divekar PA, Gupta S, Chandra S and Singh R. *Phoolghobhi ki vaigyanik kheti (In Hindi).* Extension Folder No. 17/2018.
24. Singh BK, Karmakar P, Divekar PA, Roy S and Gupta S. *Muli ki vaigyanik kheti (In Hindi).* Extension Folder No. 18/2018.
25. Singh BK, Singh B, Divekar P, Meena BR and Gupta S. *Pattaghobhi ki vaigyanik kheti (In Hindi).* Extension

Folder No. 16/2018.

26. Singh BK, Singh B, Halder J and Gupta S. *Gazar ki vaigyanik kheti (In Hindi).* Extension Folder No. 19/2018.
27. Singh Jagdish, Chaurasia Surya Nath Singh, Yadav RB, Singh Neeraj, Singh Rameshwar and Tripathi PC. *Dainik aahaar mein sabjiyon ki upyogita (In Hindi).* Extension Folder No. 28/2018.
28. Singh PM, Kumar Rajesh, Chaubey Tribhuwan, Manimurugan, Gupta Nakul, Singh Rameshwar and Chandrabhushan. *Sabji beej utpadan hetu buwai ke samay ka mahatva (In Hindi).* Extension Folder No. 25/2018.
29. Singh S, Singh Jagdish, Roy S, Singh AK, Singh R, Singh PK and Singh B. *Turai uprant sabjion ka prabandhan (In Hindi).* Extension Folder No. 27/2018.
30. Singh V, Bhardwaj DR, Karmakar P, Tiwari SK, Chaurasia A, Singh R, Roy S, Viswanath and Singh B. *Kundru ki vayganik kheti (In Hindi).* Extension Folder No. 09/2018.
31. Tiwari SK, Kumari S, Kumar R, Divekar PA, Singh N and Goswami A. *Mirch ki vaigyanik kheti (In Hindi).* Extension Folder No.14/2018.
32. Tiwari SK, Kumari S, Kumar R, Divekar PA, Singh N and Goswami A. *Baingan ki vaigyanik kheti (In Hindi).* Extension Folder No.13/2018.
33. Tripathi AN, Rai AB, Pandey KK, Singh N, Gupta S and Singh B. *Madhumakhi palan evam prabandhan (In Hindi).* Extension Folder No. 01/2018.
34. Tripathi AN, Rai AB, Pandey KK, Singh N, Gupta S and Singh B. *Madhumakhiyon ke rogon evam keeton ka surakhit niyantran (In Hindi).* Extension Folder No. 02/2018.
35. Tripathi AN, Rai AB, Pandey KK, Singh N, Gupta S and Singh B. *Madhumakhi palan: labhprad vyavsai (In Hindi).* Extension Folder No. 03/2018.
36. Vidyasagar, Chaurasia Surya Nath, Tiwari SK, Karkute Suhash and Tripathi PC. *Hari pattedar sabjiyon ki vaigyanik kheti (In Hindi).* Extension Folder No. 22/2018.
37. Yadav RB, Singh Jagdish, Chaurasia Surya Nath Singh, Prasad RN, Singh Neeraj, Verma SK and Gupta Sunil. *Sabjiyon ki jaivik kheti (In Hindi).* Extension Folder No. 34/2018.
38. Yadav RB, Singh Raghwendra, Singh SK, Singh Jagdish, Singh Neeraj and Singh Rameshwar. *Mitti ki urvarashakti evam utpaadakta ke liye hari khaad (In Hindi).* Extension Folder No. 29/2018.

Radio Talks: 31

TV Talks: 23

APPOINTMENTS, TRANSFERS AND PROMOTION

Appointments:

- Dr. Vidya Sagar joined the post of Scientist (Genetics & Plant Breeding) on 15.04.2017 at ICAR-IIVR.
- Dr. Rani A.T. joined the post of Scientist (Entomology) on 16.10.2017 at ICAR-IIVR.

Promotions

- Dr. C. Sellaperumal, Ex. Scientist (Nematology) promoted from Rs.15600-39100 + RGP 6000 to 15600-39100 + RGP 7000 *w.e.f.* 27.04.2016.
- Dr. M. Manjunath, Ex. Scientist (Agrilcultural Microbiology) promoted from Rs.15600-39100 + RGP 6000 to 15600-39100 + RGP 7000 *w.e.f.* 20.04.2014.
- Dr. Shubhadeep Roy, Scientist (Agricultural Extension) promoted from Rs.15600-39100 + RGP 6000 to 15600-39100 + RGP 7000 *w.e.f.* 20.04.2014.
- Dr. Y.S. Reddy, Scientist (Genetis & Plant Breeding) promoted from Rs.15600-39100 + RGP 6000 to 15600-39100 + RGP 7000 *w.e.f.* 27.04.2015.
- Sh. Y.P. Singh, Technical Officer promoted to Senior Technical Officer from 9300-34800 + RGP 4600 to Rs.15600-39100 + RGP 5400 *w.e.f.* 24.11.2015.

Transfers

- Dr. C. Sellaperumal, Scientist transferred from ICAR-IIVR, Varanasi to ICAR-IISR, Calicut, Kerala on 07.07.2017 (A/N).
- Dr. J.K. Ranjan, Senior Scientist transferred from ICAR-IIVR, Varanasi to ICAR-IARI, New Delhi on 15.07.2017 (A/N).
- Dr. M.H. Kodandaram Principal Scientist transferred from ICAR-IIVR, Varanasi to ICAR-IIPR, Kanpur on 06.07.2017 (A/N).
- Dr. Major Singh, Project Coordinator, AICRP (VC), ICAR-IIVR, Varanasi has selected as Director, ICAR-Directorate of Onion & Garlic Research, Pune on 12.04.2017 (A/N).
- Dr. Mamta Singh, Scientist transferred from ICAR-IIVR, Varanasi to ICAR-NBPGR, New Delhi on 19.09.2017 (A/N).
- Dr. Pragya, Senior Scientist transferred from ICAR-IIVR, Varanasi to ICAR-NBPGR, New Delhi on 15.07.2017 (A/N).
- Sh. D.K. Agnihotri, AF&AO transferred from ICAR-IIVR, Varanasi to ICAR-IIPR, Kanpur on 10.04.2017 (A/N).

Probation Clearance

- Dr. K. Nagendran Krishnan, Scientist (Plant Pathology) has cleared his probation period and confirmed from 01.01.2017.
- Sh. B. Rajashekar Reddy, Scientist (Vegetable Science) has cleared his probation period and confirmed from 01.01.2017.
- Sh. Keshav Kant Gautam, Scientist (Vegetable Science) has cleared his probation period and confirmed from 01.01.2017.
- Sh. Manjunath Gowda T., Scientist (Nematology) has cleared his probation period and confirmed from 01.01.2017.
- Sh. P.B. Chaukhande, Scientist (Vegetable Science) has cleared his probation period and confirmed from 01.01.2017.

Classified Abstracts of Expenditure
(2017-2018)

Indian Institute of Vegetable Research (plan)

(In Lakhs)

Sub-head	Plan	
	Provision made in RE	Expenditure
Establishment Charges	1134.60	1134.60
Wages	-	-
O.T.A.	-	-
T.A.	14.00	14.00
Other Charges (Contingency)	656.48	656.38
H.R.D.	4.42	4.42
Works	3.00	3.00
Equipment	-	-
Library	-	-
Vehicle	-	-
Annual Repairs / Maintenance	73.69	73.67
Information Technology	-	-
TSP NEH	36.90	36.90
Total	**1923.09**	**1922.97**

Revenue generation

(In Lakhs)

Particulars	Target	Revenue generation
IIVR	-	211.08

Krishi Vigyan Kendra (plan)

(In Lakhs)

KVKs	RE	Expenditure
KVK, Kushinagar	128.00	106.67
KVK, Deoria	115.00	108.00
KVK, Sant Ravidas Nagar	118.07	119.90
Total	**361.07**	**334.57**

Externally Funded Projects

(In Lakhs)

Name of project	Funding agency	Duration of projects	Total allocation	Allocation & Expenditure 2017-18	
				Allocation	Expenditure
Crop Improvement					
Genomics assisted selection of *Solanum chilense* introgression lines for enhancing drought resistance in tomato	DBT	2015-18	132.284	30.5185	21.89014
Introgression of begomo virus resistance genes in tomato (*Solanum lycopersicum* L.) using MAS and genomics approach	DBT	2014-19	75.128	14.39653	12.53249
National Innovation of Climate Resilient Agriculture (NICRA)	ICAR	2011-17	390.00	51.00	49.45
CRP on hybrid Technology (Tomato)	ICAR	2015-18	21.97	7.43	6.39
Network Project on Transgenic Crops (NPTC)	ICAR	2005-18	598.24	4.80	4.48
Evaluation of high yielding varieties/hybrids of cucurbitaceous vegetables for river bed (diara land) cultivation and standardization of their agro-techniques	UPCAR Lucknow	2015-17	24.32	8.43	2.039
CRP on Agrobiodiversity	ICAR	2015-18	30.46144	6.10	4.25
Central Sector Scheme for Protection of Plant Varieties and Farmers' Rights Authority (DUS Testing of tomato, brinjal, okra, cauliflower, cabbage, vegetable pea, French bean, bottle gourd, bitter gourd, pumpkin and cucumber)	PPVFRA	2009-18	-	29.69	25.59
Agri Business Incubator-IIVR, Varanasi	ICAR	2016-17	109.15	22.50	14.11
Zonal Technology Management Unit-IIVR, Varanasi	ICAR	2015-18	-	7.50	7.33
Cowpea Golden Mosaic Disease (CPGMD) Resistance: Agroinfectious clone development, Screening, Genetics of inheritance, Molecular tagging and mapping for CPGMD resistant genes in cowpea by using linked marker	DST-SERB	2017-20	44.71	20.77	8.35
Crop Production					
A total value chain on commercialization of value added convenience vegetable products	UPCAR Lucknow	2014-2017	17.82	1.52	2.97
Efficient water management in horticultural crops under Agri-CRP on water	ICAR	2015-2019	48.35	11.80	7.30541
Farmer FIRST Program on "Intervention of Improved Agricultural Technologies for Livelihood and Nutritional Security Adhering Local Resources and Working Knowledge of the Farmers"	ICAR	2016-2020	177.1	43.7	38.5

NHB Project on "Promotion of Vegetables for Nutritional Security in Eastern Uttar Pradesh including East Champaran, Bihar"	NHB	2015-2018	24.9	7.8	7.7
Scheduled Tribes Component (Earlier Tribal Sub Plan) for Tribal of Sonbhadra district in Uttar Pradesh (National Assignment by Department of Agricultural Research & Education, Ministry of Agriculture and Farmers Welfare, Govt. of India)	ICAR	2013 cont.	121.5 (upto 2018-19)	35.0	34.9
Crop Protection					
Synthesis and validation of sustainable and adoptable IPM technology for cucurbitecious vegetable crops	ICAR-NCIPM	2014-17	4.30	1.25	1.2162
Development and validation of Effective formulation(s) of PGPR having multicide mechanism for pest management in vegetables	UPCAR Lucknow	2014-17	24.89756	14.05206	11.8695
Agro infectious clones development for probing resistance to chilli leaf curl diseases caused by begomoviruses and devising integrated management strategy	DST under Early career Award	2016-19	34.393	15.274	10.064
Establishment of integrated beekeeping development centre (IBDC)/Centre of Excellence (COE) on beekeeping	National Bee Board	2017-20	199.50	99.75	38.24806

Personal (as on 31.03.2018)

S.N.	Category	Sanctioned Strength	Staff in Position	Vacant
SCIENTIFIC				
1.	Scientist	45	38	07
2.	Senior Scientist	12	08	04
3.	Principal Scientist	06	03	03
	TOTAL	**63**	**49**	**14**
TECHNICAL				
1.	Technician	11	10	01
2.	Senior Technician	-	-	-
3.	Technical Assistant	13	08	05
4.	Senior Technical Assistant	02	02	-
5.	Technical Officer	-	-	-
6.	Senior Technical Officer	-	-	-
7.	Assistant Chief Technical Officer	-	-	-
	TOTAL	**26**	**20**	**06**
ADMINISTRATIVE				
1.	Senior Administrative Officer	01	01	-
2.	Finance & Account Officer	01	01	-
3.	Assistant Finance & Accounts Officer	01	-	01
4.	Assistant Administrative Officer	01	01	-
5.	Assistant	05	04	01
6.	Private Secretary	01	01	-
7.	Personal Assistant	02	-	02
8.	Stenographer Gr. III	02	-	02
9.	Upper Divison Clerk (UDC)	02	02	-
10.	Lower Divison Clerk (LDC)	04	-	04
	TOTAL	**20**	**10**	**10**
SKILLED SUPPORTING STAFF (SSS)				
1.	S.S.S	16	16	-
	TOTAL	**16**	**16**	**-**
	Grand Total	**125**	**95**	**30**

Staff strength of Krishi Vigyan Kendras (as on 31.03.2018)

KVK Sargatia, Kushinagar

Sl. No.	Designation	Sanctioned strength	Staff in position	Vacant
1.	Programme Coordinator	01	01	-
2.	Subject Matter Specialist	06	06	-
3.	Farm Manager	01	01	-
4.	Programme Assistant	01	-	01
5.	Programme Assistant (Computer)	01	-	01
6.	Assistant	01	01	-
7.	Stenographer Gr.III	01	-	01
8.	T-1 (Driver)	02	02	-
9.	Skilled Supporting Staff (SSS)	02	-	02
	Total	**16**	**11**	**05**

KVK Deoria

Sl. No.	Designation	Sanctioned strength	Staff in position	Vacant
1.	Programme Coordinator	01	-	01
2.	Subject Matter Specialist	06	06	-
3.	Farm Manager	01	01	-
4.	Programme Assistant	01	01	-
5.	Programme Assistant (Computer)	01	-	01
6.	Assistant	01	-	01
7.	Stenographer Gr.III	01	-	01
8.	T-1 (Driver)	02	02	-
9.	Skilled Supporting Staff (SSS)	02	-	02
	Total	**16**	**10**	**06**

KVK Bhadohi

Sl. No.	Designation	Sanctioned strength	Staff in position	Vacant
1.	Programme Coordinator	01	01	-
2.	Subject Matter Specialist	06	04	02
3.	Farm Manager	01	01	-
4.	Programme Assistant	01	01	-
5.	Programme Assistant (Computer)	01	01	-
6.	Assistant	01	01	-
7.	Stenographer Gr.III	01	-	01
8.	T-1 (Driver)	02	02	-
9.	Skilled Supporting Staff (SSS)	02	-	02
	Total	**16**	**11**	**05**

Staff in position (as on 31.03.2018)

Sl. No.	Name	Designation	Email
1.	Dr. BijendraSingh	Director	directoriivr@gmail.com
Director's Cell			
2.	Sh. Ajayan P.	Private Secretary	ajaynair27@gmail.com
Project Coordinator Cell			
3.	Dr. A.B. Rai	I/C Project Coordinator	abraiiivr@gmail.com
	Dr. S. K Verma	Principal Scientist	skverma10@yahoo.com
4.	Dr. Ram chandra	Principal Scientist	rchandraiivr2016@gmail.com
5.	Dr. T. Chaubey	Principal Scientist	tchaubay@gmail.com
6.	Dr. B. Rajasekhar Reddy	Scientist	rajasekharhortico@gmail.com
7.	Dr. A.P.Singh	Senior Technical Officer	apsinghcuts@gmail.com
Division of Vegetable Improvement			
8.	Dr. P.M. Singh	Principal Scientist & I/C Head	pmiivr@gmail.com
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16.	Dr. Rakesh Kumar Dubey	Senior Scientist	rksdubey@gmail.com
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62.	Sh. Ajayan P.	Private Secretary	ajaynair27@gmail.com
63.	Sh. Gopi Nath	Assistant	gopiiivr@gmail.com
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80.	Sh. Sanjay Singh	Senior Technical Assistant	
81.	Sh. Rajendra Kumar	Senior Technical Assistant	-
82.	Sh. Manoj Kumar	Senior Technical Assistant	
83.	Sh. Ram Ashrey	Senior Technical Assistant	-
Skilled Supporting Staff (SSS)			
84.	Sh. Jagwat Ram	SSS	-
85.	Sh. Shiv Kumar	SSS	-
86.	Sh. Kailash Singh	SSS	-
87.	Sh. S.P. Mishra	SSS	-
88.	Sh. Naraini Singh	SSS	-
89.	Sh. S.K. Pandey	SSS	-
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91.	Sh. Ramraj	SSS	-
92.	Sh. Suresh Kumar Yadav	SSS	-
93.	Sh. Shuresh Kumar	SSS	-
94.	Sh. Virendra Prasad Gond	SSS	-
95.	Sh. Kamlesh Kumar Singh	SSS	-
96.	Sh. Anil Kumar Suman	SSS	-
97.	Sh. Ram Kunwar Chaubey	SSS	-
98.	Sh. Jata Shankar Pandey	SSS	-
99.	Sh. Shivajee Mishra	SSS	-
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Krishi Vigyan Kendra, Sargatia, Kushinagar			
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131.	Sh. Roshan Lal	Office Superintendent	roshaniivr@gmail.com
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134.	Sh. Pramod Paswan	(T-1) Driver	pramodpaswan1084@gmail.com

Annexure I

Research Advisory Committee

Dr. K. E. Lawande Former Vice Chancellor Dr. BSKKV, Dapoli (M.S.)	Chairman
Dr. S.M.S Tomar Cytogenetist Division of Genetics & Plant Breeeding IARI, Pusa, New Delhi - 110012	Member
Dr. Anil Sirohi Professor Division of Nematology IARI, Pusa, New Delhi – 110012	Member
Dr. D.P. Wasker Director (Research) Dr. VNMKVK, Parbhani-431402 (M.S.)	Member
Dr. P.S. Pandey ADG (EP&HS) ICAR, Pusa, New Delhi – 110012	Member
Dr. T. Janakiram Asstt. Director General (Hort.-II) ICAR, Krishi Anusandhan Bhawan-II, Pusa New Delhi-110012	Ex- officio Member
Dr. B. Singh Director ICAR-IIVR, Varanasi-221305	Ex- officio Member
Dr. Sudhakar Pandey Principal Scientist ICAR-IIVR, Varanasi-221305	Member Secretary

Annexure II

Institute Joint Staff Council (IJSC)

Dr.B. Singh	Director	Chairman
Dr. A.B. Rai	Head, Vegetable Protection	Nominated Member, Official Side
Dr.P.M .Singh	I/C ,Vegetable Improvement	Nominated Member, Official Side
Dr. Jagdish Singh	I/C ,Vegetable Production	Nominated Member, Official Side
Sh. S.K. Jindal	S.A.O	Nominated Member, Official Side
Dr. Rajesh Kumar	Senior Scientist & I/C F.A.O	Nominated Member, Official Side
Sh. U.N.Tiwari	A.A.O	Nominated Member, Official Side
Sh. P.C.Tripathi	Technical Officer	Elected Member, Staff Side
Sh. Gopi Nath	Assistant	Elected Member, Staff Side
Sh. Suresh Kr. Yadav	S.S.S	Elected Member, Staff Side
Sh. Rajesh Kumar Rai	Senior Technical Assistant	Elected Member, Staff Side
Sh. S.K.Gupta	U.D.C	Secretary, IJSC

Annexure III

List of Ongoing Research Projects

A. Institutional

MEGA PROGRAMME-1: INTEGRATED GENE MANAGEMENT			
Mega-Programme Leader: P.M. Singh			
Code	**Title of the project**	**P.I.**	**Co-PIs**
1.1	Management of Vegetable Genetic Resources	Shailesh K Tiwari	D.R. Bhardwaj, N Rai, Sudhakar Pandey, Rajesh Kumar, HC Prasanna, T Chaubey, BK Singh, RK Dubey, Vikas Singh, YS Reddy, P Karmakar, Jyoti Devi, KK Gautam, B.R. Reddy, H Lal (up to 31.01.2018), JK Ranjan (up to 15.07.2017), Pragya (up to 15.07.2017), Mamta Singh (up to 19.09.2017), S.K. Verma (w.e.f. 29.06.2017) and Vidyasagar (w.e.f. 25.08.2017)
1.2	Genetic Improvement of Solanaceous Vegetables	B. Singh (Till regular Head CI takes over)	N. Rai, Rajesh Kumar, JK Ranjan (up to 15.07.2017), SK Tiwari and YS Reddy **Associates:** A.B. Rai (Insects-Chilli & Tomato), M.H. Kodandaram (Insects-Brinjal) upto 06.07.2017, K.K. Pandey (Diseases), C. Sellaperumal (Nematodes) upto 07.07.2017, K. Nagendran (Viruses), Shweta Kumari (Phytoplasma)
1.3	Genetic improvement of legume vegetables.	Hira Lal (up to 31.01.2018) / N. Rai (w.e.f. 01.02.2018)	R.K. Dubey, Jyoti Devi, B.R. Reddy and Mamta Singh (up to 19.09.2017) **Associates:** P. Dibekar (Insects), A.N. Tripathi (Diseases), K. Nagendran (Viruses)
1.4	Genetic improvement of gourds.	D.R. Bhardwaj	Sudhakar Pandey, T Chaubey, RK Dubey, Vikas Singh, Pradip Karmakar and KK Gautam. **Associates:** J. Haldar (Insects), M. Gowda (nematodes), B. Meena (Diseases)
1.5	Genetic improvement of melon, pumpkin and cucumber	Sudhakar Pandey	D.R. Bhardwaj, Vikas Singh, Pradip Karmakar, KK Gautam and Mamta Singh (up to 19.09.2017) **Associates:** J. Haldar (Insects), M. Gowda (nematodes), A.N. Tripathi (Diseases), K. Nagendran (Viruses)
1.6	Genetic improvement of okra	Pradip Karmakar	B Singh, Achuit Singh, Jyoti Devi (up to 24.08.2017) and Vidyasagar (w.e.f. 25.08.2017) **Associates:** J Halder (Insects), C. Sellaperumal (Nematodes) upto 07.07.2017, K. Nagendran (Diseases/ Viruses)
1.7	Genetic improvement of cole & root crops	B.K. Singh	P. Karmakar **Associates:** M.H. Kodandaram (Insects) upto 06.07.2017, B. Meena (Diseases)

1.8	Transgenic and regeneration protocols	Achuit K Singh	YS Reddy, SG Karkute Pragya (up to 15.07.2017), and Mamta Singh (up to 19.09.2017).
1.9	Biotechnological interventions for improvement of selected vegetable crops	H.C. Prasanna	SK Tiwari, Sudhakar Pandey, Achuit Singh, Pradip Karmakar, SG Karkute, Jyoti Devi, KK Gautam and Mamta Singh (up to 19.09.2017) **Associates:** K. Nagendran
1.10	Genetic improvement of leafy vegetable	Pragya (up to 15.07.2017)/ B.K. Singh (w.e.f 16.07.2017)	BK Singh, Vidyasagar (w.e.f. 25.08.2017) **Associates:** P. Dibekar (Insects), B. Meena (Diseases)
1.11	Genetic improvement of aquatic vegetables	JK Ranjan (up to 15.07.2017) / R.K. Dubey (w.e.f. 16.07.2017)	**Associates:** P. Dibekar (Insects), B.Meena (Diseases)
1.12	Genetic improvement of baby corn and sweet corn	YS Reddy	JK Ranjan (up to 15.07.2017) **Associates:** P. Dibekar (Insects), B.Meena (Diseases)
MEGAPROGRAMME-2: SEED ENHANCEMENT IN VEGETABLES			
Mega-Programme leader: P.M. Singh			
2.1	Priming, Coating, pollination and ovule conversion studies	P.M. Singh	Rajesh Kumar, T Chaubey, DR Bhardwaj, Sudhakar Pandey, N Rai, SK Tiwari, Manimurugan, C. and Nakul Gupta **Associates:** J. Halder and A.N. Tripathi
2.2	Breeder and TL seed production of important vegetable crops	Rajesh Kumar	P.M. Singh, T Chaubey, Vikas Singh, Manimurugan, C and Nakul Gupta
2.3	Drying and storage studies including modified atmosphere storage	Manimurugan C.	Rajesh Kumar, PM Singh, Sudhir Singh, and Nakul Gupta
MEGAPROGRAMME-3: PRODUCTIVITY ENHANCEMENT THROUGH BETTER RESOURCE MANAGEMENT			
Mega-Programme Leader: R.N Prasad (upto 19.07.2017) / Jagdish Singh (w.e.f. 20.07.2017)			
3.1	Technologies for protected and off season vegetable production.	S.N.S. Chaurasia (upto 12.05.2017)/ Anant Bahadur (w.e.f. 13.05.2017)	R.N. Prasad, Raghwendra Singh (upto 12.05.2017), Sudhir Singh and D.K. Singh **Associates:** M.H. Kodandaram (upto 06.07.2017), K.K. Pandey and Manjunath Gowda T.
3.2	Precision farming in vegetable crops	R.N. Prasad	S.K. Singh
3.3	Devlopment of vegetable based cropping system for sustainability and profitability	R.N. Prasad	S.K. Singh, R.B. Yadav and Vanitha S.M.
3.4	Impact of organic and inorganic management systems on vegetable productivity, quality and soil health.	S.K. Singh	R.N. Prasad, D.K. Singh, Raghwendra Singh (upto 12.05.2017) and Sudhir Singh **Associates:** J. Halder, K K Pandey and C. Sellaperumal (upto 07.07.2017)
3.5	Improving soil health and carbon sequestration in vegetable production system through conservation tillage and residue incorporation.	S.K.Singh	Raghwendra Singh (upto 12.05.2017) and D.K. Singh

3.6	Enhancing water and nutrient use efficiency in vegetable crops	Anant Bahadur	D.K. Singh, S.N.S. Chaurasia (upto 12.05.2017) and R. N. Prasad
3.8	Performance of vegetable crops under subsurface drip irrigation system.	D.K. Singh	Anant Bahadur and SNS Chaurasia (upto 12.05.2017)
MEGA PROGRAMME-4: POST HARVEST MANAGEMENT AND VALUE ADDITION			
Mega-Programme Leader: Sudhir Singh			
4.1	Shelf life extension of vegetables.	Sudhir Singh	-
MEGA PROGRAMME-5: PRIORITIZATION OF R&D NEEDS AND IMPACT ANALYSIS OF TECHNOLOGIES DEVELOPED BY IIVR			
Mega-Programme Leader: Neeraj Singh			
5.1	Research prioritization for vegetable crops.	Shubhadeep Roy	Neeraj Singh
5.2	Impact of technologies developed by IIVR.	Neeraj Singh	Shubhadeep Roy and Vanitha S.M.
MEGA PROGRAMME-6: INTEGRATED PLANT HEALTH MANAGEMENT			
Mega-Programme Leader: A.B. Rai			
6.1	Bio-Intensive: Management of Major Insect Pests of Vegetables in the Current Scenario of Climate Change._	A.B. Rai	M.H. Kodandaram, J. Halder, Neeraj Singh, Y. Bijen Kumar, K. Nagendran and Pratap Divekar
6.2	Toxicological investigations on the novel insecticide molecules and plant origin insecticides against major insect pests of vegetables._	M.H. Kodandaram (up to 06.07.2017) Pratap Divekar (w.e.f 07.07.2017)	A.B. Rai, K.K. Pandey, J. Halder, Y. Bijen and Manjunath Gowda T.
6.3	Biological Control of major Insect Pests of Vegetable crops	Jaydeep Halder	A.B. Rai, M.H. Kodandaram, A.N. Tripathi, Pratap Divekar and Manjunath Gowda T.
6.4	Management of important fungal diseases of vegetable crops	K. K Pandey	A.N. Tripathi B.R. Meena, C. Sellaperumal (upto 07.07.2017), Shweta Kumari and A. Chaurasia (w.e.f. 28.06.2017)
6.5	Bioprospecting of microorganisms associated with vegetables against plant pathogens	A.N. Tripathi	K.K. Pandey, Shweta Kumari and A. Chaurasia (w.e.f. 28.06.2017)
6.6	Management of Important Bacterial Diseases of Vegetable Crop	A.N. Tripathi	K.K. Pandey, Y. Bijen Kumar and B.R. Meena
6.7	Development of diagnostics kits for major viruses infecting vegetable crops	K. Nagendran	K.K. Pandey, A.N. Tripathi, B.R. Meena Achuit Kumar Singh and Shweta Kumari
6.9	Management of Nematodes Infesting Major Vegetable Crops	C. Sellaperumal (up to 07.07.2017) Manjunatha Gowda T. (w.e.f. 08.07.2017)	K.K. Pandey, Jaydeep Halder and Subhadeep Roy
6.10	Dynamics of pest and diseases and development of forecasting models	A.B. Rai	K.K. Pandey, M.H. Kodandaram (up to 06.07.2017), J Halder, B.R. Meena and Pratap Divekar

B. Externally Funded

Division of Crop Improvement			
S.N.	**Title of the project**	**P.I.**	**Co-PIs**
1.	Introgression of begomovirus resistance genes in tomato (*Solanum lycopersicum* L.) using MAS and genomics approach	H. C. Prasanna	Major Singh (upto 11.04.2017)
2.	National Innovation in Climate Resilient Agriculture (NICRA)	P.M. Singh	N. Rai, Anant Bahadur, Suhas Karkute and AB Rai
3.	CRP on hybrid Technology (Tomato)	N. Rai	YS Reddy
4.	Network Project on Transgenic Crops (NPTC)	Achuit Kumar Singh	SG Karkute and K Nagendran
5.	Evaluation of high yielding varieties/hybrids of cucurbitaceous vegetables for river bed (diara land) cultivation and standardization of their agro-techniques	Sudhakar Pandey	SK Singh and Pradip Karmakar
6.	CRP on Agrobiodiversity	Shailesh K Tiwari	P. Karmakar and Vidyasagar (w.e.f. 25.08.2017)
7.	Central Sector Scheme for Protection of Plant Varieties and Farmers' Rights Authority (DUS Testing of tomato, brinjal, okra, cauliflower, cabbage, vegetable pea, french bean, bottle gourd, bitter gourd, pumpkin and cucumber)	B. Singh	T. Chaubey and Sudhakar Pandey
8.	Agri Business Incubator-IIVR, Varanasi	P.M. Singh	SK Tiwari, Shubhdeep Roy, Neeraj Singh and Sudhir Singh
9.	Zonal Technology Management Unit-IIVR, Varanasi	P.M. Singh	SK Tiwari, Shubhdeep Roy, Neeraj Singh and Sudhir Singh
10.	Genomics assisted selection of *Solanum chilense* introgression lines for enhancing drought resistance in tomato	H. C. Prasanna	Major Singh (upto 11.04.2017)
11.	Cowpea Golden Mosaic Disease (CPGMD) Resistance: Agroinfectious clone development, Screening, Genetics of inheritance, Molecular tagging and mapping for CPGMD resistant genes in cowpea by using linked marker	B. R. Reddy	Achuit Kumar Singh and K. Nagendran

Division of Crop Production			
12.	Efficient water management in horticultural crops under Agri-CRP on water.	DK Singh	Anant Bahadur and S.N.S Chaurasia (upto 12.05.2017)
13.	A total value chain on commercialization of value added convenience vegetable products	Sudhir Singh	-
14.	NHB Project on "Promotion of Vegetables for Nutritional Security in Eastern Uttar Pradesh" (2015-18).	Neeraj Singh	P.M. Singh, R.N. Prasad, D.R. Bhardwaj, Subhadeep Roy, Y.P. Singh, B. Singh
15.	Scheduled Tribes Component (Earlier Tribal Sub Plan) for Tribal of Sonbhadra district in Uttar Pradesh (National Assignment by Department of Agricultural Research & Education, Ministry of Agriculture and Farmers Welfare, Govt. of India)	B. Singh	Neeraj Singh, R.N. Prasad, Shubhadeep Roy, S.K. Singh, AK Chaturvedi, R.P Chaudhary and Abhay Kumar Singh
16.	Farmer FIRST Program on "Intervention of Improved Agricultural Technologies for Livelihood and Nutritional Security Adhering Local Resources and Working Knowledge of the Farmers"	Neeraj Singh	A.B. Rai, Jagdish Singh, R.N. Prasad, D.R. Bhardwaj, S.K. Singh, Subhadeep Roy, A.N. Tripathi
Division of Crop Protection			
17.	Synthesis and validation and sustainable and adaptable IPM technologies for cucurbitaceous vegetables.	Jaydeep Halder	C. Sellaperumal (up to 07.07.2017), K. Nagendran, Manjunatha Gowda T. (w.e.f 08.07.2017), A.N. Tripathi and Manoj Kumar Pandey
18.	Development and validation of effective formulation(s) of plant growth promoting rhizobacteria (PGPR) having multicide mechanisms for pest management in vegetables	A.B. Rai	Jaydeep Halder, C. Sellaperumal (up to 07.07.2017), B.R. Meena, Manjunatha Gowda T. and Neeraj Singh
19.	Agro infectious clones development for probing resistance to chilli leaf curl diseases caused by begomoviruses and devising integrated management strategy	K. Nagendran	Rajesh Kumar
20.	Establishment of integrated beekeeping development centre (IBDC)/Centre of Excellence (COE) on beekeeping	A.B. Rai	K.K. Pandey, Neeraj Singh, A.N. Tripathi, Jaydeep Halder, K. Nagendran, Y. Bijen Kumar, Pratap Divekar, B.R. Meena and Manjunatha Gowda T.

Annexure IV

Distinguished Visitors

Dr. Matthew Morell Director General, International Rice Research Institute, Philippines	05th August 2017
Shri S.K. Pattanayak Secretary, Department of Agriculture, Cooperation & Farmers' welfare, Ministry of Agriculture and Farmers' welfare, Govt. of India	13th August 2017
Shri Kedar Nath Singh Member of Legislative Council, Varanasi, Government of Uttar Pradesh	19th August 2017
Shri Surendra Singh Ex-Member of Legislative Assembly, Jaunpur	19th August 2017
Dr. Ramashrit Singh Ex-professor, RPCAU, Pusa, Samastipur	19th August 2017
Shri Virendra Singh Member of Parliament visited KVK, Bhadohi	25th August 2017
Shri Kalraj Mishra Member of Parliament, visited KVK, Kushinagar	29th August 2017
Shri Surya Pratap Shahi Minister of Agriculture & Farmers Welfare, Govt. of Uttar Pradesh visited KVK, Deoria	30th August 2017
Shri Ravinder Kushwaha M.P., Salempur visited KVK, Deoria	30th August 2017
Dr. AK Vyas ADG (HRM), ICAR, New Delhi	12th September 2017
Dr. Ram Sudhar Singh Former HOD (Hindi), U.P. Autonomous College, Varanasi	14th September 2017
Dr. B Rajendra Joint Secretary (Seeds), Ministry of Agriculture & Farmers Welfare, Govt. of India,	29th September 2017
Shri SN Tripathi Additional Secretary & Financial Advisor (DARE)/ICAR, New Delhi	03rd November 2017
Shri A.K. Panda Chief General Manager, NABARD, Lucknow	24th November 2017
Prof. B.K. Tripathi Director, IUCTI, Varanasi	03rd December 2017
Smt. Krishna Raj Minister of State for Agriculture and Farmers Welfare, Govt. of India	09th December 2017
Dr. Kirti Singh Chairman, World Noni Research Foundation, Former Chairman ASRB and Former Vice Chancellor, IGKVV, Raipur; NDUA&T, Faizabad & HPAU, Palampur	09th December 2017
Dr. G Kalloo Former DDG (Hort. Sci.) ICAR & Former Vice Chancellor, JNKVV, Jabalpur	09th December 2017
Dr. A.K. Singh DDG (Hort. Sci.), ICAR, New Delhi	09th December 2017
Dr. K V Peter Former Vice Chancellor, KAU, Kerala	09th December 2017

Prof. Panjab Singh Chancellor, RLB Central Agricultural University, Jhansi and President, NAAS, New Delhi	07^{th} February 2018
Dr. B.K. Paty Director (OSPM), MANAGE, Hyderabad	07^{th} February 2018
Dr. Prithvish Nag Vice-Chancellor, MG Kashi Vidyapeeth, Varanasi	20^{th} February 2018
Shri Radha Mohan Singh Union Minister for agriculture and farmers welfare, during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	23^{rd} February 2018
Shri Surya Pratap Shahi Minister of Agriculture & FW, Government of Uttar Pradesh during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	23^{rd} February 2018
Shri Surendra Narayan Singh MLA, Govt. of U.P. during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	24^{th} February 2018
Shri Anil Rajbhar Minister for Food Processing, Govt. of U.P. during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	25^{th} February 2018
Shri Ravindra Jaiswal MLA, Govt. of U.P. during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	25^{th} February 2018
Shri Chet Narayan Singh MLC, Govt. of U.P. during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	25^{th} February 2018
Shri Kedar Nath Singh MLC, Govt. of U.P. during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	25^{th} February 2018
Smt Aparajita Sonkar Zila Panchayat Adhyaksh, Varanasi during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	25^{th} February 2018
Dr. K.V. Prabhu Chairman, PPVFRA, New Delhi during North Zone Regional Farmers Fair organized at Deen Dayal Upadhyay Trade Center & Museum, Bada Lalpur, Varanasi	23^{rd} February 2018
Shri Surindra Narayan Singh MLA, Govt. of U.P.	17^{th} March 2018

भाकृअनुप
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